

A Text Book Of

CLASSICAL ELECTRODYNAMICS

T. Y. B. Sc. (PHYSICS)

Semester - III

Paper - II

(PH - 332)

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

P. S. TAMBADE

S. D. AGHAV

B. M. LAWARE

V. K. DHAS



NIRALI PRAKASHAN

CLASSICAL ELECTRODYNAMICS

For

T.Y.B.Sc. PHYSICS Paper – II (Semester – III)

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"THE WAY TO EXCELLENCE"

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: Authors

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The present book entitled "**Classical Electrodynamics**" is written as per new revised syllabus prescribed for the IIIrd Semester of T.Y.B.Sc. (Physics) of Pune University from June 2004. As per prescribed syllabus, the course is mainly divided into three main topics. However, we have divided the course into five topics namely Electrostatics, Boundary Value Problems in Electrostatics, Electrostatic Field in Dielectrics, Magnetostatics and Electrodynamics for the sake of convenience. At the end of each topic, there are number of solved problems for understanding the subject. There are also some unsolved problems at the end of each topic.

An attempt has been made to present the subject matter in a simple and lucid manner. Efforts have been made to explain the basic terms and mathematical treatment to classical electrodynamics problems in simple way.

All precautions has been taken to avoid mistakes and misprint in the book. However, it is possible that some mistakes and misprints might have passed unnoticed. Such mistakes and misprint, is brought to our notice will be thankfully acknowledged.

We are very much thankful to **Shri. Dineshbhai Furia, Shri. Jignesh Furia, Shri. M. P. Munde**, and the entire staff of **Nirali Prakashan** for publishing the book in time.

Suggestions to improve the quality of book will be gladly accepted.

1 m = 10³ mm
1 milim = 10⁻³ m

June, 2004.

Authors

SYLLABUS

1. Electrostatics

1.1 Coulomb's law, Gauss law, Electric field, Electrostatic potential.

1.2 Potential Energy of a System of Charges

1.3 Boundary Value Problems in Electrostatics, Solution of Poisson and Laplace equations in Cartesian Co-ordinate System.

1.4 Method of Image Charge :

Examples of :

(1) Point charge near an infinite grounded conducting plane.

(2) Point charge near a grounded conducting sphere.

1.5 Electric Displacement $\vec{D}$, Polarization $\vec{P}$, Electric susceptibility and Dielectric constant, Bound volume and surface charge densities.

1.6 Electric field at the exterior and interior points of dielectric.

Problems.

Magnetostatics :

Ampere's law for force between two current carrying loops. Magnetic vector potential ($\vec{A}$)

Magnetic energy in terms of $\vec{B}$, $\vec{H}$, Magnetization vector $\vec{M}$, Magnetic field at a point inside and outside the sphere placed in a magnetic field, Hysteresis and energy dissipation.

Problems.

Electrodynamics

1. Time Varying Fields : Faraday's law of induction. Generalization of Ampere's law Maxwell's equation in free space in differential and integral form. Wave equation and plane waves in free space.

2. Electromagnetic energy flow and Poynting vector, Polarization of plane wave (linear and circular).

Problems.

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| 1. Electrostatics | 1.1 - 1.33 |
| 2. Boundary Value Problems in Electrostatic Fields | 2.1 - 2.41 |
| 3. Electrostatic Fields in Dielectrics | 3.1 - 3.29 |
| 4. Magnetostatics | 4.1 - 4.48 |
| 5. Electrodynamics | 5.1 - 5.34 |

ELECTROSTATICS

INTRODUCTION

The study of electric charges at rest is known as electrostatics. In the present chapter, we shall study two fundamental laws namely Coulomb's law and Gauss's law. Even though in previous classes, we have studied Coulomb's law and Gauss's law, we shall discuss these laws here in brief. Based on Coulomb's law, the concept of electric field, electric potential will be introduced and applied in cases involving point, line, surface and volume charge distribution. The chapter also includes electric potential energy of a system of charges. Special problems are solved using Coulomb's law and Gauss's law.

1.1 COULOMB'S LAW

On the basis of experimental study, Charles Augustin Coulomb (1736-1806) formulated the fundamental law of electric force between two stationary charged particles which is known as Coulomb's law.

This law states that the force of attraction or repulsion between two point charges at rest is directly proportional to the product of the magnitude of the charges and inversely proportional to the square of distance between them and is along the line joining them.

Thus, if two point charges q_1 and q_2 are separated by a distance r , then force between them is given by,

$$F \propto q_1 q_2 \text{ and } F \propto \frac{1}{r^2}$$

Combining, we get

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\text{or } F = \beta \frac{q_1 q_2}{r^2}$$

where, β is constant called the Coulomb's constant.

The value of β depends upon the system of units we use and the nature of the medium in which two charges are situated. In S.I. units, the unit of charge is coulomb (symbol C), the unit of force is newton (N), the unit of distance is meter (m). If the charges are situated in free space (air or vacuum), then constant β is equal to $\frac{1}{4\pi\epsilon_0}$, i.e. $\beta = \frac{1}{4\pi\epsilon_0}$, where ϵ_0 is called permittivity of free space. Thus Coulomb's law in free space and in S.I. units is expressed as,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

(1.1)

... (1.1)

The value of ϵ_0 can be determined experimentally and it has been found to be,

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N-m}^2$$

$$\text{Hence, } \frac{1}{4\pi\epsilon_0} = 8.98 \times 10^9 \approx 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

If charges are situated in some other medium then Coulomb's law (in S.I. units) is expressed as,

$$F = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2}$$

where, ϵ is permittivity of the medium, in which charges are situated.

The relation between ϵ and ϵ_0 is $\epsilon = k\epsilon_0$, where k is called dielectric constant of the medium.

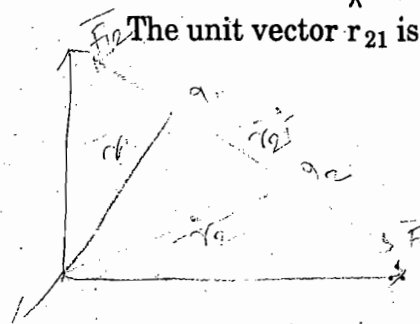
As the force and distance in equation (1.1) are vector quantities, the Coulomb's law in vector form can be expressed as,

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^3} \vec{r}_{21}$$

The force is along the line joining two charge where $\hat{r}$ is unit vector along $\vec{r}$

where, $\vec{F}_{12}$ is the force on charge q_1 due to q_2 and $\vec{r}_{21}$ is unit vector in the direction from charge q_2 to q_1 .

The force acting on charge q_1 due to charge q_2 is $\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^3} \vec{r}_{21}$ where $\vec{r}_{21}$ = unit vector from charge q_2 to q_1



$$\hat{r}_{21} = \frac{\vec{r}_{21}}{|\vec{r}_{21}|} = \frac{\vec{r}_{21}}{r}$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^3} \vec{r}_{21}$$

$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^3} \vec{r}_{21}$ In terms of position vector ... (1.2)

$$\vec{r}_{21} = \vec{r}_1 - \vec{r}_2$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$

Similarly force on charge q_2 due to q_1 is $\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_1}{r_{12}^3} \vec{r}_{12}$ where $\vec{r}_{12}$ = unit vector from charge q_1 to q_2

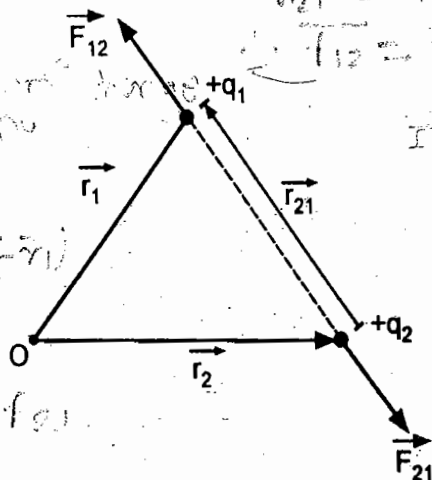


Fig. 1.1 : Electric force between two charges q_1 and q_2

Similarly, force acting on q_2 due to q_1 can be written as,

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_1}{r_{12}^3} \vec{r}_{12} \quad \dots (1.3)$$

If q_1 and q_2 are like charges, force is +ve i.e. repulsive. Nature of force will be negative i.e. attractive for q_1 and q_2 are opposite.

From Fig. 1.1, equations (1.2) and (1.3) can be expressed as,

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$

and
$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_2 - \vec{r}_1|^3} (\vec{r}_2 - \vec{r}_1)$$

Principle of Superposition : Thus, we get
$$\vec{F}_{12} = -\vec{F}_{21}$$

If we have a group of n charges $q_1, q_2, q_3, \dots, q_n$ situated at points having position vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots, \vec{r}_n$ respectively, then total force on charge q_i due to all other charges will be the vector sum of forces exerted on it due to all other charges separately (principle of superposition), i.e.,

$$\vec{F}_i = \frac{q_i}{4\pi\epsilon_0} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_j}{r_{ji}^3} \vec{r}_{ji}$$

Limitations of Coulomb's Law :

- (1) Coulomb's law is applicable in case of point charges at rest. The charged bodies, whose sizes are much smaller than distance between them, are referred as point charges.
- (2) This law holds good over the wide range from 10^{-13} meter to many kilometers. For distances less than 10^{-13} meter, it breaks down due to the predominance of nuclear force.

1.2. ELECTRIC FIELD Intensity (E):- N.B.

We have seen that like charges repel each other while unlike charges attract each other. To explain how the charges exert a force on each other, even though there is no physical contact between them, the concept of electric field or electrostatic field has been introduced by Faraday. According to this concept, an electric charge produces an electric field in its surrounding space. When another charge is brought in this region of space, it experiences an electric force.

Thus, an electric field is said to exist in the region if another charge experiences an electric force in that region.

The strength of electric field at any point in the region is described in terms of quantity called electric intensity or intensity of electric field. It is denoted by $\vec{E}$.

The intensity of an electric field at any point in an electric field is defined as the force acting on a unit positive charge placed at that point.

To determine the electric intensity at a point in an electric field, imagine a small positive test charge q_0 is placed at that point. If $\vec{F}$ is force acting on the test charge, the electric intensity at that point is given by,

$$\vec{E} = \frac{\vec{F}}{q_0} \quad \dots (1.4)$$

Electric intensity is a vector quantity and its direction is the direction of force experienced by positive test charge. Here, a test charge q_0 should be so small that it should not change the original electric field.

Therefore, equation (1.4) in correct form is expressed as,

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$$

The S.I. unit of electric intensity is newton per coulomb (N/C). However, an equivalent unit for the electric intensity is volt/meter.

The magnitude of electric intensity is given by,

$$|\vec{E}| = \frac{|\vec{F}|}{q_0}$$

Calculation of Electric Intensity:

(a) Electric intensity due to a point charge:

Consider a point charge 'q' situated in air at a point A having position vector $\vec{r}'$. The presence of this charge gives rise to an electric field in its surrounding space. Let us find electric intensity at a point P having position vector $\vec{r}$.

To find electric intensity ($\vec{E}$), imagine that a positive test charge q_0 is placed at the point P (Fig. 1.2).

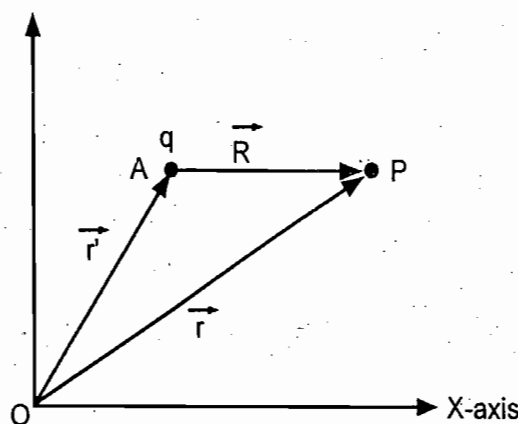


Fig. 1.2 : Electric intensity due to a point charge

Then
The force on charge q_0 is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{|\vec{R}|^2} \hat{R}$$

where, $\hat{R}$ is unit vector along AP.

From Fig. 1.2, we have,

$$\vec{R} = \vec{r} - \vec{r}'$$

and $\hat{R} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|} = \frac{\vec{R}}{R}$

Electric intensity at point P is

$$\vec{E} = \frac{\vec{F}}{q_0}$$

$$\therefore \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') \quad \text{vector form} \quad \dots (1.5)$$

This is the expression for electric intensity at point P, due to charge q . Since the charge q is positive, electric intensity is directed away from the charge. The magnitude of electric intensity at a point P is

$$\text{or } |\vec{E}| = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{r} - \vec{r}'|^2} \quad \text{magnitude form}$$

(b) Electric intensity due to a group of point charges :

If we have 'n' positive charges $q_1, q_2, \dots, q_n$ located at $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n$, then electric intensity at point P situated at $\vec{r}$ from origin O is obtained by using principle of superposition.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{|\vec{r} - \vec{r}_1|^3} (\vec{r} - \vec{r}_1) + \frac{1}{4\pi\epsilon_0} \frac{q_2}{|\vec{r} - \vec{r}_2|^3} (\vec{r} - \vec{r}_2) + \dots + \frac{1}{4\pi\epsilon_0} \frac{q_n}{|\vec{r} - \vec{r}_n|^3} (\vec{r} - \vec{r}_n)$$

or

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i (\vec{r} - \vec{r}_i)}{|\vec{r} - \vec{r}_i|^3}$$

(c) Electric field due to continuous charge distribution :

So far we have studied forces and electric field due to point charges. It is possible to have a continuous distribution along a line, on a surface or in volume as shown in Fig. 1.3 on next page.

It charge is not concentrated to a pt but is distributed over a region then such charge considered as continuous quantity,

In case of q conch charge $+$

distribution, one must define three types of charge densities

(1) Line Point charge Line charge distribution Surface charge distribution Volume charge distribution

Fig. 1.3

When we deal with continuous charge distribution, charge on an object is expressed in terms of charge density, rather than a total charge.

- (1) **Volume charge density (ρ)**: When the charge is distributed continuously in a certain volume, charge on object is expressed in terms of volume charge density (ρ). *volume charge density is defined as charge per unit volume. Then v. charge density ρ is defined as*

If δq is the net charge enclosed by the volume δV situated at r ,

$$\rho(r) = \lim_{\delta V \rightarrow 0} \frac{\delta q}{\delta V} = \frac{q}{V}$$

S.I. unit of ρ is coulomb/(meter)³ (C/m³).

- (2) **Surface charge density (σ)**: When the charge is distributed continuously over a surface, the charge on an object is expressed in terms of surface charge density (σ). *It is defined as charge per unit area. If δq is the net charge on small surface element δS of negligible thickness situated at r , then surface charge density σ is defined as*

$$\sigma(r) = \lim_{\delta S \rightarrow 0} \frac{\delta q}{\delta S} = \frac{q}{S}$$

S.I. unit of σ is coulomb/(meter)² (C/m²).

- (3) **Line charge density (λ)**: If charges are continuously distributed along a length or line, then charge on an object is expressed in terms of line charge density (λ).

It is defined as charge per unit length.

$$\lambda(r) = \lim_{\delta l \rightarrow 0} \frac{\delta q}{\delta l} = \frac{q}{l}$$

where, δl is small length element situated at r and carries a charge δq upon it.

S.I. unit of λ is coulomb/meter.

* If charges are uniformly distributed, ρ is not a function of r and is expressed as,

$$\rho = \lim_{\delta V \rightarrow 0} \frac{\delta q}{\delta V}$$

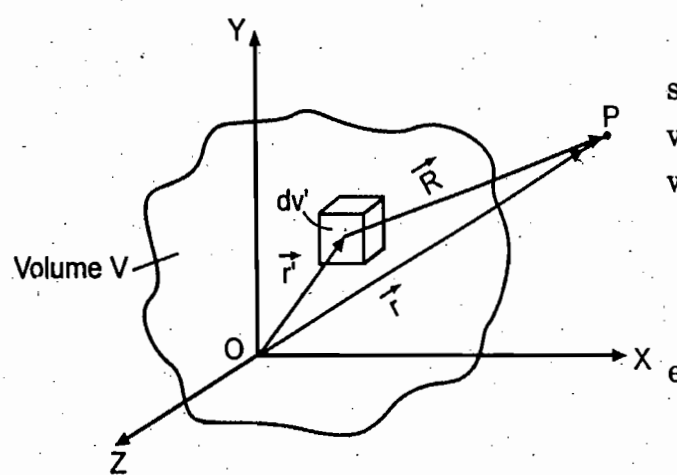
The total charge in a finite region of space having charge distributed on surface and in volume is obtained by integration of charge densities over the region i.e.

$$q = \int_V \rho dV + \int_S \sigma dS$$

In order to calculate the electric intensity or potential at any point, due to a continuous charge distribution, the charge is imagined to be divided into infinitesimal elements dq . Considering any elementary charge dq as a point charge, electric intensity at a given point is determined. Appropriate integration gives the total intensity at that point.

Expression for electric intensity, due to a volume charge distribution:

Consider a certain volume V on which charge q is continuously distributed. Let us find $\vec{E}$, at a point P having position vector $\vec{r}$, due to volume charge distribution.



Consider a small volume element dV' situated at a distance $\vec{r}'$. If $\rho(\vec{r}')$ is the volume charge density, charge associated with dV' is

$$dq = \rho(\vec{r}') dV'$$

Considering dq as a point charge, the electric intensity at point P due to it is

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{|\vec{R}|^3} \vec{R}$$

Fig. 1.4 : Continuous charge distribution

Total electric intensity at a point P , due to this given charge distribution is

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}') dV'}{|\vec{R}|^3} \vec{R} \quad \dots (1.6)$$

or
$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}') dV'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') \quad (\because \text{from Fig. 1.4, } \vec{R} = \vec{r} - \vec{r}')$$

Thus, replacing $\rho(\vec{r}') dV'$ in equation (1.6), by charge element $\sigma(\vec{r}') dS'$ or $\lambda(\vec{r}') dl'$, we get

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') dS'}{|\vec{R}|^3} \vec{R} \quad (\text{for surface charge distribution})$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_l \frac{\lambda(\vec{r}') dl'}{|\vec{R}|^3} \vec{R} \quad (\text{for linear charge distribution})$$

Line of Force - r.B

$$\rho(\vec{r}) = \frac{dq}{dV}$$

1.3 GAUSS'S THEOREM IN ELECTROSTATICS

This theorem gives a relation between the electric flux through any closed hypothetical surface (called a Gaussian surface) and the total charge enclosed by the surface).

Gauss's theorem states that the total electric flux ϕ through any closed surface is equal to $\frac{q}{\epsilon_0}$, where, q is the total charge inside the closed surface.

i.e. $\phi_E = \frac{q}{\epsilon_0}$

It can be stated in two forms:

(a) Integral form,

(b) Differential form.

(a) Integral Form:

Statement: The ele. flux thr' closed surface containing charge q is equal to charge divided by permittivity of free space.

i.e. $\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$ — (1) Integral form of Gauss law

Electric flux passing through small surface element $d\vec{S}$ is

$$d\phi_E = \vec{E} \cdot d\vec{S}$$

Now, total electric flux passing through closed surface is $\oint \vec{E} \cdot d\vec{S}$

If 'q' is the total charge enclosed by the closed surface, then,

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad \dots (1.7)$$

This equation is known as integral form of Gauss's law. If the charge is situated outside the closed surface, the total electric flux through the closed surface is zero.

i.e. $\oint \vec{E} \cdot d\vec{S} = 0$

Differential form of Gauss's Law :

We know that when a charge is distributed continuously in a certain volume, charge on object or body is expressed in terms of volume charge density. If ρ is volume charge density, then the total charge enclosed by surface enclosing the volume V is given by,

$$q = \int_V \rho dV \quad \dots (1.8)$$

Substituting equation (1.8) in equation (1.7), we get

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \int_V \rho dV$$

According to Gauss's divergent theorem, we have

$$\oint_S \vec{E} \cdot d\vec{S} = \int_V \vec{\nabla} \cdot \vec{E} dV$$

$$\therefore \int_V \vec{\nabla} \cdot \vec{E} dV = \frac{1}{\epsilon_0} \int_V \rho dV$$

$$\text{or} \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\text{i.e.,} \quad \text{div } \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\oint \vec{E} \cdot d\vec{v} = \frac{1}{\epsilon_0} \int \rho dV$$

$$\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$$

diff. form of Gauss Law ... (1.9)

This equation represents the differential form of Gauss's law.

Gauss's law is the basic theorem of electrostatics. This law is always true, but is not

always useful. Gauss's law provides an easy method of finding $\vec{E}$ due to a point charge or due to a given charge distribution. The usefulness of Gauss's law depends on our ability to evaluate $\oint \vec{E} \cdot d\vec{S}$. We can easily evaluate $\oint \vec{E} \cdot d\vec{S}$, if the given charge distribution is uniform and symmetric. It means that Gauss's law is useful, if the charge distribution is uniform and symmetric. It must be noted that, if the charge distribution is not symmetric and uniform, the Gauss's law is still obeyed but we cannot use it to determine $\vec{E}$. In such a case, Coulomb's law may be used to determine $\vec{E}$.

1.4 CURL OF $\vec{E}$

Mathematical form of Coulomb's force ($F \propto \frac{q_1 q_2}{r^2}$) is similar to that of gravitational force ($F \propto \frac{m_1 m_2}{r^2}$). That is, both forces are inversely proportional to the square of distance of separation and lie along the line joining the two particles. As gravitational force is conservative, the electric force also must be conservative. A force field $\vec{F}$ is said to be conservative if the line integral of $\vec{F}$ round any closed path is zero i.e.

$$\oint \vec{F} \cdot d\vec{l} = 0$$

Since, the force on a charge q_0 placed in an electric field $\vec{E}$ is

$$\vec{F} = q_0 \vec{E},$$

we can establish the conservative nature of electric field $\vec{E}$ by showing that,

$$\oint \vec{E} \cdot d\vec{l} = 0$$

To prove above equation, consider a charge q situated at origin O . Let us calculate the work done in moving a positive test charge q_0 from point A to B along the line joining A to B as shown in Fig. 1.5, in the field of point charge q situated at origin O . To move test charge q_0 against the direction of electric field $\vec{E}$, external work must be done on it. Let the charge

q_0 be moved from point A (situated at $\vec{r}_A$) to point B (situated at $\vec{r}_B$) under the action of $\vec{F}_{\text{ext}}$ in such a way that at every point along the path $\vec{F}_{\text{ext}} = -\vec{F}_e$ is maintained and no acceleration is imparted to the charge q_0 .

Work done to displace the charge q_0 through $d\vec{l}$ by an external force is

$$dW = \vec{F}_{\text{ext}} \cdot d\vec{l}$$

Therefore, work done to move the charge q_0 from point A to B by an external force is

$$W_{A \rightarrow B} = \int_A^B \vec{F}_{\text{ext}} \cdot d\vec{l}$$

But,
$$\vec{F}_{\text{ext}} = -\vec{F}_e = -q_0 \vec{E}$$

$$\therefore W_{A \rightarrow B} = -q_0 \int_A^B \vec{E} \cdot d\vec{l} \quad \dots (1.10)$$

It must be noted that the displacement $d\vec{l}$ is in the direction of motion of particle.

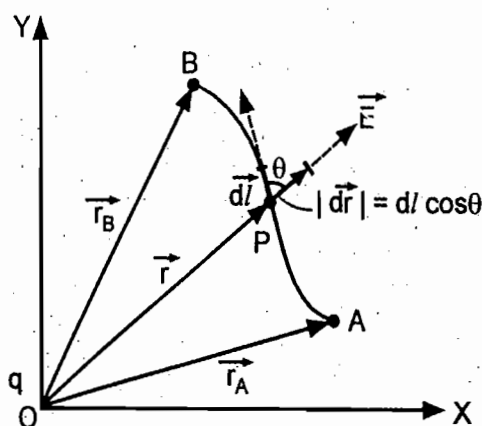


Fig. 1.5

The intensity of electric field at point P due to charge q is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad \dots (1.11)$$

Using equation (1.11) in equation (1.10), we get

$$W_{A \rightarrow B} = -q_0 \int_A^B \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \cdot d\vec{l}$$

But $\hat{r} \cdot d\vec{l} = dl \cos \theta = dr$,

$$\therefore W_{A \rightarrow B} = -\frac{qq_0}{4\pi\epsilon_0} \int_A^B \frac{dr}{r^2}$$

$$\text{or } W_{A \rightarrow B} = \frac{qq_0}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{r_A}^{r_B}$$

$$W_{A \rightarrow B} = \frac{qq_0}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right] \quad \dots (1.12)$$

By observing above equation, it is clear that work done depends only on the starting point A and the end point B and not on the path joining the point A and B.

Similarly, it can be shown that work done in bringing a charge q_0 from B to A is

$$W_{B \rightarrow A} = \frac{qq_0}{4\pi\epsilon_0} \left[\frac{1}{r_A} - \frac{1}{r_B} \right]$$

Thus, over closed path A to B and back to A, the total work done in the field of point charge q is

$$\oint dW = W_{A \rightarrow B} + W_{B \rightarrow A} = 0$$

$$\therefore \oint \vec{E} \cdot d\vec{l} = 0 \quad \dots (1.13)$$

In vector calculus, the line integral of a vector round a closed path is called circulation or the curl of the field. Thus, equation (1.13) represents that the circulation or curl of the electric field is zero i.e.,

$$\vec{\nabla} \times \vec{E} = 0 \quad \dots (1.14)$$

1.5 ELECTRIC POTENTIAL

We know that, any vector, whose curl is zero, is always equal to gradient of some scalar. Therefore, the electric field can be described not only by an electric field intensity $\vec{E}$ (vector quantity), but also by a scalar quantity, called as the electric potential ϕ .

The electric potential at any point in an electric field is defined as the work done in moving a unit positive charge from infinity to that point against the direction of electric field. Therefore, if W is work done in bringing a positive test charge q_0 from infinity to a point then potential at that point is

$$V = \frac{W}{q_0}, \quad \phi = \frac{W}{q_0}$$

If W is in joule, q_0 is in coulomb, then potential ϕ is in volt $\left(1 \text{ volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}} \right)$.

We have seen that work done to move the charge q_0 from point A to B is

$$W_{A \rightarrow B} = \frac{qq_0}{4\pi\epsilon_0} \left(\frac{1}{r_B} - \frac{1}{r_A} \right) \quad [\text{See equation (1.12)}]$$

Now work done to move the charge q_0 from infinity ($r_A = \infty$) to point B is

$$W = \frac{qq_0}{4\pi\epsilon_0} \left(\frac{1}{r_B} \right)$$

Work done per unit charge is potential. So electric potential at point B is

$$\phi_B = \frac{W}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r_B}$$

Thus, assuming that potential energy at infinity is zero, we see that potential at any point is a function of position only. So electric potential of the charge q at point P distant r from it can be expressed as,

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad \dots (1.15)$$

The electric potential at any point due to a group of point charges is obtained by finding potentials due to the individual charges and then adding them algebraically.

If we have a set of point charges $q_1, q_2, \dots, q_n$ situated at distances $r_1, r_2, \dots, r_n$ respectively from origin O, then potential at a point distant r_i from origin (Fig. 1.6) is

$$\begin{aligned} \phi(r_i) &= \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{1i}} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_{2i}} + \dots + \frac{1}{4\pi\epsilon_0} \frac{q_n}{r_{ni}} \\ \phi(r_i) &= \frac{1}{4\pi\epsilon_0} \sum_{j=1}^n \frac{q_j}{r_{ji}} \quad \dots (1.16) \end{aligned}$$

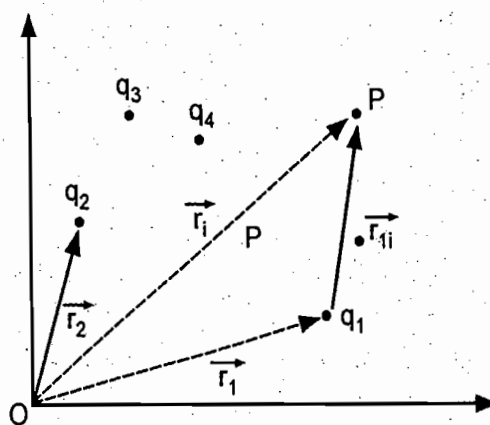


Fig. 1.6 : Potential due to a group of point charges

If there is a continuous distribution of finite quantity of charge in a finite volume V , then to calculate potential at any point due to a continuous charge distribution, the charge is imagined to be divided into infinitesimal elements dq . Considering any elementary charge dq as a point charge, potential at a given point is determined. Appropriate integration gives the total at that point.

The potential at a point due to charge dq situated at $\vec{R}$ from charge is

$$d\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{dq}{|\vec{R}|}$$

Potential due to continuous distribution of ch

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}') dV'}{|\vec{R}|} \quad (\text{for volume charge distribution})$$

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') dS'}{|\vec{R}|} \quad (\text{for surface charge distribution})$$

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_l \frac{\lambda(\vec{r}') dl'}{|\vec{R}|} \quad (\text{for linear charge distribution})$$

Potential difference: If W_A is the work done by an external force in moving a charge q_0 from infinity to point A, then potential at a point A is

$$V_A = \frac{W_A}{q_0}$$

Similarly, if W_B is the work done by an external force in moving a charge q_0 from infinity to point B, then potential at a point B is

$$V_B = \frac{W_B}{q_0}$$

$$V_B - V_A = \frac{W_B - W_A}{q_0} = \frac{W_{AB}}{q_0}$$

where, W_{AB} is the work done for moving a charge from point A to B.

Thus, the potential difference between two points is numerically equal to work done in moving a unit positive test charge from one point to the other.

Thus, from equation (1.12), we get

$$V_B - V_A = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_B} - \frac{1}{r_A} \right)$$

1.6 POTENTIAL ENERGY OF A CHARGE

We have seen that electric field exists at all the points around a charge q and every point in the space can be characterised with scalar quantity called the electric potential. The electric potential due to charge q at a point distant r from it is given by,

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Electric potential at a point is the work done in bringing a unit positive charge from infinity to that point. Instead of bringing a unit charge, if we bring a charge q' from infinity to that point, work done in bringing it is given by,

$$W = q' \times V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q'q}{r}$$

Ex: po a unit work done in bringing a unit charge from infinity to a point.

This work done is termed as the potential energy of the charge q' in the field of charge q .

- (i) If q and q' are of same nature, they are brought together against the forces of repulsion and so potential energy of q' is positive.
- (ii) If q and q' are of opposite nature, the charges move under the effect of attractive forces and potential energy is then negative.

From the above discussion, we can also say that electric potential at any point in an electric field is the potential energy of unit positive charge placed at that point.

1.7 ELECTRICAL POTENTIAL ENERGY OF A SYSTEM

The total work done in making an assembly of charges by bringing them from infinity to their respective places is known as the electrical potential energy of a system.

To obtain an expression for potential energy of a system of charges, let us consider a simple system of two charges q_1 and q_2 separated by a distance r_{12} as shown in Fig. 1.7.

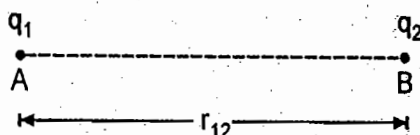


Fig. 1.7 : Potential energy of a system of two charges

Let the charges q_1 and q_2 be brought from infinity to point A and B respectively one by one. When first charge q_1 is brought to point A, there was no electric field in the region and so no work has been done in bringing it to point A. After the charge q_1 had been placed at point A, an electric field is developed in the region. Now, the potential at point B, due to the charge q_1 is $\phi = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{12}}$. Let q_1 be in space & charge q_2 be at infinity brought to distance r_{12} from q_1 , then potential energy of charge q_2 is ϕ & it is work done in bringing unit +ve charge from infinity to that pt.

According to definition of potential, this work is required to be done in order to bring a unit positive charge from infinity to point B. Therefore, the work done in bringing charge q_2 from infinity to point B will be,

$$W_{12} = q_2 \phi = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

This is work done in bringing charge q_2 in the field of q_1 . Thus, the electric potential energy U of the system of two charges q_1 and q_2 is

$$U = W_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Now, suppose a third charge q_3 is brought from infinity to a point C, which is at a distance r_{13} from charge q_1 and r_{23} from charge q_2 . The work required is

$$W = W_{13} + W_{23} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}}$$

Hence, the total potential energy of the configuration of three charges is

$$U = W_{12} + W_{13} + W_{23}$$

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$

In general, if we have N different charges in any arrangement in space, the total number of possible pairs for computing potential energy of the system are –

(12, 13, 14, $1N$), (23, 24, 25, $2N$), (34, 35, $3N$) and so on.

Thus, a general expression for the potential energy of N different charges can be written as,

$$U = \frac{1}{2} \left(\frac{1}{4\pi\epsilon_0} \right) \sum_{i=1}^n \sum_{j \neq i} \frac{q_i q_j}{r_{ij}} \quad \dots (1.17)$$

The factor $\frac{1}{2}$ included in above expression is due to the reason that the combinations $q_i q_j$ and $q_j q_i$ are to be counted as one.

From above expression it is clear that the potential energy of a system of charges, assembled together depends only upon the magnitude of charges constituting the system and their positions with respect to each other.

ILLUSTRATIVE EXAMPLES

Example 1 : Two spheres of charges $+20$ and 80 coulomb are placed 18 cm apart. Find the position of the point between them where the intensity is zero.

Solution : Fig. 1.8 shows two charges $+20$ C and $+80$ C separated by distance 18 cm.

Let P be a point, at a distance x from $+20$ C in between the two charges, where the electric intensity is zero.

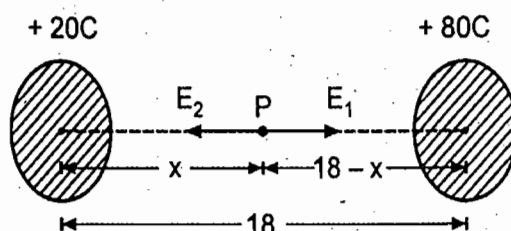


Fig. 1.8

The magnitude of $\vec{E}$ is given by,

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Therefore, the electric intensity at point P due to charge $+20$ C is

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{20}{x^2}$$

E_1 is directed from left to right as shown in Fig. 1.8.

The electric intensity at point P due to charge 80 C is

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{80}{(18-x)^2}$$

E_2 is directed from right to left as shown in Fig. 1.8.

For electric intensity to be zero at point P,

$$E_1 = E_2$$

$$\therefore \frac{1}{4\pi\epsilon_0} \frac{20}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{80}{(18-x)^2}$$

$$\text{or} \quad \frac{1}{x^2} = \frac{4}{(18-x)^2}$$

Taking square root of both sides, we get

$$\frac{1}{x} = \frac{2}{18-x}$$

$$\text{or} \quad 18-x = 2x$$

$$\therefore x = 6 \text{ cm}$$

Thus, distance of point P, where the electric intensity is zero, is at a distance 6 cm from charge +20 C.

Example 2 : Charges of 12 nanocoulomb, -12 nanocoulomb and 24 nanocoulomb are placed at the three corners P, R and S of a square of side 4 cm as shown in Fig. 1.9. Find the electric intensity at the fourth corner Q.

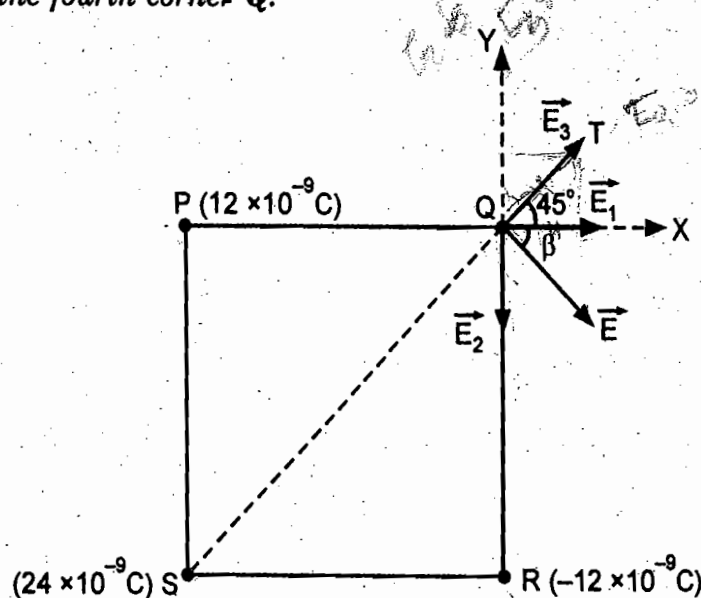


Fig. 1.9

Solution : Let E_1 , E_2 and E_3 be electric intensities produced at point Q by charges situated at P, R and S respectively.

The magnitude of electric intensity at a point due to charge q at a distance d from it is

$$|\vec{E}| = \frac{1}{4\pi\epsilon_0} \frac{q}{d^2}$$

From Fig. 1.9, we get

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{12 \times 10^{-9}}{(0.04)^2}$$

$$\text{or } E_1 = \frac{9 \times 10^9 \times 12 \times 10^{-9}}{(0.04)^2} = 6.75 \times 10^4 \text{ N/C directed along QX.}$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{-12 \times 10^{-9}}{(0.04)^2}$$

$$\text{or } E_2 = \frac{9 \times 10^9 \times 12 \times 10^{-9}}{(0.04)^2} = 6.75 \times 10^4 \text{ N/C directed along QR}$$

$$\text{and } E_3 = \frac{1}{4\pi\epsilon_0} \frac{24 \times 10^{-9}}{32 \times 10^{-4}}$$

$$\text{or } E_3 = \frac{9 \times 10^9 \times 24 \times 10^{-9}}{32 \times 10^{-4}} = 6.75 \times 10^4 \text{ directed along QT}$$

The component of E_3 along QX is

$$E_3 \cos 45 = 4.77 \times 10^4 \text{ N/C}$$

Total electric intensity along QX = $E_x = E_1 + E_3 \cos 45$

$$= 6.75 \times 10^4 + 4.77 \times 10^4$$

$$= 11.52 \times 10^4 \text{ N/C}$$

The component of E_3 along QY is

$$E_3 \sin 45 = 4.77 \times 10^4 \text{ N/C}$$

Total electric intensity along QR = $E_y = E_2 - E_3 \sin 45$

$$= 6.75 \times 10^4 - 4.77 \times 10^4$$

$$E_y = 1.98 \times 10^4 \text{ N/C}$$

The resultant intensity at point Q is

$$E = \sqrt{E_x^2 + E_y^2}$$

$$E = \sqrt{(11.52 \times 10^4)^2 + (1.98 \times 10^4)^2}$$

$$= \sqrt{136.63} \times 10^4$$

$$E = 11.68 \times 10^4 \text{ N/C}$$

This is resultant electric intensity at point Q, due to the three charges as shown in Fig. 1.9.

The angle made by resultant electric intensity with X-axis is

$$\tan \beta = \frac{E_y}{E_x} = \frac{1.98 \times 10^4}{11.52 \times 10^4}$$

$$\tan \beta = 0.171$$

$$\beta = 9.70^\circ$$

$$\sqrt{(6.75 \times 10^4)^2 + (1.98 \times 10^4)^2}$$

$$\frac{216}{3.2 \times 10^4}$$

The resultant intensity at pt Q

$$E = \sqrt{E_x^2 + E_y^2}$$

$$E_x = \text{Ele. intensity along QX}$$

$$= E_1 + E_3 \cos 45$$

$$= 6.75 \times 10^4 + (6.75 \times 10^4) \cos 45$$

$$= 6.75 + 4.77 \times 10^4$$

$$= 11.52 \times 10^4 \text{ N/C}$$

$$E_y = \text{Ele. intensity along QY}$$

$$= E_2 - E_3 \sin 45$$

$$= 6.75 \times 10^4 - [6.75 \times 10^4] \sin 45$$

$$= 6.75 - 4.77 \times 10^4$$

$$= 1.98 \times 10^4$$

$$E = \sqrt{(11.52)^2 + (1.98)^2} \times 10^4$$

$$= \sqrt{132.7104 + 3.9204} \times 10^4$$

$$= \sqrt{136.63} \times 10^4$$

$$E = 11.6889$$

$$\tan \beta = \frac{E_y}{E_x}$$

$$= \frac{1.98 \times 10^4}{11.52 \times 10^4}$$

$$\tan \beta = 0.171$$

$$\beta = \tan^{-1}(0.171)$$

$$\beta = 9.70^\circ$$

Example 3 : Find the potential at the centre of square of side 1.0 meter and having charges q , $2q$, $-3q$, $4q$ at its corners as shown in Fig. 1.10. ($q = 1.0 \times 10^{-9}$ coulomb)

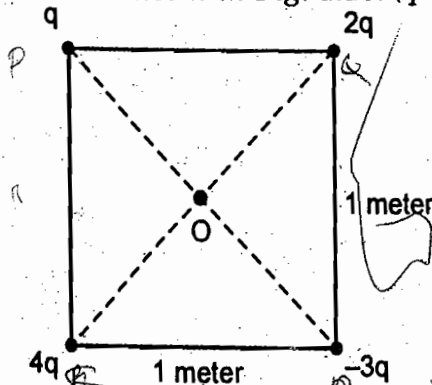


Fig. 1.10

Solution : The potential at a point O due to charge q at a distance ' r ' from it is given by,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

The potential at point O due to all the four charges is

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} \frac{q}{r} + \frac{1}{4\pi\epsilon_0} \frac{2q}{r} - \frac{1}{4\pi\epsilon_0} \frac{3q}{r} + \frac{1}{4\pi\epsilon_0} \frac{4q}{r} \\ &= \frac{q}{4\pi\epsilon_0 r} [1 + 2 - 3 + 4] \\ &= \frac{4q}{4\pi\epsilon_0 r} \\ &= \frac{9 \times 10^9 \times 4 \times 1.0 \times 10^{-9}}{0.71} \\ &= 50.70 \text{ volt} \end{aligned}$$

Example 4 : Three charges q , $-2q$ and $4q$ are placed at the corners of an equilateral triangle having length of each side 1.0 meter. Compute the potential energy of the structure.

(Assume, $q = 1 \times 10^{-6}$ coulomb)

Solution :

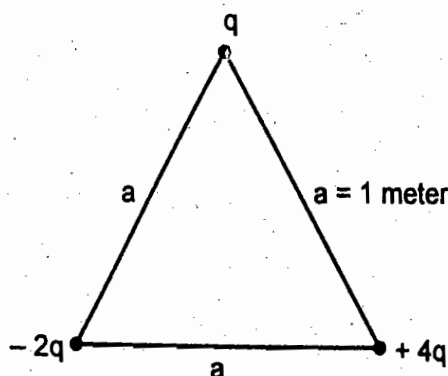


Fig. 1.11

The energy of a pair of charges q_1 and q_2 separated by distance r_{12} is given by,

$$\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

The total energy of the system of three charges is the algebraic sum of the energies of all pairs of charges.

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q(-2q)}{a} + \frac{q(4q)}{a} + \frac{(-2q)(4q)}{a} \right]$$

$$U = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} (-2 + 4 - 8)$$

But,

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

$$q = 1 \times 10^{-6} \text{ C}$$

$$a = 1.0 \text{ meter}$$

$$U = - \frac{9 \times 10^9 \times (1 \times 10^{-6})^2 \times 6}{1}$$

$$U = -54 \times 10^{-3} \text{ joule}$$

Example 5 : Find the work required to put the four charges together as shown in the following Fig. 1.12.

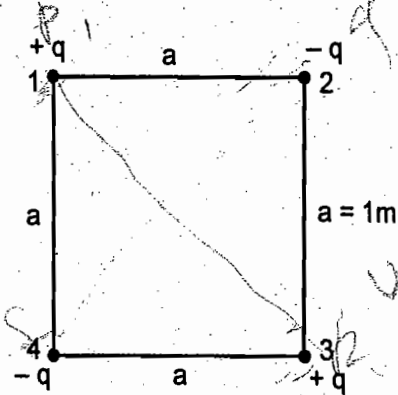


Fig. 1.12

(Assume $q = 1 \times 10^{-6}$ coulomb)

Solution : Work required is same as potential energy of the system.

Potential energy of a system of four charges q_1, q_2, q_3, q_4 is given by,

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \frac{q_3 q_4}{r_{34}} \right]$$

Potential energy of a given configuration is

$$\therefore U = \frac{1}{4\pi\epsilon_0} \left[\frac{(q)(-q)}{a} + \frac{qq}{a\sqrt{2}} + \frac{q(-q)}{a} + \frac{(-q)(+q)}{a} + \frac{(-q)(-q)}{a\sqrt{2}} + \frac{(q)(-q)}{a} \right]$$

$$\text{or } U = \frac{q^2}{4\pi\epsilon_0} \left[-1 + \frac{1}{\sqrt{2}} - 1 - 1 + \frac{1}{\sqrt{2}} - 1 \right] \frac{1}{a}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-4 + \frac{2}{\sqrt{2}} \right)$$

But, $q = 1 \times 10^{-6}$ coulomb, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$, $a = 1.0$ meter

$$U = \frac{9 \times 10^9 \times (1 \times 10^{-6})^2}{1} (-4 + \sqrt{2})$$

$$= -23.27 \times 10^{-3} \text{ joule}$$

The negative sign indicates that negative work would have to be done to assemble this system starting with the charges separated and at rest at infinity. In other words, -23.27×10^{-3} joule amount of work is to be done to dismantle the system removing the charges to an infinite separation from each other.

Example 6 : Find the magnitude and direction of an electric intensity at a point distant z above the mid-point between two equal charges '+q', a distant d apart (Fig. 1.13). Comment on the result when $z \gg d$.

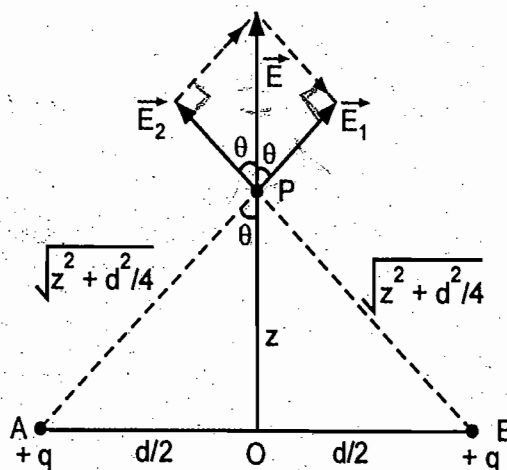


Fig. 1.13

Solution : The magnitude of electric intensity $|\vec{E}|$ at point P due to charge +q situated at point A is

$$|\vec{E}_1| = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(\sqrt{z^2 + \frac{d^2}{4}}\right)^2}$$

or

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z^2 + \frac{d^2}{4}\right)}$$

The component of E_1 along z-axis is $E_{1z} = E_1 \cos \theta$

$$\therefore E_{1z} = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z^2 + \frac{d^2}{4}\right)} \cos \theta$$

The magnitude of electric intensity $|\vec{E}_2|$ at point P due to charge +q situated at point B is

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z^2 + \frac{d^2}{4}\right)}$$

The component of E_2 along z-axis is

$$E_{2z} = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z^2 + \frac{d^2}{4}\right)} \cos \theta$$

The resultant intensity at point P due to charges $+q$ situated at A and $+q$ at B is

$$E = E_{1z} + E_{2z} = \frac{2q \cos \theta}{4\pi\epsilon_0 \left(z^2 + \frac{d^2}{4} \right)}$$

($\because$ The component perpendicular to z-axis cancels out)

But, from Fig. 1.13,

$$\cos \theta = \frac{z}{\sqrt{z^2 + \frac{d^2}{4}}}$$

$$\therefore E = \frac{2qz}{4\pi\epsilon_0 \left(z^2 + \frac{d^2}{4} \right)^{3/2}}$$

The resultant intensity at P is directed along the positive direction of z-axis. If $z \gg d$, then electric intensity at P is given by,

$$E \approx \frac{2qz}{4\pi\epsilon_0 z^3}$$

or

$$E \approx \frac{1}{4\pi\epsilon_0} \frac{2q}{z^2}$$

Thus, when $z \gg d$, the charges appear to be concentrated at centre O.

Note : If point charge $+q$ is situated at A and $-q$ at B in Fig. 1.14, then it can be shown that electric intensity at point P is

$$E = \frac{1}{4\pi\epsilon_0} \frac{2q \frac{d}{2}}{\left(\frac{d^2}{4} + z^2 \right)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{qd}{\left(\frac{d^2}{4} + z^2 \right)^{3/2}}$$

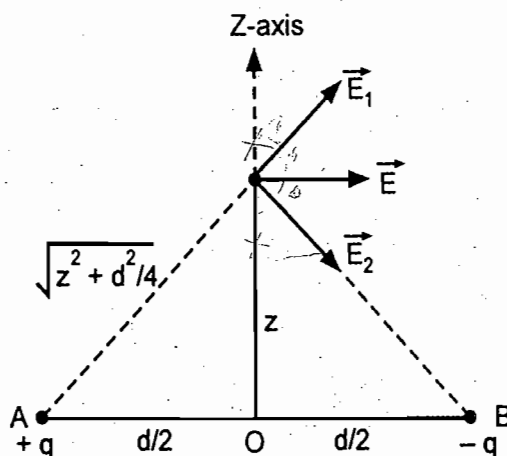


Fig. 1.14

Example 7 : Find the electric field at a distance d above the midpoint of a straight wire of length $2L$, which carries a uniform line charge λ .

Solution : Consider a straight charged wire of length $2L$ having uniform charge density λ . We have to calculate the electric intensity at a point P, distant d from the midpoint of straight wire as shown in Fig. 1.15. For convenience, let us take the x-axis along the wire or line of charge and let the y-axis passes through point P.

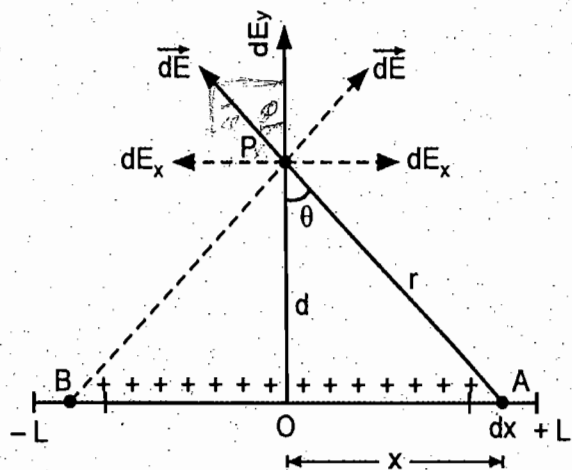


Fig. 1.15

Consider a small element of uniformly charged wire at point 'A', situated at a distance r from the point P. Let the length of small element be dx and it is at a distance x from the point O.

Since λ is the line charge per unit length, the element dx carries a charge $dq = \lambda dx$. The charge dq sets up electric field at a point P. Treating the small element as a point charge, the magnitude of electric intensity dE at P due to this charge is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

But $dq = \lambda dx$ and from Fig. 1.15,

$$r^2 = x^2 + d^2$$

$$\therefore dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{x^2 + d^2} \quad \dots (1.18)$$

The direction of $d\vec{E}$ is along the line joining the element (at A) with the point P, pointing outward as shown in Fig. 1.15. Now the vector $d\vec{E}$ can be resolved into two components $d\vec{E}_x$ and $d\vec{E}_y$.

The magnitudes of $d\vec{E}_x$ and $d\vec{E}_y$ are,

$$dE_y = dE \cos \theta \text{ and } dE_x = dE \sin \theta$$

If we consider an element dx of wire at point B at the same distance x but on other side of O, it will also produce $d\vec{E}$ at point P. This $d\vec{E}$ can also be resolved into two components,

$\vec{dE}_x$ and $\vec{dE}_y$. From Fig. 1.15, it is clear that the component dE_x due to the two elements are oppositely directed and cancel each other, while the component dE_y is in the same direction and hence these are added up. This happens for all such pairs of elements. The resultant intensity at a point P is therefore given by,

$$\vec{E} = \int d\vec{E}_y$$

The magnitude of $\vec{E}$ is

$$E = 2 \int_0^L dE \cos \theta \quad \dots (1.19)$$

From Fig. 1.15,

$$\cos \theta = \frac{d}{\sqrt{x^2 + d^2}} \quad \dots (1.20)$$

Using equations (1.18) and (1.20) in equation (1.19), we get

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} 2\lambda d \int_{x=0}^{x=L} \frac{dx}{(x^2 + d^2)^{3/2}} \quad \dots (1.21) \\ &= \frac{2\lambda d}{4\pi\epsilon_0} \left[\frac{x}{d^2 \sqrt{x^2 + d^2}} \right]_0^L \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{d \sqrt{L^2 + d^2}} \end{aligned}$$

This is the expression for electric intensity at a point P, distant d from the uniformly charged wire of length $2L$. The direction of $\vec{E}$ is radially outward.

Case (i) : Electric intensity at a large distance from the centre of wire is

For $d \gg L$,

$$E \approx \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{d^2}$$

Thus, from above equation, from far away, the uniformly charged wire looks like a point charge $q = 2\lambda L$. So the field reduces to that of point charge $\frac{q}{4\pi\epsilon_0 d^2}$.

Case (ii) : If the wire is infinitely long i.e. $L \rightarrow \infty$, equation may be written as,

$$E = \frac{1}{4\pi\epsilon_0} 2\lambda d \int_0^\infty \frac{dx}{(x^2 + d^2)^{3/2}}$$

From Fig. 1.15, we have

$$x = d \tan \theta$$

$$dx = d \sec^2 \theta d\theta$$

When $x \rightarrow 0$, then $\theta \rightarrow 0$ and when $x \rightarrow \infty$, then $\theta \rightarrow \frac{\pi}{2}$

$$E = \frac{2\lambda d}{4\pi\epsilon_0} \int_0^{\pi/2} \frac{d \sec^2 \theta d\theta}{(d^2 \tan^2 \theta + d^2)^{3/2}}$$

$$= \frac{2\lambda d}{4\pi\epsilon_0} \int_0^{\pi/2} \frac{d \sec^2 \theta d\theta}{d^3 \sec^3 \theta}$$

$$= \frac{2\lambda}{4\pi\epsilon_0 d} \int_0^{\pi/2} \cos \theta d\theta$$

$$= \frac{2\lambda}{4\pi\epsilon_0 d} [\sin \theta]_0^{\pi/2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{d}$$

This is the expression for electric intensity at point P, which is at a perpendicular distance d from the infinitely long uniformly charged wire.

Example 8 : A square of side ' d ' as shown in Fig. 1.16 carries a surface charge density with its boundaries : $\sigma(x, y) = \sigma_0 xy$

Calculate the total charge on the square.

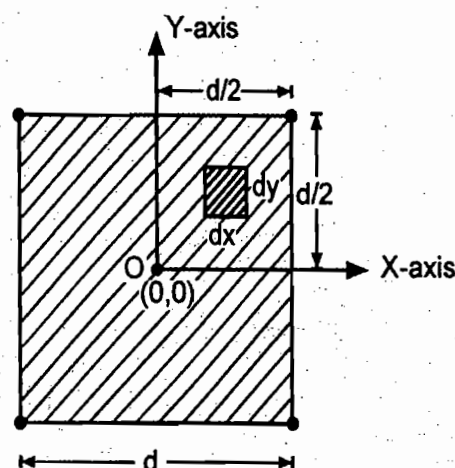


Fig. 1.16

Solution : Consider a small surface element having length dx and breadth dy as shown in Fig. 1.16.

Surface area of small surface element = $dx dy$

Surface charge density, $\sigma = \frac{dq}{ds}$

$$\therefore dq = \sigma ds = \sigma_0 xy ds$$

Integrating, we get

$$q = \int_{-d/2}^{+d/2} \int_{-d/2}^{+d/2} \sigma_0 xy dx dy$$

$$q = \sigma_0 \int_{-d/2}^{+d/2} x \left[\frac{y^2}{2} \right]_{-d/2}^{+d/2} dx$$

$$q = \sigma_0 \int_{-d/2}^{+d/2} x \left[\frac{d^2}{8} - \frac{d^2}{8} \right] dx$$

$$q = 0$$

Thus, the total charge on the square = 0.

Example 9 : A circular ring of radius r (Fig. 1.17) has a line charge density $\lambda = \lambda_0 \cos^2 \theta$ along its circumference. Calculate the total charge on the ring.

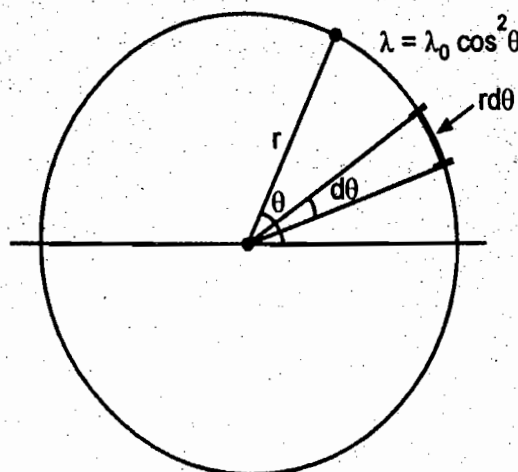


Fig. 1.17

Solution : Consider a small length element $dl = r d\theta$, of the ring.

Charge on small length element is

$$dq = \lambda r d\theta$$

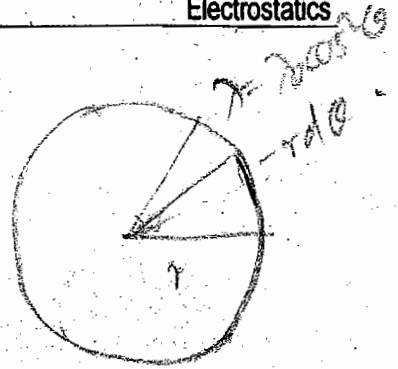
But, $\lambda = \lambda_0 \cos^2 \theta$

$\therefore$

$$dq = \lambda_0 \cos^2 \theta r d\theta$$

Integrating, we get

$$\begin{aligned} q &= \lambda_0 r \int_0^{2\pi} \cos^2 \theta d\theta \\ &= \lambda_0 r \int_0^{2\pi} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \end{aligned}$$



Handwritten notes:

$$\begin{aligned} dl &= r d\theta \\ dq &= \lambda dl \\ &= \lambda_0 \cos^2 \theta r d\theta \end{aligned}$$

$$\begin{aligned}
 &= \frac{\lambda_0 r}{2} \left\{ \int_0^{2\pi} d\theta + \int_0^{2\pi} \cos 2\theta d\theta \right\} \\
 &= \lambda_0 r \pi + 0 \\
 &= \lambda_0 r \pi
 \end{aligned}$$

Thus, the total charge on the ring is $\lambda_0 r \pi$.

Example 10 : A long cylinder carries a charge density given by $\rho(r) = kr$, where r is distance from the axis of cylinder and k is some constant. Find the electric intensity at a point inside the cylinder.

Solution : Let AB be a part of the charged cylinder of radius 'd', which carries a charge density $\rho = kr$ as shown in Fig. 1.18.

To find the electric intensity at a point P (inside the cylinder) at a distance s from the axis of cylinder, draw a Gaussian cylinder of length l , radius ' s ' and the point P is on the surface (Fig. 1.18).

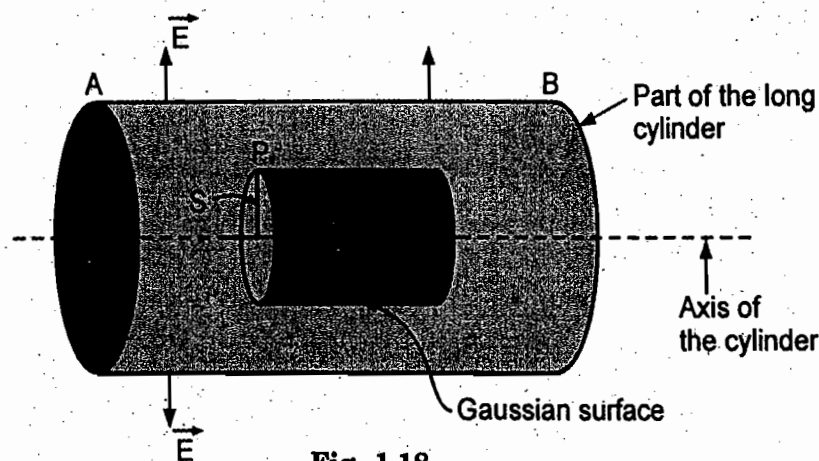


Fig. 1.18

For the Gaussian surface, Gauss's law states,

$$\phi = \frac{q_{\text{enc.}}}{\epsilon_0}$$

or

$$\oint \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc.}}}{\epsilon_0} \quad \dots (1.22)$$

(a) To use Gauss's law, let us first calculate charge enclosed by the Gaussian surface of length ' l ' and radius ' s '.

The charge in a small volume element dV is

$$dq = \rho dV$$

Therefore, total charge in a volume enclosed by Gaussian surface is

$$q_{\text{enc.}} = \int \rho dV$$

The volume element in cylindrical co-ordinate system is

$$dV = r dr d\phi dz$$

$$\begin{aligned}
 \therefore q_{\text{enc.}} &= \int_V r \, dr \, d\phi \, dz \, k r \\
 &= \int_0^s \int_0^{2\pi} \int_0^l k r^2 \, dr \, d\phi \, dz \\
 &= 2\pi l k \int_0^s r^2 \, dr \\
 &= \frac{2}{3} \pi k l s^3 \quad \dots (1.23)
 \end{aligned}$$

Thus, charge enclosed by the Gaussian surface $= \frac{2}{3} \pi k l s^3$

(b) Let us calculate $\oint_s \vec{E} \cdot d\vec{a}$

Since there is no electric field perpendicular to the plane faces of the Gaussian surface, the electric flux through these end faces is zero.

The electric intensity at every point of the curved portion of the Gaussian surface is the same and it is perpendicular to the surface. Therefore, the flux over the Gaussian surface (curved portion only) is given by,

$$\begin{aligned}
 \phi &= \int \vec{E} \cdot d\vec{a} = \int E \, da \cos \theta \\
 &= \int E \, da, \text{ since } \theta = 0 \\
 &= E \int da \\
 &= E 2\pi s l \quad \dots (1.24)
 \end{aligned}$$

Using equations (1.23) and (1.24) in equation (1.22), we get

$$E 2\pi s l = \frac{2}{3} \frac{\pi k l s^3}{\epsilon_0}$$

$$\therefore E = \frac{1}{3\epsilon_0} k s^2$$

$$\text{or } \vec{E} = \frac{1}{3\epsilon_0} k s^2 \hat{s}$$

Example 11 : Find the electric field inside and outside a sphere which carries a charge density proportional to the distance from the origin, $\rho = kr$, where k is some constant.

Solution : Consider a sphere having centre 'O' and radius 'a' as shown in Fig. 1.19 on next page.

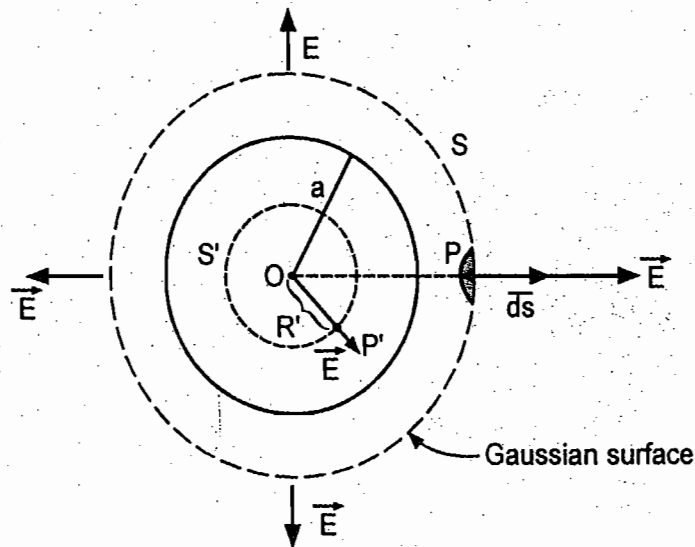


Fig. 1.19

Case (i) : At an external point :

Consider any point such as P outside the sphere at a distance 'R' from centre O. To find E at a point P, draw a Gaussian surface S in the form of concentric sphere of radius R. The electric intensity is radially outward and it has same magnitude at all points on the Gaussian surface.

Flux passing through small surface element $\vec{ds} = E ds \cos \theta$

But the angle between $\vec{E}$ and normal to the surface element of area ds is zero, i.e. $\theta = 0$.

$$\therefore d\phi = E ds$$

Total flux passing through Gaussian surface is

$$\begin{aligned} \phi &= \oint_S E ds \\ \phi &= E 4\pi R^2 \end{aligned} \quad \dots (1.25)$$

According to Gauss's theorem,

$$\phi = \frac{q_{\text{enc.}}}{\epsilon_0} \quad \dots (1.26)$$

where, $q_{\text{enc.}}$ is the total charge enclosed by Gaussian surface.

Let us calculate $q_{\text{enc.}}$ or total charge on the sphere.

The charge in a small volume element dV of the sphere is

$$dq = \rho dV \quad \left(\because \rho = \frac{dq}{dV} \right)$$

Volume element in spherical polar co-ordinate system is

$$dV = r^2 \sin \theta dr d\theta d\phi$$

$$\therefore dq = \rho r^2 \sin \theta dr d\theta d\phi$$

Total charge on the sphere of radius a is

$$q = k \int_0^a \int_0^\pi \int_0^{2\pi} r^3 \sin \theta \, dr \, d\theta \, d\phi$$

$$q = k \left(\int_0^a r^3 \, dr \right) \left(\int_0^\pi \sin \theta \, d\theta \right) \left(\int_0^{2\pi} d\phi \right)$$

$$= k \frac{a^4}{4} \times 2 \times 2\pi$$

$$q_{\text{enc.}} = \pi k a^4 \quad \dots (1.27)$$

Thus, total charge on the sphere is $q = \pi k a^4$.

Using equations (1.25) and (1.27) in equation (1.26), we get

$$\therefore E 4\pi R^2 = \frac{\pi k a^4}{\epsilon_0}$$

$$\text{or} \quad E = \frac{k}{4\epsilon_0} \left(\frac{a^2}{R} \right)^2$$

This equation gives electric intensity at a point P situated at a distance R from centre O .

Case (ii) : At an internal point :

Let us find the electric intensity at a point P' inside the charged sphere at a distance R' from the centre O . To find electric intensity at point P' , draw a Gaussian surface of radius R' (Fig. 1.19).

Flux passing through Gaussian surface S' is

$$(1.25) \quad \phi = \oint_{S'} E \, ds = E 4\pi R'^2$$

(1.26) Total charge enclosed by the surface S' is

$$q = k \int_0^{R'} \int_0^\pi \int_0^{2\pi} r^3 \sin \theta \, dr \, d\theta \, d\phi$$

$$= \pi k R'^4$$

Thus, total charge enclosed by the Gaussian surface is $\pi k R'^4$.

$$\therefore E 4\pi R'^2 = \pi k R'^4$$

$$E = \frac{k R'^2}{4\epsilon_0}$$

This is the expression for electric intensity, due to charged sphere (with $\rho = Kr$) at an internal point situated at a distance R' from the centre O of sphere.

E is directed away from the charge and is along the line joining given point and centre O .

Example 12 : Obtain an expression for the electric intensity near the surface of metallic conductor.

Solution : Consider a metallic conductor of any shape as shown in Fig. 1.20. The charge given to a conductor resides on its outer surface. The electric intensity at any point outside the conductor is always perpendicular to the surface. If this were not so, the electric intensity will have a component parallel to the surface and it will produce movement of charges along the surface. But as the charges are at rest, the electric intensity outside the surface must be everywhere perpendicular to the surface. As there is no field inside the conductor, the intensity at any point inside the conductor must be zero.

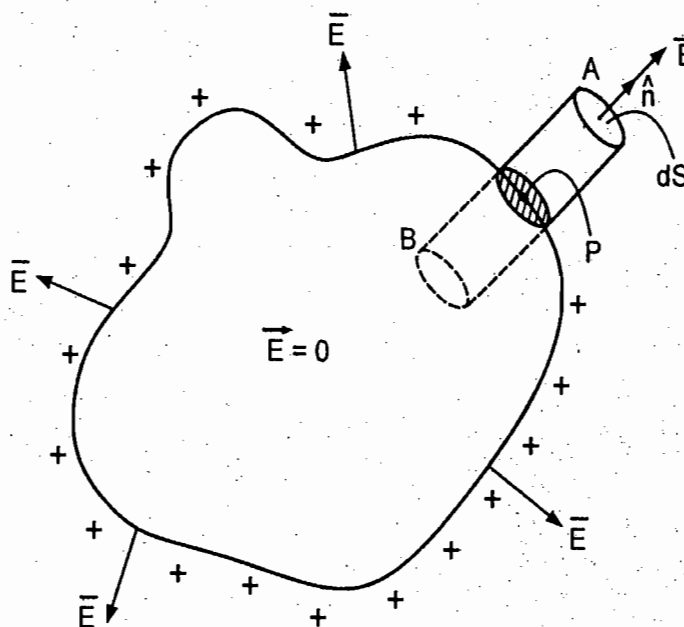


Fig. 1.20

Imagine a Gaussian surface in the form of cylinder having a small cross-sectional area dS , drawn with its axis perpendicular to conductor having one end inside the conductor and the other end i.e. A outside the conductor.

Since the field outside the conductor is perpendicular to its surface, the electric flux will pass through the end surface at A only. The electric flux does not pass through the curved sides of the cylinder. Hence, the total flux over the cylinder is flux passing through end surface A.

$$\begin{aligned} \text{Flux through surface A} &= \vec{E} \cdot \vec{dS} \\ \phi &= E dS \end{aligned}$$

$$(\because \theta = 0)$$

According to Gauss's theorem

$$\phi = \frac{q}{\epsilon_0 k}$$

If σ is surface charge density, then charge on area dS is σdS .

$$\therefore E dS = \frac{\sigma dS}{\epsilon_0 k}$$

$$\text{or } E = \frac{\sigma}{\epsilon_0 k}$$

This is expression for magnitude of the electric intensity at any point just outside the metallic conductor. It should be noted that for metallic conductor having an irregular shape, curvature is different at different points on the surface. So σ is different at different points and therefore, intensity just outside the conductor has different values at different places. For plane conductor, intensity just outside the conductor is constant at all points

and is $E = \frac{\sigma}{\epsilon_0 k}$.

QUESTIONS

1. State and explain Coulomb's law in electrostatics.
2. What do you mean by electric field or electrostatic field ? Define electric intensity at a point in electric field and obtain an expression for the electric intensity due to a point charge at any point.
3. Explain various types of charge distribution.
4. Explain the terms : surface charge density, linear charge density and volume charge density.
5. Explain the concept of electric potential and hence define electric potential at a point.
6. Obtain an expression for electric intensity and potential at any point due to continuous distribution of charge throughout the volume of an object.
7. State Gauss's law and obtain its differential form.
8. State Gauss's law. What are advantages of Gauss's law over Coulomb's law ? Give the limitations of this law.
9. Show that $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$
10. Show that $\vec{\nabla} \times \vec{E} = 0$.
11. Define potential energy of a system of charges and obtain an expression for the potential energy of a system of charges.

EXAMPLES

1. Obtain an expression for electric intensity at an external point and at an internal point due to uniformly charged conducting sphere of radius 'a'.
2. A charge + q is uniformly distributed throughout a non-conducting spherical volume of radius 'a' meter. Obtain an expression for the potential at distance 'r' from the centre of sphere, where $r < a$.

3. A sphere of radius 'a' is uniformly charged so as to have a charge per unit volume ρ . Calculate the energy of a sphere.
4. Find the electric field at a distance d above the centre of circular loop of radius 'a' (Fig. 1.21), which carries a uniform line charge λ .

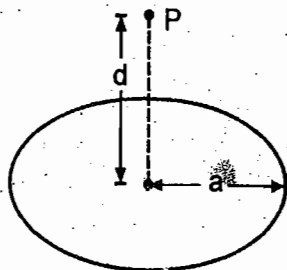


Fig. 1.21

5. A thin circular disc of radius 'a' is charged uniformly so as to have a charge per unit area σ . Calculate the electric intensity at an axial point at a distance d from the disc.
6. Show that each ion in an infinite line of equally spaced alternate negative and positive ions carrying charges of equal magnitude has equal potential energy.
7. Eight negative charges each having value q coulomb are kept at the corners of a cube of side 'a' meter, at the centre of which a positive charge of + 2q coulomb is placed. What is the electric potential energy of the system ?
8. The potential function is given by,

$$V(y, z) = \frac{10}{y^2 + z^2}$$

where, V is in volts and y, z are in cm. Find –

- (a) the expression for the gradient of the potential. (b) the value of the gradient at the point (2, 2) and (c) magnitude of electric intensity at this point the intensity of electric field at this point.
9. A sphere of radius 'a' is filled with positive charge of volume charge density ρ which varies with a distance r from the centre of sphere but is constant for a given 'r'. Find the total charge in the sphere.

if, (a) $\rho = \rho_0 \frac{a}{r}$

(b) $\rho = \rho_0 \left(1 - \frac{r^2}{a^2}\right)$

10. Calculate the electric field at a point distant 10^{-11} meter from the centre of gold nucleus ($Z = 79$). If the nuclear radius is 5×10^{-15} meter, what would be the electric field at a point on the nuclear surface ?

(Hint : Charge on gold nucleus + Ze and

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

ANSWERS

1. $E_{\text{int}} = 0$ (Since the sphere is conducting, charges reside on the surface of sphere).
 $E_{\text{ext}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$, where, q is total charge on the surface of sphere.
2. $V = \frac{q}{4\pi\epsilon_0} \left(\frac{3a^2 - r^2}{2a^3} \right)$
3. $W = \frac{4\pi\rho^2}{15\epsilon_0} a^5$
4. $E = \frac{1}{4\pi\epsilon_0} \frac{ad}{(a^2 + d^2)^{3/2}}$
5. $E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{d}{(a^2 + d^2)^{1/2}} \right]$
6. P.E. of positive ion = P.E. of negative ion = $-\frac{q^2}{2\pi\epsilon_0 d} \log_e 2$,
 where, d is spacing between any two adjacent ions.
7. P.E. of system = $3.96 \times 10^{-10} \times \frac{q^2}{a}$ joules.
8. (a) $\text{grad } V = -\frac{20}{(y^2 + z^2)^2} (y \vec{j} + z \vec{k})$
 (b) $\text{grad } V(2, 2) = -\frac{5}{8} (\vec{j} + \vec{k})$
 (c) The magnitude of $E = 0.88$ volts/cm.
9. (a) $2\pi\rho_0 a^3$
 (b) $\frac{8\pi}{15} \rho_0 a^3$
10. $E = 1.14 \times 10^{15}$ newton/coulomb.
 $E_{\text{surface}} = 4.5 \times 10^{21}$ newton/coulomb.

BOUNDARY VALUE PROBLEM IN ELECTROSTATIC FIELDS

INTRODUCTION

In the present chapter, we shall discuss some of the techniques generally used to solve electrostatic problems. The problems in electrostatics which are solved by considering boundary conditions are known as boundary value problems.

An electric field can be completely described by giving either charge distribution or electric intensity at all points or the electric potential at all points in the given region. We have studied that if any one of these is known, the other can be calculated. But in some cases, if we do not know the charge distribution in all space, instead we are given the values of potential on certain surfaces or charge distribution in a certain region of limited space, in such a case to obtain electric potential at any point, we need to solve Laplace or Poisson's equations. While solving these equations the specified values of potential on various surfaces provide us necessary boundary conditions. This chapter also includes method of electrical images used to solve some typical boundary value problems without using differential equation.

2.1 POISSON AND LAPLACE EQUATIONS

The Poisson's and Laplace's equations are result of combining Gauss's law (differential form) with the gradient of electric field.

The differential form of Gauss's law is

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \dots (2.1)$$

where, ρ is volume charge density

But,

$$\vec{E} = -\vec{\nabla} \phi \quad \dots (2.2)$$

where, ϕ is electric potential.

Using equation (2.2) in equation (2.1), we get

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0} \quad \dots (2.3)$$

This equation is known as Poisson's equation.

The solution of this equation gives field or potential in the region where charges are present.

If the region of electric field is free from charges i.e. $\rho = 0$, then equation (2.3) becomes,

$$\nabla^2 \phi = 0 \quad \dots (2.4)$$

This equation is known as Laplace's equation.

The solution of this equation gives field or potential in the charge free region.

2.2 LAPLACE'S EQUATION IN CARTESIAN CO-ORDINATES AND ITS SOLUTION

Laplace's equation is

$$\nabla^2 \phi = 0$$

In Cartesian co-ordinates, $\phi = \phi(x, y, z)$

$$\text{and } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\therefore \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad \dots (2.5)$$

This equation is known as Laplace's equation in cartesian co-ordinate system.

To obtain the solution of Laplace's equation, let us first obtain ordinary differential equations from the above partial differential equation (equation 2.5), using the method of separation of variables.

Let $\phi(x, y, z) = X(x) Y(y) Z(z)$

where, X is a function of x only, Y is a function of y only and Z is a function of z only.

Using $\phi = XYZ$ in equation (2.5), we get

$$\frac{\partial^2}{\partial x^2} XYZ + \frac{\partial^2}{\partial y^2} XYZ + \frac{\partial^2}{\partial z^2} XYZ = 0$$

$$\text{or } YZ \frac{\partial^2}{\partial x^2} X + XZ \frac{\partial^2}{\partial y^2} Y + XY \frac{\partial^2}{\partial z^2} Z = 0$$

Dividing above equation throughout by XYZ, we get

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0$$

Since, each term in above equation contains a different independent variable and all three terms cancel, each term must be equal to some constant. So we get, following three equations.

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1, \quad \dots (2.6)$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \dots (2.7)$$

$$\text{and } \frac{1}{Z} \frac{d^2 Z}{dz^2} = C_3 \quad \dots (2.8)$$

where, C_1 , C_2 and C_3 are constants with

$$C_1 + C_2 + C_3 = 0 \quad \dots (2.9)$$

Equation (2.9), suggests that at least one of these constants must be positive and one other must be negative.

While solving problem, which constants are positive and which are negative will be decided by the physical situation of the given problem.

Assuming C_1 positive, C_2 and C_3 negative and setting $C_2 = -k^2$ and $C_3 = -l^2$, we get

$$C_1 = k^2 + l^2$$

Using $C_1 = k^2 + l^2$, $C_2 = -k^2$ and $C_3 = -l^2$ in equations (2.6), (2.7) and (2.8), we get

$$\frac{1}{X} \frac{d^2 X}{dx^2} = (k^2 + l^2)$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -k^2$$

$$\text{and } \frac{1}{Z} \frac{d^2 Z}{dz^2} = -l^2$$

Thus, Laplace's equation in Cartesian co-ordinates can be separated into the following three ordinary differential equations.

$$(i) \quad \frac{d^2 X}{dx^2} - (k^2 + l^2) X = 0 \quad \dots (2.10)$$

The solution of this differential equation may be written as,

$$X(x) = A e^{\sqrt{k^2 + l^2} x} + B e^{-\sqrt{k^2 + l^2} x} \quad \dots (2.11)$$

$$(ii) \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \quad \dots (2.12)$$

The solution of this equation may be written as,

$$Y(y) = C \sin ky + D \cos ky \quad \dots (2.13)$$

$$(iii) \quad \frac{d^2 Z}{dz^2} + l^2 Z = 0 \quad \dots (2.14)$$

The solution of this equation may be written as,

$$Z(z) = E \sin lz + F \cos lz \quad \dots (2.15)$$

where, A, B, C, D, E and F are arbitrary constants.

Thus, we get first solution exponential and last two solutions oscillatory. Equation (2.9) suggests that out of three variables, at least for one variable, we must have trigonometric behaviour (oscillatory) and at least one other variable have exponential.

The solution of Laplace's equation is product of these solutions shown by equations (2.11), (2.13) and (2.15).

$$\text{i.e. } \phi = X(x) Y(y) Z(z)$$

$$\phi = \left(A e^{\sqrt{k^2 + l^2} x} + B e^{-\sqrt{k^2 + l^2} x} \right) (C \sin ky + D \cos ky) (E \sin lz + F \cos lz) \quad \dots (2.16)$$

The constants involved in above equation can be obtained from the boundary conditions to be satisfied by potential ϕ .

The solution shown by equation (2.16) is one particular solution of Laplace's equation. There may be other values of constant k, l which could give valid solutions. Thus, the general solution of Laplace's equation may be written as,

$$\phi = \sum_{r,s,t} \left(A_t e^{\sqrt{k_r^2 + l_s^2} x} + B_t e^{-\sqrt{k_r^2 + l_s^2} x} \right) (C_r \sin k_r y + D_r \cos k_r y) (E_s \sin l_s z + F_s \cos l_s z)$$

It must be noted that while solving problems, if the given surface S , on which the boundary conditions are specified, coincides with a co-ordinate surfaces of cartesian system*, then only a solution can be obtained by this technique in cartesian system. (In cartesian co-ordinate system, co-ordinate surfaces are planes parallel to the co-ordinate axes).

In two dimensional problem, Laplace's equation in Cartesian co-ordinates may be written as,

$$\frac{\partial^2 \phi(x, y)}{\partial x^2} + \frac{\partial^2 \phi(x, y)}{\partial y^2} = 0$$

Assuming $\phi(x, y) = X(x) Y(y)$, we get

$$\frac{\partial^2 XY}{\partial x^2} + \frac{\partial^2 XY}{\partial y^2} = 0$$

$$\text{or} \quad Y \frac{\partial^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} = 0$$

Dividing above equation throughout by XY , we get

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0$$

Since each term in above equation contains a different independent variable and all terms cancels, we can write,

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \text{ and } \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2$$

$$\text{with} \quad C_1 + C_2 = 0$$

One of these constants is positive, the other is negative.

Let $C_1 = k^2$ hence, $C_2 = -k^2$

$$\therefore \quad \frac{1}{X} \frac{d^2 X}{dx^2} = k^2 \quad \text{or} \quad \frac{d^2 X}{dx^2} - k^2 X = 0$$

$$\text{and} \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -k^2 \quad \text{or} \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0$$

*

If given surface S , on which boundary conditions are specified, is in the form of sphere centered at origin or plane through the origin, then solution of Laplace's equation in spherical polar co-ordinate system must be used. For the surface S , on which the boundary conditions are specified, coincides with a co-ordinate surfaces of cylindrical co-ordinate surfaces (like circular cylinders about z -axis) then solution of Laplace's equation in cylindrical co-ordinate system must be used.

**

If we choose $C_1 = -k^2$ and $C_2 = k^2$, we get,

$$X(x) = A \sin kx + B \cos kx$$

$$Y(y) = C e^{ky} + D e^{-ky}$$

The solution of above differential equations may be written as,

$$X(x) = A e^{kx} + B e^{-kx}$$

$$Y(y) = C \sin ky + D \cos ky$$

$$\therefore \phi = (A e^{kx} + B e^{-kx}) (C \sin ky + D \cos ky)$$

In one dimensional problem, Laplace's equation may be written as,

$$\frac{\partial^2 \phi(x)}{\partial x^2} = 0$$

$$\text{or} \quad \frac{\partial^2 \phi}{dx^2} = 0$$

The general solution of above equation is

$$\phi(x) = C_1 x + C_2$$

2.3 SOLUTION OF POISSON'S EQUATION

The Poisson's equation is

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}$$

Since this equation is inhomogeneous, we cannot use the method of separation of variable to obtain its solution. The solution of Poisson's equation can be obtained by use of Green's function. Before obtaining the solution of Poisson's equation, let us study Green's function.

Green's Function :

Let τ be a differential operator and $g(x)$ be a continuous function such that,

$$\tau f(x) = \frac{g(x)}{\epsilon_0} \quad \dots (2.17)$$

This is an inhomogeneous equation.

If there exists a unique solution for each $g(x)$, then we can write equation for $f(x)$, using inverse operator τ^{-1} .

$$\therefore f(x) = \tau^{-1} \frac{g(x)}{\epsilon_0} \quad \dots (2.18)$$

Here, the operator τ^{-1} is inverse operator to differentiation provided the associated boundary conditions remain same.

If τ is differential, τ^{-1} is integral.

The solution shown by equation (2.18) can also be written as,

$$f(x) = \int \tau^{-1} \delta(x-x') \frac{g(x')}{\epsilon_0} dx' \quad \dots (2.19)$$

where, $\delta(x-x')$ is Dirac Delta function.

In equation (2.19), $g(x) = \delta(x - x')$ is called as the Green's function for operator τ and the given boundary conditions.

This is the solution of equation (2.17). Hence, the Green's function $G(x - x')$ must satisfy the equation

$$\tau G(x, x') = \frac{\delta(x - x')}{\epsilon_0} \quad \dots (2.20)$$

having the same boundary conditions as on the function $f(x)$. Therefore, we can use inverse operator as,

$$G(x, x') = \tau^{-1} \frac{\delta(x - x')}{\epsilon_0}$$

Therefore, the solution shown by equation (2.19) may be written as,

$$f(x) = \int \tau^{-1} \frac{\delta(x - x')}{\epsilon_0} g(x') dx' = \int G(x, x') g(x') dx'$$

Thus, we get

$$f(x)^* = \int G(x, x') g(x') dx' \quad \dots (2.21)$$

So, the problem is reduced to integral form with proper Green's function for a given operator. The solution can be obtained by solving the integral.

Now, we shall apply this method to solve Poisson's equation,

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}$$

The potential due to a charge q at a distance r from it is given by,

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

When the charge is distributed on volume V then,

$$q = \int_V \rho dV$$

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \frac{\int_V \rho dV}{r} \quad \dots (2.22)$$

* For simplicity, we have considered one dimensional case, for three-dimensional problem, we may

$$\text{write } f(r) = \int G(\vec{r}, \vec{r}') g(r') dr'$$

$$\text{where, } G(\vec{r}, \vec{r}') = \tau^{-1} \frac{\delta(\vec{r} - \vec{r}')}{\epsilon_0}$$

Since volume charge density ρ is independent of point at which potential is to be calculated, above equation can be written as,

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho dV}{r} \quad \dots (2.23)$$

Using Laplacian operator on both sides of above equation, we get

$$\nabla^2 \phi(r) = \frac{1}{4\pi\epsilon_0} \nabla^2 \left(\int_V \frac{\rho}{r} dV \right)$$

As ρ is independent of r and derivative and integration are interchangeable, we can write above equation as,

$$\nabla^2 \phi(r) = \frac{1}{4\pi\epsilon_0} \int_V \rho \nabla^2 \left(\frac{1}{r} \right) dV \quad \dots (2.24)$$

Let us consider the term $\nabla^2 \left(\frac{1}{r} \right)$. Using radial part of Laplacian operator, we get

$$\nabla^2 \left(\frac{1}{r} \right) = \frac{1}{r} \frac{d^2}{dr^2} \left(r \cdot \frac{1}{r} \right) = \frac{1}{r} \frac{d^2}{dr^2} (1) = 0$$

This is valid for $r \neq 0$.

At $r = 0$, the expression is undefined.

Nevertheless, the vanishing of $\nabla^2 \left(\frac{1}{r} \right)$ for $r \neq 0$ means that equation (2.24) can be written in the form,

$$\nabla^2 \phi(r) = \frac{1}{4\pi\epsilon_0} \rho C \quad \dots (2.25)$$

where,

$$C = \int_V \nabla^2 \left(\frac{1}{r} \right) dV$$

To evaluate the constant C , let us use the following limiting procedure.

$$C = \lim_{a \rightarrow 0} \int_V \nabla^2 \left(\frac{1}{\sqrt{r^2 + a^2}} \right) dV$$

But,

$$\begin{aligned} \nabla^2 \left(\frac{1}{\sqrt{r^2 + a^2}} \right) &= \frac{1}{r} \frac{d^2}{dr^2} \left(\frac{r}{\sqrt{r^2 + a^2}} \right) \\ &= -\frac{3a^2}{(r^2 + a^2)^{5/2}} \end{aligned}$$

$\therefore$

$$C = \lim_{a \rightarrow 0} \int_V \left(-\frac{3a^2}{(r^2 + a^2)^{5/2}} \right) dV$$

But, in spherical polar co-ordinate system,

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

$$\begin{aligned} \therefore C &= \lim_{a \rightarrow 0} \int \int \int \left(-\frac{3a^2}{(r^2 + a^2)^{5/2}} \right) r^2 \sin \theta \, dr \, d\theta \, d\phi \\ &= \lim_{a \rightarrow 0} \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} d\phi \int_0^\infty -\frac{3a^2 r^2}{(r^2 + a^2)^{5/2}} \, dr \end{aligned}$$

To calculate above integral, let $r = ax$ and so $dr = adx$.

Using $r = ax$, above equation can be written as,

$$C = -12\pi \int_0^\infty \frac{x^2 \, dx}{(x^2 + 1)^{5/2}}$$

But,
$$\int_0^\infty \frac{x^2 \, dx}{(x^2 + 1)} = \frac{1}{3}$$

$$\therefore C = -4\pi$$

Using this value in equation (2.25), we get

$$\nabla^2 \phi = \frac{\rho}{4\pi\epsilon_0} (-4\pi) = -\frac{\rho}{\epsilon_0}$$

Thus, value of C establishes that the Laplacian of equation (2.23) does indeed satisfy the Poisson's equation.

The singular nature of Laplacian of $\frac{1}{r}$ can be exhibited formally in terms of Dirac delta function. Since $\nabla^2 \left(\frac{1}{r} \right) = 0$ for $r \neq 0$ and its volume integral is -4π , we can write the formal equation $\nabla^2 \left(\frac{1}{r} \right)$ equal to $-4\pi \delta(r)$.

i.e.,
$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta(r)$$

or more generally,
$$\nabla^2 \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = -4\pi \delta(\vec{r} - \vec{r}') \quad \dots (2.26)$$

Comparing this equation with equation (2.20), we have $\frac{1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$ is a Green's function of the operator ∇^2 and is equal to the potential due to unit charge.

Now, the solution of Poisson's equation should satisfy certain boundary conditions. For this we have to convert Poisson's equation into integral form.

Poisson's equation can be converted into integral form by using Green's theorem.

Green's theorem is,

$$\int_V [\phi \nabla^2 \psi - \psi \nabla^2 \phi] dV = \int_S \left[\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right] dS \quad \dots (2.27)$$

where, $\frac{\partial}{\partial n}$ is the normal derivative at the surface S directed outwards from inside the volume V and ϕ and ψ are arbitrary scalar functions.

We choose $\psi = \frac{1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$ and ϕ , a scalar potential satisfying Poisson's equation.

Using $\psi = \frac{1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$ in equation (2.27), we get

$$\begin{aligned} \int_V \left\{ \phi \nabla^2 \left(\frac{1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|} \right) - \frac{1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|} \nabla^2 \phi \right\} dV \\ = \int_S \left\{ \frac{-1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|} \frac{\partial \phi}{\partial n'} + \phi \frac{\partial}{\partial n'} \left(\frac{1}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|} \right) \right\} dS \end{aligned}$$

$$\begin{aligned} \text{or } \int_V \left\{ \phi \nabla^2 \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) - \frac{1}{|\vec{r} - \vec{r}'|} \nabla^2 \phi \right\} dV \\ = \int_S \left\{ \frac{-1}{|\vec{r} - \vec{r}'|} \frac{\partial \phi}{\partial n'} + \phi \frac{\partial}{\partial n'} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \right\} dS \end{aligned}$$

Using equation (2.26) and Poisson's equation in above equation, we get

$$\begin{aligned} \int_V \left\{ -4\pi \phi \delta \left(\vec{r} - \vec{r}' \right) + \frac{1}{|\vec{r} - \vec{r}'|} \frac{\rho(\vec{r}')}{\epsilon_0} \right\} dV \\ = \int_S \left\{ \frac{-1}{|\vec{r} - \vec{r}'|} \frac{\partial \phi}{\partial n'} + \phi \frac{\partial}{\partial n'} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \right\} dS \end{aligned}$$

If r lies within the volume V, we get

$$\begin{aligned} \phi(r) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV' \\ + \frac{1}{4\pi} \int_S \left\{ \frac{1}{|\vec{r} - \vec{r}'|} \frac{\partial \phi}{\partial n'} - \phi \frac{\partial}{\partial n'} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \right\} dS' \quad \dots (2.28) \end{aligned}$$

This is integral form of Poisson's equation which can be solved by using boundary conditions.

2.4 THE METHOD OF ELECTRICAL IMAGES

We have seen that the boundary value problems in electrostatics may be solved by using Laplace's equation. The method of electrical images is used to solve boundary value problems without using differential equations. It is applicable for some typical problems. Before we see the method in detail, let us understand what is an electrical image charge? How is it defined? and then we see certain electrostatic problems that are solved by using the method of electrical images.

A charge placed near the conducting body (or dielectric body) induces charge on it. So while calculating electrical intensity or potential at any point near conductor, we must consider actual charge as well as induced charges. As we do not know the induced charge

density over the body, it is very difficult to calculate $\vec{E}$ or V at any point near the conductor. The method of electrical images makes it possible because the whole of the induced charges can be replaced by a single point charge called as imaginary charge. The position and value (magnitude and sign) of imaginary charge is such that, the imaginary charge together with the actual specified charge reduces potential to fixed value at all points on the conductor. This imaginary point charge is referred as electric image of the original charge.

The concept of electrical images can be understood by considering following simplest problem of point charge near an infinite grounded conducting plane. Let the point charge '+q' be placed at a perpendicular distance 'd' from the plane surface 'S' of a grounded conducting plate of infinite extent [Fig. 2.1 (a)]. Since the plate is grounded, the potential ϕ at any point on it is zero. The charge '+q' near the conducting plate induces charge on it.

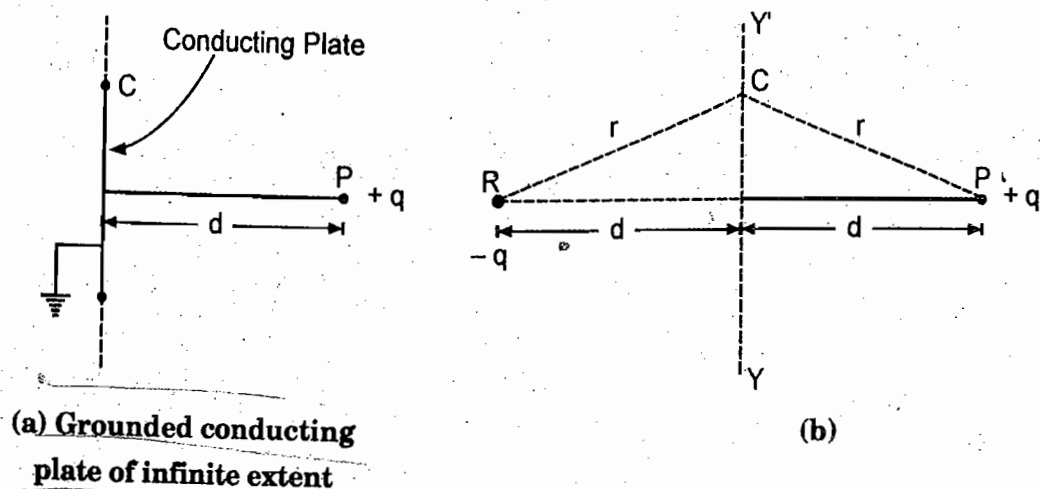


Fig. 2.1

In order to find the potential and field at any point in the region on the right hand side of the plate, the actual and induced charges must be taken into accounts.

Let us imagine that a charge '- q' is placed exactly at distance 'd' behind the plane surface of the plate. Since the charge - q and + q are at equal distance from every point of

the plane surface, the potential at any point* on the surface due to these charges is zero. It is obvious that, if we remove the grounded conducting plate and the charge $-q$ is placed at R (Fig. 2.1 b), every point on the plane YY' which was previously occupied by the conducting plate will be still at zero potential. Thus, the introduction of charge $-q$ at R and removal of the plate, does not affect the flux on right hand side of the plane. So the condition of problem remains unaltered by removal of the plate and replace it by charge $-q$ at point R. Thus, it is clear that potential (or field) produced by $+q$ (placed at point P) and $-q$ (placed at point R) is same as potential (or field) produced by charge $+q$ at P and induced charge on conducting plate. Therefore, the field or potential produced at any point due to charge $-q$ is equivalent to induced charges on the plate. From Fig. 2.1 (b), the charge ' $-q$ ' appears an exact image of actual charge. So from optical analogy, charge ' $-q$ ' is called image charge. **Therefore, an electrical image may be defined as a point charge (or a set of point charges) on one side of a surface, such that the distribution of the field remains the same as on the other side with the actual electrification of the surface.** The actual electrification is then ignored and the images are used instead of plane to find the field.

The necessary conditions, while solving electrostatic problem by using method of electrical images, are following :

- (i) The electric flux through any closed surface in the original field must remain same.
- (ii) The potential of the conductor must remain same.
- (iii) The potential at infinite distant points must remain the same.

The procedural steps for solving a particular electrostatic problem are following :

- (1) Without violating the boundary conditions, determine the correct magnitude and position of the image charge. This can be done simply by inspection of geometry in the problem.
- (2) The potential and intensity, at any point are calculated using the given values of charge and image charge. The presence of the conductor is ignored at this step.
- (3) The normal component of electric intensity at the surface of the conductor is calculated.
- (4) Surface density of induced charge is calculated from the value of normal component of electric intensity at a point on the surface of conductor,

$$\left(\text{Use the relation } E = \frac{\sigma}{\epsilon_0} \right)$$

* The potential due to charge $+q$ at a distance r from it is given by $\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$. The point, say C, on the plane (Fig. 2.1 b) is at a distance r from charge $+q$ as well as from charge $-q$. So $\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r} - \frac{1}{4\pi\epsilon_0} \frac{q}{r} = 0$. Similarly, it can be shown for other points also.

- (5) Using $\int_S \sigma dS$, calculate the total induced charge on the conductor.
- (6) The force between the conductor and point charge is determined by Coulomb's law.

Before solving the problems in electrostatics using method of electrical images, let us discuss the basic electrostatic properties of ideal conductors. The basic electrostatic properties of ideal conductors are following: **N.B.**

- (i) Electric intensity at any point inside the conductor is always zero.

A conductor contains large number of free electrons. When it is kept in an external electric field, the electrons move against the direction of field and get distributed in such a way that the electric field inside the conductor is zero.

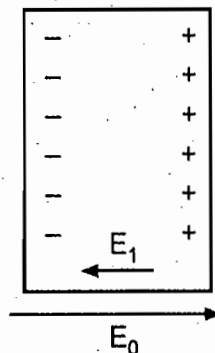


Fig. 2.2 (a)

In Fig. 2.2 (a), the induced charges produce a field of their own $\vec{E}_1$, and is in the opposite direction to that of $\vec{E}_0$. The charges will continue to flow until the induced charges cancel off the original field.

Thus, at any point inside the conductor,

$$\vec{E}_1 + \vec{E}_0 = 0$$

(ii) The electric intensity at any point just outside the conductor is perpendicular to the surface. If this were not so, the intensity will have a component parallel to the surface and it will produce the movement of charges along the surface. However, as charges are at rest, the electric intensity just outside the surface must be perpendicular to it.

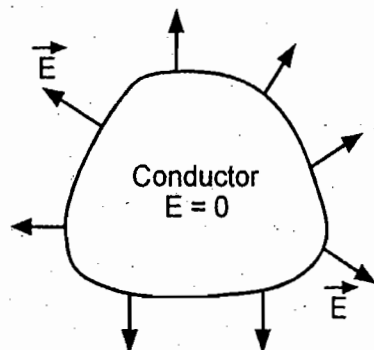


Fig. 2.2 (b)

(iii) $\rho = 0$, inside the conductor. The differential form of Gauss's law is $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$.

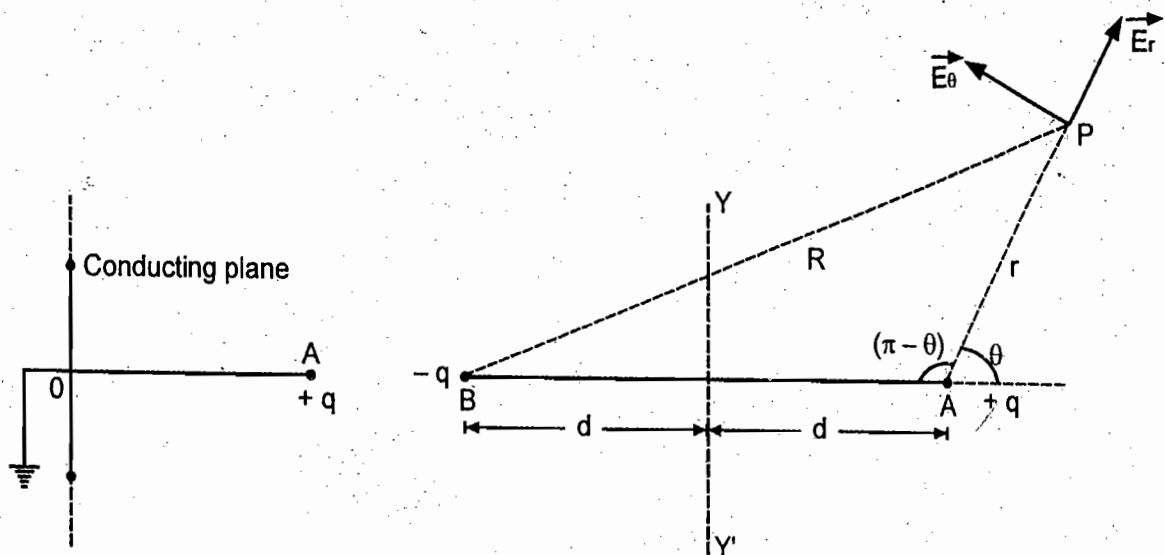
Since, $E = 0$ inside the conductor, ρ is always zero.

(iv) Any net charge resides on the surface.

(v) A conductor is equipotential.

2.5 A POINT CHARGE NEAR GROUNDED CONDUCTING PLANE

Consider a point charge '+q' placed at a perpendicular distance 'd' from the grounded conducting plane of infinite extent as shown in Fig. 2.3 (a). We have seen that, the above problem remains unaltered, if we remove the grounded plate and replaced it by a charge -q (called as image charge) at a perpendicular distance 'd' on left side from YY' plane as shown in Fig. 2.3 (b).



(a) A point charge near grounded conducting infinite plane (b) A point charge q and image charge -q at points A and B respectively

The potential at any pt on plane due to +q & -q is zero. This boundary cond. is satisfied.

Since the conducting plane is grounded, the potential at any point on the plane is zero. We can verify this by showing that potential at a point on the plane YY' is zero.

If point P is on the plane YY', then charges +q and -q are at equal distance from point P.

Let it be 'r'. The potential at point P on the plane, due to charge +q and -q is,

$$\phi = \frac{q}{4\pi\epsilon_0 r} - \frac{q}{4\pi\epsilon_0 r} = 0$$

Thus, the boundary condition is satisfied.

$$\vec{E} = \frac{\rho}{\epsilon_0}$$

(i) **Potential at point P:**

Let us calculate the potential at a point P [Fig. 2.3 (b)] due to charge '+q' and image '- q' (ignore the presence of conductor).

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r} - \frac{1}{4\pi\epsilon_0} \frac{q}{R} \quad \dots (2.29)$$

From Fig. 2.3 (b), we have,

$$R^2 = r^2 + (2d)^2 - 4rd \cos(\pi - \theta)$$

or

$$R = \sqrt{r^2 + 4d^2 + 4rd \cos \theta} \quad \dots (2.30)$$

$$[\because \cos(\pi - \theta) = -\cos \theta]$$

where θ is angle between lines BA and AP.

Using equation (2.30) in equation (2.29), we get

$$\phi = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{1}{(r^2 + 4d^2 + 4rd \cos \theta)^{1/2}} \right] \quad \dots (2.31)$$

This is expression for potential at point P.

(ii) **Electric intensity at a point P:**

The component of electric intensity at point P in the direction of AP i.e. r is,

$$\begin{aligned} E_r &= -\frac{\partial \phi}{\partial r} \\ &= -\frac{\partial}{\partial r} \left[\frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{(r^2 + 4d^2 + 4rd \cos \theta)^{1/2}} \right) \right] \\ \text{or} \quad E_r &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r^2} - \frac{(r + 2d \cos \theta)}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right] \quad \dots (2.32) \end{aligned}$$

The component of electric intensity at point P in the direction perpendicular to r is,

$$\begin{aligned} E_\theta &= -\frac{1}{r} \frac{\partial \phi}{\partial \theta} \\ &= -\frac{1}{r} \frac{\partial}{\partial \theta} \left[\frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{(r^2 + 4d^2 + 4rd \cos \theta)^{1/2}} \right) \right] \\ &= \frac{q}{4\pi\epsilon_0 r} \frac{\partial}{\partial \theta} \left[\frac{1}{(r^2 + 4d^2 + 4rd \cos \theta)^{1/2}} \right] \quad (\because \frac{\partial}{\partial \theta} \frac{q}{4\pi\epsilon_0 r} = 0) \\ &= \frac{q}{4\pi\epsilon_0 r} \left[\frac{-2rd \sin \theta}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right] \\ &= \frac{q}{4\pi\epsilon_0 r} \left[\frac{2d \sin \theta}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right] \end{aligned}$$

or

$$E_\theta = \frac{q}{4\pi\epsilon_0} \left[\frac{2d \sin \theta}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right] \quad \dots (2.33)$$

The resultant electric intensity at point P can be calculated by using,

$$E = \sqrt{E_r^2 + E_\theta^2}$$

Let us find the resultant intensity at point P normal to the plane YY'. To find E normal to the plane YY' consider Fig. 2.4.

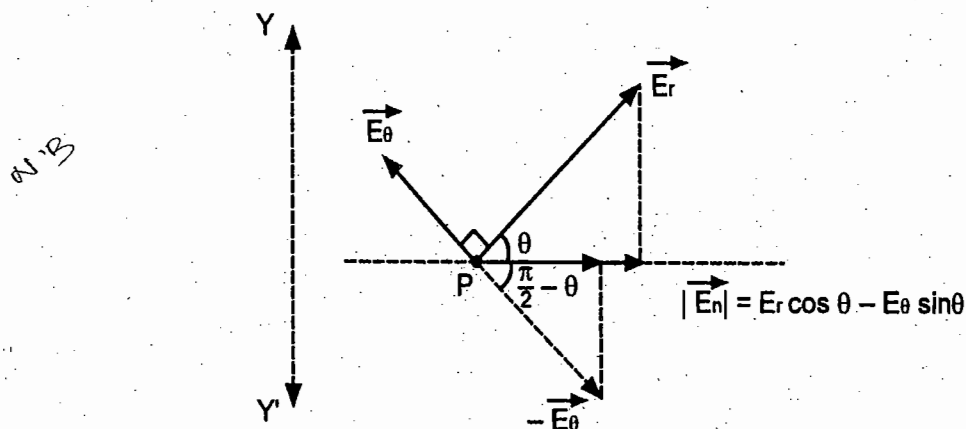


Fig. 2.4 : The direction of $\vec{E}_r$, $\vec{E}_\theta$ and $\vec{E}_n$ at point P

$$E_{\text{normal}} = E_r \cos \theta - E_\theta \sin \theta$$

Using equations (2.32) and (2.33), above equation may be written as,

$$E_n = \frac{q \cos \theta}{4\pi\epsilon_0} \left[\frac{1}{r^2} - \frac{(r + 2d \cos \theta)}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right] - \left[\frac{2qd \sin^2 \theta}{4\pi\epsilon_0 (r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{\cos \theta}{r^2} - \frac{(r \cos \theta + 2d \cos^2 \theta + 2d \sin^2 \theta)}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right]$$

or

$$E_n = \frac{q}{4\pi\epsilon_0} \left[\frac{\cos \theta}{r^2} - \frac{r \cos \theta + 2d}{(r^2 + 4d^2 + 4rd \cos \theta)^{3/2}} \right]$$

From Fig. 2.3 (b), we have, $R = (r^2 + 4d^2 + 4rd \cos \theta)^{1/2}$

$$\therefore E_n = \frac{q}{4\pi\epsilon_0} \left[\frac{\cos \theta}{r^2} - \frac{r \cos \theta + 2d}{R^3} \right] \quad \dots (2.34)$$

If point P is on conducting plane i.e. $R = r$, the normal component of electric intensity at a point on the conducting plane is,

$$E_n \text{ at } YY' = \frac{q}{4\pi\epsilon_0} \left[\frac{\cos \theta}{r^2} - \frac{r \cos \theta + 2d}{r^3} \right]$$

$$\therefore E_n \text{ at } YY' = -\frac{q}{4\pi\epsilon_0} \frac{2d}{r^3} \quad \dots (2.35)$$

(iii) Surface charge density at any point on the conducting plane :

If ' σ ' is the surface charge density on infinite conducting plane, the electric intensity at any point such as 'P' on the plane is given by,

$$E_n = \frac{\sigma}{\epsilon_0}$$

Using above equation in equation (2.35), we get

$$-\frac{qd}{2\pi\epsilon_0 r^3} = \frac{\sigma}{\epsilon_0}$$

$$\text{or} \quad \sigma = -\frac{qd}{2\pi r^3} \quad \dots (2.36)$$

This is expression for surface charge density on the conducting plane.

(iv) **Total induced charge on the plane :**

To calculate total induced charge on the conducting plane, consider a small surface element dS in the form of annular ring between two concentric circles with radii x and $x + dx$ drawn with O as a centre as shown in Fig. 2.5.

N.B.

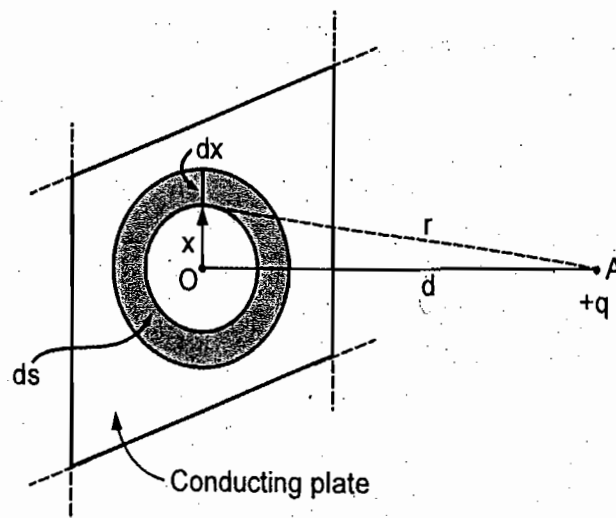


Fig. 2.5

Area of annular ring is,

$$dS = \pi (x + dx)^2 - \pi x^2$$

$$\text{or} \quad dS = 2\pi x dx$$

Induced charge on surface element dS of conducting plate $= \sigma dS = \sigma 2\pi x dx$

(2.34)

ensity

$$\begin{aligned} \text{Total induced charge on the plane} &= \int_S \sigma dS \\ &= \int_0^\infty \frac{-qd}{2\pi r^3} (2\pi x dx) \end{aligned}$$

The limit is taken from 0 to ∞ because as the conducting plane is of infinite extent, the ring starts from $x = 0$ and ends at $x = \infty$.

(2.35)

ensity

From Fig. 2.5,

$$\begin{aligned} r^2 &= x^2 + d^2 \\ &= -qd \int_0^\infty \frac{x}{r^3} dx \end{aligned}$$

$\therefore$

$$\text{Total induced charge on the plane} = -qd \int_0^\infty \frac{x}{(x^2 + d^2)^{3/2}} dx$$

$$\begin{aligned}
 &= -qd \left[-\frac{1}{(x^2 + d^2)^{1/2}} \right]_0^\infty \\
 &= -qd \left(\frac{1}{d} \right) \\
 &= -q \quad \dots (2.37)
 \end{aligned}$$

Thus, the total induced charge on the conducting plane is $-q$.

(v) **Force of attraction between the charge and the plane :**

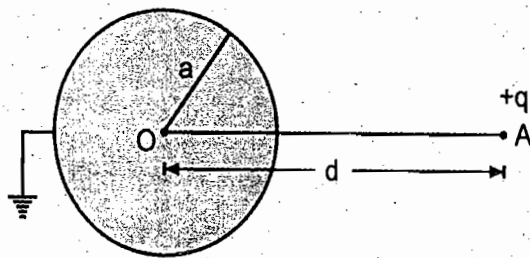
The force of attraction between charge $+q$ at A and induced charge on the plane is the same as that between $+q$ at A and image charge $-q$ at B and is given by,

$$F = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \quad \dots (2.38)$$

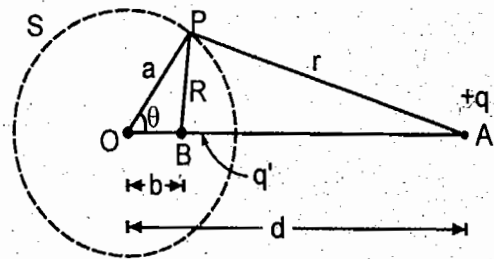
Negative sign in the above equation indicates that the force is attractive in nature.

2.6 POINT CHARGE NEAR GROUNDED CONDUCTING SPHERE

Consider a grounded conducting sphere of radius ' a '. Let the point charge ' $+q$ ' be placed at a distance ' d ' from the centre O of the sphere as shown in Fig. 2.6 (a).



(a) Grounded conducting sphere in the vicinity of charge $+q$



(b) Sphere is removed and is replaced by image charge q' at point B

Fig. 2.6

Since the sphere is grounded, its potential is zero. The charge ' $+q$ ' kept near the sphere induces charge on the sphere. We know that the induced charge can be replaced by image charge. The position, magnitude and sign of image charge is such that this charge together with charge $+q$ must satisfy the boundary conditions. Let us locate position, magnitude and sign of the image charge such that this charge together with charge $+q$ makes the potential at any point of the surface S equal to zero. Let the image charge q' be situated at point B inside the surface at a distance ' b ' from centre O. In accordance with boundary condition, the potential at any point such as 'P' on the surface S due to charge q and q' is zero. [Fig. 2.6 (b)].

$$\begin{aligned}
 \text{Therefore,} \quad \phi_P &= \frac{q}{4\pi\epsilon_0 r} + \frac{q'}{4\pi\epsilon_0 R} = 0 \\
 \text{or} \quad \frac{q}{r} + \frac{q'}{R} &= 0 \quad \dots (2.39)
 \end{aligned}$$

where, r is the distance of point P from charge $+q$ and R is the distance of point P from charge q' .

Equation (2.39) can be written as,

$$qR = -q'r$$

On squaring both sides, we get

$$q^2 R^2 = q'^2 r^2$$

From Fig. 2.6 (b), $R^2 = a^2 + b^2 - 2ab \cos \theta$

and $r^2 = a^2 + d^2 - 2ad \cos \theta$

$$\therefore q^2 (a^2 + b^2 - 2ab \cos \theta) = q'^2 (a^2 + d^2 - 2ad \cos \theta) \quad \dots (2.40)$$

Here ' θ ' is angle between line OP and line OA.

Since equation (2.40) is valid for values of θ , equating coefficient of $\cos \theta$, we get

$$-q^2 (2ab) = (-2ad) q'^2$$

$$\text{or} \quad b = d \left(\frac{q'^2}{q^2} \right)$$

$$\text{Let} \quad \alpha = \frac{q'^2}{q^2} \text{ i.e. } q'^2 = \alpha q^2 \quad \dots (2.41)$$

$$\therefore b = \alpha d \quad \dots (2.42)$$

The remaining terms of the equation (2.40) may be expressed as,

$$q^2 (a^2 + b^2) = q'^2 (a^2 + d^2) \quad \dots (2.43)$$

Using equations (2.41) and (2.42) in equation (2.43) we get

$$q^2 (a^2 + \alpha^2 d^2) = \alpha q^2 (a^2 + d^2)$$

$$\text{or} \quad a^2 + \alpha^2 d^2 = \alpha a^2 + \alpha d^2$$

$$\text{or} \quad d^2 \alpha^2 - (a^2 + d^2) \alpha + a^2 = 0$$

Above equation is quadratic in α , hence, we get

$$\begin{aligned} \alpha &= \frac{(a^2 + d^2) \pm \sqrt{(a^2 + d^2)^2 - 4d^2 a^2}}{2d^2} \\ &= \frac{(a^2 + d^2) \pm (a^2 - d^2)}{2d^2} \\ &= \frac{a^2}{d^2} \text{ or } 1 \end{aligned}$$

Thus, we get $\alpha = 1$ or $\alpha = \frac{a^2}{d^2}$.

If $\alpha = 1$, from equation (2.41) and (2.42), we get

$$q'^2 = q^2 \text{ and } b = d$$

$$\text{or} \quad q' = \pm q \text{ and } b = d$$

The equation $b = d$, suggests that image charge is outside the sphere but image charge must be inside the surface of sphere. So the solution $\alpha = 1$ is discarded.

Using $\alpha = \frac{a^2}{d^2}$ in equations (2.41) and (2.42), we get

$$q'^2 = \frac{a^2}{d^2} q^2 \quad \text{and} \quad b = \frac{a^2}{d^2} d$$

$$\text{or} \quad q' = \pm \frac{a}{d} q \quad \text{and} \quad b = \frac{a^2}{d}$$

According to boundary condition, the potential at point P due to charge q and q' is zero. So to satisfy it, we must choose q' negative. Therefore, we get

$$q' = -\frac{a}{d} q \quad \dots (2.44)$$

and

$$b = \frac{a^2}{d} \quad \dots (2.45)$$

Thus, we get position and value of image charge.

(i) Potential at an exterior point P:

Ignoring the presence of conductor, let us calculate the potential ϕ at point P due to charge q and image charge q' .

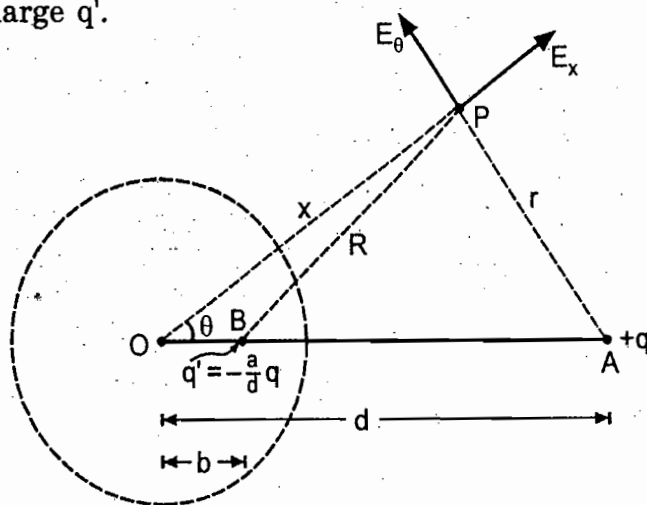


Fig. 2.7 (a)

The potential at point P due to q and q' is,

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r} + \frac{1}{4\pi\epsilon_0} \frac{q'}{R}$$

But,

$$q' = -\frac{a}{d} q$$

$$\therefore \phi = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{a}{dR} \right] \quad \dots (2.46)$$

From Fig. 2.7 (a), using cosine rule, we get

$$r = (x^2 + d^2 - 2xd \cos \theta)^{1/2} \quad \dots (2.47)$$

$$R = (x^2 + b^2 - 2xb \cos \theta)^{1/2}$$

But, $b = \frac{a^2}{d}$

$$\therefore R = \left(x^2 + \frac{a^4}{d^2} - \frac{2xa^2}{d} \cos \theta \right)^{1/2} \quad \dots (2.48)$$

Using equations (2.47) and (2.48) in equation (2.46), we get

$$\phi = \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{(x^2 + d^2 - 2xd \cos \theta)^{1/2}} - \frac{a/d}{\left(x^2 + \frac{a^4}{d^2} - \frac{2a^2}{d} x \cos \theta \right)^{1/2}} \right\} \quad \dots (2.49)$$

This is expression for potential at point P.

(ii) **Electric intensity at point P:**

The component of electric intensity at point in the direction of OP is,

$$E_x = -\frac{\partial \phi}{\partial x} = -\frac{\partial}{\partial x} \left\{ \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(x^2 + d^2 - 2xd \cos \theta)^{1/2}} - \frac{a/d}{\left(x^2 + \frac{a^4}{d^2} - \frac{2a^2}{d} x \cos \theta \right)^{1/2}} \right] \right\}$$

or
$$E_x = -\frac{q}{4\pi\epsilon_0} \left[\frac{d \cos \theta - x}{(x^2 + d^2 - 2xd \cos \theta)^{3/2}} - \frac{\frac{a}{d} \left(\frac{a^2}{d} \cos \theta - x \right)}{\left(x^2 + \frac{a^4}{d^2} - \frac{2a^2}{d} x \cos \theta \right)^{3/2}} \right] \quad \dots (2.50)$$

The component of electric intensity at point P in the direction perpendicular to OP is,

$$\begin{aligned} E_\theta &= -\frac{1}{x} \frac{\partial \phi}{\partial \theta} \\ E_\theta &= -\frac{1}{x} \frac{q}{4\pi\epsilon_0} \frac{\partial}{\partial \theta} \left\{ \frac{1}{(x^2 + d^2 - 2xd \cos \theta)^{1/2}} - \frac{a/d}{\left(x^2 + \frac{a^4}{d^2} - \frac{2a^2}{d} x \cos \theta \right)^{1/2}} \right\} \\ &= -\frac{q}{4\pi\epsilon_0 x} \left\{ \frac{-xd \sin \theta}{(x^2 + d^2 - 2xd \cos \theta)^{3/2}} - \frac{\frac{a}{d} \left(-\frac{a^2}{d} x \sin \theta \right)}{\left(x^2 + \frac{a^4}{d^2} - \frac{2a^2}{d} x \cos \theta \right)^{3/2}} \right\} \\ &= \frac{q}{4\pi\epsilon_0} \left\{ \frac{d \sin \theta}{(x^2 + d^2 - 2xd \cos \theta)^{3/2}} - \frac{\frac{a^3}{d^2} \sin \theta}{\left(x^2 + \frac{a^4}{d^2} - \frac{2a^2}{d} x \cos \theta \right)^{3/2}} \right\} \quad \dots (2.51) \end{aligned}$$

The resultant electric intensity at point P can be calculated by using,

$$E = \sqrt{E_x^2 + E_\theta^2}$$

(iii) Potential and intensity on the surface of sphere :

When the point P is on the surface of sphere, we have $x = a$.

Using $x = a$, equation (2.49) may be written as,

$$\phi = \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{(a^2 + d^2 - 2ad \cos \theta)^{1/2}} - \frac{a/d}{\left(a^2 + \frac{a^4}{d^2} - \frac{2a^3}{d} \cos \theta\right)^{1/2}} \right\}$$

or
$$\phi = \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{(a^2 + d^2 - 2ad \cos \theta)^{1/2}} - \frac{1}{(a^2 + d^2 - 2ad \cos \theta)^{1/2}} \right\} = 0$$

Thus, potential at any point on the surface of sphere is zero.

When the point P is on the surface of sphere, the component E_θ becomes zero and so electric intensity at P is,

$$E = E_x \quad \left(\because E = \sqrt{E_x^2 + E_\theta^2} \right)$$

Using $x = a$ in equation (2.50), we get

$$E = E_x = \frac{q}{4\pi\epsilon_0} \left\{ \frac{a - d \cos \theta}{(a^2 + d^2 - 2ad \cos \theta)^{3/2}} + \frac{\frac{a}{d} \left(\frac{a^2}{d} \cos \theta - a \right)}{\left(a^2 + \frac{a^4}{d^2} - \frac{2a^3}{d} \cos \theta \right)^{3/2}} \right\}$$

Let us simplify second term of above equation,

$$\begin{aligned} \frac{\frac{a^3}{d^2} \cos \theta - \frac{a^2}{d}}{\left(a^2 + \frac{a^4}{d^2} - \frac{2a^3}{d} \cos \theta \right)^{3/2}} &= \frac{\frac{a^3}{d^3} \left(d \cos \theta - \frac{d^2}{a} \right)}{\left(a^2 + \frac{a^4}{d^2} - \frac{2a^3}{d} \cos \theta \right)^{3/2}} \\ &= \frac{d \cos \theta - \frac{d^2}{a}}{\frac{d^3}{a^3} \left(a^2 + \frac{a^4}{d^2} - \frac{2a^3}{d} \cos \theta \right)^{3/2}} \\ &= \frac{d \cos \theta - \frac{d^2}{a}}{\left[\frac{d^2}{a^2} \left(a^2 + \frac{a^4}{d^2} - \frac{2a^3}{d} \cos \theta \right) \right]^{3/2}} \\ &= \frac{d \cos \theta - \frac{d^2}{a}}{[d^2 + a^2 - 2ad \cos \theta]^{3/2}} \end{aligned}$$

$$\therefore E = \frac{q}{4\pi\epsilon_0} \left\{ \frac{a - d \cos \theta + d \cos \theta - \frac{d^2}{a}}{(a^2 + d^2 - 2ad \cos \theta)^{3/2}} \right\}$$

or
$$E = \frac{q}{4\pi\epsilon_0 a} \frac{(a^2 - d^2)}{(a^2 + d^2 - 2ad \cos \theta)^{3/2}} \quad \dots (2.52)$$

$E = E_n$ is the normal to the surface.

Now, the normal component of electric intensity is given by,

$$E = E_n = \frac{\sigma}{\epsilon_0} \quad \dots (2.53)$$

where, σ is the surface density of charge induced on the sphere.

Equating equations (2.52) and (2.53), we get

$$\sigma = \frac{q}{4\pi a} \frac{(a^2 - d^2)}{(a^2 + d^2 - 2ad \cos \theta)^{3/2}}$$

$$\text{or} \quad \sigma = \frac{q}{4\pi a} \frac{(a^2 - d^2)}{r^3} \quad \dots (2.54)$$

where, r is the distance of point P from external charge $+q$.

Since, $d > a$, above expression shows that σ is negative as expected.

(iv) Total induced charge :

If σ is the surface density of induced charge, total induced charge on the surface of sphere is

$$q' = \int_S \sigma dS$$

Let the sphere be divided into number of rings as shown in Fig. 2.7 (b).

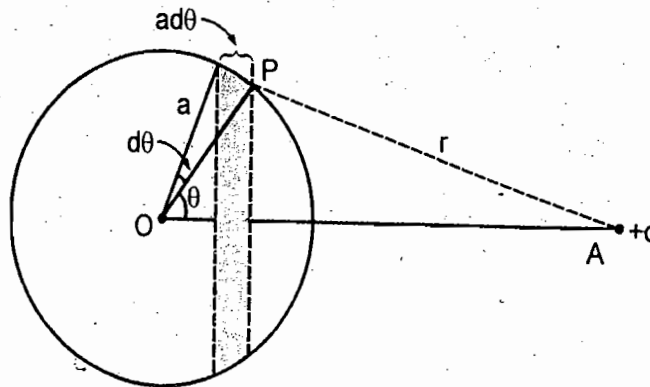


Fig. 2.7 (b)

Radius of the ring = $a \sin \theta$

Thickness of the ring = $a d\theta$

Surface area of the ring = $2\pi a \sin \theta a d\theta$
 $= 2\pi a^2 \sin \theta d\theta$

$$\begin{aligned} \text{Total induced charge } q' &= \int_0^\pi \frac{q}{4\pi a} \frac{(a^2 - d^2)}{r^3} 2\pi a^2 \sin \theta d\theta \\ &= -\frac{q}{2} (d^2 - a^2) a \int_0^\pi \frac{\sin \theta d\theta}{r^3} \end{aligned}$$

From Fig. 2.7 (b),

$$\begin{aligned} r^2 &= a^2 + d^2 - 2ad \cos \theta \\ \therefore 2r dr &= + 2ad \sin \theta d\theta \\ \text{or } \sin \theta d\theta &= \frac{r dr}{ad} \end{aligned}$$

$$\begin{aligned} \therefore \text{Total induced charge } q' &= -\frac{q}{2}(d^2 - a^2) a \int_{d-a}^{d+a} \frac{r dr}{r^3 ad} \\ &= -\frac{q(d^2 - a^2)}{2d} \int_{d-a}^{d+a} \frac{dr}{r^2} \\ &= -\frac{q(d^2 - a^2)}{2d} \left[-\frac{1}{r} \right]_{d-a}^{d+a} \\ &= -q \frac{(d^2 - a^2)}{2d} \left[-\frac{1}{(d+a)} + \frac{1}{(d-a)} \right] \\ &= -\frac{qa}{d} \quad \dots (2.55) \end{aligned}$$

Thus, total induced charge $q' = -\frac{qa}{d}$. This shows that the total induced charge is equal to image charge that we have taken to calculate the field and potential.

(v) **The force between the conducting sphere and the point charge + q kept at point A:**

$$\text{Induced charge} = \text{Image charge} = -\frac{qa}{d}$$

$$\text{Charge kept at point A} = q$$

$$\text{Distance between two charges} \left(d - \frac{a^2}{d} \right)$$

$\therefore$ The force between sphere and point charge

$$\begin{aligned} &= \frac{1(-q^2 a)}{4\pi\epsilon_0 d \left(d - \frac{a^2}{d} \right)^2} \\ F &= \frac{1}{4\pi\epsilon_0} \frac{(-q^2 a) d}{(d^2 - a^2)^2} \quad \dots (2.56) \end{aligned}$$

The negative sign shows that the force is attractive.

2.7 INSULATED CONDUCTING SPHERE NEAR A POINT CHARGE

Consider a point charge '+ q' kept at point A, near an insulated conducting sphere of radius 'a'. Let the point charge '+ q' be at a distance 'd' from the centre 'O' of the sphere (Fig. 2.8).

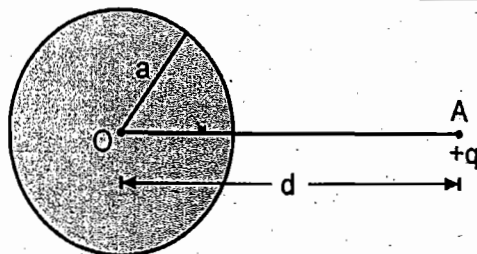


Fig. 2.8

To find the position of image charges, we first assume that conducting sphere is grounded; hence potential at any point on its surface is zero. The charge $+q$ near the sphere induces charge q' and is uniformly distributed over it. In previous example we have seen that the induced charge can be replaced by image charge $-\frac{qa}{d}$ (ignoring the presence of conductor) at point B (Fig. 2.7). Now we insulate the sphere by disconnecting it from ground. Since the sphere is insulated, it must also satisfy the following two conditions :

- (i) The potential of the sphere is uniform all over the surface since the sphere is conducting.
- (ii) No induction takes place on insulated sphere. So induced charge is zero.

To satisfy above two conditions, second image charge of magnitude $+\frac{qa}{d}$ is assumed at centre O. The charge $+\frac{qa}{d}$ at centre O together with $-\frac{qa}{d}$ provides zero charge on the surface and also keeps the surface S (Fig. 2.9) at constant potential.

The potential at any point on surface S is

$$\begin{aligned}\phi &= \frac{\text{charge}}{\text{capacity}} \\ &= \frac{\frac{qa}{d}}{4\pi\epsilon_0 a}\end{aligned}$$

(Since the charge $+q$ and $-\frac{qa}{d}$ together produces zero potential on the surfaces. So only charge $\frac{qa}{d}$ is considered).

$$= \frac{q}{4\pi\epsilon_0 d}$$

Thus, ϕ is constant for a fixed value of 'd'. Hence, both the conditions are satisfied.

The potential at any point outside the sphere can be calculated by considering three charges : $+q$ at point A, image charge $-\frac{qa}{d}$ at B and second image charge $+\frac{qa}{d}$ at centre O. (See. Fig. 2.9).

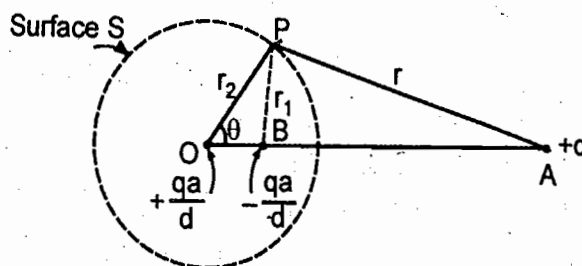


Fig. 2.9

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The potential at point P is,

$$\phi = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} - \frac{\frac{qa}{d}}{r_1} + \frac{\frac{qa}{d}}{r_2} \right]$$

The components of electric intensity E_x and E_θ can be calculated by using the relations,

$$E_x = -\frac{\partial\phi}{\partial x} \quad \text{and} \quad E_\theta = -\frac{1}{x} \frac{\partial\phi}{\partial\theta}$$

(In this problem, we have used symbol r_2 instead of x).

Surface density of charge at point P:

Let σ_p be surface density of charge at point P. The surface density (σ_p) at point P
= Surface density at point P due to charges

$$\left(q, -\frac{qa}{d} \right) + \text{Surface density at point P due to charge } \frac{qa}{d} \quad \dots (2.57)$$

The surface area of spherical surface = $4\pi a^2$

$$\therefore \text{Surface density at point P due to charge } +\frac{qa}{d} = \frac{\text{charge}}{\text{area}} = \frac{q}{4\pi ad} \quad \dots (2.58)$$

We have seen that (section 2.6), surface density at point P due to charges $\left(q, -\frac{qa}{d} \right)$ is,

$$= \frac{-q(d^2 - a^2)}{4\pi a r^3} \quad \checkmark \quad \dots (2.59)$$

Using equations (2.58) and (2.59) in equation (2.57), we get

$$\begin{aligned} \sigma_p &= \frac{-q(d^2 - a^2)}{4\pi a r^3} + \frac{q}{4\pi ad} \\ \sigma_p &= \frac{-q}{4\pi a} \left[\frac{(d^2 - a^2)}{r^3} - \frac{1}{d} \right] \quad \dots (2.60) \end{aligned}$$

In this equation d , a and q are constants. So surface charge density (σ) at a point depends only on r , i.e. the position of the point on the sphere from point A.

Case (a) : When the position of point P is very close to point A i.e. $r \approx d - a$, then equation (2.60) can be expressed as,

$$\begin{aligned} \sigma_p &= -\frac{q}{4\pi a} \left[\frac{(d^2 - a^2)}{(d - a)^3} - \frac{1}{d} \right] \\ \text{or} \quad \sigma_p &= -\frac{q}{4\pi} \left[\frac{(3d - a)}{d(d - a)^2} \right] \quad \dots (2.61) \end{aligned}$$

Since, $d > a$, the term in the square bracket is positive and thus, σ_p is negative.

Case (b) : When the position of points is away from point A i.e. $r \approx d + a$, then equation (2.60) can be expressed as,

$$\sigma_p = -\frac{q}{4\pi a} \left[\frac{(d^2 - a^2)}{(d + a)^3} - \frac{1}{d} \right]$$

$$\text{or} \quad \sigma_p = \frac{q}{4\pi} \left[\frac{3d+a}{d(d+a)^2} \right] \quad \dots (2.62)$$

The term in the square bracket is positive and thus, σ_p is positive.

Thus, from above two cases it is clear that when the point P of surface of sphere goes away from external point charge q , induced charges changes from negative to positive. The surface density of induced charges will be zero when,

$$0 = \frac{-q}{4\pi a} \left[\frac{d^2 - a^2}{r^3} - \frac{1}{d} \right]$$

$$\text{or} \quad \frac{d^2 - a^2}{r^3} = \frac{1}{d}$$

$$\therefore r^3 = d(d^2 - a^2)$$

$$\text{or} \quad r^3 = d(d+a)(d-a)$$

This equation represents equation of circle known as line of zero electrification. On right hand side of this circle, the electrification is negative because the points on the sphere are close to external charge $+q$, while on left hand side of this circle, the electrification is positive.

The force between point charge '+q' and sphere :

While calculating electrical effect of insulated conducting sphere at an external points, we ignore the presence of sphere and replace it by image charges as shown in Fig. 2.9. The force on charge $+q$ is sum of force on it due to charge $+\frac{qa}{d}$ and due to charge $-\frac{qa}{d}$.

Let F_1 be the force between charge $+\frac{qa}{d}$ and charge $+q$. These two charges are separated by distance 'd' (See Fig.2.9).

$$\therefore F_1 = \frac{1}{4\pi\epsilon_0} \frac{\frac{a}{d} q^2}{d^2}$$

The force F_1 is repulsive in nature.

Let F_2 be the force between charge $-\frac{qa}{d}$ and charge $+q$. These two charges are separated by distance $\left(d - \frac{a^2}{d}\right)$.

$$\therefore F_2 = \frac{1}{4\pi\epsilon_0} \frac{-\frac{a}{d} q^2}{\left(d - \frac{a^2}{d}\right)^2}$$

$$F_2 = -\frac{1}{4\pi\epsilon_0} \frac{q^2 ad}{(d^2 - a^2)^2}$$

The force F_2 is attractive in nature.

The resultant force is,

$$\begin{aligned} F &= F_1 + F_2 \\ \therefore F &= \frac{1}{4\pi\epsilon_0} \frac{aq^2}{d^3} - \frac{1}{4\pi\epsilon_0} \frac{q^2 ad}{(d^2 - a^2)^2} \\ F &= -\frac{q^2 a^3}{4\pi\epsilon_0 d^3} \left[\frac{2d^2 - a^2}{(d^2 - a^2)^2} \right] \quad \dots (2.63) \end{aligned}$$

This is expression for force between insulated conducting sphere and charge '+q'.

2.8 THE SPHERE IS CHARGED AND INSULATED AND PLACED NEAR A POINT CHARGE + q

This problem can be solved in a similar manner as that of section 2.7 by putting a charge '+Q' uniformly distributed over the surface of conducting insulated sphere. For an external field this charge '+Q' can be considered to be concentrated at centre O. So by following same procedure adopted in section 2.7, it can be shown that the field and potential

at an external point can be represented as due to a charge +q at point A, image charge $\left(-\frac{qa}{d}\right)$

at point B, charge $\left(Q + \frac{qa}{d}\right)$ at centre O as shown in Fig. 2.10.

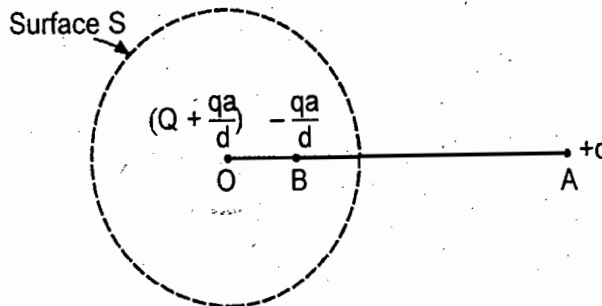


Fig. 2.10

The potential at point P is,

$$\phi_P = \frac{q}{4\pi\epsilon_0 r} + \frac{\left(-\frac{qa}{d}\right)}{4\pi\epsilon_0 r_1} + \frac{1}{4\pi\epsilon_0} \frac{\left(Q + \frac{qa}{d}\right)}{r_2} \quad \dots (2.64)$$

where, r, r_1, r_2 are distances of point P from charge +q, $-\frac{qa}{d}$ and $Q + \frac{qa}{d}$ respectively.

SOLVED EXAMPLES

Example 1 : Two infinite parallel surfaces at $x = 0$ and $x = a$ are maintained at potential ϕ_0 and ϕ_a respectively. Obtain the variation of the potential between the plates and in the outside region.

Solution : Figure 2.11 shows two infinite parallel surfaces at $x = 0$ and $x = a$ maintained at potential ϕ_0 and ϕ_a respectively.

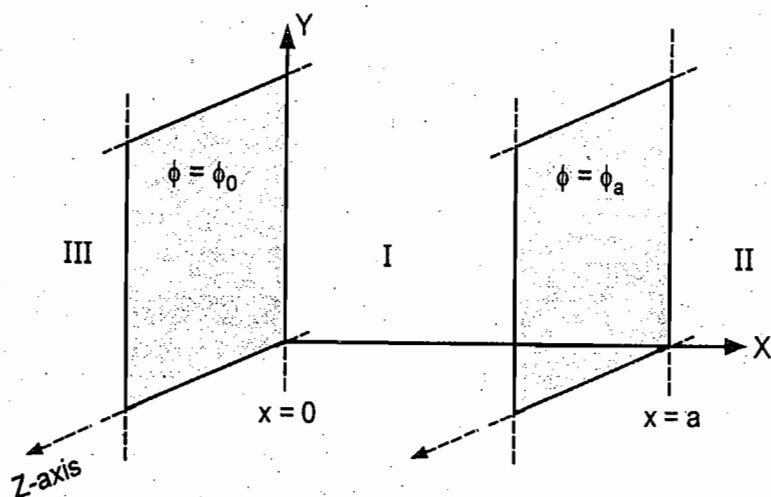


Fig. 2.11

In the given problem, there are no free charges involved. So we can use Laplace's equation to obtain the solution.

Laplace's equation is $\nabla^2\phi = 0$

In the given problem, surface involved on which potentials are specified are parallel to co-ordinate axes y and z. So we can use cartesian co-ordinate system to obtain the solution of Laplace's equation. Laplace's equation in cartesian co-ordinate system is,

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

The plates are of infinite extent in y and z direction (Fig. 2.11). Since both the plates are at constant potential, ϕ cannot depend on y and z co-ordinates. So $\frac{\partial\phi}{\partial x} = \frac{\partial\phi}{\partial y} = 0$. The Laplace's equation can be written as,

$$\frac{\partial^2\phi(x)}{\partial x^2} = 0$$

$$\text{or} \quad \frac{\partial^2\phi}{\partial x^2} = 0 \quad (\because x \text{ is only variable, } \phi = \phi(x))$$

The general solution of this equation is,

$$\phi(x) = C_1x + C_2 \quad \dots (2.65)$$

From Fig. 2.11, the entire space can be divided into three distinct regions.

(i) Region I : $0 < x < a$

(ii) Region II : $x > a$ and

(iii) Region III : $x < 0$

(i) In region I, the boundary conditions are –

$$\phi(0) = \phi_0 \text{ at } x=0$$

$$\text{and } \phi(a) = \phi_a \text{ at } x=a$$

At $x=0$, equation (2.65) may be written as,

$$\phi(0) = C_2$$

$$\therefore C_2 = \phi_0$$

At $x = a$, equation (2.65) may be written as,

$$\phi(a) = C_1 a + C_2$$

Using $C_2 = \phi_0$ and $\phi(a) = \phi_a$, we get

$$\phi_a = C_1 a + \phi_0$$

$$\text{or } C_1 = \left(\frac{\phi_a - \phi_0}{a} \right)$$

Putting the values of C_1 and C_2 in equation (2.65), we get

$$\phi(x) = \left(\frac{\phi_a - \phi_0}{a} \right) x + \phi_0$$

This equation represents equation of straight line with slope $\left(\frac{\phi_a - \phi_0}{a} \right)$. The variation of potential w.r.t. x is shown in Fig. 2.12.

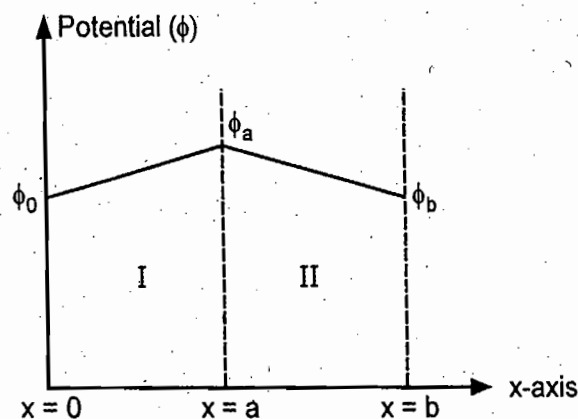


Fig. 2.12 : Variation of potential w.r.t. x

(ii) In region II, the boundary condition is,

$$\phi(a) = \phi_a \text{ at } x = a$$

We have only one boundary condition, which is insufficient to determine $\phi(x)$ in the region II. At $x = a$, equation (2.65) can be written as,

$$\phi_a = C_1 a + C_2$$

$$\text{or } C_2 = \phi_a - C_1 a$$

Using value of C_2 in equation (2.65), we get

$$\phi(x) = C_1 x + (\phi_a - C_1 a)$$

$$\text{or } \phi(x) = C_1 (x - a) + \phi_a \quad \dots (2.66)$$

This equation shows that $|\phi(x)| \rightarrow \infty$ as $x \rightarrow \infty$.

The alternative to obtain the potential in region II is to keep the potential function $\phi(x)$ [shown by equation (2.66)] as it is or assume some constant potential say ϕ_b on imaginary plane at $x = b$. Thus, the boundary conditions are -

$$\phi(a) = \phi_a \text{ at } x = a$$

$$\text{and } \phi(b) = \phi_b \text{ at } x = b$$

At $x = a$, equation (2.65) is written as,

$$\phi_a = C_1 a + C_2$$

$$\text{or } C_2 = \phi_a - C_1 a$$

At $x = b$, equation (2.65) is written as,

$$\phi_b = C_1 b + C_2$$

Using value of C_2 , we get

$$\phi_b = C_1 b + \phi_a - C_1 a$$

or

$$C_1 = \frac{(\phi_b - \phi_a)}{(b - a)}$$

Now the equation for C_2 can be written as,

$$C_2 = \phi_a - \frac{(\phi_b - \phi_a)}{(b - a)} a$$

Using values of C_1 and C_2 in equation (2.65), we get

$$\phi(x) = \left[\frac{\phi_b - \phi_a}{b - a} \right] x + \phi_a - \left[\frac{\phi_b - \phi_a}{b - a} \right] a$$

or

$$\phi(x) = (\phi_b - \phi_a) \frac{(x - a)}{(b - a)} + \phi_a \quad \dots (2.67)$$

From this equation, the variation of ϕ w.r.t. x in region II is as shown in Fig. 2.12. It can be shown that, the situation in region III is similar to that in region II.

Example 2 : Find the field intensity and potential between anode and grounded cathode of a working vacuum diode (Assume uniform space charge in between anode and cathode).

Solution : The vacuum diode is as shown in Fig. 2.13.

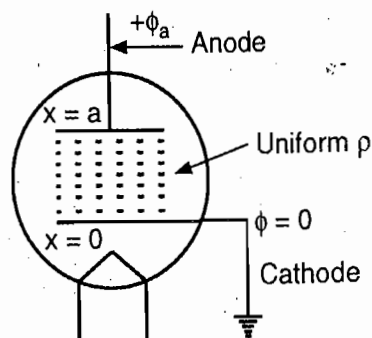


Fig. 2.13

The anode and grounded cathode are separated by distance a and in between them let ρ be the uniform volume charge density. The Poisson's equation in this region is,

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}$$

In the given problem, since both anode and cathode are at constant potential and ϕ varies along the x -axis, above equation in cartesian co-ordinate system may be written as,

$$\frac{d^2 \phi}{dx^2} = -\frac{\rho}{\epsilon_0}$$

where,

$$\phi = \phi(x)$$

Integrating above equation, we get

$$\frac{d\phi}{dx} = -\frac{\rho}{\epsilon_0} x + C_1$$

Again integrating above equation, we get

$$\phi(x) = -\frac{\rho}{\epsilon_0} \frac{x^2}{2} + C_1 x + C_2$$

... (2.68)

The values of constants C_1 and C_2 can be obtained using boundary conditions.

The boundary conditions are : $\phi = 0$, at $x = 0$ and $\phi = \phi_a$, at $x = a$

Using $\phi = 0$ at $x = 0$ in equation (2.68), we get

$$C_2 = 0$$

Using $\phi = \phi_a$ at $x = a$ in equation (2.68), we get

$$\phi_a = -\frac{\rho}{\epsilon_0} \frac{a^2}{2} + C_1 a + C_2$$

Using $C_2 = 0$, we get

$$\phi_a = -\frac{\rho a^2}{2\epsilon_0} + C_1 a$$

or

$$C_1 a = \phi_a + \frac{\rho a^2}{2\epsilon_0}$$

$\therefore$

$$C_1 = \left(\frac{\phi_a}{a} + \frac{\rho a}{2\epsilon_0} \right)$$

Using values of C_1 and C_2 in equation (2.68), we get

$$\phi(x) = -\frac{\rho}{2\epsilon_0} x^2 + \left(\frac{\phi_a}{a} + \frac{\rho a}{2\epsilon_0} \right) x$$

... (2.69)

We know that,

$$\vec{E} = -\vec{\nabla} \phi$$

where,

$$\vec{\nabla} = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

In this example,

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial z} = 0$$

$$\therefore \vec{E} = -i \frac{\partial}{\partial x} \phi$$

... (2.70)

Using equation (2.69) in equation (2.70), we get

$$\vec{E} = -i \frac{\partial}{\partial x} \left[-\frac{\rho}{2\epsilon_0} x^2 + \left(\frac{\phi_a}{a} + \frac{\rho a}{2\epsilon_0} \right) x \right]$$

or

$$\vec{E} = i \left(\frac{\rho}{\epsilon_0} x - \frac{\phi_a}{a} - \frac{\rho a}{2\epsilon_0} \right)$$

The magnitude of electric intensity is,

$$|\vec{E}| = \frac{\rho}{\epsilon_0} x - \left(\frac{\phi_a}{a} + \frac{\rho a}{2\epsilon_0} \right)$$

This represents equation of straight line. The variation of E and ϕ between cathode and anode are shown in Fig. 2.14 (b) and 2.14 (a) respectively.

8)

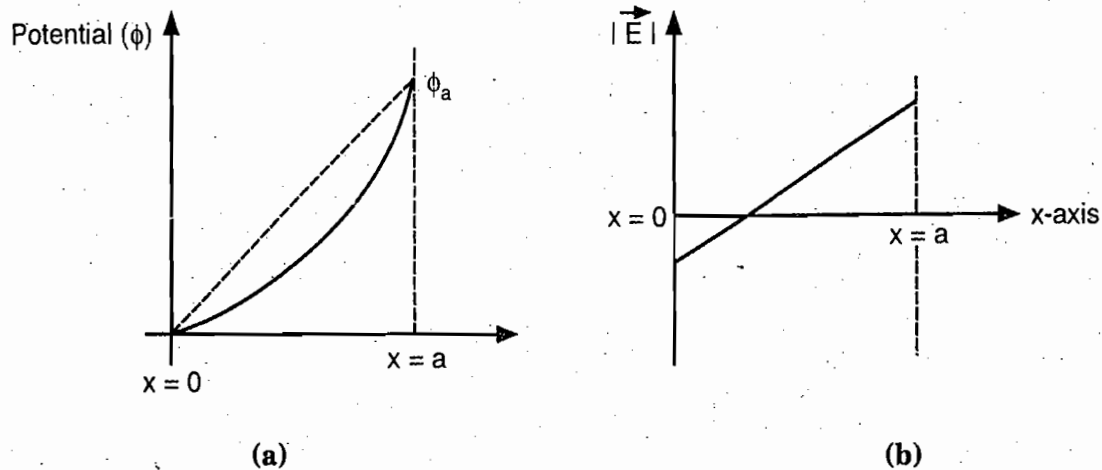


Fig. 2.14

Example 3 : Two semi-infinite grounded metal plates lie parallel to XZ plane, one at $y = 0$ and other at $y = b$. The left end, at $x = 0$ is closed off with an infinite conducting strip insulated from the two plates and maintained at a constant potential ϕ_0 . Find the solution for potential inside the enclosure.

Solution : Fig. 2.15 shows two semi-infinite grounded metal plates parallel to XZ plane. These plates are separated by a distance 'b'. The left end, at $x = 0$, is closed off with a conducting strip insulated from the two plates and maintained at a constant potential ϕ_0 . We have to find the potential at any point between the plates.

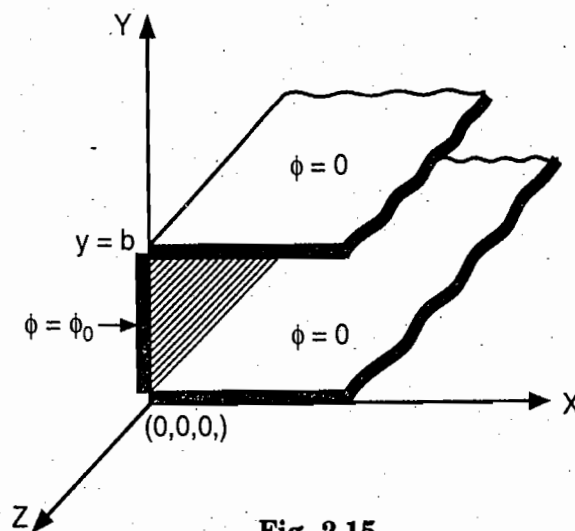


Fig. 2.15

This problem can be solved by applying Laplace's equation in cartesian co-ordinate system.

Laplace's equation in cartesian co-ordinate system is,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

Since the configuration is independent of z co-ordinate i.e. $\frac{\partial \phi}{\partial z} = 0$, the Laplace's equation is written as,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

where, $\phi = \phi(x, y)$

We know that solution of this equation is,

$$\phi = X(x) Y(y)$$

where, $X(x) = A e^{kx} + B e^{-kx}$ $\phi(0) = A + B = \phi_0$... (2.71)

and $Y(y) = C \sin ky + D \cos ky$... (2.72)

The boundary conditions involved in the problem are* -

(i) $\phi = 0$ at $y = 0$,

(ii) $\phi = 0$ at $y = b$,

(iii) $\phi = \phi_0$ at $x = 0$ and

(iv) $\phi \rightarrow 0$ when $x \rightarrow \infty$

According to the first boundary condition $\phi = 0$ at $y = 0$, we have $\phi = 0$ when $Y = 0$.

Using $Y = 0$ and $y = 0$ in equation (2.72), we get

$$0 = C \times 0 + D \times 1$$

or $D = 0$

Using $D = 0$, equation (2.72) may be written as,

$$Y(y) = C \sin ky \quad \dots (2.73)$$

According to boundary condition $\phi = 0$ at $y = b$, we have $\phi = 0$ when $Y = 0$.

Using $Y = 0$ and $y = b$ in equation (2.73), we get

$$0 = C \sin kb$$

In above equation $C \neq 0$ hence $\sin kb = 0$

$$\therefore kb = n\pi$$

or $k = \frac{n\pi}{b}$

Applying boundary condition $\phi \rightarrow 0$ when $x \rightarrow \infty$ to equation (2.71), we get $A = 0$.

Thus, we get $D = 0$, $k = \frac{n\pi}{b}$ and $A = 0$.

Using above values, equations (2.71) and (2.72) can be written as,

$$X(x) = B e^{-\frac{n\pi}{b} x}$$

$$Y(y) = C \sin \frac{n\pi}{b} y$$

* The equation of lower plane is $y = 0$, upper plane is $y = b$ and the equation of plane connecting these two planes is $x = 0$.

But, $\phi = X(x) Y(y)$

$$\therefore \phi = BC e^{-\frac{n\pi}{b}x} \sin\left(\frac{n\pi}{b}y\right) \quad \dots (2.74)$$

This equation satisfies all the boundary conditions except boundary condition (iii).

The most general solution may be written as,

$$\phi(x, y) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{b}y\right) e^{-\left(\frac{n\pi}{b}\right)x} \quad \dots (2.75)$$

where, C_n is another arbitrary constant.

To determine the coefficients C_n , let us use boundary condition $\phi = \phi_0$ at $x = 0$.

Using $\phi = \phi_0$ at $x = 0$, equation (2.75) can be written as,

$$\phi(0, y) = \phi_0 = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{b}y\right) \quad \dots (2.76)$$

The right hand side of above equation is Fourier series. To obtain C_n , multiply above equation by $\sin \frac{m\pi}{b}y$ and integrate from 0 to b .

$$\therefore \int_0^b \phi_0 \sin \frac{m\pi}{b}y dy = \sum_{n=1}^{\infty} C_n \int_0^b \sin\left(\frac{n\pi}{b}y\right) \sin \frac{m\pi}{b}y dy$$

We know that,

$$\int_0^b \sin \frac{n\pi}{b}y \sin \frac{m\pi}{b}y dy = \begin{cases} 0 & \text{if } m \neq n \\ b/2 & \text{if } m = n \end{cases}$$

Thus, all the terms in the series drop out, except $m = n$.

$$\therefore \int_0^b \phi_0 \sin \frac{n\pi y}{b} dy = C_n \frac{b}{2} \quad (\because m = n)$$

or

$$\begin{aligned} C_n &= \frac{2}{b} \int_0^b \phi_0 \sin \frac{n\pi y}{b} dy \\ &= \frac{2}{b} \phi_0 \left[\cos \frac{n\pi y}{b} \left(-\frac{b}{n\pi} \right) \right]_0^b \\ &= \frac{2\phi_0}{b} \left(-\frac{b}{n\pi} \right) \left[\cos \frac{n\pi b}{b} - 1 \right] \\ &= \frac{2\phi_0}{n\pi} (1 - \cos n\pi) \end{aligned}$$

If n is even, $1 - \cos n\pi = 0$ and if n is odd,

$$1 - \cos n\pi = 2$$

$\therefore C_n = \frac{4\phi_0}{n\pi}$, if n is odd, otherwise zero.

Using $C_n = \frac{4\phi_0}{n\pi}$, equation (2.75) can be written as,

$$\phi(x, y) = \frac{4\phi_0}{\pi} \sum_{n=1,3,5}^{\infty} \frac{1}{n} \sin \frac{n\pi}{b} y e^{-\frac{n\pi}{b} x}$$

This is required solution for potential inside the enclosure at a point having co-ordinates (x, y) .

Example 4 : An infinitely long rectangular metal pipe is grounded, but one end is maintained at a constant potential ϕ_0 . Find the solution for potential inside the pipe.

Solution : Consider an infinitely long rectangular metal pipe as shown in Fig. 2.16. Let the length of sides in y and z directions are a and b respectively. Here all the surfaces of pipe are at zero potential except the surface at $x = 0$. This surface is at a constant potential ϕ_0 .

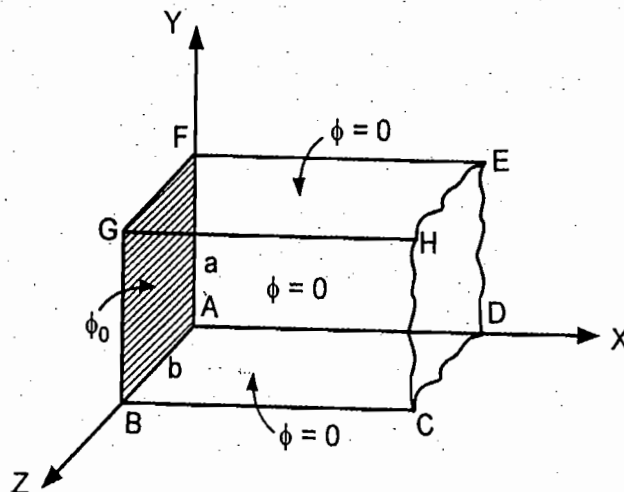


Fig. 2.16

This problem can be solved by using Laplace's equation in cartesian co-ordinate system.

The Laplace's equation is,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

... (2.77)

The boundary conditions are *

- (i) $\phi = 0$ at $y = 0$,
- (ii) $\phi = 0$ at $y = a$,
- (iii) $\phi = 0$ at $z = 0$,
- (iv) $\phi = 0$ at $z = b$,
- (v) $\phi = \phi_0$ at $x = 0$ and
- (vi) $\phi \rightarrow 0$ when $x \rightarrow \infty$.

* The equation of plane ABCD is $y = 0$, EFGH is $y = a$, ABGF is $x = 0$, ADEF is $z = 0$ and GBCH is $z = b$.

The general solution of Laplace's equation is,

$$\phi(x, y, z) = X(x) Y(y) Z(z)$$

where, $X(x) = A e^{\sqrt{k^2 + l^2} x} + B e^{-\sqrt{k^2 + l^2} x}$... (2.78)

$$Y(y) = C \sin ky + D \cos ky$$
 ... (2.79)

and $Z(z) = E \sin lz + F \cos lz$... (2.80)

Using boundary condition (i) i.e. $\phi = 0$ at $y = 0$ in equation (2.79), we get

$$D = 0.$$

Using boundary condition (iii) i.e. $\phi = 0$ at $z = 0$, in equation (2.80), we get

$$F = 0$$

Using boundary condition (vi) i.e. $\phi \rightarrow 0$, when $x \rightarrow \infty$, in equation (2.78), we get

$$A = 0$$

Substituting $A = D = F = 0$ in equations (2.78), (2.79) and (2.80), we get

$$X(x) = B e^{-\sqrt{k^2 + l^2} x}$$
 ... (2.81)

$$Y(y) = C \sin ky$$
 ... (2.82)

and $Z(z) = E \sin lz$... (2.83)

Using boundary condition (ii) i.e. $\phi = 0$ at $y = a$ in equation (2.82), we get

$$0 = C \sin ka \quad (\because \phi = 0 \text{ when } y = a)$$

But, $C \neq 0$, hence, $\sin ka = 0$.

$$\therefore ka = n\pi$$

or $k = \frac{n\pi}{a}$... (2.84)

where, n is positive integer.

Similarly, using boundary (iv) i.e. $\phi = 0$ at $z = b$ in equation (2.83), we can obtain

$$l = \frac{m\pi}{b}$$
 ... (2.85)

where, m is positive integer.

Now substituting $k = \frac{n\pi}{a}$, $l = \frac{m\pi}{b}$ in equations (2.81), (2.82) and (2.83), we get

$$X(x) = B e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x}$$

$$Y(y) = C \sin \frac{n\pi}{a} y$$

$$Z(z) = E \sin \frac{m\pi}{b} z$$

$$\therefore \phi(x, y, z) = B C E e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x} \sin \frac{n\pi}{a} y \sin \frac{m\pi}{b} z$$

This equation satisfies all the boundary conditions except boundary condition (v). The most general solution may be written as,

$$\phi(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \sin \left(\frac{n\pi}{a} \right) y \sin \left(\frac{m\pi}{b} \right) z e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x} \quad \dots (2.86)$$

Using boundary condition (v) i.e. $\phi = \phi_0$ at $x = 0$ in above equation, we get

$$\phi(0, y, z) = \phi_0(y, z) = \sum_{n,m=1}^{\infty} C_{n,m} \sin \left(\frac{n\pi}{a} \right) y \sin \left(\frac{m\pi}{b} \right) z$$

The right hand side of above equation is Fourier series. To obtain $C_{n,m}$, multiply above equation by $\sin \left(\frac{n'\pi}{a} \right) y \sin \left(\frac{m'\pi}{b} \right) z$ where, n' and m' are arbitrary positive integers and integrate between the proper limits.

Thus,

$$\begin{aligned} \phi_0(y, z) \int_0^a \sin \frac{n'\pi}{a} y \, dy \int_0^b \sin \frac{m'\pi}{b} z \, dz &= \sum_{n,m=1}^{\infty} C_{n,m} \int_0^a \sin \left(\frac{n\pi}{a} \right) y \sin \left(\frac{n'\pi}{a} \right) y \, dy \\ &\quad \times \int_0^b \sin \left(\frac{m\pi}{b} \right) z \sin \left(\frac{m'\pi}{b} \right) z \, dz \end{aligned}$$

$$\text{But, } \int_0^a \sin \left(\frac{n\pi}{a} \right) y \sin \left(\frac{n'\pi}{a} \right) y \, dy = \begin{cases} 0, & \text{if } n \neq n' \\ \frac{a}{2}, & \text{if } n = n' \end{cases}$$

$$\text{and } \int_0^b \sin \left(\frac{m\pi}{b} \right) z \sin \left(\frac{m'\pi}{b} \right) z \, dz = \begin{cases} 0, & \text{if } m \neq m' \\ \frac{b}{2}, & \text{if } m = m' \end{cases}$$

Thus, all the terms in the series drop out, except $m = m'$ and $n = n'$.

$$\therefore \phi_0(x, y) \int_0^a \sin \frac{n\pi y}{a} \, dy \int_0^b \sin \frac{m\pi z}{b} \, dz$$

$$\begin{aligned}
 &= C_{n,m} \frac{ab}{4} \\
 \text{or} \quad C_{n,m} &= \frac{4}{ab} \phi_0 \int_0^a \sin \frac{n\pi y}{a} dy \int_0^b \sin \frac{m\pi z}{b} dz \\
 &= \begin{cases} \frac{16\phi_0}{nm\pi^2}, & \text{if } n \text{ and } m \text{ are odd} \\ 0 & \text{if } n \text{ and } m \text{ are even} \end{cases}
 \end{aligned}$$

Using, $C_{n,m} = \frac{16\phi_0}{nm\pi^2}$ in equation (2.86), we get

$$\phi(x, y, z) = \frac{16\phi_0}{\pi^2} \sum_{n,m=1,3,5,\dots}^{\infty} \frac{1}{nm} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi z}{b}\right) e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x}$$

This is required expression for potential inside the pipe at a point having co-ordinates (x, y, z) .

Example 5 : Point charges $+q$ and $-2q$ are kept near an infinite grounded conducting plane at a perpendicular distance ' $3d$ ' and ' d ' respectively from it. Find the force on the charge $+q$.

Solution : Using the method of electrical images the given problem can be replaced by set of four charges consisting of two image charges $-q$ and $2q$ and actual charges $+q$ and $-2q$ as shown in Fig. 2.17. (The presence of grounded conducting plane is neglected).

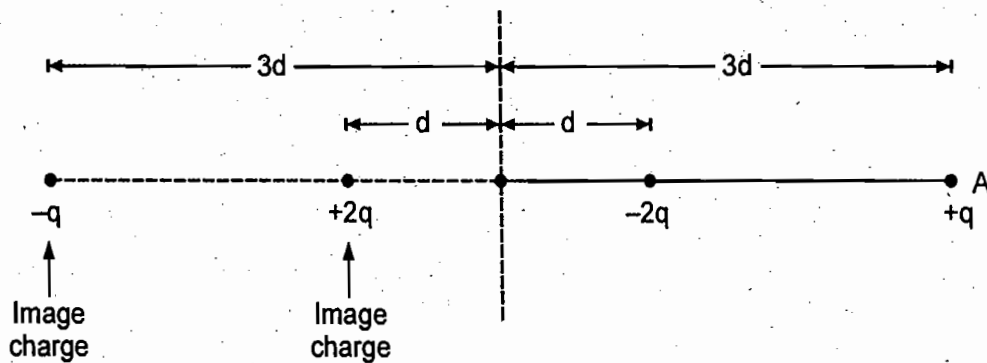


Fig. 2.17

The force on charge $+q$ at point A due to all other charges

$$\begin{aligned}
 &= \text{Force exerted by charge } -2q \text{ on } +q \\
 &+ \text{Force exerted by charge } +2q \text{ on } +q \\
 &+ \text{Force exerted by charge } -q \text{ on } +q \\
 F &= F_1 + F_2 + F_3
 \end{aligned}$$

$$\begin{aligned}
 F &= \frac{1}{4\pi\epsilon_0} \frac{q(-2q)}{(2d)^2} + \frac{1}{4\pi\epsilon_0} \frac{q(2q)}{(4d)^2} + \frac{1}{4\pi\epsilon_0} \frac{q(-q)}{(6d)^2} \\
 &= \frac{q^2}{4\pi\epsilon_0} \left[\frac{-2}{4d^2} + \frac{2}{16d^2} - \frac{1}{36d^2} \right] \\
 &= \frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2} \left[\frac{-2}{1} + \frac{1}{2} - \frac{1}{9} \right] \\
 &= \frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2} \left(\frac{-29}{18} \right) \\
 &= -\frac{1}{4\pi\epsilon_0} \frac{29q^2}{72d^2}
 \end{aligned}$$

This is total force on charge + q due to charge - 2q and conducting plane.

QUESTIONS

1. Obtain Poisson's and Laplace's equations.
2. Obtain the solution of Laplace's equation in cartesian co-ordinate system.
3. Obtain the solution of Poisson's equation.
4. Explain the method of electrical images for the solution of problems in electrostatics.
5. What are necessary conditions while solving electrostatic problems by using method of electrical images.
6. Outline the basic principles of electrical images. A point charge '+ q' is placed at a perpendicular distance d from infinite grounded conducting plane, obtain an expression for surface density of induced charges on the plane.
7. A point charge '+q' is placed at a distance 'd' from the centre of grounded conducting sphere of radius 'a'. Find the position of the image charge and show that surface charge density of induced charges is $\sigma = \frac{q}{4\pi a} \left(\frac{a^2 - d^2}{r^3} \right)$, where r is the distance of point on the sphere from external charge + q. Also calculate the force between the conducting sphere and point charge + q.
8. A point charge '+ q' is placed at a distance 'd' from the centre of grounded conducting sphere of radius a. Using method of electrical images, obtain an expression for electric intensity at any point on the surface of sphere.
9. What is meant by an electrical image ? A point charge q is kept at a distance 'd' from the centre of grounded conducting sphere of radius 'a'. Show that induced charge on sphere is $q' = -q \frac{a}{d}$.

10. A point charge + q is kept at a distance 'd' from the centre of insulated conducting sphere of radius 'a'. Obtain an expression for the surface density of induced charges at a point on the surface of sphere. Comment on your result.
11. A point charge + q is kept at a distance d from the centre of insulated, uncharged conducting sphere of radius 'a'. Find the rise in potential due to the presence of charge and show that there is an attractive force between the external charge and the sphere equal to,
$$-\frac{1}{4\pi\epsilon_0} \frac{q^2 a^3}{d^3} \left[\frac{(2d^2 - a^2)}{(d^2 - a^2)^2} \right]$$
12. The sphere is charged and insulated and placed near a point charge + q. Find the potential at any exterior point.

PROBLEMS

1. Two infinite parallel surfaces at $x = 0$ and $x = a$ are maintained at potentials ϕ_0 and ϕ_a respectively. Obtain the variation of field between the plates and in outside regions.
2. A rectangular metal pipe, running parallel to z -axis (from $-\infty$ to $+\infty$) has three grounded sides, at $y = 0$, $y = b$, and $x = 0$. The fourth side at $x = a$ is maintained at a constant potential ϕ_0 (y). Obtain a general formula for the potential within the pipe.
3. Two infinitely long grounded metal plates at $x = 0$ and $x = a$ are connected at $y = \pm b$ by metal strips maintained at a constant potential ϕ_0 as shown in Fig. 2.18. Find the solution for potential inside the enclosure.

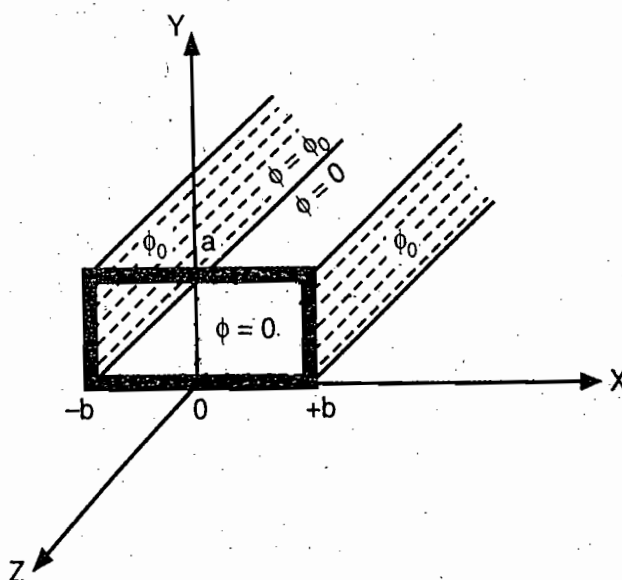


Fig. 2.18

5. A conducting rectangular box (sides of length a , b and c) of which all the sides except one are grounded and the remaining side is at a potential $\phi_0(x, y)$ as shown in Fig. 2.19. Find the solution for potential inside the rectangular box.

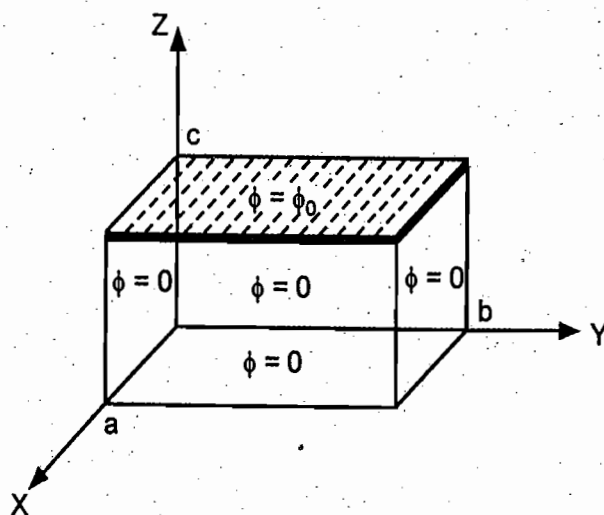


Fig. 2.19

6. A cubical box having length of each side 'a' of which all the sides except one are grounded and the remaining side is at a potential ϕ_0 as shown in Fig. 2.20. Find the solution for potential inside the cube.

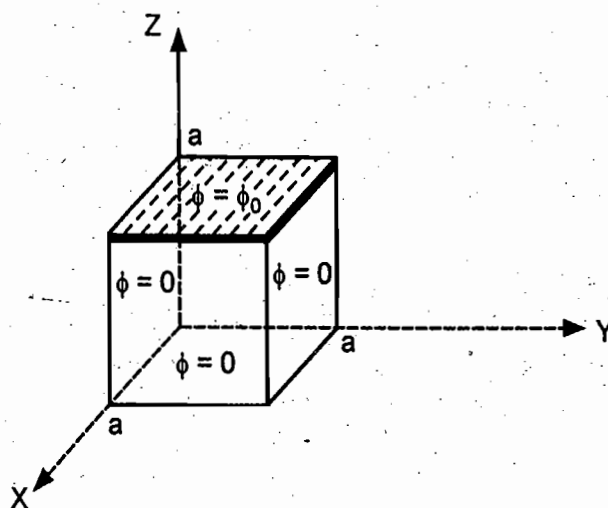


Fig. 2.20

7. A point charge $+q$ is placed at a distance 'd' from the centre of grounded conducting sphere of radius 'a'. If the sphere is not earthed, it would have certain constant potential, say ϕ_0 , instead of zero potential. To solve this problem what image charge should you use and where should you put it?
8. Using Laplace's equation in cartesian co-ordinate, calculate the potential at any point between the plates of a parallel plate condenser and the surface charge density on a plate of a condenser.

*The mag of molecule p
Part 1 p=260 E
atomic or molecular*

ELECTROSTATIC FIELD IN DIELECTRICS

INTRODUCTION

So far we have studied electric field or potential produced by a given charge distribution in air or vacuum. In the present chapter, we shall study electric field in the presence of dielectric or insulating materials like glass, mica, water, etc. We know about conductors that these are the substances which contain large number of free electrons. When a conductor is placed in an electric field, the electrons move against the direction of field and gets distributed in such a way that net electric field inside it is zero. In insulating or dielectric materials, by contrast, positive or negative charges in each atom or molecule are bound together by a strong force and hence these charges are not free to move from one place to another. When such a material is kept in an external electric field, the charges undergo minute displacement, giving rise to an electric displacement which is proportional to the applied electric field. The atoms or molecules of dielectric material are of two types : (i) Polar molecule and (ii) Non-polar molecule.

(i) **Polar molecule** : A polar molecule is a molecule in which the effective centre of positive charge does not coincide with effective centre of negative charge. The molecule thus has permanent dipole moment. Some common examples of polar molecules are HCl, CO, H₂O, NH₃, etc.

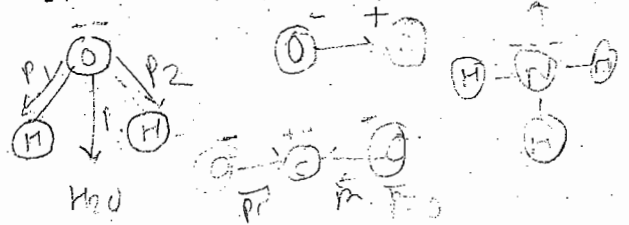
(ii) **Non-polar molecules** : A non-polar molecule is a molecule in which centre of positive charges exactly coincide with centre of negative charges. The molecule thus does not have permanent dipole moment e.g. H₂, N₂, O₂, CO₂, etc.

When non-polar molecule is placed in an external electric field $\vec{E}$, the electric field causes a force to be exerted on each charged centres. Positive charged centre is pushed along the direction of field and negative charged centre opposite to the direction of the field. Thus, the molecule or atom acquires an induced electric dipole moment along the direction of field. The magnitude of the induced dipole moment in the atom or molecule is proportional to the applied field i.e.

$$\vec{p} \propto \vec{E} \quad \text{or} \quad \vec{p} = \alpha \epsilon_0 \vec{E}$$

where, α is constant called atomic or molecular polarisability. α has dimensions of volume and its unit is m³.

(3.1)



In a dielectric material of polar molecules, the dipole moments of individual atoms or molecules are randomly oriented. When such a molecule is placed in an external electric field, there is a tendency for the permanent dipole to align itself along the direction of field direction.

3.1 ELECTRIC POLARISATION OF MATTER

In the introductory section of this chapter, we have studied the effect of an external electric field on an individual atom or molecule. Now, let us study the effect of an external electric field on a piece of dielectric material.

In a dielectric material of polar molecules, each molecule possesses net dipole moment. The dipole moments of molecules are randomly oriented giving the dipole moment of the material zero [Fig. 3.1 (a)]. When such a dielectric is placed in an external electric field, the dipole moments of molecules tend to be aligned along the direction of field as shown in [Fig. 3.1 (b)]. Due to this the material acquires a resultant dipole moment in the direction of the field. If the external applied field is increased, more and more molecules get aligned along the direction of field and hence the net dipole moment increases. As soon as external field is removed, the dipole moments of molecules get distributed at random, so that the net dipole moment of the material becomes zero again.

Now, let us study the effect of external electric field on a dielectric material of non-polar molecules.

Without external electric field, the molecules of the non-polar dielectric does not possess individual magnetic dipole moments. Therefore, the dielectric material has no electric dipole moment. However, when an external electric field is applied, the centre of negative charges of each molecule or atom is slightly displaced in the opposite direction of the field, while the centre of positive charges is equally displaced in the direction of the field. [Fig. 3.2 (b)]. Thus, each molecule acts as an electric dipole and its dipole moment is called induced dipole moment. Due to this the material acquires a net electric dipole moment in the direction of external electric field. The induced dipole moment increases with the increase in external electric field. As soon as the external field is removed, centres of positive and negative charges coincide i.e. the dipole moment of each molecule becomes zero and so the dipole moment, material becomes zero again.

In short, any dielectric material (polar or non-polar) acquires an electric dipole moment in the direction of field when it is placed in external electric field. This phenomenon is known as polarisation of matter.

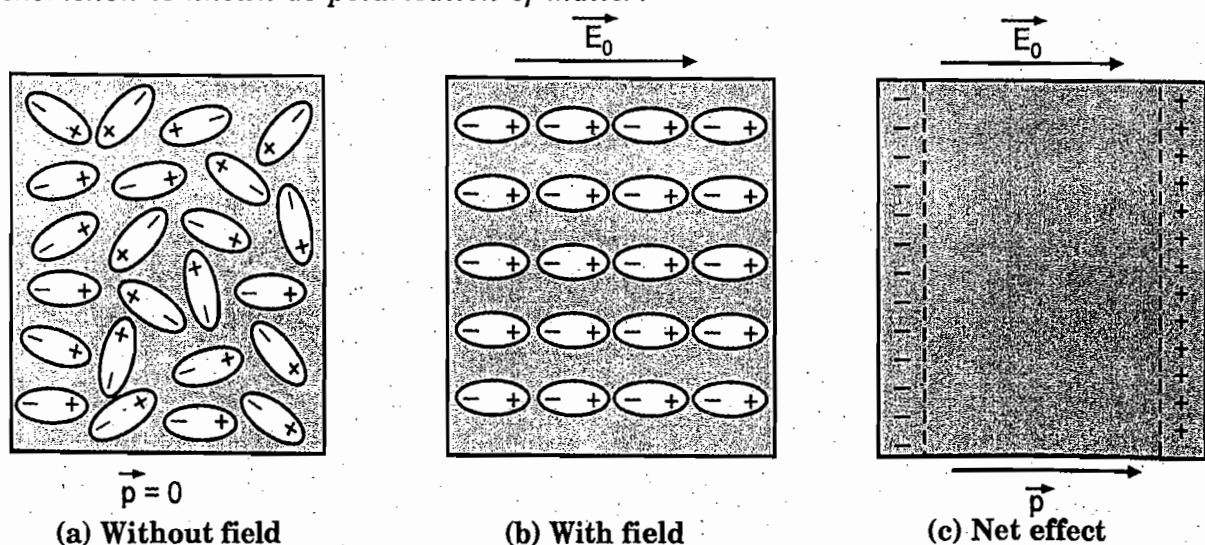


Fig. 3.1 : Polar dielectric material

Whenever an external electric field is applied across the faces of a dielectric material, there is minute displacement of charges in the material. This displacement of bound charges constitutes small current in the dielectric material.

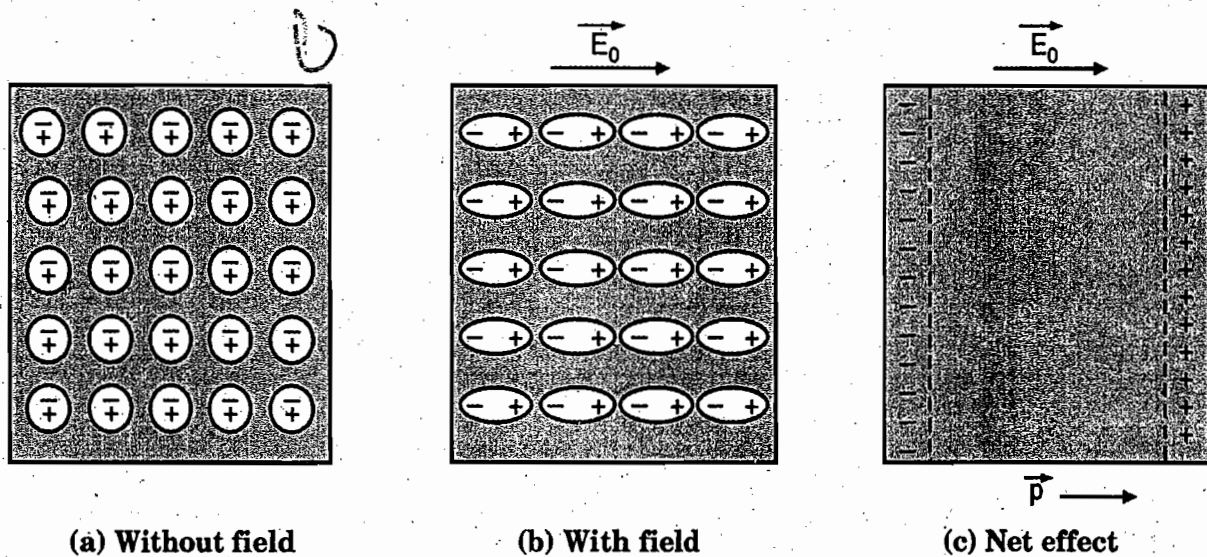


Fig. 3.2 : Non-polar dielectric material.

This current is called displacement current which lasts so long as the field is changing. When the field reaches its steady value, there is no further displacement of charges and displacement current becomes zero.

When a dielectric material is polarised, the molecular dipoles get oriented along the field direction. [Fig. 3.1 (b) or Fig. 3.2 (b)]. The surface, which is on positive side of electric field, acquires net negative charge ($-q'$) and opposite face acquires net positive charge (q') as shown in Fig. 3.3. The net charge in the interior of slab is zero because the positive charge and negative charge neutralise each other. [Fig. 3.1 (c) and 3.2 (c)].

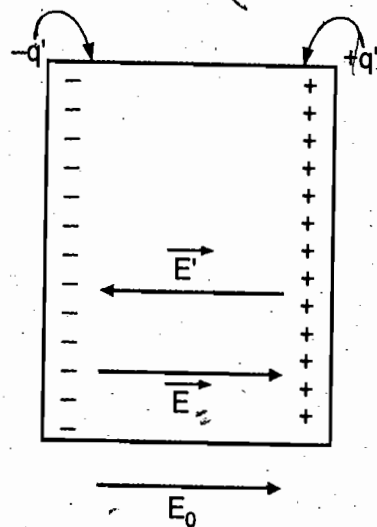


Fig. 3.3

The polarisation charges q' and $-q'$, induced on the two opposite faces of the slab, produce their own electric field E' , directed from q' to $-q'$. It is opposite to external field. The magnitude of resultant field within the dielectric is,

$$E = E_0 - E'$$

$$\vec{E} = \vec{E}_0 - \vec{E}'$$

From this equation, we conclude that, when a dielectric is placed in an external electric field, the field within the dielectric is weakened. The ratio of field without dielectric to within the dielectric is called dielectric constant (K) of material.

Therefore,
$$\frac{E_0}{E} = k$$

Electric Polarisation Vector ($\vec{P}$):

When a dielectric is placed in an external electric field, the net dipole moment is produced in the material along the direction of field and the dielectric is said to be polarised.

The net electric dipole moment per unit volume of the dielectric material is known as electric polarisation vector or polarisation density ($\vec{P}$). Since the electric dipole moment is a vector, the electric polarisation is also a vector quantity.

Mathematically, polarisation density ($\vec{P}$) is expressed as,

electric dipole moment per unit volume =
$$\vec{P} = \frac{\text{Dipole moment}}{\text{Volume}} = \frac{q' d}{V} = \frac{q' d}{Ad} = \frac{q'}{A} = \sigma_p$$

The S.I. unit of polarisation is coulomb/(meter)².

Polarisation ($\vec{P}$) at a Point:

To define the meaning of polarisation ($\vec{P}$) at a point, consider a small volume element ΔV of a dielectric material. When the dielectric material is placed in an external electric field, it gets continuously polarised. In a volume element ΔV , each molecule becomes a dipole. Let rdq' be the dipole moment of each molecule, where, r is distance between positive and negative charge of the molecule and dq' is magnitude of each charge. Thus, the volume element ΔV is characterised by an electric dipole moment,

$$\Delta \vec{p} = \int_{\Delta V} rdq'$$

The value of $\vec{P}$ at a point can be defined as the net dipole moment $\Delta \vec{p}$ of a small volume ΔV divided by the volume ΔV , with the limit taken as ΔV shrinks to zero around the point.

Thus,
$$\vec{P}(\vec{r}) = \lim_{\Delta V \rightarrow 0} \frac{\Delta \vec{p}}{\Delta V}$$

3.2 POLARISATION CHARGE DENSITIES

A polarisation in a dielectric material can give two types of densities in the material. One is surface polarisation charge density (σ_p) and other is volume polarisation charge density (ρ_p). In the absence of external dielectric field, there is no polarisation density and the dielectric is neutral everywhere.

(1) Surface polarisation charge density or surface density of bound charges (σ_p):

We have seen that when an isotropic, homogeneous dielectric material is kept in external electric field, the material gets polarised. One face of the material acquires net negative charge and opposite face acquires net positive charge (Fig. 3.4). The charges

developed across the faces are of equal magnitude. Hence, the net charge or average charge density is zero. Such a polarisation is called uniform polarisation. Consider a dielectric material of cross-sectional area 'A' and thickness 'd' kept in an external electric field E_0 as shown in Fig. 3.4.

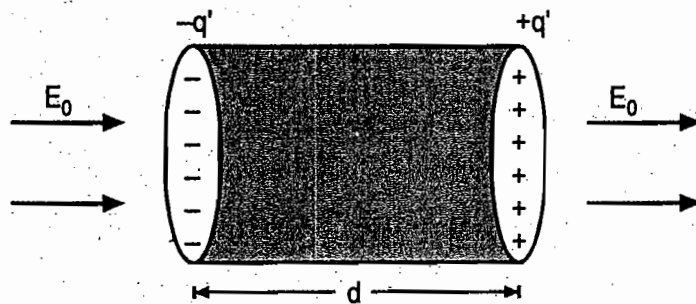


Fig. 3.4

The volume of the slab = Ad

The magnitude of electric polarisation = $\frac{\text{Dipole moment}}{\text{Volume}} = \frac{p}{V}$

But, dipole moment, $p = q'd$

$\therefore$ Polarisation, $P = \frac{q'd}{Ad}$

or $P = \frac{q'}{A}$... (3.1)

Thus, polarisation can also be defined as the induced charge per unit area i.e. surface density of bound charges in dielectric material.

$\therefore$ $P = \frac{q'}{A} = \sigma_p$... (3.2)

Above equation can be generalised by considering the case when face of dielectric material is not perpendicular to P .

Let the normal to the surface area 'S' makes an angle ' θ ' with the direction of polarisation vector $\vec{P}$ (Fig. 3.5).

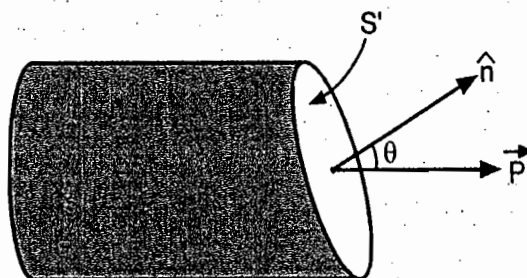


Fig. 3.5

From Fig. 3.5, the component of $\vec{P}$ normal to the surface 'S' is $P \cos \theta = \vec{P} \cdot \hat{n}$, where, $\hat{n}$ is unit vector along normal to the surface 'S'. Now equation (3.2) may be written as,

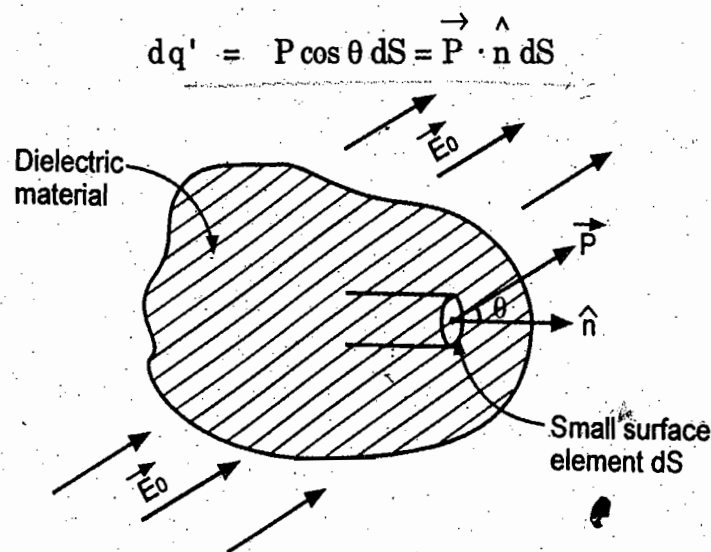
$$\sigma_p = \vec{P} \cdot \hat{n} \quad \dots (3.3)$$

This is required expression for surface polarisation charge density.

(2) Volume polarisation charge density (ρ_p):

Instead of taking homogeneous, isotropic dielectric material, if the material is inhomogeneous, non-isotropic, then dielectric constant (k) and polarisation ($\vec{P}$) are not constant i.e. k and $\vec{P}$ varies from point to point inside the material (non-uniform polarisation). This means that, more charges can be moved into some particular region than away from it. We then get a volume density of charge.

When an unpolarised slab of dielectric material is kept in electric field, certain amount of charge (dq') will pass through a given surface element ' dS ' due to polarisation process. The amount of charge passing through dS will be equal to the product of component of P along the normal to dS and the magnitude of surface area dS (Fig. 3.6) i.e.,

**Fig. 3.6**

The total charge q' crossing surface area ' S ' (which encloses volume V) by polarisation process is,

$$q' = \oint_S \vec{P} \cdot \hat{n} dS$$

where, $\hat{n}$ is unit vector normal to surface element dS and S is surface that encloses volume V .

An equal excess charge of opposite sign is left behind when some charge crosses the surface. Let q' be the charge left behind inside the surface ' S '.

$$\therefore q' = - \oint_S \vec{P} \cdot \hat{n} dS \quad \text{--- (3.4)}$$

If ρ_p is the density of the polarisation charges within the volume, we have,

$$q' = \int_V \rho_p dV \quad \text{--- (3.5)}$$

Equating equations (3.4) and (3.5), we get

$$-\int_S \vec{P} \cdot \hat{n} dS = \int_V \rho_p dV$$

According to Gauss' divergence theorem,

$$\int_S \vec{P} \cdot \hat{n} dS = \int_V \vec{\nabla} \cdot \vec{P} dV$$

$$\therefore \int_V \vec{\nabla} \cdot \vec{P} dV = - \int_V \rho_p dV$$

or

$$\rho_p = - \vec{\nabla} \cdot \vec{P} \quad \dots (3.6)$$

Thus, volume charge density ρ_p is equal to negative divergence of polarisation.

In uniform polarisation process, which occurs in homogeneous isotropic material, the value of $\vec{P}$ is constant throughout the material. Hence $\vec{\nabla} \cdot \vec{P}$ is zero and so there is no volume charge density anywhere in the dielectric. However, in a non-uniform polarisation $\vec{P}$ is not constant i.e. it varies from point to point inside the material and so volume charge density can appear in the dielectric given by relation,

$$\rho_p = - \vec{\nabla} \cdot \vec{P}$$

3.3 RELATION BETWEEN $\vec{D}$, $\vec{E}$ AND $\vec{P}$

Differential form of Gauss's law is

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

where, ρ is volume charge density of free charges.

An equivalent law for dielectric medium can be written by considering volume polarisation charge density (ρ_p) in addition to ρ . Thus,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho + \rho_p}{\epsilon_0}$$

But, $\rho_p = - \vec{\nabla} \cdot \vec{P}$

$$\therefore \vec{\nabla} \cdot \epsilon_0 \vec{E} = \rho - \vec{\nabla} \cdot \vec{P}$$

$$\text{or, } \vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho$$

Let us introduce a new vector field $\vec{D}$ such that,

$$\epsilon_0 \vec{E} + \vec{P} = \vec{D}$$

Thus, Gauss's law in dielectric may be written as.

$$\vec{\nabla} \cdot \vec{D} = \rho$$

(Differential form of Gauss's law in dielectrics) $\dots (3.7)$

where, $\vec{D}$ is called electric displacement vector and is given by,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \dots (3.8)$$

The integral form of Gauss's law in dielectrics may be written as,

$$\oint_S \vec{D} \cdot d\vec{S} = q_{\text{enc}}$$

where, q_{enc} is total free charge inside Gaussian surface.

Thus, Gauss's law in dielectrics can be stated as "the surface integral of displacement vector $\vec{D}$ over a closed surface is equal to the free charges enclosed within the surface".

3.4 DIELECTRIC CONSTANT AND DISPLACEMENT VECTOR

If the gap between plates of parallel plate condenser is filled with dielectric, its capacity increases. The increase in capacity of a capacitor with a dielectric with respect to capacity of same capacitor without dielectric (i.e. vacuum) measures a constant known as dielectric constant or relative permittivity, it is denoted by k . Thus,

$k = \frac{C}{C_0}$ Dielectric const $k = \frac{\text{Capacity with dielectric}}{\text{Capacity with vacuum}}$ $C = \epsilon A/d$
 $\therefore k = \frac{C}{C_0}$ $C_0 = \frac{\epsilon_0 A}{d}$... (3.9)

But,

$$C = \frac{\epsilon A}{d}$$

and

$$C_0 = \frac{\epsilon_0 A}{d}, \text{ for parallel plate capacitor}$$

Therefore, $k = \frac{\epsilon}{\epsilon_0}$... (3.10)

where, ϵ_0 is permittivity of free space and ϵ is permittivity of dielectric medium. Since $C > C_0$, $k > 1$. Thus, the dielectric constant may be defined as the ratio of the absolute permittivity of the dielectric to that of the free space.

Relative permittivity (or dielectric constant) is also defined as,

$$k = \frac{\text{Electric field in vacuum}}{\text{Electric field in dielectric}} = \frac{|\vec{E}_0|}{|\vec{E}|}$$

Since the electric intensity inside a dielectric depends upon the medium, we define new electrical vector which depends upon magnitude of charge and its displacement but is independent of medium. This vector is termed as electric displacement vector $\vec{D}$ and is given by,

$$\vec{D} = \frac{1}{4\pi r^2} q \hat{r} = \frac{1}{4\pi r^3} q \vec{r}$$

or,
$$\vec{D} = \epsilon \left(\frac{1}{4\pi\epsilon} \frac{q}{r^3} \vec{r} \right) \quad \vec{D} =$$

$$\therefore \vec{D} = \epsilon \vec{E} \quad \dots (3.11)$$

($\because E = \frac{1}{4\pi\epsilon} \frac{q}{r^3} \vec{r}$; field in dielectric)

S.I. unit of electric displacement vector is coulomb/(meter)².

The relation between D, E and P is,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \text{--- relation between } \vec{D}, \vec{E} \text{ \& } \vec{P}$$

Since $\vec{E}$ and $\vec{P}$ are vectors directed along same direction, electric displacement vector $\vec{D}$ is also directed along the direction of $\vec{E}$. Thus, in terms of vector form, the relation between $\vec{D}$, $\vec{E}$ and $\vec{P}$ may be written as,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

We must note the following points about $\vec{D}$, $\vec{E}$ and $\vec{P}$

1. Electric displacement vector $\vec{D}$ is connected with the free charge or true charge only, so it is not altered by the introduction of dielectric.
2. $\vec{P}$ is connected with polarisation (bound) charges or induced charges only. It is represented by lines directed from the negative induced charge to the positive induced charge present within the dielectric.
3. $\vec{E}$ is connected with all charges that are actually present, whether free or bound.

We know the relation between $\vec{D}$, $\vec{E}$ and $\vec{P}$ is,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

The relation between $\vec{D}$ and $\vec{E}$ is,

$$\vec{D} = \epsilon \vec{E}$$

(See equation 3.11)

or
$$\vec{D} = \epsilon_0 k \vec{E}$$

$$\therefore \epsilon_0 k \vec{E} = \epsilon_0 \vec{E} + \vec{P}$$

$$\text{or } \vec{P} = \epsilon_0 \vec{E} (k - 1)$$

This is the required relation between $\vec{P}$ and $\vec{E}$.

3.5 ELECTRIC SUSCEPTIBILITY AND DIELECTRIC CONSTANT

For many dielectric materials, it has been experimentally found that the magnitude of polarisation is directly proportional to intensity of the electric field at a given point of dielectric, provided $\vec{E}$ is not too strong i.e. we have,

For linear dielectrics

$$\vec{P} \propto \vec{E}$$

or

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad \dots (3.12)$$

The constant of proportionality, χ_e is called the electric susceptibility of the medium.

The value of χ_e depends on the microscopic structure of the substance and also on external conditions such as temperature.

A material which obeys equation (3.12) is called linear dielectrics.

The relation between $\vec{D}$, $\vec{E}$ and $\vec{P}$ is

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$= \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \epsilon_0 \vec{E} (1 + \chi_e) \quad \dots (3.13)$$

Since ϵ_0 , χ_e are constants, $\vec{D}$ is also proportional to $\vec{E}$. *For air $\chi_e = 0$ $K=1$*

Using the relation $\vec{D} = \epsilon \vec{E}$ in equation (3.13), we get, *for dielectric $K = 1 + \chi_e$*

$$\epsilon \vec{E} = \epsilon_0 (1 + \chi_e) \vec{E}$$

or

$$\frac{\epsilon}{\epsilon_0} = 1 + \chi_e$$

The constant $\frac{\epsilon}{\epsilon_0}$ is called relative permittivity (ϵ_r) or dielectric constant of the material. Thus, dielectric constant of a material is given by,

$$k = 1 + \chi_e \quad \dots (3.14)$$

3.6 ELECTRIC FIELD AT AN EXTERIOR POINT OF DIELECTRIC MATERIAL

Consider a finite piece of dielectric material that is polarised. The polarised material gives rise to an electric field in its surrounding space. Our aim is to calculate the field at a point Q, which is outside the dielectric material and situated at a distance 'r' from the origin O (Fig. 3.7).

Let the volume of the dielectric be divided into small elementary volumes. Let us consider one such volume element of volume $\Delta V'$ around point P of dielectric material.

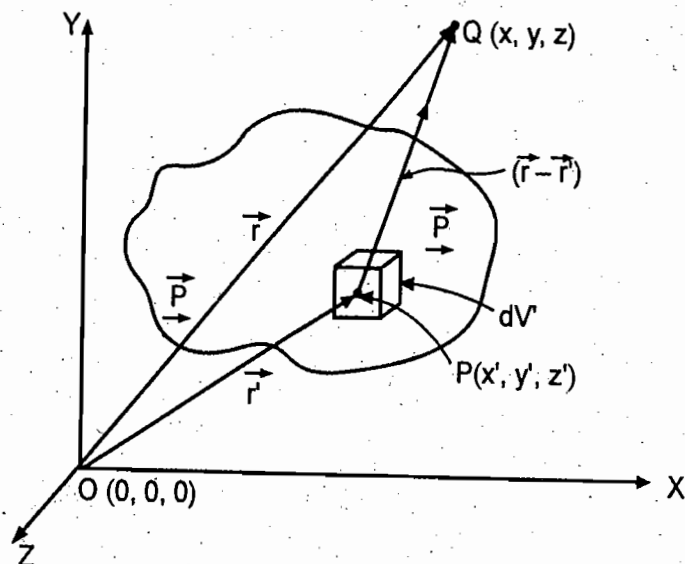


Fig. 3.7

Each volume element of dielectric material is characterised by a dipole moment. The dipole moment of small volume element $\Delta V'$ is,

$$\Delta \vec{p} = \vec{P} \Delta V' \quad \left[\because \text{Polarisation } \vec{P} = \frac{\text{dipole moment } (\Delta \vec{p})}{\text{Volume } (\Delta V')} \right]$$

where, $\vec{P}$ is polarisation vector.

We know that, potential at a point due to dipole is given by,

$$\phi(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{|\vec{r}|^3} \quad \dots (3.15)$$

where, r is the distance of a point from the centre of dipole and $\vec{p}$ is dipole moment.

Let the point P be situated at $\vec{r}'$ from the origin O and $\vec{r}'$ be very large as compared with the dimensions of $\Delta V'$.

The dipole moment $\Delta \vec{p}$ completely determines $\Delta V'$'s contribution to the potential at point Q . Referring equation (3.15), the potential at point Q due to dipole of dipole moment $\Delta \vec{p}$ is,

$$\Delta \phi(\vec{r}) = \frac{\Delta \vec{p} \cdot (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} \quad \dots (3.16)$$

But,

$$\Delta \vec{p} = \vec{P}(\vec{r}') \Delta V'$$

$$\Delta \phi(\vec{r}) = \frac{\vec{P}(\vec{r}') \cdot (\vec{r} - \vec{r}') \Delta V'}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} \quad \dots (3.17)$$

$(\vec{r} - \vec{r}')$ is a vector directed away from $\Delta V'$, whose magnitude is given by,

$$|\vec{r} - \vec{r}'| = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2} \quad \dots (3.18)$$

The total potential due to whole dielectric material at point Q ($\vec{r}$) is obtained by integrating equation (3.17).

$$\therefore \phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\vec{P}(\vec{r}') \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dV'$$

where, V' is total volume of the dielectric material. If $|\vec{r} - \vec{r}'|$ is given by equation (3.18), then by elementary vector calculus, we have,

But
$$\vec{\nabla}' \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = + \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \quad \dots (3.19)$$

The $\vec{\nabla}'$ operator involves derivatives with respect to the primed co-ordinates. If we take into account equation (3.18) and equation (3.19), we get $\vec{\nabla}' = -\vec{\nabla}$, where $\vec{\nabla}$ operator involves derivatives with respect to unprimed co-ordinates.

$$\text{Thus, } \frac{\vec{P}(\vec{r}') \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} = \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{|\vec{r} - \vec{r}'|} \right)$$

$$\text{Therefore, } \phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{V'} \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) dV' \quad \dots (3.20)$$

Now, using the vector identity,

$$\vec{\nabla}' \cdot (\phi \vec{A}) = \phi \vec{\nabla}' \cdot \vec{A} + \vec{A} \cdot \vec{\nabla}' \phi$$

where, ϕ is scalar function and $\vec{A}$ is vector function.

Taking $\phi = \frac{1}{|\vec{r} - \vec{r}'|}$ and $\vec{A} = \vec{P}$, we get

$$\vec{\nabla}' \cdot \left(\frac{\vec{P}}{|\vec{r} - \vec{r}'|} \right) = \frac{1}{|\vec{r} - \vec{r}'|} \vec{\nabla}' \cdot \vec{P} + \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{|\vec{r} - \vec{r}'|} \right)$$

$$\therefore \vec{P} \cdot \vec{\nabla}' \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = \vec{\nabla}' \cdot \left(\frac{\vec{P}}{|\vec{r} - \vec{r}'|} \right) - \frac{1}{|\vec{r} - \vec{r}'|} \vec{\nabla}' \cdot \vec{P} \quad \dots (3.21)$$

Using equation (3.21) in (3.20), we get

$$\begin{aligned} \phi(r) &= \frac{1}{4\pi\epsilon_0} \left\{ \int_{V'} \vec{\nabla}' \cdot \left(\frac{\vec{P}}{|\vec{r} - \vec{r}'|} \right) dV' + \int_{V'} \frac{(-\vec{\nabla}' \cdot \vec{P})}{|\vec{r} - \vec{r}'|} dV' \right\} \\ \therefore \phi(r) &= \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\vec{P} \cdot \hat{n} dS'}{|\vec{r} - \vec{r}'|} + \frac{1}{4\pi\epsilon_0} \int_{V'} -\frac{\vec{\nabla}' \cdot \vec{P}}{|\vec{r} - \vec{r}'|} dV' \dots (3.22) \end{aligned}$$

By Gauss's divergence theorem

where, in first term, volume integral of $\vec{\nabla}' \cdot \left(\frac{\vec{P}}{|\vec{r} - \vec{r}'|} \right)$ has been replaced by a surface

integral through the application of divergence theorem. In equation (3.22), $\hat{n}$ is the outward normal unit vector to the surface element dS' .

The quantities $\vec{P} \cdot \hat{n}$ and $-\vec{\nabla}' \cdot \vec{P}$ in equation (3.22) are two scalar functions obtained from polarization $\vec{P}$. The quantity $\vec{P} \cdot \hat{n}$ has dimensions of charge per unit area and the quantity $-\vec{\nabla}' \cdot \vec{P}$ has dimensions of charge per unit volume. Thus, we write,

But $\sigma_p = \vec{P} \cdot \hat{n}$ *surface polarisation charge density*

and

$\rho_p = -\vec{\nabla}' \cdot \vec{P}$ *volume charge density*

σ_p is called polarization surface charge density and ρ_p is called polarization volume charge density. Thus, the potential due to dielectric material is,

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \left[\int_{S'} \frac{\sigma_p dS'}{|\vec{r} - \vec{r}'|} + \int_{V'} \frac{\rho_p dV'}{|\vec{r} - \vec{r}'|} \right] \quad \dots (3.23)$$

The intensity of electric field due to polarized volume may be expressed as,

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \left[\int_{S'} \frac{\sigma_p (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dS' + \int_{V'} \frac{\rho_p (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dV' \right] \quad \dots (3.24)$$

$$(\because \vec{E} = -\nabla \phi)$$

The total charge of a dielectric body is,

$$q_p = \int_{V'} (-\vec{\nabla}' \cdot \vec{P}) dV' + \int_{S'} \vec{P} \cdot \hat{n} dS' \quad \dots (3.25)$$

It must equal zero, since the dielectric, as a whole is electrically neutral.

3.7 ELECTRIC FIELD DUE TO A POINT CHARGE +q PLACED IN AN INFINITE DIELECTRIC

Consider a point charge +q placed inside an infinite dielectric. Let us calculate the electric field at a point P, which is at a distance r from the charge.

Let us draw a spherical surface (Gaussian surface) passing through point P and with the centre at charge +q. From Gauss's law, we have,

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

$$\text{or } k \oint E dS \cos \theta = \frac{q}{\epsilon_0}$$

$$\text{But, } \theta = 0$$

$$\therefore k E \oint dS = \frac{q}{\epsilon_0}$$

$$k E 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\text{or } E = \frac{q}{4\pi\epsilon_0 k r^2}$$

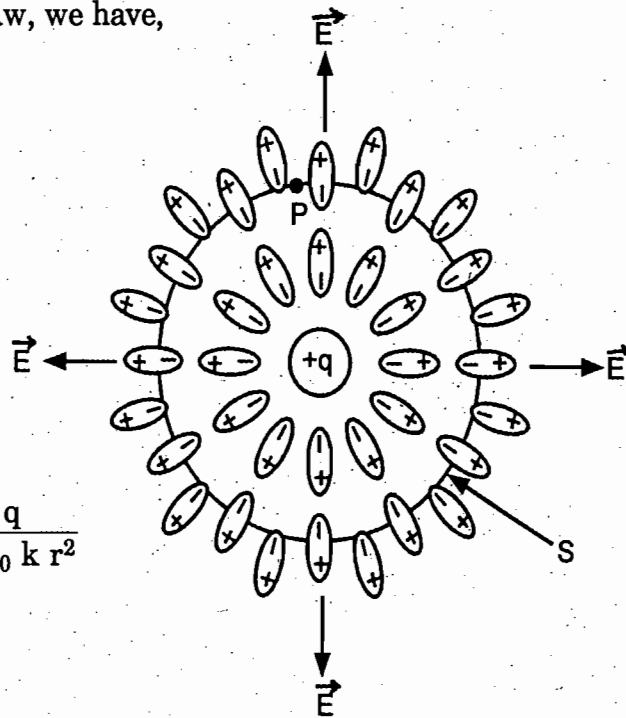


Fig. 3.8

This is expression for magnitude of electric intensity due to charge +q at point P. The field is radially away from the charge +q.

It must be noted that q is the total free charge inside the Gaussian surface.

In vector form,
$$\vec{E} = \frac{1}{4\pi k \epsilon_0} \frac{q}{r^3} \vec{r}$$

Thus, the electric field is smaller by a factor k than the electric field, if the dielectric medium would be absent.

The electric field at point P is $\frac{q}{4\pi k \epsilon_0 r^2}$ and it is due to free charge q and polarisation charges induced in the dielectric medium. Since the charge q is positive, negative charges inside the atoms or molecules are shifted inward i.e. towards charge +q. This produces an induced charge $-q \left(1 - \frac{1}{k}\right)$ on the surface of cavity in the dielectric in which charge q is situated. The effective charge is, therefore,

$$q - q \left(1 - \frac{1}{k}\right) = \frac{q}{k} \text{ and hence the field at point P is } \frac{q}{4\pi\epsilon_0 k r^2}.$$

3.8 THE ELECTRIC FIELD INSIDE A DIELECTRIC

Calculation of electric field inside a dielectric from microscopic point of view is complicated. Because if the point, where electric field is to be calculated, is near to negative charge such as electrons, the field is gigantic whereas a short distance away, it may be small or a point in totally different direction. Furthermore, an instant later, as the atom

moves, about its mean position, the field will have altered entirely. Thus, calculation of true microscopic field inside the material is impossible. However, it is possible to calculate the macroscopic field inside the material. Before obtaining an expression for field inside a polarised dielectric material, let us define electric field precisely.

The electric field is the force per unit charge on a test charge embedded in the dielectric in the limit where the test charge is so small that it does not itself change the charge distribution inside the polarised dielectric material.

We know that electric field in a dielectric material as well as in free space or vacuum is conservative and hence it is derivable from a scalar potential, i.e.

$$\vec{\nabla} \times \vec{E} = 0$$

or $\oint \vec{E} \cdot d\vec{l} = 0$, So, $E = -\nabla \phi$

To apply this equation, let us create a needle shaped cavity inside the dielectric material as shown in Fig. 3.9. Consider a rectangular loop ABCD of small width such that segment AB lies in a cavity and segment CD lies in the dielectric material.

We have $\oint_{ABCD} \vec{E} \cdot d\vec{l} = 0$

Since the length of segments AD and BC are very small, above equation can be written as,

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_C^D \vec{E} \cdot d\vec{l} = 0$$

Let $AB = CD = l$

$\therefore E_v l - E_d l = 0$

or $E_{vt} = E_{dt}$

where, the subscript v and d refers to vacuum and dielectric respectively. The subscript t refers for tangential component. If the needle shaped cavity is oriented along the direction of $\vec{E}$, then $E_{dt} = E_d$ and by symmetry the field in the cavity is along the direction of cavity, i.e. $E_{vt} = E_v$.

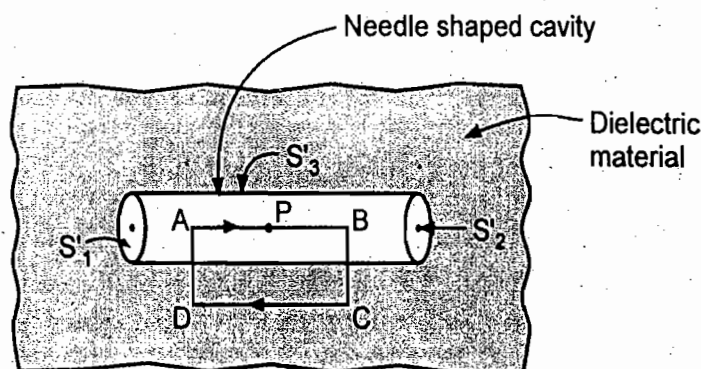


Fig. 3.9

From above discussion, it can be concluded that the electric field in a dielectric material is equal to the electric field inside the cavity (situated in dielectric material) provided, the axis of cavity is oriented along the direction of external field.

Thus, the problem of calculating electric field inside dielectric reduces to problem of calculating the electric field inside a needle shaped cavity. But the electric field calculated at a point in the cavity is an electric field at an external point. Therefore electric field inside the dielectric may be determined by the result of section 3.6 i.e. equation 3.23.

Thus, potential at a point P, situated at the centre of cavity, due to polarised dielectric material is,

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\int_{S'_0 + S'} \frac{\sigma_p dS'}{|\vec{r} - \vec{r}'|} + \int_{V'_0 - V'_1} \frac{\rho_p dV'}{|\vec{r} - \vec{r}'|} \right] \dots (3.26)$$

In above equation V'_0 is total volume of dielectric, V'_1 is volume of cavity, so $V'_0 - V'_1$ represents the volume of dielectric excluding the cavity.

In surface integral, ' S'_0 ' is exterior surface of dielectric material and $S' = S'_1 + S'_2 + S'_3$ are the surfaces of cavity as shown in Fig. 3.9. If the needle shaped cavity is thin, S'_1 and S'_2 are negligibly small and hence can be neglected. Further more, σ_p on the curved surface of cavity is zero. So only the exterior surface contribute to the surface integral of equation (3.26).

In a similar way, the contribution to the volume integral is small and can be neglected.

Thus equation (3.26 or 3.23) gives the potential $\phi(\vec{r})$ regardless of whether the point situated at $\vec{r}$ is located inside or outside the dielectric. So the potential at point P is,

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\int_{S'} \frac{\sigma_p dS'}{|\vec{r} - \vec{r}'|} + \int_{V'} \frac{\rho_p dV'}{|\vec{r} - \vec{r}'|} \right]$$

The electric field at point 'P' situated at $\vec{r}$ may be calculated as minus of gradient of above equation or equation (3.23).

The electric field at point P is therefore,

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\int_S \frac{\sigma_p (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dS' + \int_V \frac{\rho_p (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dV' \right]$$

ILLUSTRATIVE EXAMPLES

Example 1 : A parallel plate capacitor of plate area $A = 100 \text{ cm}^2$ and separation $d = 1.5 \text{ cm}$ is charged by a potential of 60 volts. Then the battery is disconnected and a dielectric slab of thickness $b = 0.8 \text{ cm}$ and dielectric constant $k = 5$ inserted. Calculate the three vectors E , D and P in the dielectric.

Solution :

For a capacitor

$$A = 100 \text{ cm}^2$$

$$d = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}$$

$$\phi_0 = 60 \text{ volt} \quad k = 5$$

The electric intensity between two plates without dielectric is,

$$E_0 = \frac{\phi_0}{d} = \frac{60 \text{ volt}}{1.5 \times 10^{-2} \text{ m}} = 40 \times 10^2 \text{ volt/m}$$

When the dielectric slab of $k = 5$ is inserted, the electric intensity inside the dielectric becomes,

$$E = \frac{E_0}{k} = \frac{40 \times 10^2}{5} = 8 \times 10^2 \text{ volt/m}$$

The electric displacement vector is given by,

$$D = k \epsilon_0 E$$

$$D = 5 \times 8.85 \times 10^{-12} \times 8 \times 10^2$$

$$D = 3.54 \times 10^{-8} \text{ C/m}^2$$

The electric polarization in the dielectric is given by,

$$P = (k - 1) \epsilon_0 E$$

$$P = (5 - 1) \times 8.85 \times 10^{-12} \times 8 \times 10^2$$

$$= 4 \times 8.85 \times 8 \times 10^{-10}$$

$$P = 2.83 \times 10^{-8} \text{ C/m}^2$$

Example 2 : Find the magnitudes of $\vec{D}$ and $\vec{P}$ for a dielectric material in which $E = 0.20 \text{ MV/m}$ and $\chi_e = 4$. (Given $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$).

Solution :

Given : (i) Intensity of electric field $E = 0.20 \times 10^6 \text{ V/m}$

(ii) Electric susceptibility $\chi_e = 4$

(iii) $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ or $\epsilon_0 = \frac{10^{-9}}{36\pi}$

(iv) k or $\epsilon_r = 1 + \chi_e = 5$.

Formulae :

$$(i) \quad \vec{P} = \chi_e \epsilon_0 \vec{E}$$

$$(ii) \quad \vec{D} = \epsilon \vec{E} \text{ or } \vec{D} = \epsilon_0 \epsilon_r \vec{E}$$

$$\left(\because \frac{\epsilon}{\epsilon_0} = \epsilon_r \right)$$

Magnitude of electric displacement vector D is given by,

$$\begin{aligned} D &= \epsilon_0 \epsilon_r E \\ &= \frac{10^{-9}}{36\pi} \times 5 \times 0.20 \times 10^6 \\ &= 8.8 \times 10^{-6} \text{ C/m}^2 \end{aligned}$$

Magnitude of electric polarisation vector P is given by,

$$\begin{aligned}
 P &= \chi_e \epsilon_0 E \\
 &= 4 \times \frac{10^{-9}}{36\pi} \times 0.20 \times 10^6 \\
 &= 7.07 \times 10^{-3} \text{ C/m}^2
 \end{aligned}$$

Handwritten notes: $k = 1 + \chi_e$, $D = \epsilon_0 E$, $E = \frac{D}{\epsilon_0}$

Example 3 : Find the magnitude of polarisation $\vec{P}$ in a dielectric material with $k = 2.8$, if $D = 3.5 \times 10^{-7} \text{ C/m}^2$ (Assume material is homogeneous and isotropic).

Solution :

Given : (i) Dielectric constant, $k = 2.8$

(ii) Electric displacement, $D = 3.5 \times 10^{-7} \text{ C/m}^2$

Formula : $|\vec{P}| = \epsilon_0 \chi_e |\vec{E}|$

Using $k = 1 + \chi_e$ and $D = \epsilon_0 k E$, above equation may be written as,

$$\begin{aligned}
 |\vec{P}| &= \epsilon_0 (k - 1) \frac{D}{\epsilon_0 k} \\
 \text{or } P &= \frac{(k - 1)}{k} D \\
 &= \frac{(2.8 - 1)}{2.8} \times 3.5 \times 10^{-7} \\
 P &= 2.25 \times 10^{-7} \text{ C/m}^2
 \end{aligned}$$

Handwritten notes: $P = \chi_e E$, $P = \epsilon_0 \chi_e \left[\frac{D}{\epsilon_0 k} \right]$, $= \frac{(k-1)D}{k}$

Example 4 : Two point charges in a dielectric medium having $k = 5.2$ interact with a force of $8.6 \times 10^{-3} \text{ N}$. What could be the force if the charges were in free space ?

Solution :

Given : (i) Dielectric constant, $k = 5.2$.

(ii) Force between the two charges when situated in dielectric material $F = 8.6 \times 10^{-3} \text{ N}$.

When two point charges are situated in free space, force between two charges is given by,

$$F_{\text{air}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

When two point charges are situated in dielectric medium with dielectric constant k , force between two charges is given by,

$$\begin{aligned}
 F &= \frac{1}{4\pi\epsilon_0 k} \frac{q_1 q_2}{r^2} \\
 \therefore F_{\text{air}} &= F \times k \\
 &= 8.6 \times 10^{-3} \times 5.2 \\
 &= 4.4 \times 10^{-2} \text{ newton}
 \end{aligned}$$

Handwritten notes: $F_{\text{air}} = F \times \frac{1}{k}$, $F = \frac{F_{\text{air}}}{k}$

Example 5 : A metallic sphere of radius 'a' carries a charge +q. It is surrounded out to radius 'b', by homogeneous, isotropic dielectric material of permittivity ϵ as shown in Fig. 3.10. Find the potential at the centre. Calculate the polarisation and hence find the surface and volume bound charges.

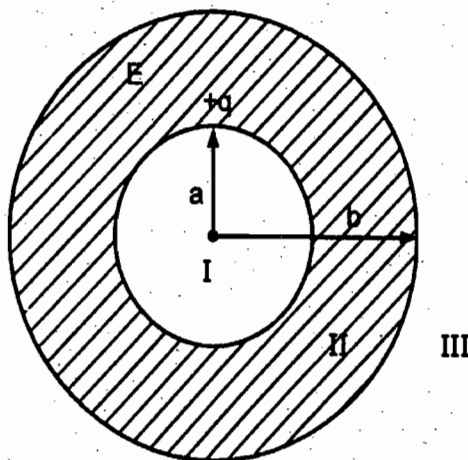


Fig. 3.10

Solution : When a charge is given to conducting or metallic sphere, it gets uniformly distributed over the surface of sphere. From the Fig. 3.10, there are three regions.

(i) **Region I ($r < a$) :** This region is inside the sphere and radius 'a'.

To find electric intensity at any point such as P, inside the metal sphere, draw a Gaussian surface passing through that point [Fig. 3.11]. According to Gauss's law,

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{q}{k\epsilon_0},$$

we get,

$$\mathbf{E} = 0$$

($\because$ Free charge enclosed by Gaussian surface = 0)

Thus, in region I for all points $r < a$, we have $\mathbf{E} = \mathbf{P} = \mathbf{D} = 0$, because the Gaussian surface (sphere) drawn through any point in this region encloses no charge.

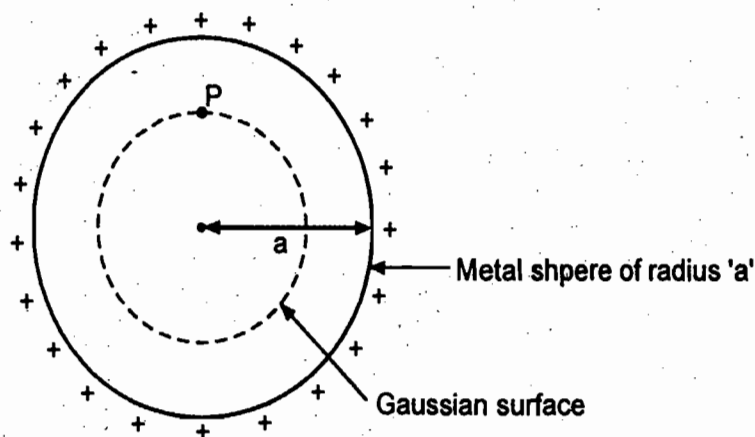


Fig. 3.11

(ii) **Region II ($a < r < b$) :** This region contains a homogeneous, isotropic material of permittivity ϵ .

Let us find the electric intensity at point P in this region. The Gaussian surface passing through point P encloses a charge +q (Fig. 3.12). Therefore,

$$\oint \vec{D} \cdot d\vec{S} = q$$

(Gauss's law in dielectrics)

$$\text{or} \quad \oint \mathbf{D} \cdot d\mathbf{S} \cos \theta = q$$

$$\text{or} \quad D 4\pi r^2 = q \quad (\because \theta = 0 \text{ and surface area} = 4\pi r^2)$$

$$\therefore D = \frac{1}{4\pi} \frac{q}{r^2}$$

$$\text{But, } D = \epsilon E$$

$$\therefore E = \frac{1}{4\pi\epsilon} \frac{q}{r^2}$$

Thus, we get,

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{q}{r^2} \hat{r} \text{ for } a < r < b \quad \dots (3.27)$$

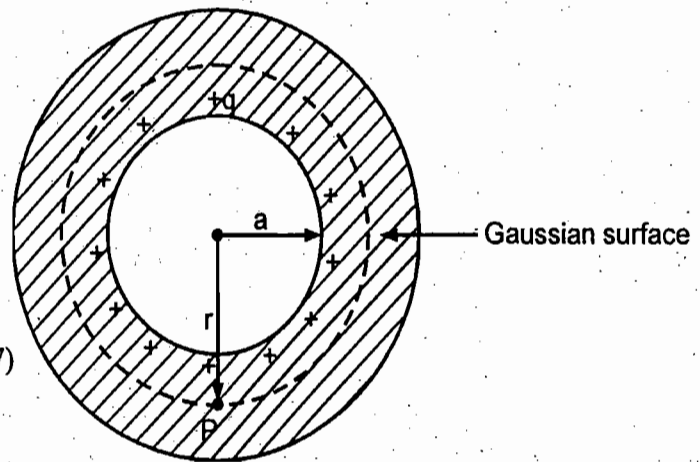


Fig. 3.12

(iii) **Region III ($r > b$)** : This region is again free space of permittivity ϵ_0 .

It can be shown that in this region i.e. for $r > b$, we have,

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Potential at the centre is,

$$\begin{aligned} V &= - \int_{r=\infty}^{r=0} \vec{E} \cdot d\vec{r} \\ &= - \int_{\infty}^b \vec{E} \cdot d\vec{r} - \int_b^a \vec{E} \cdot d\vec{r} - \int_a^0 \vec{E} \cdot d\vec{r} \\ &= - \int_{\infty}^b \left(\frac{q}{4\pi\epsilon_0 r^2} \right) dr - \int_b^a \left(\frac{q}{4\pi\epsilon r^2} \right) dr - \int_a^0 (0) dr \\ &= - \frac{q}{4\pi} \left[\frac{1}{\epsilon_0} \left(-\frac{1}{r} \right)_{\infty}^b + \frac{1}{\epsilon} \left(-\frac{1}{r} \right)_b^a \right] \\ V &= \frac{q}{4\pi} \left[\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b} \right] \end{aligned}$$

This is expression for potential at the centre of metal sphere.

Polarisation in dielectric is given by,

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

Using equation (3.27) in above equation, we get

$$\vec{P} = \epsilon_0 \chi_e \frac{1}{4\pi\epsilon} \frac{q}{r^2} \hat{r}$$

where, χ_e is electric susceptibility and ϵ is permittivity of dielectric medium.

Surface density of polarisation charges is,

(πr^2)

$$\sigma_p = \vec{P} \cdot \hat{n}$$

$$= \begin{cases} \frac{\chi_e \epsilon_0 q}{4\pi \epsilon b^2} \text{ at the outer surface} \\ \frac{\chi_e \epsilon_0 q}{4\pi \epsilon a^2} \text{ at the inner surface} \end{cases}$$

Volume density of polarisation charges is,

face

$$\rho_p = -\vec{\nabla} \cdot \vec{P} = 0$$

Example 6 : Calculate the field and potential inside and outside the uniformly polarised solid sphere of radius R .

Solution : Consider a uniformly polarised solid sphere of radius R . Let the sphere be polarised with polarisation density P and is directed along Y -axis. Since the sphere is uniformly polarised, the volume polarisation charge density is zero.

Consider a thin cylindrical column with edge surface area δS [Fig. 3.13 (a)]. From Fig. 3.13 (a), the component of polarisation P in the direction of unit vector normal to the surface element dS will be $P \cos \theta$.

The charge on the upper edge surface of area $\delta S = P \cos \theta \delta S$... (3.28)

$$\left(\because \text{Polarisation} = \frac{\text{Charge}}{\text{Area}} \right)$$

The charge on the lower edge surface of area $\delta S = -P \cos \theta \delta S$.

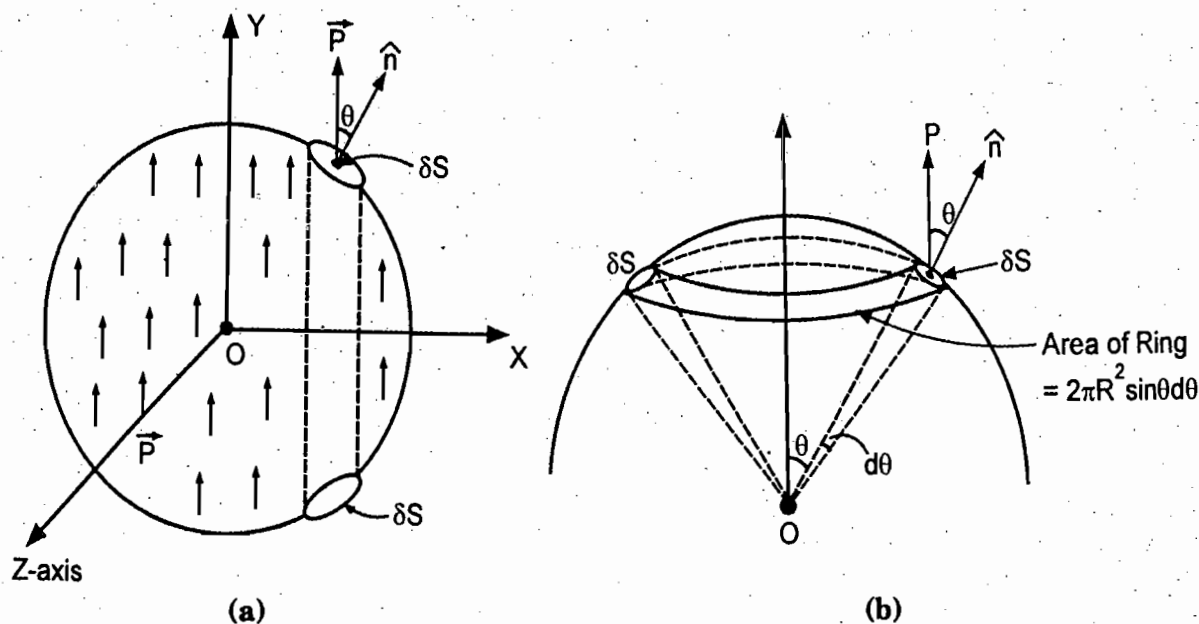


Fig. 3.13

To calculate the charge on the upper half sphere, consider a ring [Fig. 3.13 (b)] of radius $R \sin \theta$ and width $R d\theta$.

$$\text{Area of the ring} = 2\pi R \sin \theta R d\theta$$

From equation (3.28), charge on the ring $= 2\pi R^2 \sin \theta d\theta P \cos \theta$

Total charge on the upper half sphere can be obtained by integrating above equation between the limit $\theta = 0$ to $\theta = \frac{\pi}{2}$.

$$\begin{aligned}
 \therefore \text{Total charge on upper half sphere} &= \int_0^{\pi/2} 2\pi R^2 \sin \theta \cos \theta d\theta \\
 &= \pi R^2 P \int_0^{\pi/2} \sin 2\theta d\theta \\
 &= \pi R^2 P \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2} \\
 &= \pi R^2 P
 \end{aligned}$$

Thus, total charge on the upper half sphere = $+\pi R^2 P$.

In a similar manner, it can be shown that the total charge on the lower half sphere = $-\pi R^2 P$.

To find out field and potential, let us consider that sphere is composed of two identical spheres, one having positive charge of volume charge density $\frac{P}{\delta y}$ * and other having negative charge of volume charge density $-\frac{P}{\delta y}$ (Fig. 3.14). The centres of these two spheres are δy apart from each other. In the overlap region, net charge density is zero.

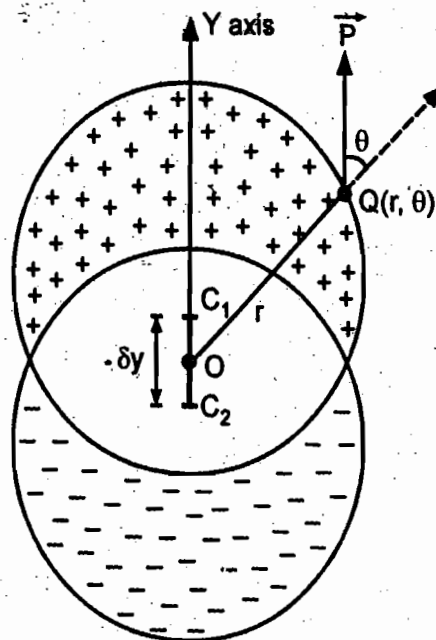


Fig. 3.14

* Polarisation $P = \frac{\text{Charge}}{\text{Area}}$

$\therefore \frac{P}{\delta y} = \frac{\text{Charge}}{\text{Area} \cdot \delta y}$

or $\frac{P}{\delta y} = \frac{\text{Charge}}{\text{Volume}}$

$\therefore \text{Volume charge density} = \frac{P}{\delta y}$

The total charge inside the upper sphere = $\left(\frac{4\pi}{3} R^3\right) \frac{P}{\delta y}$ and is concentrated at the centre C_1 . The total charge inside the lower sphere = $-\left(\frac{4\pi}{3} R^3\right) \frac{P}{\delta y}$ and is concentrated at C_2 . Thus, we have a system of dipole, i.e. two charges of equal magnitude and separated by distance δy as shown in Fig. 3.15.

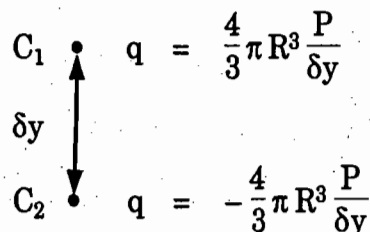


Fig. 3.15

$$\begin{aligned} \text{The dipole moment of the dipole} &= \frac{4\pi}{3} R^3 \frac{P}{\delta y} \delta y \\ &= \left(\frac{4}{3} \pi R^3\right) P \end{aligned}$$

Let us calculate the field and potential at any external point such as $Q(r, \theta)$.

We know that potential due to the dipole at point $Q(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$

$$\text{In this case, dipole moment } p = \left(\frac{4}{3} \pi R^3\right) P$$

$$\begin{aligned} \therefore \text{Potential at point } Q \text{ is, } \phi &= \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3} \pi R^3 P \cos \theta}{r^2} \\ \text{or } \phi &= \frac{R^3 P \cos \theta}{3 \epsilon_0 r^2} \end{aligned}$$

Let us calculate electric intensity at point Q .

$$\begin{aligned} E_r &= -\frac{\partial \phi}{\partial r} = \frac{2R^3 P \cos \theta}{3\epsilon_0 r^3} \\ E_\theta &= -\frac{1}{r} \frac{\partial \phi}{\partial \theta} = \frac{R^3 P \sin \theta}{3\epsilon_0 r^3} \end{aligned}$$

The resultant magnitude of electric intensity is given by,

$$E = \sqrt{E_r^2 + E_\theta^2}$$

If the point Q is on the surface of sphere, then $r = R$ and we get,

$$\phi = \frac{P R \cos \theta}{3\epsilon_0} = \frac{P y}{3\epsilon_0}$$

where, $R \cos \theta$ is component of R along y i.e. it is y co-ordinate at point Q .

Thus, potential on the surface of sphere = $\frac{P y}{3\epsilon_0}$.

The equation $\phi = \frac{P y}{3\epsilon_0}$ gives the electric potential at every point on the boundary of the region or on the surface of sphere. Since inside region is charge free, ϕ must satisfy Laplace's equation.

Thus,

$$\phi_{in} = \frac{P y}{3\epsilon_0}$$

This is expression for potential at a point inside the sphere. Since polarisation is along Y-axis, we have $(E_x)_{in} = (E_z)_{in} = 0$.

$$\therefore (E_y)_{in} = -\frac{\partial \phi}{\partial y}$$

or

$$E_y = -\frac{P}{3\epsilon_0}$$

Thus, in general,

$$E_{in} = -\frac{P}{3\epsilon_0}$$

Thus, electric field inside the sphere remains constant throughout and points opposite to the direction of polarisation as shown in Fig. 3.16.

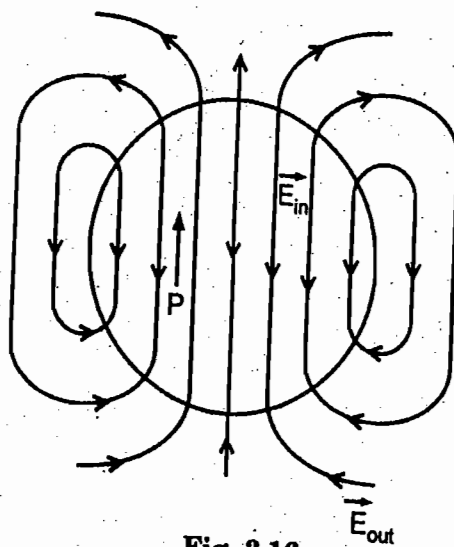


Fig. 3.16

Example 7 : A long wire, having uniform linear charge density λ , is surrounded by rubber insulation out to a radius a . Find the electric displacement.

Solution : Fig. 3.17 shows a part of long wire carrying uniform linear charge density λ and is surrounded by rubber insulation out to radius 'a'.

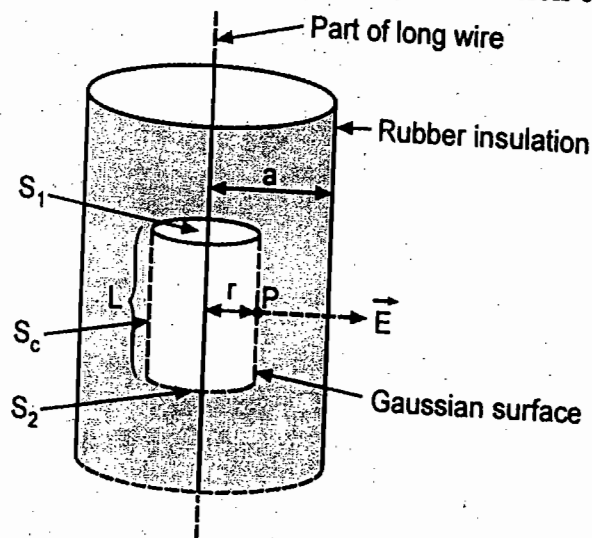


Fig. 3.17

To find the electric intensity at a point P situated at perpendicular distance r from the axis of the wire, a cylinder of radius r and length L is drawn co-axial to wire. This cylinder (shown by dotted line) is Gaussian surface and the point P is on its surface.

Let $\vec{E}$ be the electric field intensity at point P.

Since there is no electric field perpendicular to the plane faces of the Gaussian surface, flux passing through these faces (S_1 and S_2) is zero.

Flux through the Gaussian surface (curved surface S_c) is given by,

$$\int \vec{D} \cdot d\vec{S} = q_{\text{enc.}}$$

But, $q_{\text{enc.}} = \lambda L$

$$\therefore \int D dS \cos \theta = q_{\text{enc.}}$$

$$\therefore D \int dS = q_{\text{enc.}} \quad (\because \theta = 0)$$

$$\therefore D 2\pi r L = \lambda L$$

or $D = \frac{\lambda}{2\pi r}$

In vector form, $\vec{D} = \frac{\lambda}{2\pi r} \hat{r}$

It must be noted that this formula holds within the insulation and outside it.

The electric field at point P is,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\therefore \vec{E} = \frac{1}{\epsilon_0} (\vec{D} - \vec{P})$$

Since, we do not know $\vec{P}$, electric field $\vec{E}$ cannot be determined.

Electric field outside the insulation can be determined because outside the insulation

region,

$$\vec{P} = 0$$

$$\therefore \vec{E} = \frac{1}{\epsilon_0} \vec{D}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} \quad \text{for } r > a$$

Example 8 : A sphere of radius 'a' is centered at the origin and has a uniform surface charge density σ_s . Using Gauss's law, find $\vec{D}$ inside and outside the hollow sphere.

Solution : Consider a hollow sphere of radius 'a' meter [Fig. 3.18].

Let us find $\vec{D}$ at a point P situated inside the sphere i.e. $r < a$.

To find electric field at point P, draw a Gaussian surface through point P. The Gaussian surface is sphere shown by dotted line in Fig. 3.18.

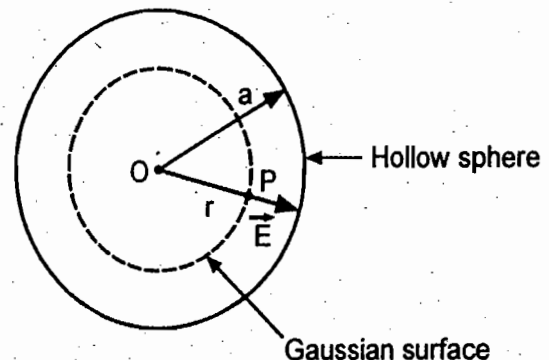


Fig. 3.18

The charge enclosed by Gaussian surface is zero.

$$\therefore \int \vec{D} \cdot d\vec{S} = 0$$

Thus, we get $D = 0$

So for all points inside the hollow sphere, $D = 0$

Let us find $\vec{D}$, at a point P situated outside the sphere i.e. $r > a$.

$$\text{Total charge on the sphere} = \sigma_s 4\pi a^2$$

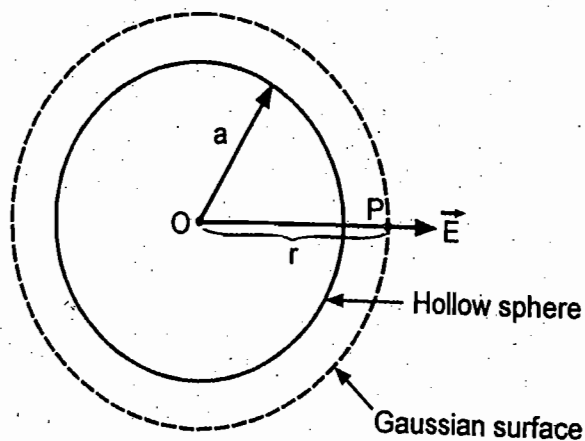


Fig. 3.19

Flux passing through Gaussian surface is,

$$\begin{aligned} \phi_E &= \int \vec{D} \cdot d\vec{S} \\ &= \int D dS \cos \theta \\ &= D \int dS \quad (\because \theta = 0) \\ &= D 4\pi r^2 \end{aligned}$$

According to Gauss's theorem,

$$D 4\pi r^2 = q_{\text{enc}}$$

$$\text{But, } q_{\text{enc}} = \sigma_s 4\pi a^2$$

$$\therefore D 4\pi r^2 = \sigma_s 4\pi a^2$$

$$\text{or } D = \sigma_s \left(\frac{a}{r}\right)^2$$

$$\text{In vector form, } \vec{D} = \sigma_s \left(\frac{a}{r}\right)^2 \hat{r}$$

We can calculate electric field also.

$$D = \epsilon_0 E \quad (\because P = 0)$$

$$\therefore \vec{E} = \frac{\sigma_s}{\epsilon_0} \left(\frac{a}{r}\right)^2 \hat{r}$$

Example 9 : A parallel plate capacitor of area 0.5 m^2 and plate separation 3 cm is filled with two dielectric slabs as shown in Fig. 3.20. The thickness of upper slab is 2 cm with dielectric constant 6 and lower slab is of thickness 1 cm and with dielectric constant 12 .

If a potential of 200 V is applied to the capacitor, find,

- (a) Polarization P ,
 (b) Electric displacement D and
 (c) Electric field E .

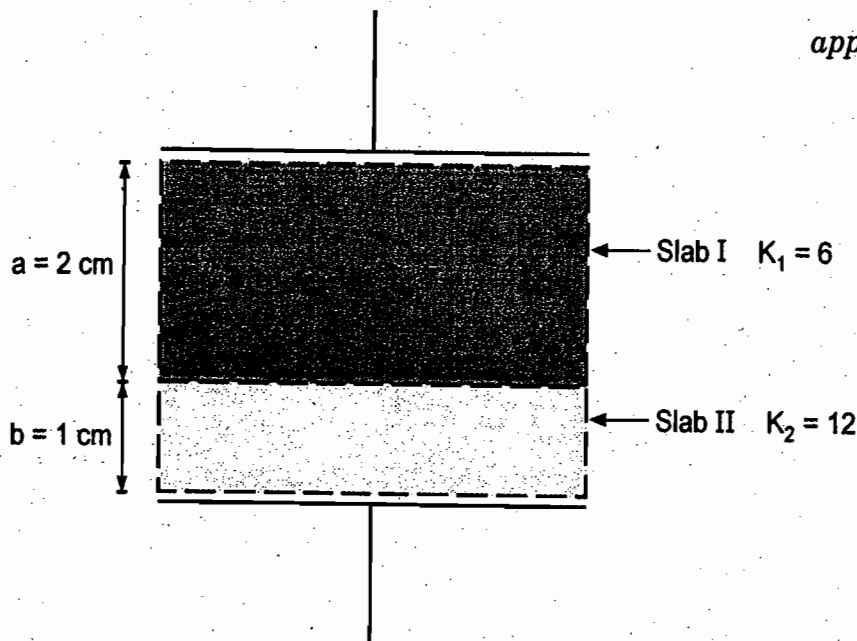


Fig. 3.20

Solution : The electric displacement is given by $D = k \epsilon_0 E$

Since it is dependent on free charges only, we have

$$\begin{aligned} D_1 &= D_2 \\ \text{But, } D_1 &= k_1 \epsilon_0 E_1 \\ \text{and } D_2 &= k_2 \epsilon_0 E_2 \\ \therefore E_1 &= \frac{k_2}{k_1} E_2 = \frac{12}{6} E_2 = 2E_2 \end{aligned}$$

Potential V is $V = E_1 \cdot a + E_2 \cdot b = 2 E_2 a + E_2 b = E_2 (2a + b)$

$$\therefore E_2 = \frac{V}{(2a + b)}$$

$$\text{and } D_2 = \frac{k_2 \epsilon_0 V}{(2a + b)}$$

$$(1) \quad D_1 = D_2 = \frac{8.85 \times 10^{-12} \times 12 \times 200}{(2 \times 20 + 10) \times 10^{-3}} = 4.25 \times 10^{-7} \text{ C/m}^2$$

$$(2) \quad E_1 = 2 E_2 = \frac{2 \times 200}{(2 \times 20 + 10) \times 10^{-3}} = 8 \times 10^3 \text{ V/m}$$

$$\therefore E_2 = 4 \times 10^3 \text{ V/m}$$

$$\begin{aligned} \text{Polarization, } P_1 &= D_1 - \epsilon_0 E_1 = 4.25 \times 10^{-7} - 8.85 \times 10^{-12} \times 8 \times 10^3 \\ &= 3.54 \times 10^{-7} \text{ C/m}^2 \end{aligned}$$

$$\begin{aligned} P_2 &= D_2 - \epsilon_0 E_2 = 4.25 \times 10^{-7} - 8.85 \times 10^{-12} \times 4 \times 10^3 \\ &= 3.90 \times 10^{-7} \text{ C/m}^2 \end{aligned}$$

Example 10 : Calculate the induced dipole moment per unit volume of Helium gas if placed in an electric field of 6000 V/cm. The molecular polarizability of helium is $\frac{21}{9} \times 10^{-41}$ farad-m² and the density of helium is 20.6×10^{25} molecules/m³.

Solution : Electric field is

$$E = 6000 \text{ V/cm} = 6 \times 10^5 \text{ V/m}$$

$$\alpha = \frac{21}{9} \times 10^{-41} \text{ farad-m}^2$$

Dipole moment is,

$$p = \alpha E$$

$$p = 14 \times 10^{-36} \text{ C-m}$$

Number of helium molecules

$$= 20.6 \times 10^{25} \text{ per m}^3$$

$\therefore$ Induced dipole moment per unit volume

$$p' = np$$

$$= 2.88 \times 10^{-9} \text{ C/m}^2$$

QUESTIONS

1. What do you mean by polar and non-polar molecules ?
2. Explain polarisation of matter.
3. Define electric polarisation and show that,

$$\sigma_p = \vec{P} \cdot \hat{n} \text{ and } \rho_p = -\vec{\nabla} \cdot \vec{P}$$

4. State Gauss's law in dielectrics and obtain its differential form.

5. Explain $\vec{D}$, $\vec{E}$ and $\vec{P}$ vectors and obtain relation between them.

6. Find relation between $\vec{D}$, $\vec{E}$ and $\vec{P}$. Hence explain the physical significance of three vectors.

7. What is meant by electric susceptibility ? Show that,

$$k = 1 + \chi_e$$

8. Define dielectric constant and show that electric displacement vector $\vec{D}$ is given by the relation,

$$\vec{D} = \epsilon \vec{E}$$

9. Obtain an expression for electric field at an exterior point of dielectric material.
10. Obtain an expression for electric field at a point inside the dielectric material.
11. Define the following terms :
 - (i) Polar and non-polar molecules,
 - (ii) Polarisation at a point,
 - (iii) Electric susceptibility,
12. Obtain an expression for electric field due to a point charge + q placed in an infinite dielectric.

PROBLEMS

1. A thin dielectric rod of cross-section 'a' extends along Z-axis from $z = 0$ to $z = L$. The polarisation is along its length, and is given by $P_z = Az^2 + B$. Find the volume density of polarisation charge and the surface polarisation charge on each end.
(Hint : Take $\vec{P}_z = (Az^2 + B)\hat{k}$, Use $\rho = -\vec{\nabla} \cdot \vec{P}$ and $\sigma = \vec{P} \cdot \hat{n}$)
(Ans. $\rho_p = -2Az$, on ends : $a(AL^2 + B)$, $-Ba$)
2. Show that the force between two charges separated by a distance 'r' is reduced by a factor $\frac{\epsilon_0 E}{\epsilon_0 E + P}$ due to the presence of dielectric.
3. The permittivity of diamond is $1.46 \times 10^{-10} \text{ C}^2/\text{N-m}^2$, calculate the dielectric constant of diamond. (Ans. $k = 16.5$)
4. Calculate the flux passing through a 4 m^2 area that is normal to the X-axis at $x = 3 \text{ cm}$. Given $\vec{D} = 10x \hat{a}_x$.
(Hint : Use formula $d\phi = \vec{D} \cdot d\vec{S}$) (Ans. 120 C)
5. Find the charge in the volume defined by $0 \leq y \leq 1 \text{ m}$, $0 \leq z \leq 1 \text{ m}$ and $0 \leq x \leq 1$, if $\rho = 10x^2y \mu \text{ C/m}^3$
(Hint : $dQ = \rho dV$)
$$Q = \int_0^1 \int_0^1 \int_0^1 10x^2y \, dx \, dy \, dz \quad (\mu \text{C}).$$
6. Find the magnitudes of $\vec{D}$ and $\vec{P}$ for a dielectric material in which $E = 0.10 \text{ MV/m}$ and $\chi_e = 5$.
(Ans. $D = 5.295 \times 10^{-6} \text{ C/m}^2$
 $P = 4.41 \times 10^{-6} \text{ C/m}^2$)
7. Determine the polarisation $\vec{P}$ in a dielectric material with $\epsilon_r = 3$, if $\vec{D} = 4 \times 10^7 \hat{a}_x \text{ C/m}^2$. (Ans. $2.66 \times 10^7 \hat{a}_x$)
8. Determine the value of E in a material for which electric susceptibility is 4 and $\vec{P} = 3 \times 10^{-7} \hat{a}_x \text{ C/m}^2$. (Ans. $8.49 \times 10^3 \hat{a}_x$)
9. A dielectric rod in the shape of a right circular cylinder of length l and radius 'a' is polarised in the direction of its length. If the polarisation is uniform and of magnitude P , find the electric field resulting from this polarisation at a point on the axis of the rod.

$$(\text{Ans. } E_z = \frac{P}{2\epsilon_0} \left[\frac{\frac{1}{2}l - z}{\sqrt{\left(\frac{1}{2}l - z\right)^2 + a^2}} + \frac{\frac{1}{2}l + z}{\sqrt{\left(\frac{1}{2}l + z\right)^2 + a^2}} \right])$$

MAGNETOSTATICS

INTRODUCTION

We have seen that an electrostatic field is produced by stationary or static charges. Now, if these charges are moving with constant velocity, static magnetic field is produced. A definite link between electric and magnetic field was established by Hans Christian Oersted (Danish Physicist 1771 to 1851). He discovered that an electric current flowing through a conductor produces magnetic fields. Before the discovery of this effect, electricity and magnetism were treated as separate branches. But, it was found that source of all magnetic fields is electric current, whether it be the current in a wire or the current produced by motion of charges within the atoms or molecules of a material. Experiments by Biot and Savart, Amper and others helped to establish the laws of magnetism as we know them today. In first three chapters of the present book, we have seen that static electric fields

are characterised by $\vec{E}$ or $\vec{D}$. We now focus our attention on static magnetic fields which are characterised by $\vec{B}$ or $\vec{H}$. Like electric potential (scalar quantity) in electrostatics, we shall study magnetic vector potential in this chapter. This chapter also includes magnetisation of matter which is parallel to polarisation in an electric field. Furthermore, this chapter includes magnetic energy, hysteresis and energy dissipation, etc.

Concept of Magnetic Field, Magnetic Induction and Biot-Savart's Law :

Magnetic Field :

The space around a magnet or a current carrying conductor is defined as a region of magnetic field where the magnetic effects are observed. The existence of magnetic field is detected by the mechanical force experienced by other magnet or a current carrying conductor (i.e. moving charges) when kept in this region. A magnetic field is said to exist in the given region if a moving test charge experiences a sideways deflecting force in that region.

The magnetic induction at a point is described or expressed in terms of quantity called magnetic induction ($\vec{B}$).

Magnetic Induction ($\vec{B}$) :

To define magnetic induction at a point in a magnetic field, consider a test charge q_0 moving with velocity $\vec{v}$ through a point in the region of magnetic field. If the moving charge q_0 is acted upon by a sideways force $\vec{F}$, the magnetic induction $\vec{B}$ at the given point is defined by the relation,

$$\vec{F}_B = q_0 \vec{v} \times \vec{B}$$

The magnitude of the force is given by,

$$F_B = q_0 v B \sin \theta$$

where, θ is the angle between $\vec{v}$ and $\vec{B}$.

Therefore, the magnitude of magnetic induction is,

$$B = \frac{F_B}{q_0 v \sin \theta}$$

S.I. unit of B is $\frac{N}{C} \frac{1}{m/s}$.

This S.I. unit is given the name weber/(meter)² or wb/m². It is also called tesla (T).

$$1 \text{ tesla} = \frac{\text{wb}}{\text{m}^2} = \frac{N}{C} \cdot \frac{1}{m/s}$$

Biot-Savart's Law :

A current carrying conductor (or wire) produces a magnetic field around it. The magnetic induction, at any point, in this field can be obtained with the help of Biot-Savart's law.

This law can be stated as follows :

The magnetic induction at a point due to an element of a conductor carrying a current is directly proportional to :

- (i) the current in the element,
- (ii) the length of the element,
- (iii) the sine of angle between the element and the line joining the centre of the element to the point,

and is inversely proportional to

- (iv) the square of the distance of the point from the centre of the element.

To express this law mathematically, consider a small element of length ' dl ' of a conductor carrying current I (Fig. 4.1). Take a point P at a distance r from the centre of this element. Let ' θ ' be the angle between the direction of current element ($I d\vec{l}$) and the line joining the centre of the element to the point P .

If 'dB' is magnetic induction at a point P due to this small element, then according to Biot-Savart's law,

$$dB \propto I, dB \propto dl, dB \propto \sin \theta \text{ and } dB \propto \frac{1}{r^2}$$

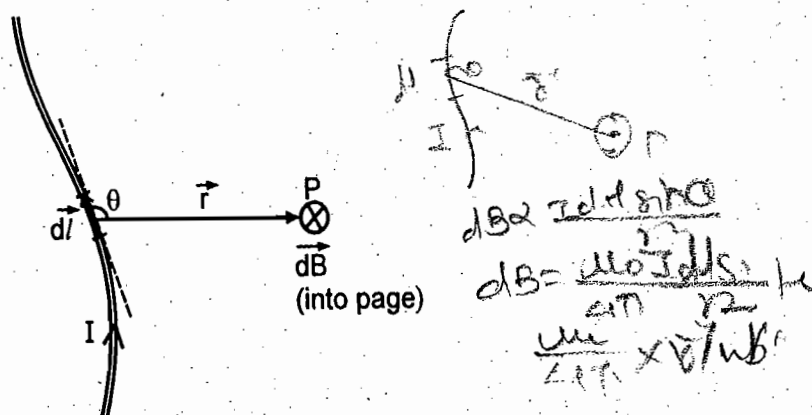


Fig. 4.1 : Biot-Savart's law (⊗ symbol indicates that dB is directed into page).
Combining all these terms, we get,

$$dB \propto \frac{I dl \sin \theta}{r^2}$$

$$\text{or} \quad dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2} \text{ tesla} \quad \dots (4.1)$$

where, μ_0 is called the permeability of free space

$$\mu_0 = 4\pi \times 10^{-7} \text{ wb/A-m}$$

$$\text{and } \frac{\mu_0}{4\pi} = 10^{-7} \text{ wb/A-m}$$

In vector form, equation (4.1) can be expressed as,

$$\vec{dB} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3} \quad \dots (4.2)$$

The direction of $\vec{dB}$ is that of cross product $I d\vec{l} \times \vec{r}$. In Fig. 4.1, direction of $\vec{dB}$ is perpendicular to the plane containing $d\vec{l}$ and $\vec{r}$ i.e. perpendicular to the plane of the page and pointing inward.

4.1 CURRENT DENSITY ($\vec{J}$)

If there are 'n' charges per unit volume at a point, each carrying a charge 'e' and moving with a mean velocity $\vec{v}$, then the quantity of charge passing per unit area perpendicular to the mean velocity per unit time, is,

$$\vec{J} = ne \vec{v}$$

$\vec{J}$ is called the current density and it is defined as the amount of charge passing per second through unit area of a plane normal to the line of flow.

If dS is the area element in the material, the amount of charge passing through that area in one second is $\vec{J} \cdot d\vec{S}$.

The net charge passing the surface per unit time is called electric current I and is given by,

$$I = \frac{dQ}{dt} = \int_S \vec{J} \cdot d\vec{S}$$

Let a closed surface S encloses a volume V . If ρ is the volume density of charge, the net charge enclosed by the volume is $\int_V \rho dV$. Since the electric charge is conserved, the net

flow of charge out of this volume must be equal to the rate of decrease of the total charge inside the volume.

$$\text{Therefore,} \quad \int_S \vec{J} \cdot d\vec{S} = -\frac{d}{dt} \int_V \rho dV \quad \dots (4.3)$$

$$\therefore \quad \int_S \vec{J} \cdot d\vec{S} = - \int_V \frac{d\rho}{dt} dV \quad \dots (4.4)$$

By divergence theorem, we have,

$$\int_S \vec{J} \cdot d\vec{S} = \int_V (\nabla \cdot \vec{J}) dV$$

Thus, equation (4.4) becomes,

$$\int_V (\nabla \cdot \vec{J}) dV = - \int_V \frac{\partial \rho}{\partial t} dV$$

$$\therefore \quad \int_V \left(\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} \right) dV = 0 \quad \dots (4.5)$$

This integral must be zero for any arbitrary volume V . Hence, the integrand must vanish, i.e.

$$\boxed{\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0} \quad \dots (4.6)$$

This is known as *equation of continuity* and is based on the law of conservation of charge.

$$\text{For steady state,} \quad \frac{\partial \rho}{\partial t} = 0$$

$$\text{Therefore,} \quad \nabla \cdot \vec{J} = 0 \quad \dots (4.7)$$

This equation is valid in the region where there is no creation of charges or absorption of charges.

4.2 FORCE ON A CURRENT ELEMENT : AMPERE'S FORCE LAW

Consider a charge ' q ' moving with velocity ' $\vec{v}$ ' through a point in magnetic field of induction $\vec{B}$. The force $\vec{F}$ experienced by a charge q is

$$\vec{F} = q \vec{v} \times \vec{B} \quad \dots (4.8)$$

The force experienced by small charge dq moving with velocity $\vec{v}$ in magnetic field of induction $\vec{B}$ is

$$d\vec{F} = dq \vec{v} \times \vec{B}$$

But, current $I = \frac{dq}{dt}$

$$\therefore d\vec{F} = I dt \vec{v} \times \vec{B}$$

Let in time dt , charge dq travels along the length $d\vec{l}$ of the conductor, then

$$\vec{v} = \frac{d\vec{l}}{dt}$$

$$\therefore d\vec{F} = I dt \frac{d\vec{l}}{dt} \times \vec{B}$$

$$\text{or } d\vec{F} = I d\vec{l} \times \vec{B} \quad \dots (4.9)$$

This is termed as Ampere's force law. $I d\vec{l}$ is called current element.

4.3 MAGNETIC FORCE BETWEEN TWO CURRENT CARRYING LOOPS

Consider two closed current carrying loops as shown in Fig. 4.2. Let I_1 be current flowing through loop 1 and I_2 be current flowing through loop 2.

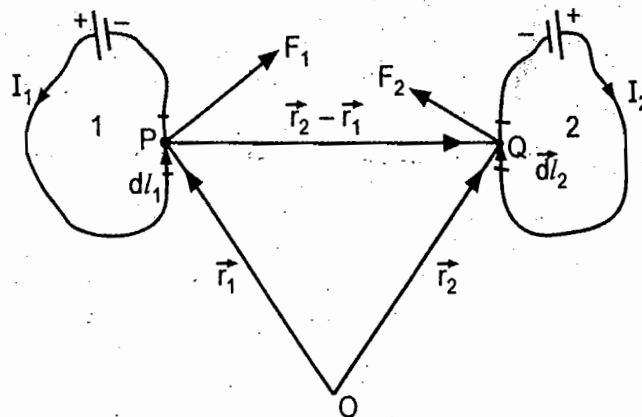


Fig. 4.2

Using Biot-Savart's law, the magnetic induction at the point Q due to current element $I_1 dl_1$ is given by,

$$d\vec{B}_1 = \frac{\mu_0}{4\pi} \frac{I_1 d\vec{l}_1 \times (\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^3}$$

Therefore, the total magnetic induction at point Q due to entire loop 1 is,

$$\vec{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint_1 \frac{d\vec{l}_1 \times (\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^3} \quad \dots (4.10)$$

The force experienced by the element $d\vec{l}_2$ due to this field is given by Ampere's force law, which is

$$d\vec{F}_2 = I_2 d\vec{l}_2 \times \vec{B}_1 \quad \dots (4.11)$$

Thus, the force on whole loop 2 due to magnetic induction $\vec{B}_1$ of first loop is,

$$\vec{F}_2 = \oint_2 I_2 d\vec{l}_2 \times \vec{B}_1$$

$$\therefore \vec{F}_2 = \frac{\mu_0}{4\pi} I_1 I_2 \oint_2 \oint_1 \frac{d\vec{l}_2 \times [d\vec{l}_1 \times (\vec{r}_2 - \vec{r}_1)]}{|\vec{r}_2 - \vec{r}_1|^3} \quad \dots (4.12)$$

Mathematical formulation of Ampere's force law

Similarly, the force exerted by the loop 2 on loop 1 is

$$\vec{F}_1 = \frac{\mu_0}{4\pi} I_1 I_2 \oint_1 \oint_2 \frac{d\vec{l}_1 \times [d\vec{l}_2 \times (\vec{r}_1 - \vec{r}_2)]}{|\vec{r}_1 - \vec{r}_2|^3} \quad \dots (4.13)$$

It should be noted that equation (4.13) is obtained by interchanging suffix 1 by 2 and vice-versa from equation (4.12). But, direction of $(\vec{r}_2 - \vec{r}_1)$ is opposite to that of $(\vec{r}_1 - \vec{r}_2)$. Hence, the cross product in equation (4.12) has a sign opposite to that of cross product in equation (4.13). Change in sequence of integration $\oint_2 \oint_1$ to $\oint_1 \oint_2$ does not affect the magnitude of the result. Hence, we conclude that

$$\vec{F}_2 = -\vec{F}_1 \quad \text{or} \quad \vec{F}_1 = -\vec{F}_2$$

Thus, the force between the current loops due to their magnetic interaction are same in magnitude but opposite in direction. This shows that it is in conformity with Newton's law.

Equations (4.12) and (4.13) are the mathematical formulation of Ampere's force law.

4.4 DIVERGENCE OF A MAGNETIC INDUCTION

To find out divergence of a magnetic induction, consider a current loop carrying current I as shown in Fig. 4.3.

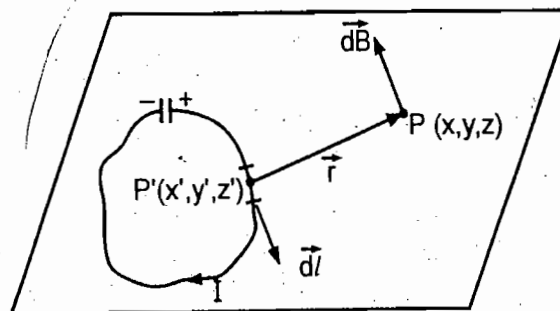
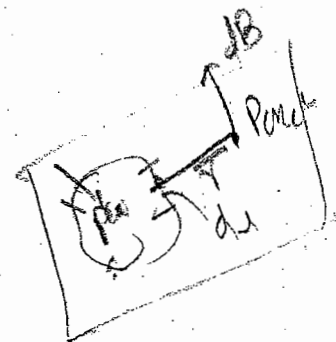


Fig. 4.3



The magnetic induction $\vec{dB}$ at point P (x, y, z) due to small current element $I \vec{dl}$ situated at P' (x', y', z') is

$$\vec{dB} = \frac{\mu_0}{4\pi} \frac{I \vec{dl} \times \vec{r}}{r^3}$$

The total magnetic induction $\vec{B}$ at point P due to entire current loop is

$$\vec{B} = \frac{\mu_0 I}{4\pi} \oint \frac{\vec{dl} \times \vec{r}}{r^3}$$

Taking divergence of $\vec{B}$, we get

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0 I}{4\pi} \vec{\nabla} \cdot \oint \frac{\vec{dl} \times \vec{r}}{r^3}$$

By interchanging differentiation and integration operation, we get

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0 I}{4\pi} \oint \vec{\nabla} \cdot \left(\frac{\vec{dl} \times \vec{r}}{r^3} \right)$$

We know the identity

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0 I}{4\pi} \oint \left\{ \left(\frac{\vec{r}}{r^3} \right) \cdot (\vec{\nabla} \times \vec{dl}) - \vec{dl} \cdot \left(\vec{\nabla} \times \frac{\vec{r}}{r^3} \right) \right\}$$

Since $\vec{dl}$ is not a function of (x, y, z), we have,

$$\vec{\nabla} \times \vec{dl} = 0$$

$$\vec{\nabla} \cdot \vec{B} = -\frac{\mu_0 I}{4\pi} \oint \left\{ \vec{dl} \cdot \left(\vec{\nabla} \times \frac{\vec{r}}{r^3} \right) \right\}$$

Since

$$\vec{\nabla} \left(\frac{1}{r} \right) = -\frac{\vec{r}}{r^3}$$

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0 I}{4\pi} \oint \left[\vec{dl} \cdot \vec{\nabla} \times \vec{\nabla} \left(\frac{1}{r} \right) \right]$$

We know that curl of gradient of a scalar function is always zero i.e.

$$\vec{\nabla} \times \vec{\nabla} \left(\frac{1}{r} \right) = 0$$

$$\therefore \vec{\nabla} \cdot \vec{B} = 0$$

Thus, divergence of a magnetic induction $\vec{B}$ is always zero i.e. magnetic monopole does not exist.

4.5 MAGNETIC VECTOR POTENTIAL ($\vec{A}$)

We know that electrostatic field intensity $\vec{E}$ at any point can be derived from the electric potential 'V' at that point by the equation

$$\vec{E} = -\vec{\nabla} V$$

In a similar manner, magnetic induction $\vec{B}$ can be expressed as,

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$(\because \vec{\nabla} \cdot \vec{B} = 0)$$

where, $\vec{A}$, by analogy, is called as *magnetic vector potential*.

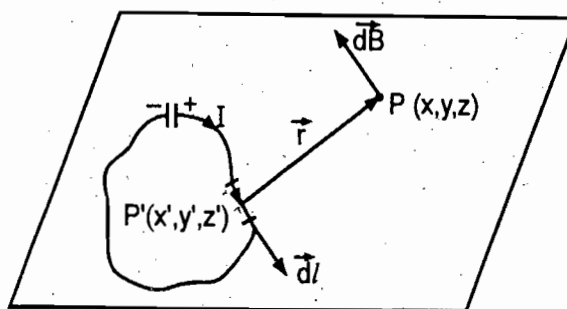


Fig. 4.4

To obtain expression for $\vec{A}$, consider the current carrying loop as shown in Fig. 4.4

According to Biot and Savart's law, magnetic induction $d\vec{B}$ at point P (x, y, z) due to current element $I d\vec{l}$ at the point P' (x', y', z') is given by,

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$$

Magnetic induction at point P due to entire current loop is

$$\vec{B} = \frac{\mu_0}{4\pi} I \oint \frac{d\vec{l} \times \vec{r}}{r^3}$$

We know that,

$$\vec{\nabla} \left(\frac{1}{r} \right) = -\frac{\vec{r}}{r^3}$$

$$\vec{E} = -\vec{\nabla} V$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

A-mmm

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{dl \sin \theta}{r^2}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \oint \frac{d\vec{l} \times \vec{r}}{r^3}$$

where, $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$

$$\therefore \vec{B} = -\frac{\mu_0 I}{4\pi} \oint d\vec{l} \times \vec{\nabla} \left(\frac{1}{r} \right) \quad \dots (4.14)$$

Let us use vector identity

$$\vec{\nabla} \times (f \vec{A}) = f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} f)$$

or

$$\vec{A} \times (\vec{\nabla} f) = f(\vec{\nabla} \times \vec{A}) - \vec{\nabla} \times (f \vec{A})$$

Replacing $\vec{A}$ by $d\vec{l}$ and f by $\frac{1}{r}$ we have,

$$d\vec{l} \times \left(\vec{\nabla} \frac{1}{r} \right) = \frac{1}{r} (\vec{\nabla} \times d\vec{l}) - \vec{\nabla} \times \left(\frac{1}{r} d\vec{l} \right)$$

Substituting above expression in equation (4.14), we get,

$$\vec{B} = \frac{\mu_0 I}{4\pi} \left\{ \oint \left(\vec{\nabla} \times \frac{d\vec{l}}{r} \right) - \oint \frac{1}{r} (\vec{\nabla} \times d\vec{l}) \right\}$$

As $d\vec{l}$ is function of (x', y', z') and not the function of (x, y, z) hence we have,

$$\vec{\nabla} \times d\vec{l} = 0$$

$$\therefore \vec{B} = \frac{\mu_0 I}{4\pi} \oint \vec{\nabla} \times \frac{d\vec{l}}{r}$$

The equation can be written as,

$$\vec{B} = \frac{\mu_0 I}{4\pi} \vec{\nabla} \times \oint \frac{d\vec{l}}{r}$$

$$\text{or} \quad \vec{B} = \vec{\nabla} \times \frac{\mu_0 I}{4\pi} \oint \frac{d\vec{l}}{r} \quad \dots (4.15)$$

The vector quantity $\frac{\mu_0 I}{4\pi} \oint \frac{d\vec{l}}{r}$ is called magnetic vector potential and it is denoted by $\vec{A}$. Thus, magnetic induction is given by curl of vector potential. If the current is distributed with current density $\vec{J}$ such that $I = \int \vec{J} \cdot d\vec{S}$ then $\vec{A}$ can be written as,

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J} dV'}{r} \quad (\text{Since } d\vec{S} \cdot d\vec{l} = dV')$$

4.6 THE DIVERGENCE OF MAGNETIC VECTOR POTENTIAL $\vec{A}$

The magnetic vector potential at a point is given by,

(4)

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J}}{r} dV'$$

where, J is current density at source point $P' (x', y', z')$ and $\vec{A}$ is magnetic vector potential at point P (See Fig. 4.5).

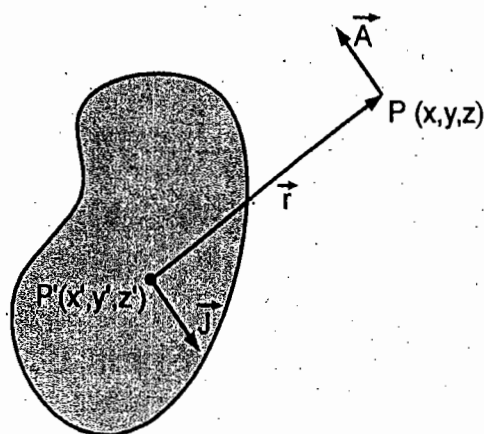


Fig. 4.5

In Fig. 4.5, $P' (x', y', z')$ is called source point and $P (x, y, z)$ is called field point. The volume element dV' , current density J , density ρ are at source point (x', y', z') and $\vec{A}$ is at field point (x, y, z) . The operator $\vec{\nabla}$ involves derivative at field point (x, y, z) and $\vec{\nabla}'$ involves derivative at source point (x', y', z') .

Taking divergence of $\vec{A}$, we get,

$$\vec{\nabla} \cdot \vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \vec{\nabla} \cdot \left(\frac{\vec{J}}{r} \right) dV'$$

$$\text{But, } \vec{\nabla} \cdot \left(\frac{\vec{J}}{r} \right) = \frac{1}{r} \vec{\nabla} \cdot \vec{J} + \vec{J} \cdot \nabla \left(\frac{1}{r} \right)$$

$$\therefore \vec{\nabla} \cdot \vec{A} = \frac{\mu_0}{4\pi} \left[\int_{V'} \frac{1}{r} (\vec{\nabla} \cdot \vec{J}) dV' + \int_{V'} \vec{J} \cdot \vec{\nabla} \left(\frac{1}{r} \right) dV' \right]$$

Since J is a function of (x', y', z') and $\vec{\nabla}$ is function of (x, y, z) ,

$$\vec{\nabla} \cdot \vec{J} = 0$$

$$\therefore \vec{\nabla} \cdot \vec{A} = \frac{\mu_0}{4\pi} \int_V \vec{J} \cdot \vec{\nabla} \left(\frac{1}{r} \right) dV' \quad \dots (4.16)$$

where, $r = [(x - x')^2 + (y - y')^2 + (z - z')^2]^{1/2}$

We know that,

$$\begin{aligned} \vec{\nabla} \left(\frac{1}{r} \right)^* &= -\frac{\vec{r}}{r^3} \\ \text{and } \nabla' \left(\frac{1}{r} \right) &= \frac{\vec{r}}{r^3} \\ \therefore \nabla \left(\frac{1}{r} \right) &= -\nabla' \left(\frac{1}{r} \right) \quad \dots (4.17) \end{aligned}$$

Using equation (4.17) in equation (4.16), we get

$$\vec{\nabla} \cdot \vec{A} = -\frac{\mu_0}{4\pi} \int_V \vec{J} \cdot \nabla' \left(\frac{1}{r} \right) dV'$$

We know that

$$\nabla' \cdot \left(\frac{\vec{J}}{r} \right) = \frac{1}{r} (\vec{\nabla}' \cdot \vec{J}) + \vec{J} \cdot \nabla' \left(\frac{1}{r} \right)$$

$$\begin{aligned} * \quad \vec{\nabla} \left(\frac{1}{r} \right) &= \vec{i} \frac{\partial}{\partial x} [(x - x')^2 + (y - y')^2 + (z - z')^2]^{-1/2} + \vec{j} \frac{\partial}{\partial y} [(x - x')^2 + (y - y')^2 + (z - z')^2]^{-1/2} \\ &\quad + \vec{k} \frac{\partial}{\partial z} [(x - x')^2 + (y - y')^2 + (z - z')^2]^{-1/2} \\ &= -\frac{1}{2} [(x - x')^2 + (y - y')^2 + (z - z')^2]^{-3/2} \\ &\quad \left\{ \vec{i} \frac{\partial}{\partial x} [(x - x')^2 + (y - y')^2 + (z - z')^2] + \vec{j} \frac{\partial}{\partial y} [(x - x')^2 + (y - y')^2 + (z - z')^2] \right. \\ &\quad \left. + \vec{k} \frac{\partial}{\partial z} [(x - x')^2 + (y - y')^2 + (z - z')^2] \right\} \\ &= -\frac{1}{2r^3} \left\{ 2(x - x') \vec{i} + 2(y - y') \vec{j} + 2(z - z') \vec{k} \right\} \\ &= -\frac{\vec{r}}{r^3} \end{aligned}$$

In a similar manner, it can be shown that $\nabla' \left(\frac{1}{r} \right) = +\frac{\vec{r}}{r^3}$

6) or
$$\vec{J} \cdot \nabla' \left(\frac{1}{r} \right) = \vec{\nabla}' \cdot \left(\frac{\vec{J}}{r} \right) - \frac{1}{r} (\vec{\nabla}' \cdot \vec{J})$$

$$\therefore \vec{\nabla} \cdot \vec{A} = -\frac{\mu_0}{4\pi} \left\{ \int_{V'} \vec{\nabla}' \cdot \frac{\vec{J}}{r} dV' - \int_{V'} \frac{1}{r} (\vec{\nabla}' \cdot \vec{J}) dV' \right\}$$

According to Gauss's divergence theorem, we can write

$$\int_{V'} \vec{\nabla}' \cdot \frac{\vec{J}}{r} dV' = \int_S \frac{\vec{J}}{r} \cdot d\vec{S}'$$

7)
$$\therefore \vec{\nabla} \cdot \vec{A} = -\frac{\mu_0}{4\pi} \left\{ \int_{S'} \frac{\vec{J}}{r} \cdot d\vec{S}' - \int_{V'} \frac{1}{r} (\vec{\nabla}' \cdot \vec{J}) dV' \right\}$$

Everywhere on the surface S' , $\vec{J}$ is either zero or tangential. So,

$$\vec{J} \cdot d\vec{S} = 0$$

$$\therefore \vec{\nabla} \cdot \vec{A} = -\frac{\mu_0}{4\pi} \int_{V'} \frac{1}{r} (\vec{\nabla}' \cdot \vec{J}) dV'$$

From the equation of continuity, we have,

$$\vec{\nabla}' \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\therefore \vec{\nabla} \cdot \vec{A} = -\frac{\mu_0}{4\pi} \int_{V'} \frac{1}{r} \frac{\partial \rho}{\partial t} dV'$$

Above equation can also be written as,

$$\vec{\nabla} \cdot \vec{A} = -\frac{\mu_0}{4\pi} \frac{\partial}{\partial t} \int_{V'} \frac{\rho}{r} dV' \quad \dots (4.18)$$

where, ρ is volume charge density.

We know that, potential produced by continuous charge distribution is

$$\phi = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho dV'}{r}$$

or,
$$\epsilon_0 \phi = \frac{1}{4\pi} \int_{V'} \frac{\rho}{r} dV' \quad \dots (4.19)$$

Using equation (4.19) in equation (4.18), we get

$$\vec{\nabla} \cdot \vec{A} = -\mu_0 \epsilon_0 \frac{\partial \phi}{\partial t} \quad \dots (4.20)$$

where, ϕ is electric potential.

This is expression for divergence of magnetic vector potential.

If, $\phi = \text{constant}$

we get

$$\vec{\nabla} \cdot \vec{A} = 0$$

4.7 AMPERE'S CIRCUITAL LAW

This law states that, the line integral of magnetic induction $\vec{B}$ around a closed curve (or loop) is equal to μ_0 times the net current 'I' through the area bounded by the curve.

i.e.
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad \dots (4.21)$$

where, μ_0 is the permeability of vacuum (or free space). The law indicates that the magnetic field of a current is non-conservative. Equation (4.21) is integral form of Ampere's law.

The current I in terms of current density J can be written as,

$$I = \oint_S \vec{J} \cdot d\vec{S}$$

$$\therefore \oint_C \vec{B} \cdot d\vec{l} = \mu_0 \oint_S \vec{J} \cdot d\vec{S}$$

Using Stoke's theorem, we have,

$$\oint_C \vec{B} \cdot d\vec{l} = \oint_C (\vec{\nabla} \times \vec{B}) \cdot d\vec{S}$$

$$\therefore \oint_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{S} = \mu_0 \oint_S \vec{J} \cdot d\vec{S}$$

$$\text{or } \oint_S [\vec{\nabla} \times \vec{B} - \mu_0 \vec{J}] \cdot d\vec{S} = 0$$

As this is true for any arbitrary surface element, we have

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \dots (4.22)$$

Equation (4.22) is differential form of Ampere's law. If the closed curve or loop does not enclose the current carrying conductor, then, we have,

$$\oint \vec{J} \cdot d\vec{l} = 0$$

$$\therefore \nabla \times \vec{B} = 0$$

Equation (4.22) is differential form of Ampere's law. Maxwell modified Ampere's

law (will be discussed in Chapter V) replacing $\vec{J}$ by $\vec{J} + \frac{\partial \vec{D}}{\partial t}$.

$$\therefore \nabla \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \quad \dots (4.23)$$

This equation is modified Ampere's law.

4.8 ENERGY DENSITY IN A MAGNETIC FIELD

Any current distribution can be considered as being built up step-by-step. We know that current passing through the conductor produces a magnetic field around it. Therefore, an increasing current through conductor produces an increasing field around the conductor. This time varying magnetic field induces back e.m.f. (Law of electromagnetic induction). The induced back e.m.f. exerts a force on the electric charges constituting the current flow. At any instant, work must be done by external voltage against the back emf to maintain the current. This work, which is provided by external agent (such as battery) is stored as a magnetic energy around the conductor.

To obtain an expression for energy density (U) in a magnetic field, we need a system which produces a uniform field B in a given volume. Such a system is long solenoid. (A solenoid is a long wire wound closely in the form of helix and carries an electric steady current).

In a long, closely wound solenoid, we can make the assumption that the magnetic field which is produced is confined to the cylindrical volume inside the solenoid. And the field is uniform over the cross-section; i.e. B has almost constant magnitude at all points (except end points) inside the solenoid. It is reasonable to expect that energy density should depend only on B. Hence, in a region where B is uniform, the magnetic energy density can also be expected to be uniform.

Consider a long solenoid of radius 'a' and carrying an electric current 'I'.

The magnetic induction B at points inside the solenoid is,

$$B = \mu_0 n I \quad \dots (4.24)$$

where, n is number of turns per unit length, I is current flowing through the wire of solenoid and μ_0 is magnetic permeability of free space.

The self-inductance of unit length of solenoid is

$$L^* = \pi a^2 \mu_0 n^2$$

* The magnetic induction at an interior point is $B = \mu_0 n I$. Magnetic flux over the cross-section is $\phi = \mu_0 n I \cdot \pi a^2$, where πa^2 is cross-sectional area of solenoid. Since n is number of turns per unit length, the total flux linkage per unit length = $\pi a^2 \mu_0 n^2 I$. Self-inductance is flux linkage per unit current.

Therefore, $L = \pi a^2 \mu_0 n^2$

The energy stored per unit length of solenoid is $\frac{1}{2} LI^2$

$$\therefore \frac{1}{2} LI^2 = \frac{1}{2} \pi a^2 \mu_0 n^2 I^2 \quad \dots (4.25)$$

To show that energy stored per unit length of solenoid is $\frac{1}{2} LI^2$, consider a circuit consisting of inductive coil of unit length having self-inductance L , resistor of resistance R and battery of e.m.f. E volt as shown in Fig. 4.6.

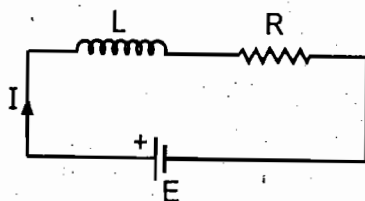


Fig. 4.6

Let 'I' be the current flowing in the circuit at time instant t .

$$\text{Voltage across inductor} = L \frac{dI}{dt}$$

$$\text{Voltage across resistor} = IR$$

Using Kirchhoff's law, we get,

$$L \frac{dI}{dt} + IR = E$$

Work done by e.m.f. (E) of a battery in moving a small amount of charge dQ through circuit is,

$$dW = E dQ$$

$$\text{or} \quad dW = E dt \frac{dQ}{dt}$$

$$\text{But,} \quad \frac{dQ}{dt} = I$$

$$\therefore dW = EI dt$$

$$\text{or} \quad \frac{dW}{dt} = EI = LI \frac{dI}{dt} + RI^2 \quad \left(\because E = L \frac{dI}{dt} + IR \right)$$

Initially at time $t = 0$, current flowing through circuit is zero, and at time T , let the current flowing is I_T .

Therefore, the total work done by e.m.f. of battery in time interval T is,

$$\begin{aligned} W &= \int_0^T EI dt \\ &= \int_0^T \left(LI \frac{dI}{dt} + RI^2 \right) dt \end{aligned}$$

$$\begin{aligned}
 &= L \int_0^T I dI + R \int_0^T I^2 dt \\
 &= \frac{1}{2} L I_T^2 + R \int_0^T I^2 dt \quad \dots (4.26)
 \end{aligned}$$

The term $\frac{1}{2} L I_T^2$ is the energy stored in the inductance in time T and the term $R \int_0^T I^2 dt$ is the energy dissipated as heat in the resistance.

The energy $\frac{1}{2} L I^2$ can be considered to be stored in the magnetic field of solenoid.

From equation (4.25),

$$\text{Energy stored per unit length of solenoid} = \frac{1}{2} L I^2 = \frac{1}{2} \pi a^2 \mu_0 n^2 I^2$$

Energy stored per unit volume of solenoid

$$\begin{aligned}
 &= \frac{\frac{1}{2} L I^2}{\pi a^2} = \frac{1}{2} \mu_0 n^2 I^2
 \end{aligned}$$

Energy stored per unit volume is called energy density (U).

$$\therefore U = \frac{1}{2} \frac{(\mu_0 n I)^2}{\mu_0}$$

But,

$$B = \mu_0 n I$$

$\therefore$

$$U = \frac{1}{2} \frac{B^2}{\mu_0} \quad \dots (4.27)$$

or

$$U = \frac{1}{2} \frac{\vec{B} \cdot \vec{B}}{\mu_0}$$

Since $\vec{B} = \mu_0 \vec{H}$ (for free space), above equation may be expressed as,

$$U = \frac{1}{2} \vec{B} \cdot \vec{H} \quad \dots (4.28)$$

Note that the above relation is true only when B is linearly proportional to H .

Though above expression is derived for the case of a solenoid, it is quite general.

4.9 MAGNETISATION OF MATTER

In the chapter, Dielectrics, we have seen that when a dielectric slab or material is kept in external electric field, it gets polarised. In the process of polarisation, equal and opposite charges are induced on opposite faces of the dielectric material and due to this, material acquires a resultant dipole moment. The dipole moment per unit volume is called polarisation or polarisation density.

We come across a similar situation in magnetism. If a magnetic material is kept in external magnetic field, it gets magnetised and a material acquires a resultant magnetic dipole moment. The magnetic dipole moment per unit volume is called magnetisation. Let us study, how magnetisation occurs in material ?

We know that a given material is composed of atoms. Each atom consists of electrons those move in orbits around the nucleus as well as spin about their own axes. The circulating electrons are equivalent to current loops. Such circulating currents were first postulated by Ampere and are called Amperian currents or Atomic currents. We know that any current carrying element of very small dimension can be considered as magnetic dipole. So in a given material, each current loop can be considered as a magnetic dipole and hence it has magnetic dipole moment.

In the absence of external magnetic field, the loops (or magnetic dipoles) are oriented at random so that the vector sum of their magnetic dipole moments is zero and the material does not possess any magnetic field.

When the material is kept in external field, the magnetic dipole moments (or magnetic moments) of the atoms more or less align themselves with the direction of external field so that the net magnetic moment is not zero and the material is said to be *magnetised*.

Depending upon the behaviour of substances under external magnetic field, substances are classified broadly under three categories. These are as follows :

1. Diamagnetic substances : In this type of substances, magnetisation produced in the substance is proportional to the applied magnetic field but it is directed opposite to that of direction of applied magnetic field.

In a non-uniform magnetic field, such substances experience a force in the direction of decreasing field strength. e.g. Cu, Pb, graphite, etc.

2. Paramagnetic substances : In this type of substances, magnetisation produced in the substance is proportional to the applied magnetic field and directed along the same direction as that of applied magnetic field.

In a non-uniform magnetic field, such substances experience a force in the direction of increasing field strength.

e.g. platinum, aluminium, etc.

These materials when taken out of applied field, their magnetic field vanishes.

3. Ferromagnetic substances : Magnetisation produced in the substance is not proportional to applied magnetic field but directed along the same direction as that of external magnetic field.

These materials retain their magnetic field even after the applied magnetic field is removed.

e.g. iron, nickel and cobalt and all their alloys, etc.

Definition of Magnetic Dipole Moment :

Any current carrying element of very small dimension (Fig. 4.7) can be considered as magnetic dipole.

The magnetic dipole moment is given by,

$$\vec{m} = I\vec{A}$$

$$\vec{m} = I\vec{A}$$

where, I is the current flowing through a loop of area A .

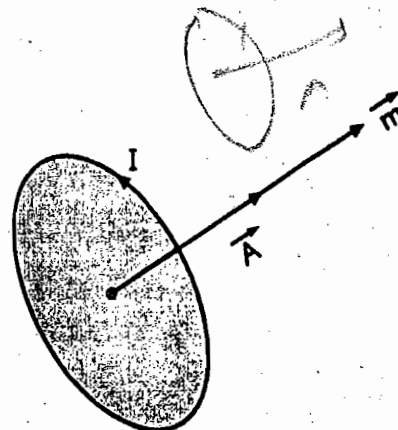


Fig. 4.7

S.I. unit of magnetic dipole moment is Ampere-(meter)².

Magnetisation ($\vec{M}$) :

In the material substances, the magnetic dipoles are randomly oriented, but if the material is placed in external magnetic field, alignment of dipole takes place and the material is said to be magnetised. The effect of atomic magnet is described by a quantity called magnetisation. ($\vec{M}$). It is defined as the magnetic dipole moment per unit volume.

If there are N atoms in a given volume Δv , and the i^{th} atom has a magnetic moment $\vec{m}_i$, we define magnetisation as,

$$\vec{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{i=1}^N \vec{m}_i}{\Delta v}$$

any other val

S.I. unit of $\vec{M}$:

Dimensionally, above equation can be written as,

$$\begin{aligned} \text{Magnetisation} &= \frac{\text{Magnetic dipole moment}}{\text{Volume}} \\ &= \frac{\text{Current} \times \text{Area}}{\text{Volume}} = \frac{\text{Amp. met}^2}{(\text{met})^3} \end{aligned}$$

Thus, S.I. unit of magnetisation is Ampere/meter.

To derive a relation between magnetisation current density $\vec{J}_m$ and magnetisation $\vec{M}$, consider two small elements, each of volume $\delta x \delta y \delta z$ located close to each other in the direction of x -axis. Let $\vec{M}$ is magnetisation vector in the first volume element at a point (x, y, z) .

Component of magnetic moment associated with the first volume element in the z -direction is $M_z \delta x \delta y \delta z$.

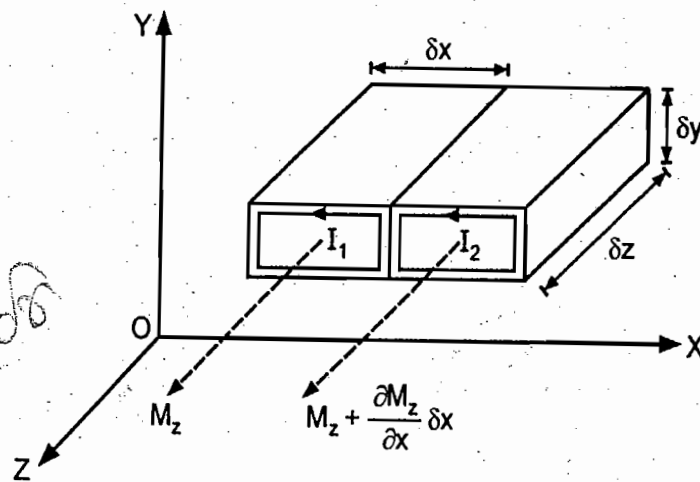


Fig. 4.8

We know that, magnetic moment is product of current passing through the loop and area under the loop.

So, $M_z \delta x \delta y \delta z = I_1 \delta x \delta y$

where, I_1 is current flowing in the loop parallel to xy plane.

$\therefore I_1 = M_z \delta z$

The current in adjacent loop at $(x + \delta x, y, z)$ is

$$I_2 = I_1 + \frac{\partial I_1}{\partial x} \delta x$$

Substituting $I_1 = M_z \delta z$ in above equation, we get

$$I_2 = M_z \delta z + \frac{\partial M_z}{\partial x} \delta x \delta z$$

Hence, net upward current in the interface between two volume elements is,

$$I_1 - I_2 = -\frac{\partial M_z}{\partial x} \delta x \delta z$$

In a similar manner, if we consider two adjacent volume elements in the yz -plane, then net upward current in the interface between the two volume elements is

$$\frac{\partial M_x}{\partial z} \delta x \delta z$$

Thus, the total current in y -direction is

$$I_y = \left(\frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x} \right) \delta x \delta z$$

This equation can be written in terms of magnetisation current density $(J_m)_y$ as,

$$(J_m)_y = \left(\frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x} \right)$$

$$\text{Since, } (J_m)_y = \frac{I_y}{\delta x \delta z}$$

Similarly, it can be shown that,

$$(J_m)_x = \left(\frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z} \right)$$

and $(J_m)_z = \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right)$

Hence, $\vec{J}_m = \text{Curl } \vec{M}$

or $\vec{J}_m = \nabla \times \vec{M}$

Thus, incomplete cancellation of electronic circuit produce magnetisation currents. Note that magnetisation current flow exists only if $\vec{M}$ is varying.

$$\vec{J}_m = 0 \text{ if } \vec{M} \text{ is uniform}$$

Magnetic Intensity ($\vec{H}$):

When a magnetic material is kept in magnetic field, it gets magnetised. The capability of the magnetic field to magnetise a material is expressed by $\vec{H}$, called the magnetic intensity of the field.

Magnetic intensity ($\vec{H}$) is a vector defined by the relation,

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$

where, μ_0 is permeability of free space.

The above equation may be written as,

$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

This is the relation between three magnetic vectors.

From the above discussion, it is clear that magnetisation vector $\vec{M}$ is analogous to electric polarisation. The Amperian current in the material corresponds to induced charges in the dielectric. We have three electric vectors $\vec{E}$, $\vec{D}$ and $\vec{P}$ in electrostatics and the relation between them is $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$. In a similar manner we have three magnetic vectors $\vec{B}$, $\vec{H}$ and $\vec{M}$ in magnetostatics and the relation between them is,

$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

4.10 MAGNETIC SUSCEPTIBILITY AND RELATIVE PERMEABILITY

Magnetic Susceptibility:

For most magnetic materials, the magnetisation $\vec{M}$ is always proportional to the magnetic field intensity $\vec{H}$ that is,

$$\vec{M} \propto \vec{H}$$

or $\vec{M} = \chi_m \vec{H}$

... (4.29)

The constant χ_m is called as the magnetic susceptibility.

From above,
$$\chi_m = \frac{|\vec{M}|}{|\vec{H}|} \quad \dots (4.30)$$

Thus, magnetic susceptibility is defined as the ratio of magnetisation M to the magnetic field intensity H . It is pure number and has no unit.

Using χ_m we can classify the magnetic materials.

For free space or vacuum $\chi_0 = 0$.

For paramagnetic materials, χ_m is positive and less than one.

For diamagnetic materials, χ_m , is negative and less than one.

For ferromagnetic materials, χ_m is positive and greater than one but not constant.

Magnetic Permeability:

The relation between $\vec{B}$, $\vec{H}$ and $\vec{M}$ is,

$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

We know that,

$$\vec{M} = \chi_m \vec{H}$$

$$\vec{B} = \mu_0 (\vec{H} + \chi_m \vec{H})$$

$$\vec{B} = \mu_0 (1 + \chi_m) \vec{H}$$

$(1 + \chi_m)$ is constant and is called as relative permeability denoted by μ_r .

So, $1 + \chi_m = \mu_r$

Hence, above equation becomes,

$$\vec{B} = \mu_0 \mu_r \vec{H}$$

As μ_0 and μ_r are constants, so let

$$\mu_0 \mu_r = \mu$$

is called as magnetic permeability of the material.

$\therefore$
$$\vec{B} = \mu \vec{H} \quad \dots (4.32)$$

Thus, magnetic permeability may be defined as the ratio of magnetic induction B to the magnetic intensity H .

One can also classify the magnetic materials from the value of relative permeability.

For diamagnetic substances, $\mu_r < 1$.

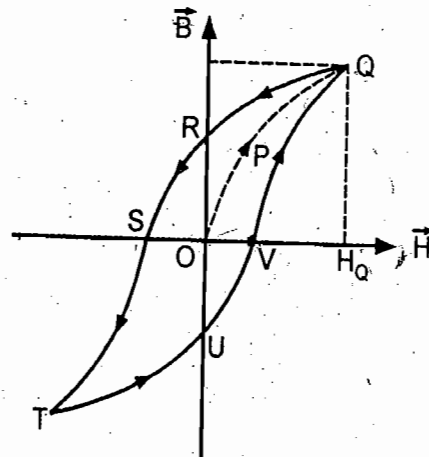
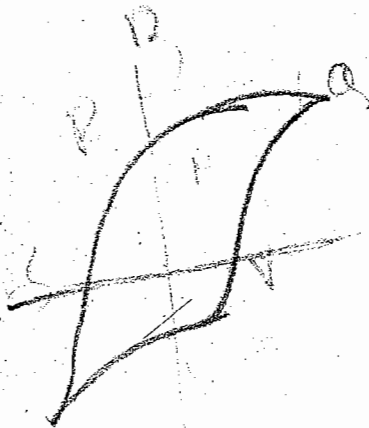
For paramagnetic substances, $\mu_r > 1$.

For ferromagnetic substances, $\mu_r \gg 1$.

4.11 HYSTERESIS, THE B - H CURVE

To study behaviour of ferromagnetic material, let us keep ferromagnetic substance such as iron in external magnetic field. By using proper experimental arrangement, intensity of magnetic field (H) around the substance can be varied. For different values of H , B is measured. The variation in B with variation in H is as shown in Fig. 4.9.

- (i) The point O in figure represents, initially substance is unmagnetised and there is no external field i.e. $H = 0$. In this situation, the magnetic dipole of atoms of the substance are randomly oriented. So B is zero.



B is mag. induction
 $B = \frac{F_B}{90 \times 10^9} \frac{N}{C \cdot m}$
 OR - retentivity
 QRSTUVQ - hysteresis curve

coercive

Fig. 4.9

- (ii) As H is increased, B also increases and we get the curve OPQ . From the curve OPQ , it is clear that B is not proportional to H and for large values of H , the dotted curve tends to become almost parallel to H indicating saturation effect.
- (iii) If H is now decreased, the specimen does not follow the path QO , but it follows the path QR . Thus, B lags behind H . For $H = 0$, B still has value OR . This magnetic flux density remaining in the specimen in the absence of any external magnetic field is called "residual magnetism". This is because during the region OPQ , atomic magnetic dipoles have been aligned in particular position and they get locked in that state and it takes a reverse field H to go back to original value of B i.e. zero.

The power of retaining this magnetism (OR in figure) even in the absence of external magnetic field is called "retentivity" or "remanence" of the specimen.

- (iv) Now, if we reduce H below zero, i.e. increase of H in reverse direction, then B becomes zero at the point S . The value of the magnetic intensity OS is called "coercive force" or "coercivity" of the specimen.

Thus, coercivity is a measure of the magnetic intensity required to destroy the residual magnetism of the specimen.

- (v) When H is increased beyond OS , the specimen is increasingly magnetised in the opposite direction and we get curve ST . The curve ST becomes almost parallel to H axis indicating a saturation effect.
- (vi) If H gradually brought back from its negative maximum value, through zero to its original maximum value H_Q , a similar curve $TUVQ$ is obtained and the cycle is complete.

From the figure, at points 'R' and 'U' where the specimen is magnetised in the absence of external magnetic field, it is said to be "permanent magnet".

From the figure, it is clear that B always lags behind H . The lagging of B behind H is called "*Hysteresis*". The closed curve QRSTUVQ representing a cycle of magnetisation of the specimen is known as "hysteresis curve" of the specimen.

It is to be noted that main features of hysteresis loop or curve are common to all ferromagnetic materials, but the actual shape of the loop can show considerable variation.

Demagnetisation :

The $B - H$ curve (Fig. 4.9) shows that magnetic flux remains in the specimen, when magnetising force is removed or made zero. The specimen can be demagnetized by taking it through a series of hysteresis loop of diminishing amplitudes. (See Fig. 4.10).

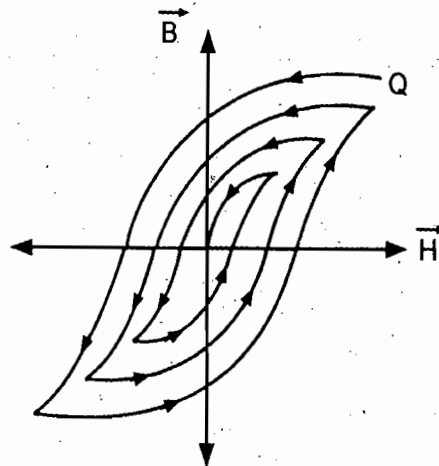


Fig. 4.10 : Method of demagnetising a specimen

Starting from Q, the specimen goes through the cycles of smaller and smaller size until no magnetism remain in the specimen. Another way to demagnetize a specimen is to heat the specimen to a sufficiently high temperature so that the ferromagnetic properties of the material vanishes and magnetisation of the specimen becomes zero in the absence of external magnetic field.

4.12 HYSTERESIS LOSS

A ferromagnetic material consists of molecular magnets forming a closed chain. When we magnetise the specimen by placing it in an external magnetic field, some energy is spent in aligning the molecular magnet in the direction of the field. Now, when magnetic field is removed, the material retain some magnetisation (hysteresis effect). Therefore, the energy supplied to material during magnetisation is not fully recovered. The balance of energy left in the material is lost as heat. This is called "hysteresis loss". Let us calculate hysteresis loss or energy loss per unit volume of the material through a complete cycle of magnetisation.

If current ' i ' is passing through the coil, then flux passing through single turn is BA where B is magnetic flux density in the material.

Total flux associated with coil is NBA .

i.e. $\phi_B = NBA$

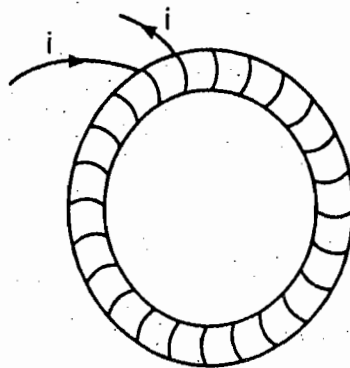


Fig. 4.11

Consider a ferromagnetic material in the form of ring of cross-sectional area 'A' and circumferential length l . Let a coil of 'N' turns be wound uniformly on the ring (Fig. 4.11).

Magnetic intensity H produced by current 'i' in the material is

$$H = \frac{Ni}{l}$$

$$\therefore i = \frac{Hl}{N} \text{ amp}$$

Current 'i' is called magnetising current. If 'i' is increased gradually, magnetic flux associated with coil or material changes. This changing magnetic flux induces back e.m.f 'E' in the coil (Law of electromagnetic induction).

$$E = -\frac{d\phi_B}{dt}$$

$$E = -\frac{d}{dt} NBA$$

$$E = -NA \frac{dB}{dt}$$

Negative sign in above equation indicates that back e.m.f. opposes the increase in the current. Work done dW' by current 'i' against the back e.m.f. is

$$dW' = -E i dt$$

$$dW' = NA \frac{dB}{dt} \frac{Hl}{N} dt$$

$$dW' = HAl dB \text{ in joule}$$

This work done or increase in the energy is stored in the form of magnetic field.

Volume of the material in the form of ring is Al .

$$\therefore \text{Increase in energy per unit volume} = \frac{HAl dB}{Al}$$

$$\therefore dW = HdB$$

When the magnetic flux density in the material is increased from B_1 to B_2 , then, energy required is

$$W = \int_{B_1}^{B_2} H dB \text{ in joule/meter}^3$$

Consider the B-H curve as shown in the Fig. 4.12.

Let us take extremely small area in the form of strip near the point 'b' on the curve. The area of the strip abcd is $ab \times ad$. We can represent H by ab and dB by ad . So area of strip is $H dB$. Hence, increase in energy per unit volume of material from the state V to Q will be the sum of all such areas i.e. area under curve $OVQWO$.

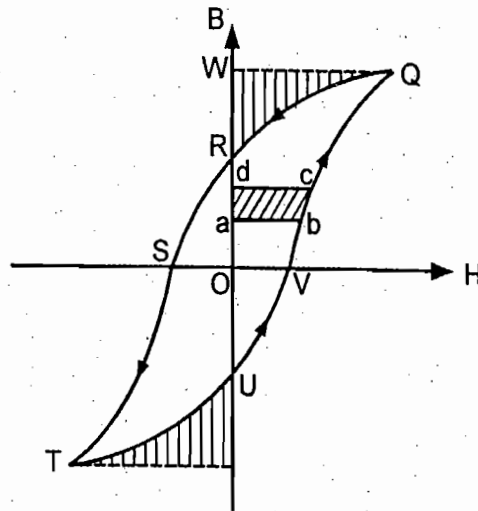


Fig. 4.12

Along the curve QR , H as well as B reduces. So energy equal to area $QWRQ$ per unit volume is restored back. Thus, net increase in energy per unit volume in taking material from state V to R is equal to area $OVQRO$. Thus, if whole cycle is considered, then increase in energy per unit volume is equal to area under the $B-H$ curve.

i.e. energy dissipated per cycle of magnetisation

$$= \oint H dB \text{ (Joule/meter}^3\text{)} = \text{area of } B-H \text{ curve}$$

4.13 BOUNDARY VALUE PROBLEMS INVOLVING MAGNETIC MATERIALS

(Magnetic field inside and outside the sphere placed in magnetic field)

In this section, we shall study calculation of magnetic field inside and outside the magnetic material in which no transport current exists. This is identical to the dielectric material with no external charge density.

With $J = 0$, the equations for $\vec{\nabla} \cdot \vec{B}$ and $\vec{\nabla} \times \vec{H}$ can be written as,

$$\vec{\nabla} \cdot \vec{B} = 0,$$

$$\text{and } \vec{\nabla} \times \vec{H} = 0$$

Since curl of $\vec{H}$ is zero, it can be expressed as negative gradient of a scalar field, i.e.,

$$\vec{H} = -\vec{\nabla} \phi^*$$

where, ϕ^* is called magnetic scalar potential. There are two types of materials for which the magnetic field calculation becomes simple boundary value problems. These two types are : (i) linear magnetic material for which $B = \mu H$ and (ii) a uniformly magnetised

material for which $\vec{\nabla} \cdot \vec{M} = 0$.

In both cases, the operation $\vec{\nabla} \cdot \vec{B} = 0$ can also be written as,

$$\vec{\nabla} \cdot \vec{H} = 0$$

But,

$$\vec{H} = -\vec{\nabla} \phi^*$$

$\therefore$

$$\nabla^2 \phi^* = 0$$

This is Laplace's equation.

Thus, the solution of magnetic problem can be obtained by finding a solution of Laplace's equation which satisfy boundary conditions. While solving the problem, $\vec{H}$ can be calculated by using the relation $\vec{H} = -\vec{\nabla} \phi^*$. Once we know, $\vec{H}$, $\vec{B}$ can be obtained by using the relation $\vec{B} = \mu \vec{H}$ or $\vec{B} = \mu_0 (\vec{H} + \vec{M})$. To illustrate the usefulness of this method, let us find the magnetic field at a point inside and outside the sphere in a uniform magnetic field.

Consider a sphere of linear magnetic material of radius 'a' and magnetic permeability μ placed in a uniform magnetic field, B_0 . We have to determine how the magnetic field is modified by the presence of sphere and also we have to determine the magnetic field inside and outside the sphere. For simplicity, let us take the origin of our co-ordinate system at the centre of the sphere and the direction of magnetic field as polar direction (z-axis). Then from the symmetry of the problem, it is clear that potential will be independent of azimuthal angle ϕ and may be expressed as a sum of zonal harmonics.

The solution of Laplace's equation in spherical polar co-ordinate system is

$$\phi^*(r, \theta) = \sum_{l=0}^{\infty} [A_l r^l + C_l r^{-(l+1)}] P_l(\cos \theta) \quad \dots (4.33)$$

where, $P_0(\cos \theta) = 1$, $P_1(\cos \theta) = \cos \theta$ and so on.

Let ϕ_1^* represent the potential at a point outside the sphere i.e. in free space having permeability μ_0 and ϕ_2^* represent potential at a point inside the sphere having permeability μ .

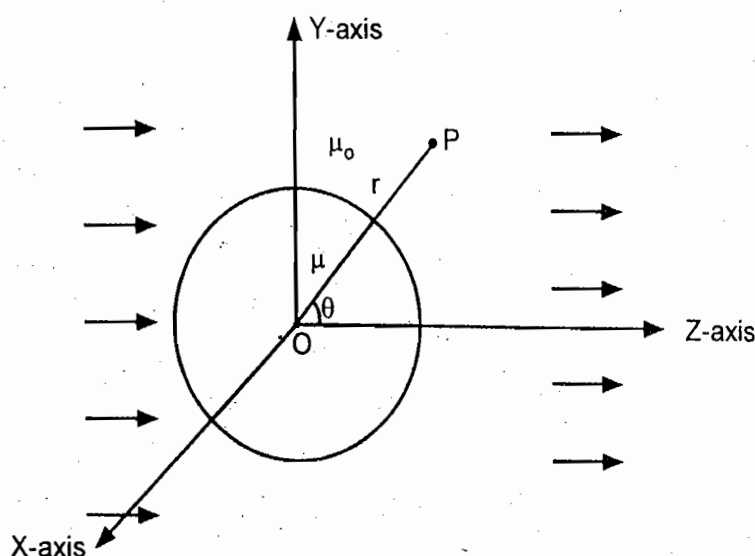


Fig. 4.13

The potential ϕ must satisfy the following boundary conditions.

(i) At distances far away from the sphere, (i.e. when $r \rightarrow \infty$), the magnetic field retains its uniform character i.e. $\vec{B} = B_0 \vec{k}$ and $\phi_1^* = -\left(\frac{B_0}{\mu_0}\right) r \cos \theta$.

(ii) $\phi_2^*(r, \theta)$ must be finite at $r = 0$.

(iii) At interface i.e. at $r = a$,

$$H_{1\theta} = H_{2\theta}$$

$$\text{or } -\frac{1}{r} \frac{\partial \phi_1^*}{\partial \theta} = -\frac{1}{r} \frac{\partial \phi_2^*}{\partial \theta}$$

(iv) At interface i.e. at $r = a$,

$$B_{1r} = B_{2r}$$

$$\text{or } \mu_0 \frac{\partial \phi_1^*}{\partial r} = \mu \frac{\partial \phi_2^*}{\partial r}$$

All the boundary conditions are satisfied for $l = 1$. Thus, equation (4.33) for inside and outside region can be written as

$$\phi_1^*(r, \theta) = A_1 r \cos \theta + C_1 r^{-2} \cos \theta, \quad \dots (4.34)$$

for free space i.e. region outside the sphere and

$$\phi_2^*(r, \theta) = A_2 r \cos \theta + C_2 r^{-2} \cos \theta, \quad \dots (4.35)$$

for the region inside the sphere.

Boundary condition (i) suggests that when

$$r \rightarrow \infty, \vec{B} = B_0 \vec{k}$$

$$\text{and } \phi_1^* = -\left(\frac{B_0}{\mu_0}\right) r \cos \theta \quad \dots (4.36)$$

Comparing equation (4.36) with (4.34), we get

$$\boxed{A_1 = -\frac{B_0}{\mu_0}} \quad \dots (4.37)$$

Boundary condition (ii) suggests that the potential at origin i.e. $r = 0$ must be finite. To satisfy this condition, no term of the form $C_2 r^{-2}$ should be included in the expression of ϕ_2 because it gives infinite contribution to the potential.

Thus, $C_2 = 0$

Now using $A_1 = -\frac{B_0}{\mu_0}$ and $C_2 = 0$, equations (4.34) and (4.35) can be written as

$$\phi_1^*(r, \theta) = -\frac{B_0}{\mu_0} r \cos \theta + C_1 r^{-2} \cos \theta \quad \dots (4.38)$$

$$\text{and } \phi_2^*(r) = A_2 r \cos \theta \quad \dots (4.39)$$

We have to determine B_1 and A_2 .

eld

According to boundary condition (iii)

$$\frac{1}{r} \frac{\partial \phi_1^*}{\partial \theta} = \frac{1}{r} \frac{\partial \phi_2^*}{\partial \theta} \text{ for } r = a.$$

$$\therefore \frac{1}{r} \frac{\partial \phi_1^*}{\partial \theta} = \frac{1}{r} \left[+ \frac{B_0}{\mu_0} r \sin \theta - C_1 r^{-2} \sin \theta \right]$$

$$\text{and} \quad \frac{1}{r} \frac{\partial \phi_2^*}{\partial \theta} = \frac{1}{r} [-A_2 r \sin \theta]$$

$$\therefore \frac{B_0}{\mu_0} \sin \theta - \frac{C_1 \sin \theta}{a^3} = -A_2 \sin \theta \quad (\because r = a) \quad \dots (4.40)$$

According to boundary condition (iv),

$$\mu_0 \frac{\partial \phi_1^*}{\partial r} = \mu \frac{\partial \phi_2^*}{\partial r} \text{ for } r = a$$

$$\therefore \mu_0 \frac{\partial \phi_1^*}{\partial r} = -B_0 \cos \theta - \frac{2\mu_0 C_1 \cos \theta}{r^3}$$

$$\text{and} \quad \frac{\partial \phi_2^*}{\partial r} = \mu A_2 \cos \theta$$

$$\therefore -B_0 \cos \theta - \frac{2\mu_0 C_1 \cos \theta}{a^3} = \mu A_2 \cos \theta \quad (\because r = a) \quad \dots (4.41)$$

Solving equations (4.40) and (4.41) simultaneously, we get

$$A_2 = \frac{-3 B_0}{(\mu + 2 \mu_0)}$$

and

$$C_1 = \left[\left(\frac{\mu}{\mu_0} \right) - 1 \right] \frac{B_0 a^3}{(\mu + 2 \mu_0)}$$

The magnetic field inside the sphere is,

$$\vec{B}_2 = B_{2r} \hat{a}_r + B_{2\theta} \hat{a}_\theta \quad \dots (4.42)$$

where,

$$B_{2r} = -\mu \frac{\partial \phi_2^*}{\partial r} \text{ and } B_{2\theta} = -\frac{\mu}{r} \frac{\partial \phi_2^*}{\partial \theta}$$

But,

$$\phi_2^* = A_2 r \cos \theta \quad (\text{See equation (4.39)})$$

or

$$\phi_2^* = -\frac{3 B_0}{(\mu + 2 \mu_0)} r \cos \theta \quad \left(\because A_2 = -\frac{3 B_0}{\mu + 2 \mu_0} \right)$$

 $\therefore$

$$\begin{aligned} B_{2r} &= -\mu \left[\frac{\partial}{\partial r} \left(\frac{-3 B_0}{(\mu + 2 \mu_0)} r \cos \theta \right) \right] \\ &= \frac{3 B_0}{\left(1 + \frac{2 \mu_0}{\mu} \right)} \cos \theta \quad \dots (4.43) \end{aligned}$$

$$\begin{aligned} B_{2\theta} &= -\frac{\mu}{r} \frac{\partial \phi_2^*}{\partial \theta} \\ &= -\frac{\mu}{r} \left[\frac{3 B_0 r \sin \theta}{(\mu + 2 \mu_0)} \right] \end{aligned}$$

$$B_{2\theta} = - \frac{3 B_0 \sin \theta}{\left(1 + \frac{2\mu_0}{\mu}\right)} \quad \dots (4.44)$$

Using equations (4.43) and (4.44) in equation (4.42), we get

$$\begin{aligned} \vec{B}_2 &= \frac{3 B_0}{\left(1 + \frac{2\mu_0}{\mu}\right)} \cos \theta \hat{a}_r - \frac{3 B_0 \sin \theta}{\left(1 + \frac{2\mu_0}{\mu}\right)} \hat{a}_\theta \\ \text{or } \vec{B}_2(r, \theta) &= \frac{3 B_0}{\left(1 + \frac{2\mu_0}{\mu}\right)} (\cos \theta \hat{a}_r - \sin \theta \hat{a}_\theta) \\ |\vec{B}_2| &= \frac{3 B_0}{\left(1 + \frac{2\mu_0}{\mu}\right)} \end{aligned}$$

This is magnetic induction at a point inside the sphere and it is constant at all the points inside the sphere.

The magnetic field outside the sphere is

$$\vec{B}_1 = B_{1r} \hat{a}_r + B_{1\theta} \hat{a}_\theta \quad \dots (4.45)$$

where,

$$B_{1r} = -\mu_0 \frac{\partial \phi_1^*}{\partial r} \quad \dots (4.46)$$

We have,

$$\phi_1^* = -\frac{B_0}{\mu_0} r \cos \theta + C_1 r^{-2} \cos \theta \quad (\text{See equation 4.38})$$

But,

$$C_1 = \left(\frac{\mu}{\mu_0} - 1\right) \frac{B_0 a^3}{(\mu + 2\mu_0)}$$

$\therefore$

$$\phi_1^* = -\frac{B_0}{\mu_0} r \cos \theta + \left(\frac{\mu}{\mu_0} - 1\right) \frac{B_0 a^3}{\mu_0 \left(2 + \frac{\mu}{\mu_0}\right)} r^{-2} \cos \theta \quad \dots (4.47)$$

Using equation (4.47) in equation (4.46), we get

$$\begin{aligned} B_{1r} &= -\mu_0 \frac{\partial}{\partial r} \left[-\frac{B_0}{\mu_0} r \cos \theta + \frac{\left(\frac{\mu}{\mu_0} - 1\right) B_0 a^3 r^{-2} \cos \theta}{\mu_0 \left(2 + \frac{\mu}{\mu_0}\right)} \right] \\ &= B_0 \cos \theta + \frac{2 \left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \frac{a^3}{r^3} \cos \theta \\ &= B_0 \cos \theta + \frac{2 \left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \left(\frac{a}{r}\right)^3 \cos \theta \quad \dots (4.48) \end{aligned}$$

$$\begin{aligned}
 B_{1\theta} &= -\frac{\mu_0}{r} \frac{\partial \phi_1^*}{\partial \theta} \\
 &= -\frac{\mu_0}{r} \frac{\partial}{\partial \theta} \left[-\frac{B_0}{\mu_0} r \cos \theta + \frac{\left(\frac{\mu}{\mu_0} - 1\right) B_0 a^3 r^{-2} \cos \theta}{\mu_0 \left(2 + \frac{\mu}{\mu_0}\right)} \right] \\
 B_{1\theta} &= -B_0 \sin \theta + \frac{\left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \frac{a^3}{r^3} \sin \theta \quad \dots (4.49)
 \end{aligned}$$

Using equations (4.48) and (4.49) in equation (4.45), we get

$$\begin{aligned}
 \vec{B}_1 &= B_0 \cos \theta \hat{a}_r + \frac{2\left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \left(\frac{a}{r}\right)^3 \cos \theta \hat{a}_r \\
 &\quad - B_0 \sin \theta \hat{a}_\theta + \frac{\left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \left(\frac{a}{r}\right)^3 \sin \theta \hat{a}_\theta \\
 &= B_0 (\cos \theta \hat{a}_r - \sin \theta \hat{a}_\theta) + \frac{\left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \left(\frac{a}{r}\right)^3 [\sin \theta \hat{a}_\theta + 2 \cos \theta \hat{a}_r]
 \end{aligned}$$

or

$$\vec{B}_1(r, \theta) = B_0 \hat{k} + \frac{\left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \left(\frac{a}{r}\right)^3 [2 \cos \theta \hat{a}_r + \sin \theta \hat{a}_\theta]$$

This is expression for magnetic induction at a point outside the sphere. The magnitude of magnetic induction is

$$|\vec{B}| = B_0 + \frac{\left(\frac{\mu}{\mu_0} - 1\right)}{\left(\frac{\mu}{\mu_0} + 2\right)} B_0 \left(\frac{a}{r}\right)^3 [\sqrt{3 \cos^2 \theta + 1}] \quad \dots (4.50)$$

Analogy between Electric and Magnetic Fields :

Term	Electric Field	Magnetic Field
(1)	(2)	(3)
Basic law	Coulomb's Law $\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^3} \vec{r}$	Biot and Savart's law $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$
Force law	$\vec{E}$ is defined from equation $\vec{F} = q\vec{E}$	$\vec{B}$ is defined from equation $\vec{F} = q\vec{V} \times \vec{B}$
Source element	Charge dQ	Current element $I d\vec{l}$
Field intensity	Electric field strength $E = \frac{V}{l}$	Magnetic field strength $H = \frac{I}{l}$
Potential	As $\vec{E} = -\vec{\nabla} V$ Potential $V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho dV}{r}$ $V = - \int \vec{E} \cdot d\vec{r}$	As $\vec{B} = \vec{\nabla} \times \vec{A}$ A is magnetic vector potential $\vec{A} = \frac{\mu_0}{4\pi} \oint \frac{I d\vec{l}}{r}$
Dipole moment	Electric dipole moment $\vec{P} = q\vec{d}$	Magnetic dipole moment $\vec{m} = I\vec{A}$
Relation between the three vectors	$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ where, $\vec{D}$ is electric displacement vector associated with free charges, $\vec{E}$ is electric field strength associated with all charges i.e. free charges as well as induced charges and $\vec{P}$ is polarisation defined as electric dipole moment per unit volume related with induced charges.	$\vec{B} = \mu_0 (\vec{H} + \vec{M})$ $\vec{H}$ is magnetic field strength associated with true current, $\vec{B}$ is magnetic induction associated with true as well as magnetising current and $\vec{M}$ is magnetisation defined as magnetic dipole moment per unit volume associated with magnetizing current only.
Basic relation for field	$\vec{\nabla} \times \vec{E} = 0$ $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$	$\vec{\nabla} \cdot \vec{B} = 0$ $\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$

Term (1)	Electric Field (2)	Magnetic Field (3)
Theorem/Law	Gauss's theorem in vacuum or free space : (Integral form) $\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$ (Differential form) $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ Gauss's theorem in the presence of dielectric : (Integral form) $\oint \vec{D} \cdot d\vec{S} = q$ (Differential form) $\vec{\nabla} \cdot \vec{D} = \rho$	Ampere's Law is (Integral form) : $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$ (Differential form) $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$ Ampere's law in the presence of magnetic material : $\oint \vec{H} \cdot d\vec{l} = I$ $\vec{\nabla} \times \vec{H} = \vec{J}$

Note : Electric field strength $\vec{E}$ and magnetic field strength $\vec{H}$ are not analogous but $\vec{E}$ is analogous to $\vec{B}$ and $\vec{D}$ is analogous to $\vec{H}$.

ILLUSTRATIVE EXAMPLES

Example 1 : Calculate force between two long parallel wires separated by distance d and carrying currents I_1 and I_2 in the same direction.

Solution : Consider two long parallel wires distant d apart and carrying current I_1 and I_2 respectively as shown in Fig. 4.14. Let the first wire be placed along Z-axis and passes through origin and the other be parallel to Z-axis.

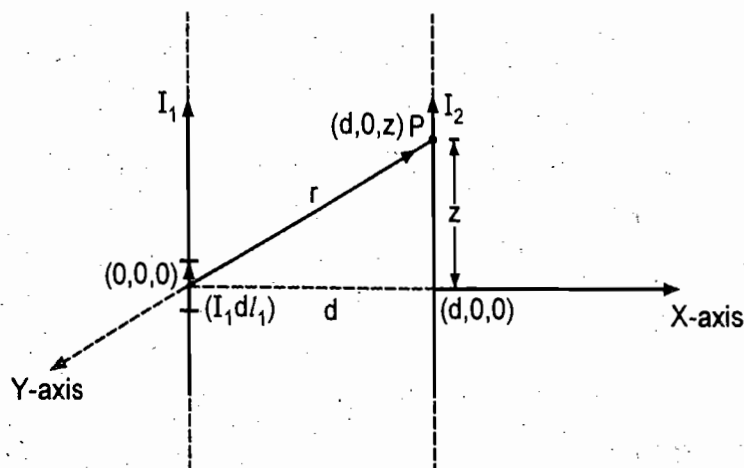


Fig. 4.14

Let us calculate the magnetic induction at point P (d, 0, z) due to the current I_1 in first wire.

Consider a length element dl_1 , of first wire at origin O. Its components are : (0, 0, dz). Let the point P be situated at distance r from the origin O. The components of r are : (d, 0, z).

The components of $\vec{dl}_1 \times \vec{r}$ are : (0, d · dz, 0).

This shows that magnetic induction dB at point P due to current element $I_1 dl_1$ is parallel to Y-axis.

Therefore, $B_x = B_z = 0$

According to Biot-Savart's law, the dB at point P due to current element $I_1 dl_1$ is

$$d\vec{B}_y = \frac{\mu_0}{4\pi} \frac{I_1 |\vec{dl}_1 \times \vec{r}|}{r^3}$$

Total magnetic induction at point P, due to 1st wire carrying current I_1 is

$$\begin{aligned} B_y &= \frac{\mu_0 I_1}{4\pi} \int_{-\infty}^{+\infty} \frac{|\vec{dl}_1 \times \vec{r}|}{r^3} \\ &= \frac{\mu_0 I_1}{4\pi} \int_{-\infty}^{+\infty} \frac{d \cdot dz}{(d^2 + z^2)^{3/2}} \end{aligned}$$

From Fig. 4.14,

$$\tan \theta = \frac{z}{d}$$

or

$$z = d \tan \theta$$

$$dz = d \sec^2 \theta d\theta$$

$$\begin{aligned} B_y^1 &= \frac{\mu_0 I_1}{4\pi} \int_{-\pi/2}^{+\pi/2} \frac{d \cdot d \sec^2 \theta d\theta}{d^3 \sec^3 \theta} \\ &= \frac{\mu_0 I_1}{4\pi d} \int_{-\pi/2}^{+\pi/2} \cos \theta d\theta \end{aligned}$$

$$B_y^1 = \frac{\mu_0 I_1}{2\pi d}$$

$\vec{B}$ is always normal to the plane containing the wire the vector $\vec{r}$, i.e. along Y-axis.

According to Ampere's force law, force $d\vec{F}_2$ acting on the element dl_2 of second wire due to magnetic induction B_y^1 of first wire is

$$d\vec{F}_2 = I_2 dl_2 \times \vec{B}_y^1$$

$\vec{dF}_2$ will be in a plane perpendicular to plane containing B_y^1 (along Y-axis) and dl_2 (along Z-axis) i.e. $d\vec{F}$ will be along X-axis.

Since B_y^1 is in positive Y-direction and $I_2 dl_2$ in the Z-direction; according to vector rule, force acting on dl_2 i.e. $d\vec{F}_2$ will be in the negative X-direction. Thus, the length dl of second wire is attracted towards the first wire.

$$\begin{aligned}\text{The magnitude of } dF_2 &= I_2 dl_2 B_y^1 \\ &= I_2 dl_2 \frac{\mu_0 I_1}{2\pi d}\end{aligned}$$

$$\text{or} \quad dF_2 = \frac{\mu_0}{4\pi} \left(\frac{2 I_1 I_2}{d} \right) dl_2$$

Force per unit length on second wire is

$$\frac{dF_2}{dl_2} = \frac{\mu_0}{4\pi} \left(\frac{2 I_1 I_2}{d} \right)$$

If instead, we find force due to second wire on first wire, it will be of same magnitude but opposite in direction i.e. $\frac{dF_1}{dl}$ will be in positive X-direction.

Thus, if current in the two wires is in the same direction, then the wires will attract each other. And if the current in the two wires is in the opposite direction, then the wires will repel each other.

Example 2 : Two long parallel wires separated by 1 cm in air carry current 50 amp. Find the force on one meter length of wire.

Solution : Formula -

Force per unit length is given by

$$\frac{dF_2}{dl} = \frac{\mu_0}{4\pi} \frac{2 I_1 I_2}{d}$$

$$\text{Given data :} \quad \frac{\mu_0}{4\pi} = 10^{-7} \text{ weber/amp.meter}$$

$$I_1 = I_2 = 50 \text{ amp}$$

$$d = 1 \text{ cm} = 1 \times 10^{-2} \text{ m}$$

$$\frac{dF_2}{dl} = \frac{\mu_0}{4\pi} \frac{2 I_1 I_2}{d}$$

$$\frac{dF_2}{dl} = \frac{10^{-7} \times 2 \times 50 \times 50}{1 \times 10^{-2}}$$

$$\frac{dF_2}{dl} = 0.05 \text{ newton/meter}$$

Force on one meter length of wire = 0.05 N/m

Example 3 : A conductor 10 m long lies along Z-direction with a current of 1 A in the $\hat{a}_z$ direction. Find the force if $\vec{B} = 0.04 \hat{a}_x$ tesla.

Solution : Force on a current element $I d\vec{l}$ is given by,

$$d\vec{F} = I d\vec{l} \times \vec{B}$$

$$\therefore \vec{F} = I \vec{l} \times \vec{B}$$

$$I = 1 \text{ amp.}, l = 10 \text{ m,}$$

$$B = 0.04 \text{ T,}$$

$$\therefore \vec{F} = 1 (10 \hat{a}_z \times 0.04 \hat{a}_x)$$

$$\vec{F} = 0.4 \hat{a}_y \text{ N}$$

Example 4 : If the current density in the wire of circular cross-section with radius 'a' is proportional to the distance from the axis, find the total current in the wire.

Solution : The current density is proportional to the distance from the axis i.e. $J \propto S$.

$$\text{or } J = kS$$

The current in the small shaded portion is

$$dI = J \cdot S dS d\phi$$

where, $(S \cdot dS \cdot d\phi)$ is area of strip as shown in Fig. 4.15.

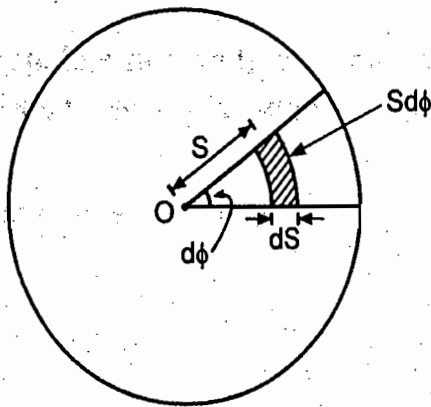


Fig. 4.15 : Cross-sectional area of wire

$$I = \int JS dS d\phi$$

$$\text{But } J = kS$$

$$\therefore I = \int_0^{2\pi} \int_0^a kS^2 dS d\phi$$

$$= 2\pi k \int_0^a S^2 dS$$

$$= 2\pi k \left[\frac{S^3}{3} \right]_0^a$$

$$= 2\pi k \frac{a^3}{3}$$

Thus, we get

$$I = 2\pi k \frac{a^3}{3}$$

Example 5 : Using the vector potential method, calculate the magnetic induction due to a long straight wire of radius 'a' carrying a steady current I.

Solution : Consider a long straight wire of radius 'a' and carrying current I as shown in Fig. 4.16.

Let the current 'I' be uniformly distributed throughout its cross-section.

As the wire is kept along Z-axis, the current density inside the wire is

$$J_z = \frac{I}{\pi a^2}$$

Since, $J_x = 0$, $J_y = 0$, we have $A_x = A_y = 0$.

Now, to obtain A_z , let us use solution for the electrostatic potential ϕ of the wire with a uniform linear charge density λ .

For points outside an infinite long wire, the potential ϕ is given by

$$\phi = -\frac{\lambda}{2\pi\epsilon_0} \log r$$

where, $r = \sqrt{x^2 + y^2}$ and λ is linear charge density.

Let ρ be the volume charge density, i.e.,

$$\rho = \frac{\text{Charge}}{\text{Volume}} = \frac{\text{Charge}}{\text{Length} \times \text{cross-sectional area}}$$

Charge contained in a unit length = $\rho \pi a^2$

But, λ is charge contained per unit length.

$$\therefore \lambda = \rho \pi a^2$$

Using $\lambda = \rho \pi a^2$, the expression for potential ϕ may be written as

$$\phi = -\frac{\rho \pi a^2}{2\pi\epsilon_0} \log r$$

The corresponding vector potential A_z can be obtained by replacing $\frac{\rho}{\epsilon_0}$ by $\mu_0 J_z$.

$$\therefore A_z = -\frac{1}{2\pi} \pi a^2 \mu_0 J_z \log r$$

But,

$$I = J_z \pi a^2$$

$\therefore$

$$A_z = -\frac{\mu_0 I}{2\pi} \log r$$

We know that

$$\vec{B} = \nabla \times \vec{A}$$

$$\therefore \vec{B}_x \vec{i} + \vec{B}_y \vec{j} + \vec{B}_z \vec{k} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Comparing coefficients of $\hat{i}$, we get

$$B_x = \left(\frac{\partial}{\partial y} A_z - \frac{\partial}{\partial z} A_y \right)$$

But,

$$A_z = -\frac{\mu_0 I}{2\pi} \log r \quad \text{and } A_y = 0$$

$\therefore$

$$B_x = -\frac{\partial}{\partial y} \frac{\mu_0 I}{2\pi} \log r \quad \text{where, } r = \sqrt{x^2 + y^2}$$

$$= -\frac{\mu_0 I}{2\pi} \frac{\partial}{\partial y} \log r$$

$$B_x = -\frac{\mu_0 I y}{2\pi r^2}$$

Similarly, we can obtain

$$B_y = -\frac{\mu_0 I x}{2\pi r^2}$$

Since, $A_x = A_y = 0$, we get $B_z = 0$.

Magnetic induction $\vec{B}$ in terms of components is

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$\therefore$

$$|\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2}$$

$$|\vec{B}| = \frac{\mu_0 I}{2\pi r^2} (x^2 + y^2)^{1/2}$$

or

$$B = \frac{\mu_0 I}{2\pi r}$$

This is magnetic induction at a perpendicular distance r from infinitely long current carrying wire.

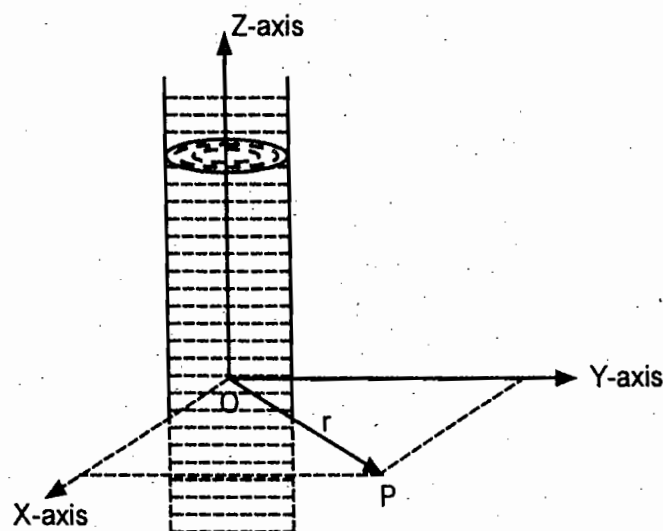


Fig. 4.16

Example 6 : Find the vector potential of an infinite solenoid having n turns per unit length, radius R and carries a current I through its winding.

Flux passing through loop is

$$\phi = \int_S \vec{B} \cdot d\vec{S}$$

But,
$$\vec{B} = \nabla \times \vec{A}$$

$$\therefore \phi = \int_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

But, by Stoke's theorem,

$$\int (\nabla \times \vec{A}) \cdot d\vec{S} = \oint \vec{A} \cdot d\vec{l}$$

$$\therefore \phi = \oint \vec{A} \cdot d\vec{l}$$

Thus, we get

$$\int \vec{B} \cdot d\vec{S} = \oint \vec{A} \cdot d\vec{l}$$

Let us find the magnetic induction at a point situated inside the solenoid at perpendicular distance S from its axis.

$$\therefore \oint \vec{A} \cdot d\vec{l} = A 2\pi S$$

and
$$\int \vec{B} \cdot d\vec{S} = \mu_0 n I (\pi S^2) \quad (\text{since } B = \mu_0 n I)$$

Thus, we get

$$A 2\pi S = \mu_0 n I \pi S^2$$

or
$$A = \frac{\mu_0 n I}{2} S \text{ for } S < R$$

In vector form,
$$\vec{A} = \frac{\mu_0 n I}{2} S \hat{\phi}$$

For the point outside the solenoid i.e. $S > R$, the flux is

$$\int \vec{B} \cdot d\vec{S} = \mu_0 n I (\pi R^2)$$

$$\therefore A 2\pi S = \mu_0 n I (\pi R^2)$$

or
$$A = \frac{\mu_0 n I}{2} \frac{R^2}{S} \text{ For } S > R$$

Example 7 : Calculate the resultant force on the current carrying rectangular loop ABCD (shown in Fig. 4.17) due to current carrying long wire.

Given : Current through long wire is $I_2 = 20 \text{ A}$

Current through loop ABCD I_1 is $= 15 \text{ A}$

$$L = 20 \text{ cm} \quad d_1 = 2 \text{ cm} \quad d_2 = 6 \text{ cm}$$

Solution : The wire and sides AB and DC are parallel.

The force on side AB due to current I_1 in the wire is,

$$F_1 = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d_1} L$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

$$I_1 = 20 \text{ A}, I_2 = 15 \text{ A}, d_1 = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$$

$$L = 20 \text{ cm} = 20 \times 10^{-2} \text{ m}$$

$$\therefore F_1 = \frac{4\pi \times 10^{-7}}{2\pi} \times \frac{20 \times 15}{2 \times 10^{-2}} \times 20 \times 10^{-2}$$

$$F_1 = \frac{2 \times 20 \times 15 \times 20}{2} \times 10^{-7}$$

$$= 6 \times 10^{-4} \text{ N}$$

F_1 is attractive and directed towards wire.

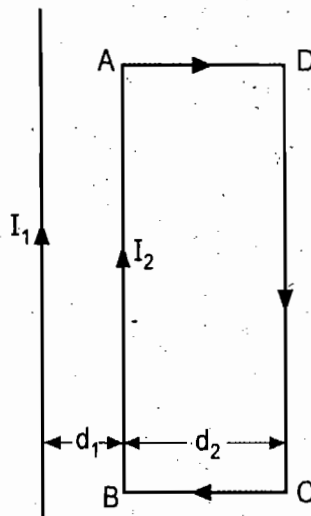


Fig. 4.17

The force on side DC due to current I_1 is

$$F_2 = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{(d_1 + d_2)} \cdot L$$

$$= \frac{4\pi \times 10^{-7}}{2\pi} \times \frac{20 \times 15 \times 20 \times 10^{-2}}{8 \times 10^{-2}}$$

$$F_2 = \frac{2 \times 20 \times 15 \times 20}{8} \times 10^{-7}$$

$$= 1.5 \times 10^{-4} \text{ N}$$

$$F_2 = 1.5 \times 10^{-4} \text{ N}$$

F_2 is repulsive and directed away from the wire.

The forces on AD and BC are equal and opposite and mutually cancel out.

Thus, the resultant force on loop ABCD is

$$F = F_1 - F_2 = 6 \times 10^{-4} - 1.5 \times 10^{-4}$$

$$F = 4.5 \times 10^{-4} \text{ N}$$

Example 8 : A square loop of wire of each side 'a' carries a current I. Find the magnetic induction B at the centre of the loop.

Solution : Let ABCD be a square loop carrying current I as shown in the Fig. 4.18. Length of each side is 'a'. Let O be the centre of the loop. Let PO be perpendicular to side AB.

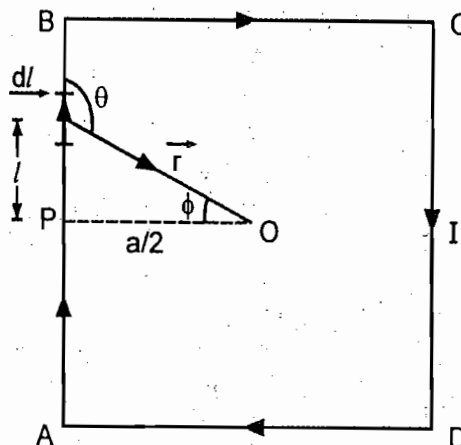


Fig. 4.18

Consider a current element $d\vec{l}$ in the side AB at a distance 'l' from foot P. Let $\vec{r}$ be the radius vector from $d\vec{l}$ to O.

Due to this element the magnetic induction $d\vec{B}$ at O is given by Biot-Savart's law.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$$

The magnitude of $d\vec{B}$ is

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

From figure, we have

$$\sin(\pi - \theta) = \frac{a/2}{r}$$

$$\text{i.e.} \quad \sin \theta = \frac{a}{2r}$$

$$\text{and} \quad r^2 = \left(\frac{a}{2}\right)^2 + l^2$$

$$\therefore \quad \sin \theta = \frac{a}{2 \left(\left(\frac{a}{2}\right)^2 + l^2 \right)}$$

$$\therefore \quad dB = \frac{\mu_0}{4\pi} \frac{I dl a}{2 \left(\frac{a^2}{4} + l^2 \right)^{3/2}}$$

The magnitude of the field due to entire side AB is

$$B_1 = \frac{\mu_0 I a}{8\pi} \int_{l=-a/2}^{l=+a/2} \frac{dl}{\left(\frac{a^2}{2} + l^2\right)^{3/2}}$$

Let $l = \frac{a}{2} \tan \phi$

$\therefore dl = \frac{a}{2} \sec^2 \phi d\phi$

Then, $B_1 = \frac{\mu_0 I a}{8\pi} \int_{-\pi/4}^{+\pi/4} \frac{\left(\frac{a}{2}\right) \sec^2 \phi d\phi}{\left(\frac{a^2}{4} \sec^2 \phi\right)^{3/2}}$

$$B_1 = \frac{\mu_0 I}{8\pi} a \int_{-\pi/4}^{+\pi/4} \cos \phi d\phi$$

$$\therefore B_1 = \frac{\mu_0 I}{\sqrt{2} \pi a}$$

The field is perpendicular to the paper and directed into it. By symmetry, the field at O due to the remaining three sides is same as B_1 and also have same direction. Therefore, total magnetic field at O is

$$B = 4B_1 = \frac{4\mu_0 I}{\sqrt{2} \pi a}$$

or

$$B = \frac{2\sqrt{2} \mu_0 I}{\pi a} \text{ wb/m}^2$$

Example 9 : Find the energy stored in a section of length 'l' of a long coaxial cable carrying current I.

Solution : Consider a coaxial cable consisting of a small central conductor of radius 'a' and a co-axial cylindrical outer cable of radius 'b' shown in Fig. 4.19.

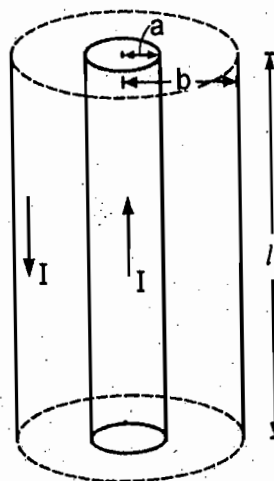


Fig. 4.19

We know that

$$\text{Energy stored} = \frac{1}{2\mu_0} \int_{\text{all space}} B^2 dV$$

$\therefore$ Energy stored per unit volume called as magnetic energy density = $\frac{B^2}{2\mu_0}$.

Using Ampere's law, the field between the cylinder is,

$$B^* = \frac{\mu_0 I}{2\pi r}$$

elsewhere it is zero.

$$\therefore \text{Energy stored per unit volume} = \frac{1}{2\mu_0} \cdot \frac{\mu_0^2 I^2}{4\pi^2 r^2}$$

$$\text{i.e. Energy stored per unit volume} = \frac{\mu_0 I^2}{8\pi^2 r^2}$$

The energy stored in a cylindrical shell of length l , radius r and thickness dr is then,

$$dW = \frac{\mu_0 I^2}{8\pi^2 r^2} 2\pi r dr l$$

To find energy stored in a section length of a co-axial cable, let us integrate from 'a' to 'b'. Then, we get

$$W = \int_a^b \frac{\mu_0 I^2}{8\pi^2 r^2} 2\pi r dr l$$

$$W = \frac{\mu_0 I^2 l}{4\pi} \ln \frac{b}{a}$$

Thus, energy stored in a section of length ' l ' of long co-axial cable carrying current I is

$$W = \frac{\mu_0 I^2 l}{4\pi} \cdot \ln \frac{b}{a}$$

Example 10 : Write expressions for the magnetic scalar and vector potentials from which the following magnetic field can be derived

$$\vec{B} = K(y \hat{i} + x \hat{j})$$

Solution : The magnetic vector potential $\vec{A}$ is given by the relation

$$\nabla \times \vec{A} = \vec{B}$$

* The current passing through central conductor of radius 'a' is I . The Amperian loop encloses only current I , so according to Ampere's Circuital Law,

$$\int \vec{B} \cdot d\vec{l} = \mu_0 I, \text{ we get}$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Therefore, in component form,

$$\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

We have,
$$\vec{B} = k(y\hat{i} + x\hat{j})$$

Therefore,

$$\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} = ky \quad \dots (4.51)$$

$$\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} = kx \quad \dots (4.52)$$

and
$$\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} = 0 \quad \dots (4.53)$$

On integration of equations, we get two possible solutions of vector potentials.

$$\vec{A}_1 = \hat{i} kx z - \hat{j} ky z + 0$$

and
$$\vec{A}_2 = \frac{k}{2}(-x^2 + y^2)\hat{k}$$

The magnetic scalar potential is given by

$$\vec{B} = -\nabla\phi$$

or
$$-\left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right) = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

$$-\frac{\partial\phi}{\partial x} = ky, \quad -\frac{\partial\phi}{\partial y} = kx \quad \text{and} \quad \frac{\partial\phi}{\partial z} = 0$$

Integrating we get the solution for the scalar potential.

$$\phi = -kxy + C$$

where, C is constant of integration.

Example 11 : Show that $\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$.

Solution : Let us calculate $\vec{\nabla} \times \vec{B}$

We have shown that the magnetic induction $\vec{B}$ can be expressed as $\vec{B} = \vec{\nabla} \times \vec{A}$, where $\vec{A}$ is magnetic vector potential.

$$\therefore \vec{\nabla} \times \vec{B} = \vec{\nabla} \times \vec{\nabla} \times \vec{A}$$

Using identity $\vec{A} \times \vec{B} \times \vec{C} = \vec{B}(\vec{A} \cdot \vec{C}) - (\vec{A} \cdot \vec{B})\vec{C}$,
we get

$$\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$$

$$\therefore \vec{\nabla} \times \vec{B} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad \dots (4.54)$$

We know that, $\vec{\nabla} \cdot \vec{A} = -\mu_0 \epsilon_0 \frac{\partial \phi}{\partial t}$... (See equation 4.20)

$$\therefore \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \vec{\nabla} \phi$$

$$\text{But, } \vec{E} = -\vec{\nabla} \phi$$

$$\therefore \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \dots (4.55)$$

Let us evaluate second term i.e. $\nabla^2 \vec{A}$ of equation (4.54).

Magnetic vector potential is given by,

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J}}{r} dV'$$

$$\therefore \nabla^2 \vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \nabla^2 \frac{\vec{J}}{r} dV'$$

Since, $\vec{J}$ is function of (x', y', z') and ∇ is function of (x, y, z) , above equation may be written as

$$\nabla^2 \vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \vec{J} \nabla^2 \left(\frac{1}{r} \right) dV' \quad \dots (4.56)$$

$$\text{Using } \vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\text{and } r = [(x - x')^2 + (y - y')^2 + (z - z')^2]^{1/2}$$

$$\text{it can be shown that } \nabla^2 \left(\frac{1}{r} \right) = 0, \text{ if } r \neq 0.$$

From this equation, it is clear that there can be no contribution to the integral from any volume element dV' except possibly at a point $P = P'$ where, r is equal to zero.

Now to find value of integral of equation (4.56) at $r = 0$, consider the small volume enclosing point P .

Since the volume taken is extremely small, J does not change appreciably within it, so J in equation (4.56) can be taken outside of integral.

Thus, we get

$$\nabla^2 \vec{A} = \frac{\mu_0 J}{4\pi} \int_{V' \rightarrow 0} \nabla^2 \left(\frac{1}{r} \right) dV'$$

According to Gauss's divergence theorem, we have

$$\int_{V'} \vec{\nabla} \cdot \vec{\nabla} \left(\frac{1}{r} \right) dV' = \int_{S'} \vec{\nabla} \left(\frac{1}{r} \right) \cdot d\vec{S}' \quad \dots (4.54)$$

$$\therefore \nabla^2 A = \frac{\mu_0 J}{4\pi} \int_{S' \rightarrow 0} \vec{\nabla} \left(\frac{1}{r} \right) \cdot d\vec{S}'$$

$$\text{But, } \vec{\nabla} \left(\frac{1}{r} \right) = -\frac{\vec{r}}{r^3}$$

$$\begin{aligned} \therefore \nabla^2 A &= -\frac{\mu_0 J}{4\pi} \int_{S' \rightarrow 0} \frac{\vec{r}}{r^3} \cdot d\vec{S}' \\ &= -\frac{\mu_0 J}{4\pi} \int_{S' \rightarrow 0} \frac{dS' \cos \theta}{r^2} \quad (\because \vec{r} \cdot d\vec{S} = r dS \cos \theta) \end{aligned}$$

$$\text{But, } \frac{dS' \cos \theta}{r^2} = d\omega$$

where, $d\omega$ is the element of solid angle subtended at point P by an element having area dS .

$$\text{But, } \int_{S' \rightarrow 0} d\omega = 4\pi$$

$$\therefore \nabla^2 A = -\frac{\mu_0 J}{4\pi} \times 4\pi$$

$$\text{or } \nabla^2 A = -\mu_0 J \quad \dots (4.57)$$

Using equations (4.55) and (4.57) in equation (4.54), we get

$$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$$

$$\text{But, } \epsilon_0 \vec{E} = \vec{D}, \text{ electric displacement vector}$$

$$\therefore \vec{\nabla} \times \vec{B} = \mu_0 \frac{\partial \vec{D}}{\partial t} + \mu_0 \vec{J}$$

$$\text{or } \vec{\nabla} \times \vec{B} = \mu_0 \left(\frac{\partial \vec{D}}{\partial t} + \vec{J} \right) \quad \dots (4.58)$$

where, $\frac{\partial \vec{D}}{\partial t}$ is displacement current density and $\vec{J}$ is current density.

If $J \gg D$, we get

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \dots (4.59)$$

Example 12 : A plastic disc of radius 'a' has a charge uniformly distributed over its surface. If the disc is rotated at an angular frequency ' ω ' about its axis, show that the magnetic field at the centre of the disc is $B = \frac{\mu_0 \omega q}{2\pi a}$.

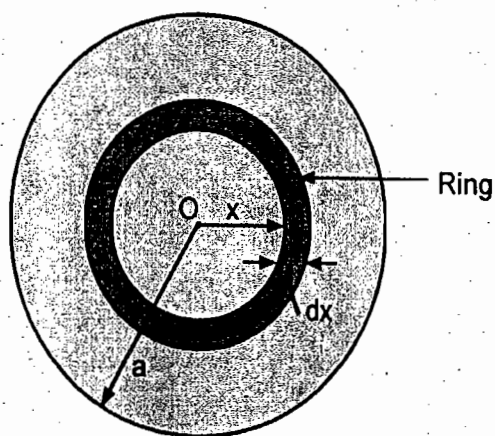
Given : Magnetic field at the centre of a ring of radius r and carrying a current I is

$$B = \frac{\mu_0 I}{2r}$$

Solution : Consider a plastic disc of radius 'a' as shown in Fig 4.20.

Let ' σ ' be surface charge density on the disc.

Let us divide the disc into large number of concentric rings with radii increasing from 0 to a.



Consider one such ring having radius x and thickness dx (See Fig. 4.20).

Surface area of the ring $= 2\pi x dx$

Charge on the ring $= \sigma 2\pi x dx$

When the disc rotate about an axis passing through centre and perpendicular to it's plane, the elementary ring is equivalent to current loop. If T is time required to complete one rotation, then the charge on the ring $2\pi x dx \sigma$ passes any point

of the loop during each time T . Therefore, the charge flowing per second through any point (i.e. current) is

$$i = \frac{2\pi x dx \sigma}{T}$$

But $\omega = \frac{2\pi}{T}$ or $T = \frac{2\pi}{\omega}$

$$\therefore i = x dx \sigma \omega$$

Magnetic field at the centre of a ring of radius r and carrying current is given as

$$B = \frac{\mu_0 I}{2r}$$

In our case, $dB = \frac{\mu_0 I}{2x}$

Using $i = x dx \sigma \omega$, in above equation, we get

$$dB = \frac{\mu_0 x dx \sigma \omega}{2x}$$

or $dB = \frac{\mu_0 \sigma \omega dx}{2}$

The magnetic field due to entire disc is

$$B = \int_0^a \frac{\mu_0 \sigma \omega}{2} dx$$

$$= \frac{\mu_0 \sigma \omega}{2} \int_0^a dx$$

or $B = \frac{\mu_0 \sigma \omega}{2} a$

If total charge on the disc is q , then,

$$\sigma = \frac{q}{\pi a^2}$$

$$\therefore B = \frac{\mu_0 \omega q}{2\pi a}$$

Thus, magnetic field at the centre of disc is

$$B = \frac{\mu_0 \omega q}{2\pi a}$$

Example 13 : The hysteresis loop for a specimen (iron) of 16 kg is equivalent to 3000 erg/cm³. Find the loss of energy per hour at a frequency of 50 cycles/sec; if density of iron is 10 gm/cm³.

Solution : Given data :

- (1) Mass of specimen = 16000 gm.
- (2) Density of specimen = 10 gm/cm³.
- (3) The energy dissipated per cm³ per hysteresis loop = 3000 erg/cm³.
- (4) Frequency i.e. number of cycles per sec. = 50 cycles/sec.

$$\text{The volume of the specimen} = \frac{\text{Mass}}{\text{Density}} = \frac{16000 \text{ gm}}{10 \text{ gm/cm}^3} = 1600 \text{ cm}^3.$$

$$\text{The energy dissipated in the iron per cycle} = 3000 \times 1600 = 4.8 \times 10^6 \text{ ergs.}$$

$$\begin{aligned} \text{Since number of cycles per second is 50, number of cycles per hour} &= 50 \times 60 \times 60 \\ &= 1.8 \times 10^5 \end{aligned}$$

$$\begin{aligned} \therefore \text{Energy loss per hour} &= 4.8 \times 10^6 \times 1.8 \times 10^5 \\ &= 8.64 \times 10^{11} \text{ ergs} \end{aligned}$$

QUESTIONS

1. State and explain Biot-Savart's law.
2. State Ampere's circuital law and show that

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$
3. Define B , H and M . Write the relation between them and give S.I. units.
4. Explain the term B , H and M and show that the magnetic field is given by curl of vector potential.

$$\vec{\nabla} \times \vec{A} = \vec{B}$$
5. Obtain an expression for divergence of a magnetic vector potential.
6. Show that

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$
7. Show that

$$\vec{\nabla} \cdot \vec{B} = 0$$
8. What is meant by magnetic energy? Show that magnetic energy density is given by expression $\frac{B^2}{2\mu_0}$.

9. What do you mean by hysteresis ? Draw hysteresis loop and explain the term remanence and coercivity.
10. State and explain Ampere's force law.
11. What is meant by hysteresis ? Show that energy dissipated per cycle of magnetisation is equal to area of B-H curve.
12. Find the magnetic field inside the uniformly magnetised sphere of radius a placed in an external magnetic field H_0 .
13. Define magnetisation $\vec{M}$ and magnetisation current density $\vec{J}_m$. Show that magnetisation current density is curl of magnetisation.
14. Find the magnetic field at a point inside and outside the sphere in a uniform magnetic field.
15. Write a note on magnetisation of matter.
16. What do you mean by magnetic susceptibility ? Show that $\mu_r = 1 + \chi_m$.

PROBLEMS

1. If $\vec{B}$ is uniform, show that $\vec{A}(\vec{r}) = -\frac{1}{2}(\vec{r} \times \vec{B})$.
2. Find the forces per unit length on two long straight, parallel conductors if each carries a current of 8 A in the same direction and separated by a distance of 20 cm.
(Ans. 640×10^{-7} N/m)
3. A spherical shell of radius ' a ', carrying a uniform surface charge ' σ ', is set spinning at angular velocity ω . Find the vector potential it produces at point P situated outside the sphere at $\vec{r}$ from the centre of sphere.
4. Hysteresis loss per cycle per unit volume for iron is 10^3 joule. Taking the density and specific heat of iron as 8 gm/cm^3 and 100 cal/kg-deg . respectively. Calculate rise in temperature per minute when the specimen is subjected to an alternating magnetic field of frequency 50 c/s.
(Given : $J = 4.2 \text{ Joule/cal}$)
5. In the Bohr model of hydrogen atom, the electron circulate around the nucleus in a circular path of radius $5.1 \times 10^{-11} \text{ m}$ at a frequency of $6.8 \times 10^{15} \text{ rev/sec}$. What value of magnetic field induction B is set up at the centre of the orbit ?
(Given : Magnetic induction on the axis of circular coil of radius ' a ' and carrying current I is

$$B = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$$

Hint : At the centre, $B = \frac{\mu_0 I}{2a}$ ($\because x=0$)

where, current $I = \frac{q}{T} = \frac{1.6 \times 10^{-19} \text{ C}}{6.8 \times 10^{15}}$

6. A 20 cm long wire carrying a current of 10 amp. is held at an angle 30° with the direction of a uniform magnetic field of strength 1 wb/m^2 . Calculate the force acting on the wire.
(Ans. 1 newton)

ELECTRODYNAMICS

INTRODUCTION

We know that electric field produced by electric charges at rest is called electrostatic field because such a field is time independent. Static electric field is conservative because any line integral of electric field vector $\vec{E}(\vec{r})$ around a closed path vanishes in such a field. There is also another type of electric field which is non-conservative as well as time dependent. Such a field is not produced by static charges. According to Faraday's Law of electromagnetic induction, when a magnetic flux linked with a closed circuit vary with time, it produces an electric field. Thus, a time varying magnetic field behaves as a source of electric field. The principle of generating electric field by changing magnetic field linked with a closed circuit is employed in transformers and inductances. Thus, according to Faraday's law, the change in magnetic field gives rise to electric field. On the similar line, Maxwell modified Ampere's law. According to Maxwell, a changing electric field in a space gives rise to a magnetic field. It means that either electric or magnetic field changes with time, the other field is induced in the space. This analogy leads to the creation of electromagnetic disturbance of time varying electric and magnetic fields. Such a disturbance propagating through the space is called electromagnetic wave.

5.1 FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

The induction of electromotive force by changing magnetic flux was first observed by Faraday and Henry in the early nineteenth century. From their pioneering experiments in the area of electromagnetic induction, it became easier to develop generator, transformer which has wide applications in electricity.

Faraday proved experimentally that when a magnetic flux linked with a closed circuit is changed with time, the e.m.f. is induced in the circuit. The magnitude of induced e.m.f. in a circuit is proportional to rate of change of flux.

$1 \text{ Volt} = 1 \text{ Wb/sec}$
 $|e| \propto \frac{d\phi_B}{dt}$
 $e = \frac{d\phi}{dt}$

In SI unit, proportionality const = 1. ϕ is weber & t is sec. e is in volt

The direction of induced e.m.f. is given by Lenz's law.

- 1) Whenever there is change in the mag. flux (current) linked with a circuit a current or emf is induced in the circuit.
- 2) The mag. of induced emf is proportion to rate of change of mag.

Lenz's law is stated that :

The direction of induced e.m.f. always opposes the change in magnetic flux that produces the induced e.m.f.

In S.I. units the constant of proportionality is unity.

Hence,

$$e = - \frac{d\phi_B}{dt} \quad \dots (5.1)$$

Consider a closed circuit C as shown in Fig. 5.1. The magnetic flux linked with this circuit is changing continuously with time.

Let $\vec{B}$ be the magnetic flux density around the circuit at an instant. The magnetic flux through area element $d\vec{S}$ is

$$d\phi_B = \vec{B} \cdot d\vec{S}$$

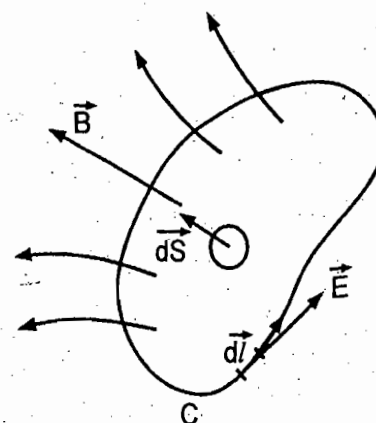


Fig. 5.1

The flux through entire circuit is

$$\phi_B = \int_S \vec{B} \cdot d\vec{S} \quad \dots (5.2)$$

As the magnetic flux is changing, an electric field is induced around the circuit. Thus, induced e.m.f. is

$$e = - \frac{d}{dt} \int \vec{B} \cdot d\vec{S}$$

or

$$e = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad \dots (5.3 (a))$$

If $\vec{E}$ is the electric field set up in the circuit, the induced e.m.f. in the closed circuit is equal to line integral of an electric field $\vec{E}$ around a closed circuit.

Therefore,

$$e = \oint \vec{E} \cdot d\vec{l} \quad \dots (5.3 (b))$$

where, $\vec{E}$ is the electric field at an element $d\vec{l}$.

By equations (5.3 (a)) and (5.3 (b)), we have

$$\oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad \text{Integral form} \quad \dots (5.4)$$

By Stoke's theorem, we have

$$\oint \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{S}$$

$$\therefore \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{S} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\therefore \int_S \left[\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} \right] \cdot d\vec{S} = 0$$

Since, this is true for all surfaces, we have

$$\boxed{\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}} \quad \text{... (5.5)}$$

Equation (5.4) is integral form of Faraday's law and equation (5.5) is differential form of Faraday's law.

5.2 THE BASIC EQUATIONS OF ELECTROMAGNETISM

In this chapter we seek to identify a basic set of equations for electromagnetism. Let us see the basic equations in electromagnetic theory so far we have studied. The integral forms of these equations are -

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad \text{(Gauss's law in electrostatics)}$$

$$\oint \vec{B} \cdot d\vec{S} = 0 \quad \text{(Gauss's law in magnetostatics)}$$

$$\oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad \text{(Faraday's law)}$$

and $\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad \text{(Ampere's law)}$

Then

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{or} \quad \vec{\nabla} \cdot \vec{D} = \rho \quad \therefore \vec{D} = \epsilon_0 \vec{E} \quad \text{... (5.6)}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{... (5.7)}$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \text{... (5.8)}$$

and $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \text{or} \quad \vec{\nabla} \times \vec{H} = \vec{J} \quad \therefore \vec{B} = \mu_0 \vec{H} \quad \text{... (5.9)}$

Above equations are called Maxwell's equations. The first three of these equations are general equations and are valid for static as well as dynamic charges. The fourth equation was derived from steady state field and not valid for time varying fields. So Maxwell modified this fourth equation so that it could be valid in time-varying fields also.

5.3 THE GENERALIZATION OF AMPERE'S LAW

In order to modify Ampere's law, Maxwell postulated that it is not only the current in the conductor that produces magnetic field but also the time varying electric field in a vacuum or medium produces magnetic field. Thus, changing electric field in a space or medium is equivalent to a current. This current flows as long as the electric field is changing and produces magnetic field same as an ordinary current in conductor. This current is known as *displacement current*.

The general form of Ampere's law is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

or
$$\oint \vec{H} \cdot d\vec{l} = I$$

$$I = \int \vec{J} \cdot d\vec{S}$$

where, $\vec{J}$ is current density,

$$\therefore \oint \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S}$$

Using Stoke's law,
$$\oint \vec{H} \cdot d\vec{l} = \oint_S (\vec{\nabla} \times \vec{H}) \cdot d\vec{S}$$

$$\int (\vec{\nabla} \times \vec{H}) \cdot d\vec{S} = \int \vec{J} \cdot d\vec{S}$$

This implies that,

$$\vec{\nabla} \times \vec{H} = \vec{J} \quad \dots (5.10)$$

Taking divergence on both sides, we get

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} \quad \dots (5.11)$$

But we know that divergence of a curl of any vector is zero.

i.e.
$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = 0$$

$$\therefore \vec{\nabla} \cdot \vec{J} = 0 \quad \dots (5.12)$$

This is true for steady state fields. However, when these fields are changing with time, the result is incompatible with the principle of conservation of charge. In general, equation of continuity is

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \quad \dots (5.13)$$

Maxwell took into account this equation and said that in time varying field, there is also conservation of charge. According to Gauss's law we have,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \text{ or } \vec{\nabla} \cdot \vec{D} = \rho$$

$$\therefore \vec{\nabla} \cdot \vec{J} + \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{D} = 0$$

$$\therefore \vec{\nabla} \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = 0$$

Maxwell replaced $\vec{J}$ by $\vec{J} + \frac{\partial \vec{D}}{\partial t}$ in equation (5.10). With this, Ampere's law gets modified to,

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \dots (5.14)$$

The introduction of the second term on the right, which is known as the displacement current, represents one of the Maxwell's major contribution to electromagnetic theory.

$$\text{or } \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \frac{\partial \vec{D}}{\partial t} \quad \dots (5.15)$$

$$\vec{D} = \epsilon_0 \vec{E} \text{ in free space,}$$

$$\therefore \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \dots (5.16)$$

The law in this form is valid for steady-state phenomenon as well as for the time varying fields. The law is compatible with the equation of continuity for time dependent fields.

The term $\vec{J}$ in equation (5.15) is called '*conduction current density*' and the term $\frac{\partial \vec{D}}{\partial t}$ is called '*displacement current density*'.

The displacement current cannot be detected directly, but the confirmation of its existence is provided indirectly by the electromagnetic theory.

5.4 MAXWELL'S EQUATIONS (DIFFERENTIAL AND INTEGRAL FORMS)

Let us now list all the four equations.

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \text{ or } \vec{\nabla} \cdot \vec{D} = \rho \quad \dots (5.17)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \dots (5.18)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \dots (5.19)$$

$$\text{and} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{or} \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \dots (5.20)$$

These equations are the fundamental equations of electromagnetic field and are known as called *Maxwell's equations*. Each of these equations represents a generalization of certain experimental observations : equation (5.17) is Gauss's law which is derived from Coulomb's law; equation (5.18) is usually said to represent the fact that the single magnetic pole never exists; equation (5.19) is the differential form of Faraday's law of electromagnetic induction; equation (5.20) represents extension of Ampere's law.

The integral forms of Maxwell's equations are

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad \text{or} \quad \oint \vec{D} \cdot d\vec{S} = q \quad (\text{Gauss's law in electrostatics}) \quad \dots (5.21)$$

$$\oint \vec{B} \cdot d\vec{S} = 0 \quad (\text{Gauss's law in magnetostatics}) \quad \dots (5.22)$$

$$\oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad (\text{Faraday's law}) \quad \dots (5.23)$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \int \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S} \quad (\text{Modified Ampere's law}) \quad \dots (5.24)$$

5.5 PHYSICAL SIGNIFICANCE OF MAXWELL'S EQUATIONS

Equation (5.21) is the Gauss's law for electricity which states that the total electric flux through a closed surface is equal to total charge enclosed by the surface upon permittivity of space.

Equation (5.22) is the Gauss' law for magnetism stating that the total magnetic flux through a closed surface is zero. This signifies that magnetic monopole does not exist. Any volume enclosed by a closed surface will contain equal and opposite magnetic poles giving a net magnetic pole-strength zero.

Equation (5.23) is the Faraday's law of electromagnetic induction, which shows that changing magnetic flux produces electric field.

Equation (5.24) is the Ampere's law modified by Maxwell, which signifies that the current in conductor as well as changing electric field in the space produces magnetic field.

5.6 WAVE EQUATION IN FREE SPACE

We know that time varying magnetic field produces electric field and vice-versa. This leads to generation of electromagnetic disturbance of time varying electric and magnetic fields. This disturbance propagates through a space called electromagnetic wave. We will now show that the Maxwell's equations lead to electromagnetic wave equation. We have Maxwell's equations

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{or} \quad \vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{or} \quad \vec{\nabla} \cdot \vec{H} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{or} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

For free space, $\rho = 0$ and $\vec{\nabla} \cdot \vec{J} = 0$

$$\therefore \vec{\nabla} \cdot \vec{E} = 0 \quad \dots (5.25)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \dots (5.26)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \dots (5.27)$$

$$\vec{\nabla} \times \vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \dots (5.28)$$

Above four equations are Maxwell's equations in free space.

Taking curl of equation (5.27), we get

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\vec{\nabla} \times \left(\frac{\partial \vec{B}}{\partial t} \right)$$

$$\therefore \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\mu_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H}) \quad \because \vec{B} = \mu_0 \vec{H} \text{ in free space}$$

According to vector product rule,

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E}$$

Above equation becomes

$$\vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H}) \quad \dots (5.29)$$

Using equations (5.25) and (5.28) in equation (5.29), we get

$$-\nabla^2 \vec{E} = -\mu_0 \frac{\partial}{\partial t} \left(\epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\therefore -\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\therefore \nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \dots (5.30)$$

Similarly, taking curl of equation (5.28) and solving it, we get

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2} \quad (5.31)$$

The equations (5.30) and (5.31) are the type of wave equations which are familiar for us.

The general wave equation is

$$\nabla^2 \phi = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} \quad \dots (5.32)$$

Comparing equations (5.30) and (5.31) with equation (5.32), we get

$$\frac{1}{c^2} = \epsilon_0 \mu_0$$

or

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

where, c is velocity of propagation.

For free space, we have

$$\mu_0 = 4\pi \times 10^{-7} \text{ Wb/A}\cdot\text{m}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

$$c = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}} \\ = 3 \times 10^8 \text{ m/sec.}$$

Thus, we can conclude that the field vectors $\vec{E}$ and $\vec{B}$ propagate as waves in vacuum (free space) with velocity of light c. This suggests that light waves are electromagnetic in nature.

5.7 SOLUTION OF WAVE EQUATION FOR PLANE WAVE IN FREE SPACE

The simplest type of wave that is a solution of equations (5.30) and (5.31) is a plane wave. An electromagnetic wave whose field vectors are the function of only one space co-ordinate and time only. Let the wave propagate along the z-axis. Thus,

$$\frac{\partial E_x}{\partial x} = 0 = \frac{\partial E_y}{\partial y} \quad \text{and} \quad \frac{\partial H_x}{\partial x} = \frac{\partial H_y}{\partial y} = 0$$

$$\text{But} \quad \frac{\partial \vec{E}}{\partial z} \neq 0 \quad \text{and} \quad \frac{\partial \vec{H}}{\partial z} \neq 0 \quad \dots (5.33)$$

$$\text{Here} \quad \left. \begin{aligned} \vec{E} &= \vec{E}(z, t) \\ \vec{H} &= \vec{H}(z, t) \end{aligned} \right\} \quad \dots (5.34)$$

But we know that $\vec{\nabla} \cdot \vec{E} = 0$ and $\vec{\nabla} \cdot \vec{H} = 0$ for free space
Therefore,

$$\frac{\partial E_z}{\partial z} = 0 \quad \text{and} \quad \frac{\partial H_z}{\partial z} = 0$$

H_z and E_z are constants in space, i.e. E_z and H_z cannot be a function of z which is the direction of propagation.

This means E_z and H_z are constants in Z-direction.

Also from equation (5.27)

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

or
$$\vec{\nabla} \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

The z-component of above equation is,

$$(\vec{\nabla} \times \vec{E})_z = -\mu_0 \frac{\partial H_z}{\partial t}$$

We have

$$(\vec{\nabla} \times \vec{E})_z = k \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) = 0 \quad \left\{ \because \frac{\partial E_y}{\partial x} = \frac{\partial E_x}{\partial y} = 0 \right\}$$

$$\therefore \mu_0 \frac{\partial H_z}{\partial t} = 0$$

i.e. H_z is constant in time, and $\hat{k}$ is unit vector along Z-axis.

Similarly, we can show that E_z is constant in time. Thus, both E_z and H_z are constants in space and time. Therefore, these are static components of fields and are not part of wave motion. So we can choose $E_z = H_z = 0$.

Therefore, we can write

$$\vec{E} = E_x \hat{i} + E_y \hat{j} \quad \dots (5.35)$$

and
$$\vec{H} = H_x \hat{i} + H_y \hat{j} \quad \dots (5.36)$$

Since $\vec{E}$ and $\vec{H}$ do not have any component along Z-axis, Z is direction of propagation. Thus, the electric and magnetic fields are perpendicular to the direction of propagation. Thus, the Maxwell's electromagnetic waves are purely transverse in nature.

Let the general solutions of plane wave are

$$\vec{E}(z, t) = \vec{E}_0 e^{i(kz - \omega t)} \quad \dots (5.37)$$

$$\vec{H}(z, t) = \vec{H}_0 e^{i(kz - \omega t)} \quad \dots (5.38)$$

The fields $\vec{E}$ and $\vec{H}$ have to satisfy Maxwell's equation,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

or
$$\vec{\nabla} \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

Using equations (5.37) and (5.38) in above equation, we get

$$\begin{aligned}\nabla \times (\vec{E}_0 e^{i(kz - \omega t)}) &= -\mu_0 \vec{H}_0 \frac{\partial}{\partial t} e^{i(kz - \omega t)} \\ \therefore \vec{\nabla} e^{i(kz - \omega t)} \times \vec{E}_0 &= i \mu_0 \omega \vec{H}_0 e^{i(kz - \omega t)} \\ \therefore i \vec{k} e^{i(kz - \omega t)} \times \vec{E}_0 &= i \mu_0 \omega \vec{H}_0 e^{i(kz - \omega t)} \quad \left\{ \because \hat{k} \cdot \vec{k} = \vec{k} \right\} \\ \therefore \vec{k} \times \vec{E} &= \mu_0 \omega \vec{H} \quad \dots (5.39)\end{aligned}$$

Hence, $\vec{H}$ perpendicular to both $\vec{k}$ and $\vec{E}$, where $\vec{k}$ is propagation constant. The transverse electromagnetic wave is represented in Fig. 5.2.

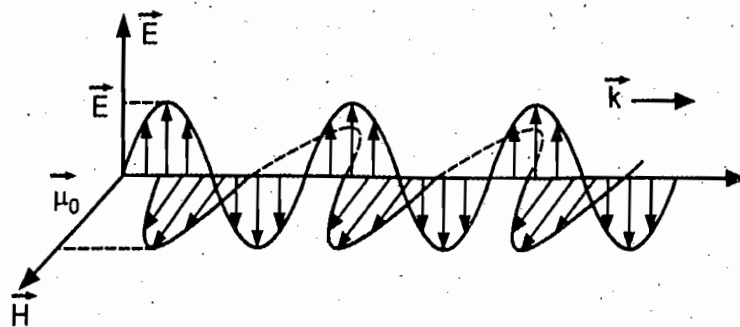


Fig. 5.2

Let $\vec{E}$ is along x-axis and $\vec{H}$ is along y-axis. Therefore, from equation (5.39), we have

$$\begin{aligned}k E_x &= \mu_0 \omega H_y \\ \frac{E_x}{H_y} &= \frac{\mu_0 \omega}{k}\end{aligned}$$

But $\frac{\omega}{k}$ = phase velocity of wave which is equal to the velocity of light 'c'.

$$\begin{aligned}\therefore \frac{E_x}{H_y} &= \mu_0 c \\ \therefore \frac{E_x}{H_y} &= \frac{\mu_0}{\sqrt{\mu_0 \epsilon_0}} \quad \left\{ \because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \right\} \\ \therefore \frac{E_x}{H_y} &= \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.7 \Omega\end{aligned}$$

The ratio $\frac{E_x}{H_y}$ is called the intrinsic impedance of space.

From above discussion we conclude that since $\vec{E}$ and $\vec{H}$ are perpendicular to direction of propagation and also mutually perpendicular to each other, the electromagnetic wave propagates in free space in direction of $\vec{E} \times \vec{H}$.

5.8 POYNTING VECTOR IN FREE SPACE

We know that plane electromagnetic wave in free space propagates in direction of vector $\vec{E} \times \vec{H}$.

Maxwell's third and fourth equations are

$$\left. \begin{aligned} \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{H} &= \frac{\partial \vec{D}}{\partial t} \end{aligned} \right\} \dots (5.40)$$

Taking scalar product on both sides of above equations (5.40) with $\vec{H}$ and $\vec{E}$ respectively and subtracting, we get

$$\begin{aligned} \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H}) &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \\ \therefore \vec{E} \cdot (\vec{\nabla} \times \vec{H}) - \vec{H} \cdot (\vec{\nabla} \times \vec{E}) &= \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} + \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \end{aligned} \dots (5.41)$$

But

$$\begin{aligned} \vec{\nabla} \cdot (\vec{E} \times \vec{H}) &= \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H}) \\ &= -[\vec{E} \cdot (\vec{\nabla} \times \vec{H}) - \vec{H} \cdot (\vec{\nabla} \times \vec{E})] \\ \therefore \vec{\nabla} \cdot (\vec{E} \times \vec{H}) &= -\left(\vec{E} \cdot \frac{\partial \vec{D}}{\partial t} + \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \right) \end{aligned} \dots (5.42)$$

But we know,

$$\vec{D} = \epsilon_0 \vec{E} \text{ and } \vec{B} = \mu_0 \vec{H}$$

Equation (5.42) gives

$$\begin{aligned} \vec{\nabla} \cdot (\vec{E} \times \vec{H}) &= -\left[\vec{E} \cdot \frac{\partial}{\partial t} (\epsilon_0 \vec{E}) + \vec{H} \cdot \frac{\partial}{\partial t} (\mu_0 \vec{H}) \right] \\ &= -\frac{1}{2} \frac{\partial}{\partial t} [\epsilon_0 E^2 + \mu_0 H^2] \\ \therefore \vec{\nabla} \cdot (\vec{E} \times \vec{H}) &= -\frac{\partial}{\partial t} \left(\frac{\epsilon_0 E^2}{2} + \frac{\mu_0 H^2}{2} \right) \end{aligned} \dots (5.43)$$

Integrating equation (5.43) over volume V bounded by surface S , we get

$$\int_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV = \int_V \frac{\partial}{\partial t} \left(\frac{\epsilon_0 E^2}{2} + \frac{\mu_0 H^2}{2} \right) dV$$

By using Gauss's divergence theorem on L.H.S. of above equation, we get

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = -\frac{\partial}{\partial t} \int_V \left(\frac{\epsilon_0 E^2}{2} + \frac{\mu_0 H^2}{2} \right) dV \quad \dots (5.44)$$

The integral on R.H.S. of equation (5.44) is the sum of electric and magnetic energies. Electrostatic field energy per unit volume is $\frac{1}{2} \epsilon_0 E^2$, and magnetostatic field energy per unit volume is $\frac{1}{2} \mu_0 H^2$ for free space. So R.H.S. is energy lost per unit time by volume V and L.H.S. must be total outward flux of energy through surface S bounding volume V . The quantity $\vec{P} = (\vec{E} \times \vec{H})$ in watt/m² is called *Poynting vector*. It represents amount of field energy passing through area of surface in unit time normal to direction of flow of energy.

So $(\vec{E} \times \vec{H})$ has dimension of $\frac{\text{energy}}{\text{area} \times \text{time}}$

The vector $(\vec{E} \times \vec{H})$ which gives rate at which energy flow across unit area of boundary is called poynting vector and is represented by $\vec{P}$.

From equation 5.44,

$$\oint_S \vec{P} \cdot d\vec{S} = -\frac{\partial}{\partial t} \int_V \left(\frac{\epsilon_0 E^2}{2} + \frac{\mu_0 H^2}{2} \right) dV = -\frac{\partial W}{\partial t}$$

The term $-\frac{dW}{dt}$ is the rate at which energy is decreasing within given volume.

where, $W = \int_V \left(\frac{\epsilon_0 E^2}{2} + \frac{\mu_0 H^2}{2} \right) dV$, is called the total electromagnetic energy in a given volume in free space.

For any arbitrary closed surface the amount of power flowing out through surface is $\oint \vec{P} \cdot d\vec{S}$

$$\therefore -\frac{dW}{dt} = \oint \vec{P} \cdot d\vec{S}$$

This is known as Poynting theorem.

Poynting Theorem:

It states that "decrease of electromagnetic energy (-ve sign) per unit time in certain volume V is equal to work done (energy) by field forces per unit time plus flux flow outwards per unit time".

5.9 FIELD ENERGY OF POYNTING THEOREM (Electromagnetic Energy Flow)

Maxwell's equations are given by,

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \dots (5.45)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \dots (5.46)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \dots (5.47)$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \dots (5.48)$$

The above four Maxwell's equations are used to derive the principle of energy is electromagnetic field.

The above Maxwell's equations (5.47) and (5.48) are Ampere's and Faraday's laws in differential form.

If we take scalar product of equations (5.47) with $\vec{H}$ and of equation (5.48) with $\vec{E}$ we get

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \quad \dots (5.49)$$

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \quad \dots (5.50)$$

Subtracting equation (5.49) from equation (5.50), we get

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) - \vec{H} \cdot (\vec{\nabla} \times \vec{E}) = \vec{J} \cdot \vec{E} + \left[\vec{E} \cdot \frac{\partial \vec{D}}{\partial t} + \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \right] \quad \dots (5.51)$$

Using vector identity,

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot (\vec{\nabla} \times \vec{H}) - \vec{H} \cdot (\vec{\nabla} \times \vec{E})$$

Hence, equation (5.51) becomes,

$$\therefore -\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{J} \cdot \vec{E} + \left[\vec{E} \cdot \frac{\partial \vec{D}}{\partial t} + \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \right] \quad \dots (5.52)$$

$$\text{But,} \quad \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} = \vec{E} \cdot \frac{\partial}{\partial t} (\epsilon \vec{E}) = \epsilon_0 \epsilon_r \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \quad (\because \epsilon = \epsilon_0 \epsilon_r)$$

$$\begin{aligned} \therefore \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} &= \frac{1}{2} \epsilon_0 \epsilon_r \frac{\partial}{\partial t} (\vec{E} \cdot \vec{E}) \\ &= \frac{1}{2} \frac{\partial}{\partial t} (\vec{E} \cdot \vec{D}) \end{aligned} \quad \dots (5.53)$$

$$\begin{aligned} \text{Similarly,} \quad \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} &= \vec{H} \cdot \frac{\partial}{\partial t} (\mu \vec{H}) \\ &= \vec{H} \cdot \mu_r \mu_0 \frac{\partial}{\partial t} (\vec{H}) \quad (\because \mu = \mu_r \mu_0) \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \mu_r \mu_0 \frac{\partial}{\partial t} (\vec{H} \cdot \vec{H}) \\
 &= \frac{1}{2} \frac{\partial}{\partial t} (\vec{H} \cdot \vec{B}) \quad \dots (5.54)
 \end{aligned}$$

where, ϵ_r and μ_r are relative permittivity and permeability respectively.

So equation (5.52) reduces to,

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) + \vec{J} \cdot \vec{E} + \frac{1}{2} \frac{\partial}{\partial t} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) = 0 \quad \dots (5.55)$$

Each term in equation (5.55) have some physical meaning. If it is multiplied by volume dV and integrated over a volume V whose enclosed surface is S , the equation (5.55) is given by,

$$\int_V \vec{J} \cdot \vec{E} dV + \int_V \frac{1}{2} \frac{\partial}{\partial t} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) dV + \int_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV = 0 \quad \dots (5.56)$$

By divergence theorem,

$$\int_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV = \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$$

So equation (5.56) gives

$$\int_V \vec{J} \cdot \vec{E} dV + \int_V \frac{1}{2} \frac{\partial}{\partial t} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) dV + \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = 0 \quad \dots (5.57)$$

Interpretation of each term on L.H.S. of equation (5.57) is given below.

1. Interpretation of $\int_V \vec{J} \cdot \vec{E} dV$:

Current distribution represented by vector $\vec{J}$ can be considered as made up of various charges q_i moving with velocity $\vec{v}_i$, so that,

$$\begin{aligned}
 \int_V \vec{J} \cdot \vec{E} dV &= \int_V \vec{J} dV \cdot \vec{E} \\
 &= \int (\vec{J} \cdot d\vec{S}) d\vec{l} \cdot \vec{E} \\
 &= \int I d\vec{l} \cdot \vec{E} \quad (\because I = \vec{J} \cdot d\vec{S}) \\
 &= \int \frac{dq_i}{dt} d\vec{l} \cdot \vec{E} \\
 &= \int dq_i (\vec{v}_i) \cdot \vec{E} \quad \left(\because \vec{v}_i = \frac{d\vec{l}}{dt} \right)
 \end{aligned}$$

$\therefore$ In general,

$$\int_V \vec{J} \cdot \vec{E} dV = \int dq_i (\vec{v}_i \cdot \vec{E}_i)$$

$$= \sum_i q_i (\vec{v}_i \cdot \vec{E}_i) \quad \dots (5.58)$$

where, $\vec{E}_i$ is electric field at position of charge q_i . Electromagnetic force on i^{th} charged particle is given by Lorentz expression.

$$\vec{F}_i = q_i (\vec{E}_i + \vec{v}_i \times \vec{B}_i)$$

So work done per unit time on charge q_i by the field is,

$$\begin{aligned} \frac{\partial W_i}{\partial t} &= \vec{F}_i \cdot \vec{v}_i \\ &= q_i (\vec{E}_i + \vec{v}_i \times \vec{B}_i) \cdot \vec{v}_i \\ &= q_i \vec{v}_i \cdot \vec{E}_i + q_i \vec{v}_i \cdot (\vec{v}_i \times \vec{B}_i) \\ &= q_i \vec{v}_i \cdot \vec{E}_i + 0 \quad [\because \vec{v}_i \cdot (\vec{v}_i \times \vec{B}_i) = 0] \\ &= q_i \vec{v}_i \cdot \vec{E}_i \end{aligned}$$

$\therefore$ Rate at which work is done by the field on charge q_i is

$$\frac{\partial W}{\partial t} = \sum_i \frac{\partial W_i}{\partial t} = \sum_i q_i \vec{v}_i \cdot \vec{E}_i \quad \dots (5.59)$$

Comparing equations (5.58) and (5.59), the first term $\int_V \vec{J} \cdot \vec{E} \, dV$ of equation (5.57)

represents the rate at which work is done by the field on charges.

2. Interpretation of $\frac{1}{2} \int_V \frac{\partial}{\partial t} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) \, dV$:

If volume V is to be arbitrarily large, surface integral in equation (5.57) vanishes by placing surface S sufficiently far away so that field cannot propagate to this distance in finite time. So equation (5.57) reduces to,

$$\begin{aligned} \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} &= 0 \\ \frac{\partial}{\partial t} \int_{\text{all space}} \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) \, dV + \frac{\partial W}{\partial t} &= 0 \\ \therefore \frac{\partial}{\partial t} \left[\left\{ \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) \, dV \right\} + W \right] &= 0 \quad \dots (5.60) \end{aligned}$$

The quantity in square bracket is conserved i.e. total energy in closed system is constant. The system consists of electromagnetic energy due to all charge particles and total P.E. of particles.

$$\therefore \text{EM energy} = U = \int_{\text{all space}} \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) dV \quad \dots (5.61)$$

We therefore, interpret L.H.S. of equation (5.60) as the rate at which energy is stored in electromagnetic field diminishes.

$\therefore$ Equation (5.60) gives

$$\frac{\partial}{\partial t} (U + W) = 0 \quad \dots (5.62)$$

3. Interpretation $\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$:

Instead of taking volume integral over all space, let us consider finite volume. So surface integral of $\vec{E} \times \vec{H}$ cannot vanish. i.e. $\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$ has some finite values. So for conservation of energy,

$$\frac{\partial}{\partial t} (V + W) = - \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} \quad \dots (5.63)$$

L.H.S. of equation (5.63) is time rate of change of energy of field and of particles contained within volume V . Thus, surface integral $\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$ is considered as energy flowing out of volume bounded by surface S per unit time. But by hypothesis, no particles are crossing surface. So vector $\vec{E} \times \vec{H} = \vec{P}$ is called Poynting vector and it is interpreted as amount of field energy passing through unit area of surface in unit time which is normal to the direction of energy flow.

4. Interpretation of energy equation :

From above discussion, the equation (5.55) in differential form can be written as,

$$\vec{J} \cdot \vec{E} + \frac{\partial U}{\partial t} + \nabla \cdot \vec{P} = 0 \quad \dots (5.64)$$

where, $U = \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B})$

If media has zero conductivity, then $\vec{J} = \sigma \vec{E} = 0$.

$$\frac{\partial U}{\partial t} + \nabla \cdot \vec{P} = 0$$

Above equation has exactly same form as that of continuity equation which expresses law of conservation of charge. Physical meaning of equations (5.64) and (5.63) is that, law of conservation of energy is electromagnetic phenomenon and this law is called Poynting theorem. According to equation (5.63), Poynting theorem can be stated as "time rate of change of EM energy within volume V plus rate at which work is done by field on charges is equal to energy flowing into system through boundary surface of volume V per unit time".

5.10 POLARIZATION OF PLANE WAVES

Plane Polarized Wave :

The electromagnetic plane waves have electric and magnetic vectors perpendicular to each other and are always perpendicular to the direction of propagation. If electromagnetic waves are propagating along the positive direction of Z-axis, then $\vec{E}$ and $\vec{H}$ can have no components in the Z-direction.

Therefore, $E_z = 0$ and $H_z = 0$

Thus, a plane electromagnetic wave propagating in free space is therefore, transverse, since it has no longitudinal components.

In practice, the electromagnetic wave consisting of combination of plane waves propagating in the same direction, with arbitrary orientations of field vectors, arbitrary magnitudes and random phases is called an **unpolarized wave**.

A superposed wave is obtained by the combination of various plane waves. The electric field vector in a superposed wave or a resultant wave lie in certain fixed direction. Such a wave is said to be **plane polarized wave** or **linearly polarized wave**.

The electromagnetic waves having vertical electric field vector is known as vertically polarized. Similarly, the electromagnetic waves having horizontal electric field vector is referred to as horizontally polarized. The vertically and horizontally polarized waves are shown in Fig. 5.3.

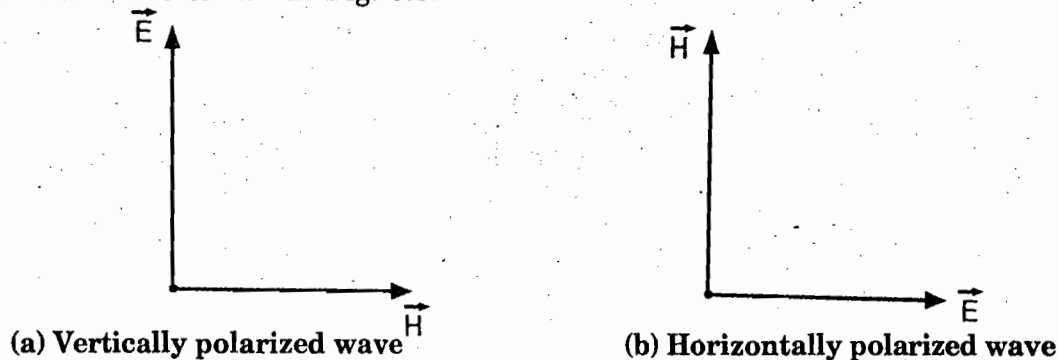


Fig. 5.3

The electric field of linearly polarized wave in Y-direction is given by equation (5.65). This wave is propagating in positive direction of Z-axis.

$$E_y = E_{0y} \sin(\omega t - kz) \quad \dots (5.65)$$

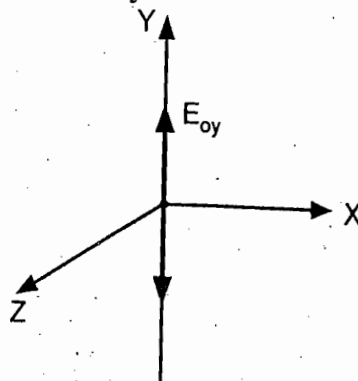


Fig. 5.4 : Linearly polarized wave in y-direction

In general, the electric field of a wave travelling in the Z-direction may have both a y-component and an x-component as shown in Fig. 5.5.

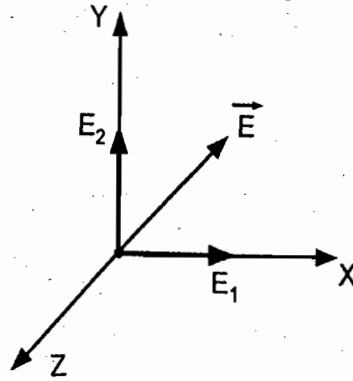


Fig. 5.5 : The electric field components

Here, E_1 is the amplitude of wave which is linearly polarized in X-direction and E_2 is the amplitude of wave which is linearly polarized in Y-direction.

5.11 CIRCULAR POLARIZATION

Consider an electromagnetic wave travelling in the positive Z-direction as shown in Fig. 5.6.

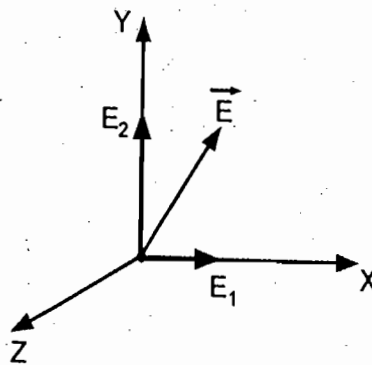


Fig. 5.6 : The electric field components in vertically and horizontally polarized wave

The electric field components in X and Y directions are,

$$E_x = E_1 \sin(\omega t - kz) \quad \dots (5.66)$$

$$\text{and} \quad E_y = E_2 \sin(\omega t - kz + \delta) \quad \dots (5.67)$$

where, δ is the time phase angle by which E_y leads E_x .

Combining equations (5.66) and (5.67) we get the instantaneous total vector field $\vec{E}$.

$$\text{Therefore,} \quad \vec{E} = E_x \hat{i} + E_y \hat{j}$$

$$\therefore \quad \vec{E} = E_1 \sin(\omega t - kz) \hat{i} + E_2 \sin(\omega t - kz + \delta) \hat{j} \quad \dots (5.68)$$

$$\text{At } z = 0, \quad E_x = E_1 \sin \omega t \quad \dots (5.69)$$

$$E_y = E_2 \sin(\omega t + \delta) \quad \dots (5.70)$$

$$\text{Therefore,} \quad E_y = E_2 (\sin \omega t \cos \delta + \cos \omega t \sin \delta) \quad \dots (5.71)$$

From equation (5.69),

$$\sin \omega t = \frac{E_x}{E_1}$$

Also,

$$\begin{aligned} \cos \omega t &= \sqrt{1 - \sin^2 \omega t} \\ &= \sqrt{1 - \frac{E_x^2}{E_1^2}} \end{aligned}$$

Putting these values of $\sin \omega t$ and $\cos \omega t$ in equation (5.71), we get

$$E_y = \frac{E_2 E_x}{E_1} \cos \delta + E_2 \sqrt{1 - \frac{E_x^2}{E_1^2}} \sin \delta$$

$$\therefore E_y - \frac{E_2}{E_1} E_x \cos \delta = E_2 \sqrt{1 - \frac{E_x^2}{E_1^2}} \sin \delta \quad \dots (5.72)$$

Squaring on both sides equation (5.72) gives, we get

$$\begin{aligned} E_y^2 + \frac{E_x^2}{E_1^2} E_2^2 \cos^2 \delta - \frac{2 E_2}{E_1} E_x E_y \cos \delta &= E_2^2 \left(1 - \frac{E_x^2}{E_1^2} \right) \sin^2 \delta \\ E_y^2 + \frac{E_x^2}{E_1^2} E_2^2 - 2 \frac{E_2}{E_1} E_x E_y \cos \delta &= E_2^2 \sin^2 \delta \quad \dots (5.73) \end{aligned}$$

Dividing equation (5.73) throughout by E_2^2 , we get

$$\frac{E_y^2}{E_2^2} + \frac{E_x^2}{E_1^2} - \frac{2 E_x E_y}{E_1 E_2} \cos \delta = \sin^2 \delta \quad \dots (5.74)$$

This is general expression of an ellipse.

It represents the elliptically polarized wave.

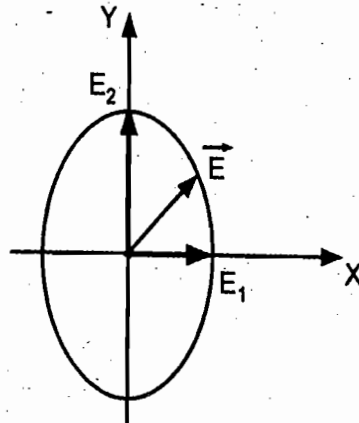


Fig. 5.7 : Elliptical polarization

If $E_1 = E_2$ and $\delta = \frac{\pi}{2}$, then equation (5.74) gives circularly polarized wave.

$$\frac{E_x^2}{E_1^2} + \frac{E_y^2}{E_1^2} = 1$$

$$E_x^2 + E_y^2 = E_1^2 \quad \dots (5.75)$$

This is the equation of a circle of radius E_1 . The equation (5.75) shows that the magnitude of E remains constant, although its direction rotates.

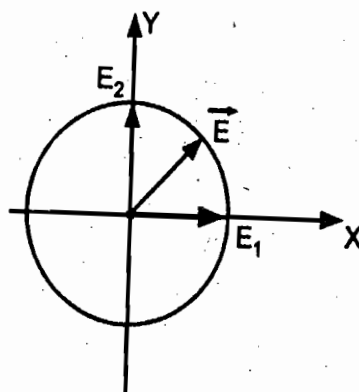


Fig. 5.8 : Circular polarization

When $\delta = +\frac{\pi}{2}$, the wave is left circularly polarized and when $\delta = -\frac{\pi}{2}$, the wave is right circularly polarized.

ILLUSTRATIVE EXAMPLES

Example 1 : Show that another point form of Faraday's law is

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t}$$

where, $\vec{A}$ is magnetic vector potential.

Solution : From Maxwell's equation,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

But

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$\therefore$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} \vec{\nabla} \times \vec{A}$$

$$\vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \frac{\partial \vec{A}}{\partial t}$$

$\therefore$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} \text{ which is the required result.}$$

Example 2 : A circular loop of radius 10 cm is located in XY plane in a $\vec{B}$ field given by $\vec{B} = (0.5 \cos 377 t) (3\hat{a}_y + 4\hat{a}_z)$. Determine the voltage induced in the loop.

Solution : Given :

$$r = 10 \text{ cm} = 0.1 \text{ m}$$

Induced e.m.f., $V = -\frac{\partial \phi}{\partial t}$

But $\phi = \int \vec{B} \cdot d\vec{S}$

$$\begin{aligned} \therefore V &= -\frac{\partial}{\partial t} \int \vec{B} \cdot d\vec{S} \\ &= -\frac{\partial}{\partial t} \left[(0.5 \cos 377 t) (3 \vec{a}_y + 4 \vec{a}_z) \cdot \pi (0.1)^2 \vec{a}_z \right] \end{aligned}$$

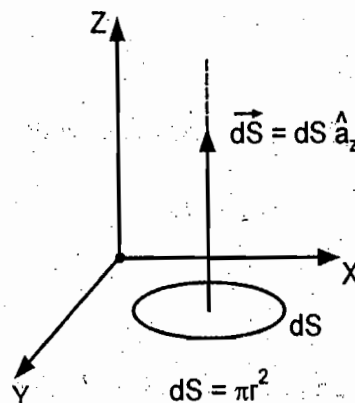


Fig. 5.9

$$\begin{aligned} \therefore V &= -\frac{\partial}{\partial t} [0.5 \times 4\pi (0.10)^2 \cos 377 t] \\ &= 0.5 \times 377 \sin 377 t \times 4\pi (0.10)^2 \\ V &= 23.69 \sin 377 t \text{ volt} \end{aligned}$$

Example 3 : An inductor is formed by winding 10 turns of a thin wire around a wooden dowel rod of radius 2 cm. A uniform sinusoidal magnetic field of magnitude 10 mwb/m² and frequency 10 kHz, is directed along the axis of the rod. Find the induced voltage in winding.

Solution : The flux through winding is

$$\begin{aligned} \phi &= BA \\ \text{Here, } B_0 &= 0.01 \text{ wb/m}^2 \\ \text{and } \vec{B} &= B_0 \sin \omega t \hat{a}_z \\ \therefore \vec{B} &= B_0 \cos \omega t \\ \vec{B} &= 0.01 \cos 2\pi ft \\ \vec{B} &= (0.01 \cos 2\pi \times 10^4 t) \end{aligned}$$

$$\begin{aligned} \phi &= BA \\ &= (0.01 \cos 2\pi \times 10^4 t) \cdot \pi (0.02)^2 \\ \phi &= 4\pi \times 10^{-6} \cos 2\pi \times 10^4 t \text{ wb} \end{aligned}$$

This flux links with winding with turns $N = 10$.

Therefore,

$$\begin{aligned}
 V &= -10 \frac{d}{dt} \phi \\
 &= -10 \frac{d}{dt} 4\pi \times 10^{-6} \cos 2\pi \times 10^4 t \\
 &= +10 \times 4\pi \times 10^{-6} \times 2\pi \times 10^4 \sin 2\pi \times 10^4 t \\
 &= 7.986 \sin 2\pi \times 10^4 t \text{ volts}
 \end{aligned}$$

Example 4 : For a material medium characterised by conductivity σ and permittivity ϵ exposed to sinusoidally varying electric field of frequency ω , obtain the ratio of the conduction current density, $|J_c|$, to the displacement current density $|J_d|$.

Solution : Let the field be

Then,

$$\begin{aligned}
 E &= E_0 e^{i\omega t} \\
 J_c &= \sigma E \\
 &= \sigma E_0 e^{i\omega t}
 \end{aligned}$$

and

$$\begin{aligned}
 -J_d &= \frac{\partial D}{\partial t} = i\omega D \quad (\because \vec{D} = \epsilon \vec{E}) \\
 &= i\omega \epsilon E \\
 -J_d &= i\omega \epsilon E_0 e^{i\omega t} \\
 \frac{|J_c|}{|J_d|} &= \frac{\sigma E_0 e^{i\omega t}}{i\omega \epsilon E_0 e^{i\omega t}} \\
 &= \frac{\sigma}{\omega \epsilon}
 \end{aligned}$$

For conductor, $|J_c| \gg |J_d|$ or $\omega \ll \frac{\sigma}{\epsilon}$

For insulator,

$$|J_c| \ll |J_d| \quad \text{or} \quad \omega \gg \frac{\sigma}{\epsilon}$$

Example 5 : For copper, $\sigma = 58 \text{ Ms/m}$ and $\epsilon = \epsilon_0$; for Teflon, $\sigma = 30 \text{ ns/m}$ and $\epsilon = 21 \epsilon_0$. Verify that at 1 MHz, copper is a good conductor and Teflon is a good insulator.

Solution :

$$\frac{\sigma}{\epsilon \omega} = \frac{58 \times 10^6}{2\pi \times 10^6 \times \frac{1}{36\pi} \times 10^{-9}} = 10^{12} \gg 1$$

Here,

$$\epsilon = \epsilon_0 = \frac{10^{-9}}{36\pi}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

and for Teflon,

$$\begin{aligned}
 \frac{\sigma}{\omega \epsilon} &= \frac{30 \times 10^{-9}}{2\pi \times 10^6 \times \frac{1}{36\pi} \times 10^{-9} \times 2.1} \\
 &= 2.57 \times 10^{-4} \ll 1
 \end{aligned}$$

Example 6 : A material has $\sigma = 10^{-2} \text{ S/m}$ and $\epsilon = 3 \epsilon_0$. At what frequency will conduction current be equal to the displacement current?

Solution : For two currents to be equal,

$$\frac{J_c}{J_d} = 1$$

$$\begin{aligned} \therefore \frac{J_c}{J_d} &= \frac{\sigma}{\omega \epsilon} = 1 \\ \therefore \sigma &= \omega \epsilon = 2\pi f \epsilon \\ f &= \frac{\sigma}{2\pi \epsilon} = \frac{10^{-2}}{2\pi \times 3 \times 10^{-9}} \\ f &= 60 \text{ MHz} \end{aligned}$$

Example 7 : A thin metal sheet of conductivity σ , is fixed in XZ plane, in a time varying magnetic field $\vec{B} = (B_m \cos \beta x \cosh \beta y \cos \omega t) \hat{a}_y$. Determine the induced electric field at a point (x, z) of the sheet.

Solution : Assuming that,

$$\vec{E} = E_z \hat{a}_z$$

From Faraday's law,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & E_z \end{vmatrix}$$

$$\vec{\nabla} \times \vec{E} = \hat{a}_x \frac{\partial}{\partial y} E_z - \hat{a}_y \frac{\partial}{\partial x} E_z + \hat{a}_z (0)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\hat{a}_x \frac{\partial}{\partial y} E_z - \hat{a}_y \frac{\partial}{\partial x} E_z = -\frac{\partial \vec{B}}{\partial t}$$

The x-equation is satisfied by choosing $E_z = E_z(x)$

Then,

$$-\hat{a}_y \frac{d}{dx} E_z = \left(\frac{\partial}{\partial t} B_y \right)_{y=0} \hat{a}_y$$

or

$$\frac{dE_z}{dx} = + \left(\frac{d}{dt} B_y \right)_{y=0}$$

$$\frac{dE_z}{dx} = -\omega B_m \cos \beta x \sin \omega t$$

$$\therefore \int dE_z = -\omega B_m \int \cos \beta x \sin \omega t dx$$

$$\therefore dE_z = -\frac{\omega B_m}{\beta} \sin \omega t \sin \beta x$$

Example 8 : Do the fields

$$\vec{E} = E_m \sin x \sin t \hat{a}_y$$

$$\text{and } \vec{H} = \frac{E_m}{\mu_0} \cos x \cos t \hat{a}_z$$

satisfy Maxwell's equations ?

Solution : For free space, $J = 0$

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\begin{aligned} \vec{\nabla} \times \vec{H} &= \epsilon_0 \frac{\partial \vec{E}}{\partial t} & \vec{D} &= \epsilon_0 \vec{E} \\ &= \epsilon_0 \frac{\partial E_y}{\partial t} \hat{a}_y \end{aligned}$$

$$\vec{\nabla} \times \vec{H} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & H_z \end{vmatrix}$$

$$= -\hat{a}_y \frac{\partial H_z}{\partial x}$$

$$-\frac{\partial}{\partial x} H_z \hat{a}_y = \epsilon_0 \frac{\partial E_y}{\partial t} \hat{a}_y$$

$$\frac{E_m}{\mu_0} \sin x \cos t = \epsilon_0 E_m \sin x \cos t$$

$$\mu_0 \epsilon_0 = 1 \text{ which is false.}$$

So given equation does not satisfy Maxwell's equation.

Example 9 : Given an electric field

$E = E_0 \sin \beta z \cos \omega t \hat{a}_z$ in free space, determine the corresponding charge density.

Solution : From Gauss's law,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{D} = \rho \quad (\because \vec{D} = \epsilon_0 \vec{E})$$

$$\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E}$$

$$= \epsilon_0 \frac{\partial}{\partial z} E_z$$

$$\rho = \epsilon_0 E_0 \beta \cos \beta z \cos \omega t$$

Example 10 : Find the magnetic field associated with $E = E_0 \cos \beta x \cos \omega t \hat{a}_z$.

Solution : From Faraday's law,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & E_z \end{vmatrix}$$

The y-component of $\vec{\nabla} \times \vec{E}$ is

$$-\frac{\partial}{\partial x} E_z \hat{a}_y = -\frac{\partial}{\partial t} B_y \hat{a}_y$$

$$-\frac{\partial}{\partial x} (E_0 \cos \beta x \cos \omega t) = -\frac{\partial}{\partial t} B_y$$

$$E_0 \beta \sin \beta x \cos \omega t = -\frac{\partial}{\partial t} B_y$$

$$dB_y = -E_0 \beta \sin \beta x \cos \omega t dt$$

$$B_y = \mu_0 H_y$$

$$B_y = \mu_0 H_y = -E_0 \beta \sin \beta x \int \cos \omega t dt$$

$$= -\frac{E_0 \beta}{\omega} \sin \beta x \sin \omega t$$

$$H_y = -\frac{\beta}{\mu_0 \omega} E_0 \sin \beta x \sin \omega t$$

Here, $\beta = k = \frac{2\pi}{\lambda}$

$$\frac{\omega}{\beta} = \frac{\omega}{k} = C = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$H_y = -\frac{\sqrt{\mu_0 \epsilon_0}}{\mu_0} E_0 \sin \beta x \sin \omega t$$

$$H_y = -\sqrt{\frac{\epsilon_0}{\mu_0}} E_0 \sin \beta x \sin \omega t$$

Example 11 : Show that the last two of Maxwell's equations are contained in first two equations (that is 4 is implicit in (1) and (3) is implicit in (2)).

Solution : Maxwell's equations are

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Faraday's law ... (1)

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{Ampere's law ... (2)}$$

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \text{Gauss's law for } \vec{D} \text{ ... (3)}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{Gauss's law for } \vec{B} \text{ ... (4)}$$

Taking the divergence of equation in (1) we get

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) = - \vec{\nabla} \cdot \left(\frac{\partial \vec{B}}{\partial t} \right)$$

$$0 = - \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{B}$$

Thus, $\vec{\nabla} \cdot \vec{B} = 0$ within static field.

Similarly, taking divergence of equation (2),

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{D} = 0$$

But $\vec{\nabla} \cdot \vec{J} = - \frac{\partial \rho}{\partial t} \quad \text{(From equation of continuity)}$

$$- \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{D} = 0$$

Hence, $\frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{D} = \frac{\partial \rho}{\partial t}$

or $\vec{\nabla} \cdot \vec{D} = \rho$ which is Maxwell's equation (3)

Example 12 : Show that $\vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$ and $\vec{H} \cdot \frac{\partial \vec{B}}{\partial t}$ respectively denotes the rate of change of energy density stored in electric and magnetic fields.

Solution : The energy densities corresponding to electric field and magnetic field are

$$\omega_e = \frac{\epsilon}{2} \vec{E} \cdot \vec{E} \quad \text{and} \quad \omega_m = \frac{\mu}{2} \vec{H} \cdot \vec{H}$$

Hence,

$$\begin{aligned} \frac{\partial}{\partial t} \omega_e &= \frac{\partial}{\partial t} \left[\frac{\epsilon}{2} \vec{E} \cdot \vec{E} \right] \\ &= \frac{1}{2} \epsilon \left[\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} + \frac{\partial \vec{E}}{\partial t} \cdot \vec{E} \right] \end{aligned}$$

$$= \frac{1}{2} \epsilon \left[2 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right]$$

$$= \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\therefore \vec{D} = \epsilon \vec{E}$$

$$\frac{\partial}{\partial t} \omega_e = \vec{E} \cdot \frac{\partial}{\partial t} (\epsilon \vec{E}) = \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

Similarly,

$$\frac{\partial}{\partial t} \omega_m = \frac{\partial}{\partial t} \left[\frac{\mu}{2} \vec{H} \cdot \vec{H} \right]$$

$$= \mu \vec{H} \cdot \frac{\partial \vec{H}}{\partial t}$$

$$= \vec{H} \cdot \frac{\partial}{\partial t} \mu \vec{H}$$

$$= \vec{H} \cdot \frac{\partial \vec{B}}{\partial t}$$

$$\therefore \vec{B} = \mu \vec{H}$$

Example 13 : The average energy per second reaching the earth from the sun is about 1350 J/sec m^2 . Assume this is a single EM wave and calculate the maximum values of E and B fields for sunlight at the earth.

Solution : The average energy per unit area per second is

$$S_{av} = \frac{1}{2} \epsilon_0 C E_0^2$$

$$E_0 = \sqrt{\frac{2 S_{av}}{\epsilon_0 C}}$$

$$= \sqrt{\frac{2 \times 1350}{8.85 \times 10^{-12} \times 3 \times 10^8}}$$

$$E_0 = 1.01 \times 10^3 \text{ V/m}$$

The maximum value of B-field is

$$B_0 = \frac{E_0}{C}$$

$$= \frac{1.01 \times 10^3}{3 \times 10^8}$$

$$B_0 = 3.4 \times 10^{-6} \text{ weber/m}^2$$

Example 14 : A plane monochromatic plane polarized em-wave is traveling eastward. The wave is polarised with E directed vertically up and down alternately. Calculate E , B

and S , provided that the amplitude of the electric field strength is 0.05 V/m and the frequency is 6 MHz .

Solution : Let the em-wave be propagated along the X-axis with its direction of polarization along Y-axis. Thus, $E_x = E_z = 0$.

The magnetic field vector $\vec{B}$ is perpendicular to the direction of propagation of wave and that of $\vec{E}$. Therefore, $\vec{B}$ will be confined along the Z-direction.

Thus, $B_x = B_y = 0$

The frequency of wave is $f = 6 \times 10^6 \text{ Hz}$

Therefore, we have $\lambda = \frac{c}{f}$

$$= \frac{3 \times 10^8}{6 \times 10^6}$$

$$= 50 \text{ m}$$

Thus, the electric field is given by,

$$\begin{aligned} E_y &= E_0 \sin(\omega t - kx) \\ &= E_0 \sin\left(2\pi ft - \frac{2\pi}{\lambda} x\right) \\ &= 0.05 \sin\left(2\pi \times 6 \times 10^6 t - \frac{2\pi}{50} x\right) \\ E_y &= 0.05 \sin(3.77 \times 10^6 t - 0.126 x) \text{ V/m} \end{aligned}$$

Since $\vec{B}$ is perpendicular to $\vec{E}$ and direction of propagation of wave, therefore,

$$\begin{aligned} B_z &= \frac{E_y}{c} \\ B_z &= 1.67 \times 10^{-10} \sin(3.77 \times 10^6 t - 0.126 x) \end{aligned}$$

or $\vec{B} = 1.67 \times 10^{-10} \sin(3.77 \times 10^6 t - 0.126 x) \hat{k} \text{ wb/m}^2$

The poynting vector $\vec{S}$ is given by,

$$\begin{aligned} \vec{S} &= \frac{\vec{E} \times \vec{B}}{\mu_0} \\ &= \frac{E_y \hat{j} \times B_z \hat{k}}{\mu_0} \\ \therefore \vec{S} &= \frac{E_y B_z}{\mu_0} \hat{i} \end{aligned}$$

$\because \hat{j} \times \hat{k} = \hat{i}$

Thus, $\vec{S}$ will have only x-component. Its magnitude is given by,

$$\begin{aligned} S &= \frac{E_y B_z}{\mu_0} \\ &= \frac{0.05 \times 1.67 \times 10^{-10}}{4\pi \times 10^{-7}} \sin^2(3.77 \times 10^6 t - 0.126 x) \\ &= 6.65 \times 10^{-6} \sin^2(3.77 \times 10^6 t - 0.126 x) \end{aligned}$$

$$S_{av} = \langle S \rangle = \frac{6.65 \times 10^{-6}}{2} \text{ W/m}^2$$

$$\langle S \rangle = 3.325 \times 10^{-6} \text{ W/m}^2$$

Hint : $\sin^2(\omega t - kx) = \frac{1}{2}$

Example 15 : Sunlight strikes earth outside its atmosphere with intensity of 2 cal/cm^2 . Calculate amplitude of electric and magnetic vectors.

Given : $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

$\mu_0 = 4\pi \times 10^{-7} \text{ cal/Am}$

to show that, $\vec{P} = \vec{E} \times \vec{H}$

Solution : $\vec{E}$ and $\vec{H}$ varies sinusoidally. So average value of $\vec{P}$ is given by,

$$\vec{P}_{av} = \vec{E}_{av} \times \vec{H}_{av}$$

$$= \frac{\vec{E}_0}{\sqrt{2}} \times \frac{\vec{H}_0}{\sqrt{2}}$$

$$\vec{P}_{av} = \frac{1}{2} (\vec{E}_0 \times \vec{H}_0) \quad (|\vec{a} \times \vec{b}| = ab \sin \theta)$$

As $\vec{E}$ and $\vec{H}$ are perpendicular to each other, $\theta = 90^\circ$

$$\therefore P_{av} = E_0 H_0 / 2 \quad \dots (5.76)$$

H_0 and E_0 are related by,

$$\sqrt{\mu_0} H_0 = \sqrt{\epsilon_0} E_0 \quad \dots (5.77)$$

$$\therefore P_{av} = \frac{1}{2} E_0 \frac{\sqrt{\epsilon_0}}{\sqrt{\mu_0}} E_0$$

$$= \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_0^2 \quad \dots (5.78)$$

$$\vec{P}_{av} = (\vec{E} \times \vec{H})_{ava.} = \text{average value of poynting vector}$$

$$P_{av} = 2 \text{ cal/cm}^2 - \text{min}$$

$$= \frac{2 \times (4.25)}{(10^{-2})^2 \text{ m}^2 \cdot 60 \text{ sec}} \quad (1 \text{ cal} = 4.2 \text{ J})$$

$$= \frac{2 \times 4.2}{60 \times 10^{-4}} \text{ J/m}^2\text{-sec.}$$

$$= 1.4 \times 10^3 \text{ J/m}^2\text{-sec.}$$

Equation (5.78) gives,

$$E_0^2 = 2 P_{av} \times \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$= 2 \times 1.4 \times 10^3 \sqrt{\frac{4 \times 3.14 \times 10^{-7}}{8.85 \times 10^{-12}}}$$

$$= 2 \times 1.4 \times 10^3 \sqrt{1.42 \times 10^5}$$

$$= 2 \times 1.4 \times 10^5 \sqrt{14.2}$$

$$= 10.55 \times 10^5$$

$$E_0^2 = 1.055 \times 10^6$$

$$E_0 = 1.03 \times 10^3 \text{ V/m}$$

From equation (5.77),

$$\begin{aligned} H_0 &= \frac{\sqrt{\epsilon_0}}{\sqrt{\mu_0}} \cdot E_0 \\ &= \sqrt{\frac{8.85 \times 10^{-12}}{4 \times 3.14 \times 10^{-7}}} \times 1.03 \times 10^3 \\ &= \sqrt{\frac{8.85}{0.4 \times 3.14} \times 10^{-3}} \times 1.03 \times 10^3 \\ &= 2.73 \text{ A/m} \end{aligned}$$

But

$$\begin{aligned} B_0 &= \mu_0 H_0 \\ &= 4 \times 3.14 \times 10^{-7} \times 2.73 \\ &= 3.42 \times 10^{-6} \text{ wb/m}^2 \end{aligned}$$

Example 16 : Using appropriate Maxwell's field equations show that for propagation of EM wave in any arbitrary direction is,

$$\vec{H} = \frac{\vec{k} \times \vec{E}}{\mu\omega} \text{ OR } \vec{E} = -\frac{\vec{k} \times \vec{H}}{\epsilon\omega}$$

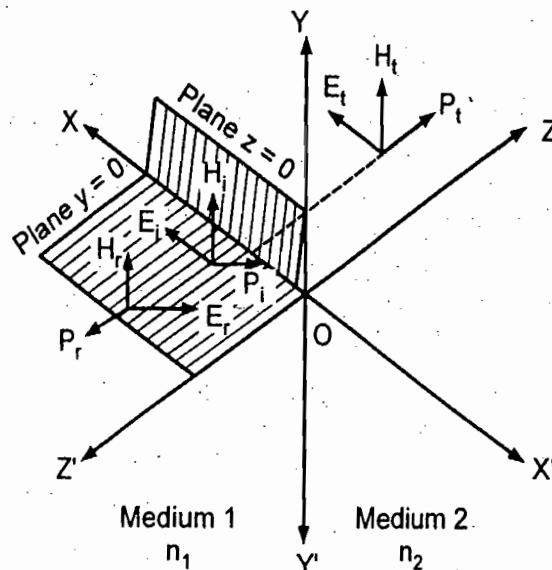


Fig. 5.10

Solution : Consider EM wave in medium 1 and medium 2 as shown in Fig. 5.10.

By applying boundary condition :

$$\left[\begin{array}{l} \text{Tangential component of electric} \\ \text{field in medium - 1} \end{array} \right] = \left[\begin{array}{l} \text{Tangential component of electric} \\ \text{field in medium - 2} \end{array} \right]$$

$$\vec{E}_i - \vec{E}_t = \vec{E}_t$$

... (5.79)

$$\text{and } \vec{H}_i + \vec{H}_r = \vec{H}_t \quad \dots (5.80)$$

At plane $Z = 0$, part of energy is reflected back in medium - 1 along normal, giving reflected wave with $\vec{P}_r$ directed opposite to $\vec{P}_i$ and remaining energy is transmitted as wave along normal in medium - 2.

Thus, $\vec{P}_t$ is in same direction as $\vec{P}_i$.

Let $z = 0$ be the plane separating two semi-infinite perfect dielectrics (1) and (2) in Fig. 5.10.

Condition that incident wave is incident normally on boundary means poynting vector $\vec{P}_i$ perpendicular to plane Z . We are choosing X-axis in direction $\vec{E}_i$. As wave is traced in direction of $\vec{E} \times \vec{H}$, this means that $\vec{H}_i$ must be in direction of Y-axis.

$$\begin{aligned} n &= \sqrt{\frac{\epsilon}{\epsilon_0} \cdot \frac{\mu}{\mu_0}} & \because \epsilon_0 = \mu_0 = 1 \\ n &= \sqrt{k \cdot k_r} \\ \left. \begin{aligned} \vec{E}_i - \vec{E}_r &= \vec{E}_t \\ \vec{H}_i + \vec{H}_r &= \vec{H}_t \\ \vec{E}_i &= n_1 \vec{H}_i \\ \vec{E}_r &= n_1 \vec{H}_r \\ \vec{E}_t &= n_2 \vec{H}_t \end{aligned} \right\} & E = E_0 [\cos x + i \sin x] \end{aligned}$$

Propagation of wave in arbitrary direction :

$$\vec{E}(r, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \dots (5.81)$$

$$\vec{H}(r, t) = \vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \dots (5.82)$$

where,

$$k = \frac{\omega}{v}$$

v = velocity,

$$\vec{H}_0, \vec{E}_0 = \text{vector constants}$$

$\vec{k}$ = propagation vector

But $\vec{E}$ and $\vec{H}$ are real. So we are interested in real part of equation (5.81) and (5.82).

$$\therefore \vec{\nabla} \cdot \vec{D} = 0 \quad \therefore \vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \left[\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right] = 0$$

$$\left[\vec{E}_0 \cdot [\hat{i} k_x + \hat{j} k_y + \hat{k} k_z] \right] e^{i(\vec{k} \cdot \vec{r} - \omega t)} = 0$$

$$\vec{k} \cdot \vec{E} = 0 \quad i \neq 0$$

... (5.83)

$$\vec{\nabla} \cdot \vec{B} = 0 \Rightarrow \vec{\nabla} \cdot \vec{H} = 0$$

$$\therefore \vec{k} \cdot \vec{H} = 0$$

... (5.84)

$\Rightarrow \vec{E}$ and $\vec{H}$ are perpendicular to propagation of vector $\vec{k}$.

Such a wave is called transverse wave.

Now,
$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{aligned} \vec{\nabla} \times \vec{E} &= \vec{\nabla} \times \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} = \vec{\nabla} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \times \vec{E}_0 \\ &= i \vec{k} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \times \vec{E}_0 = i \vec{k} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \times \vec{E}_0 \\ &= i \vec{k} \times \vec{E} = \frac{\partial \vec{B}}{\partial t} \end{aligned}$$

$$\therefore i \vec{k} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} = -\mu (-\omega \vec{H} i)$$

$$\vec{k} \times \vec{E} = \mu \omega \vec{H}$$

$$\therefore \vec{H} = \frac{\vec{k} \times \vec{E}}{\mu \omega}$$

$\Rightarrow \vec{H}$ is perpendicular to $\vec{k}$ and $\vec{E}$. So vector $\vec{E} \times \vec{H}$ points along the direction of propagation.

$\therefore$ Vectors $\vec{E}$, $\vec{H}$, $\vec{k}$ constitute right hand orthogonal set as shown in Fig. 5.11.

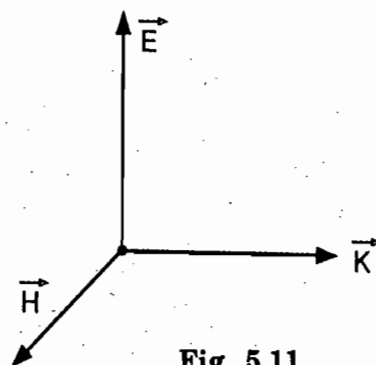


Fig. 5.11

Velocity of EM wave is given by $c_0 = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$

$$\epsilon_0 = 8.85 \times 10^{-12}, \mu_0 = 4\pi \times 10^{-7}$$

$$c_0 = 2.997 \times 10^8 \text{ m/s.}$$

This shows that light is in the form of EM radiation.

QUESTIONS

1. Show that in a charge-free non-conducting medium, Maxwell's equations lead to :

$$\nabla^2 \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

$$\text{and } \nabla^2 \vec{B} - \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

2. Explain integral and differential forms of Faraday's law of induction.
3. What is Poynting vector ? Show that the surface integral of the poynting vector measures the rate of flow of electromagnetic energy.
4. Starting from Faraday's law of electromagnetic induction, obtain the corresponding Maxwell's equations of electromagnetic field. Explain the significance of the equations.

5. State Faraday's law of electromagnetic induction and prove that $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$.

6. State Faraday's law of electromagnetic induction and prove that,

$$\oint_C \vec{E} \cdot d\vec{l} = \oint_C \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

7. Show that power transferred into electromagnetic field through the volume V is given by,

$$- \int (\vec{E} \cdot \vec{J}) dV = \frac{d}{dt} \int \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H}) dV + \int (\vec{E} \cdot \vec{H}) d\vec{S}$$

8. Using appropriate Maxwell's field equations, show that for propagation of EM wave in any arbitrary direction

$$\vec{H} = \frac{\vec{k} \times \vec{E}}{\mu\omega}$$

$$\text{or } \vec{E} = -\frac{\vec{k} \times \vec{H}}{\epsilon\omega}$$

9. Show the x-component of $\vec{E}$ at any point decreases exponentially with time.
10. Explain in detail :
- (a) Plane polarized wave.
- (b) Circularly polarized wave.

PROBLEMS

1. The average light intensity on the earth's surface is $1.3 \times 10^6 \text{ erg/cm}^2/\text{s}$. What are the peak values of E and B fields assuming that light is monochromatic.
2. Show that for a plane wave in vacuum

$$\sqrt{\frac{E}{H}} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 337 \Omega$$

3. A train is travelling along a rail, whose gauge is 144 cm, at 96 km/hour. The rails are insulated and connected by a millivoltmeter. Find the reading on the voltmeter assuming earth's magnetic field to be $4.5 \times 10^{-5} \text{ wb/m}^2$.

(Ans. 1.73 mV)

4. A rectangular loop of wire, with dimensions a, b and lying in XY plane is moving with uniform velocity v in the direction of positive Y-axis in a static magnetic vector field pointing in Z-direction and varying sinusoidally in the Y-direction as $B_z = B_0 \sin ky$. Calculate induced e.m.f. in moving loop.

$$\left(\text{Ans. } -2 B_0 a v \sin \frac{kb}{2} \cos kv t \right)$$

5. A laser beam has a diameter of 10^{-3} m . Calculate the amplitude of magnetic field in the beam in the free space if power of laser is 1 watt. (Ans. 10^{-4})
6. In a material for which $\sigma = 5.0 \text{ S/m}$ and $\epsilon_r = 1$, the electric intensity is $E = 250 \sin 10 t$. Find conduction and displacement current densities and the frequency at which they have equal magnitude.

$$[\text{Ans. } J_c = \sigma E = 1250 \times 10^{10} t$$

$$J_D = \frac{\partial D}{\partial t} = 22.1 \cos 10^{10} t$$

$$J_D = J_c \text{ at } f = 8.99 \times 10^{10} \text{ Hz}].$$

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